



FORMAN
CHRISTIAN
COLLEGE
(A CHARTERED UNIVERSITY)

BACHELOR'S

DEGREE PROGRAM CATALOG

2026-2027

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Message from the
RECTOR

Message from the Rector

Welcome to Forman Christian College (A Chartered University)!

Choosing a university is an important decision, ranking among the most important you will ever make. Partly, it involves choices about what you choose to study. More importantly, it also involves choices about what kind of person you want to become. At Forman, we believe a university education should prepare students not simply for their first job but for a lifetime of learning, leadership, and service.

For more than 160 years, Forman has offered an education grounded in academic excellence, intellectual curiosity, character development, and a commitment to service. Our four-year Bachelor's Degree Program reflects this longstanding tradition while preparing students to engage thoughtfully and confidently with a rapidly changing world.

At FCCU, we combine the depth of specialized study with the breadth of a liberal arts education, exposing students to a range of disciplines and perspectives. As part of this commitment, our curriculum encourages students to ask meaningful questions and explore different fields of knowledge while also thinking critically, creatively, and independently about the major issues of our time. Regardless of whether you choose to study the sciences, social sciences, humanities, management, or another field, you will be encouraged to connect knowledge across disciplines and apply what you learn to real-world challenges.

Your education at Forman will also extend well beyond the classroom. Our faculty members are committed to teaching and mentoring while academic advisors and student support services are available to help you navigate your university journey. Through student societies, sports, cultural activities, leadership opportunities, research, internships, and community engagement, you will have many opportunities to discover your interests, develop your strengths, and learn from experiences in ways that go far beyond your formal coursework.

Our campus provides an attractive environment in which learning and community join together in meaningful and sometimes unexpected ways. Our buildings range from the historic to the modern and our beautiful 122-acre garden campus includes sports fields, green spaces, and a botanical garden, combining to provide a distinctive setting in which students can study, collaborate, create, and form lasting friendships. Our community also brings students, faculty and staff from diverse backgrounds together, providing opportunities to listen, learn, understand different perspectives, and grow together.

We also recognize that the world our students will enter following their graduation is changing rapidly. Advances in technology including artificial intelligence; evolving workplaces; global challenges; new forms of communication -- collectively, all of these together are transforming the ways in which we live and work. In this changing environment, knowledge alone is not enough. Rather, our graduates must learn to communicate effectively, understand others, adapt to change, make thoughtful and ethical decisions, and use their knowledge responsibly.

These qualities are central to the Forman experience. In particular, we seek to develop graduates who are confident in their abilities, thoughtful in their decisions, respectful in their conduct, and conscious of their responsibilities. Looking ahead, we hope that your years at Forman will challenge you intellectually, broaden your horizons, strengthen your character, and inspire you to make a meaningful difference in your families, communities, and country.

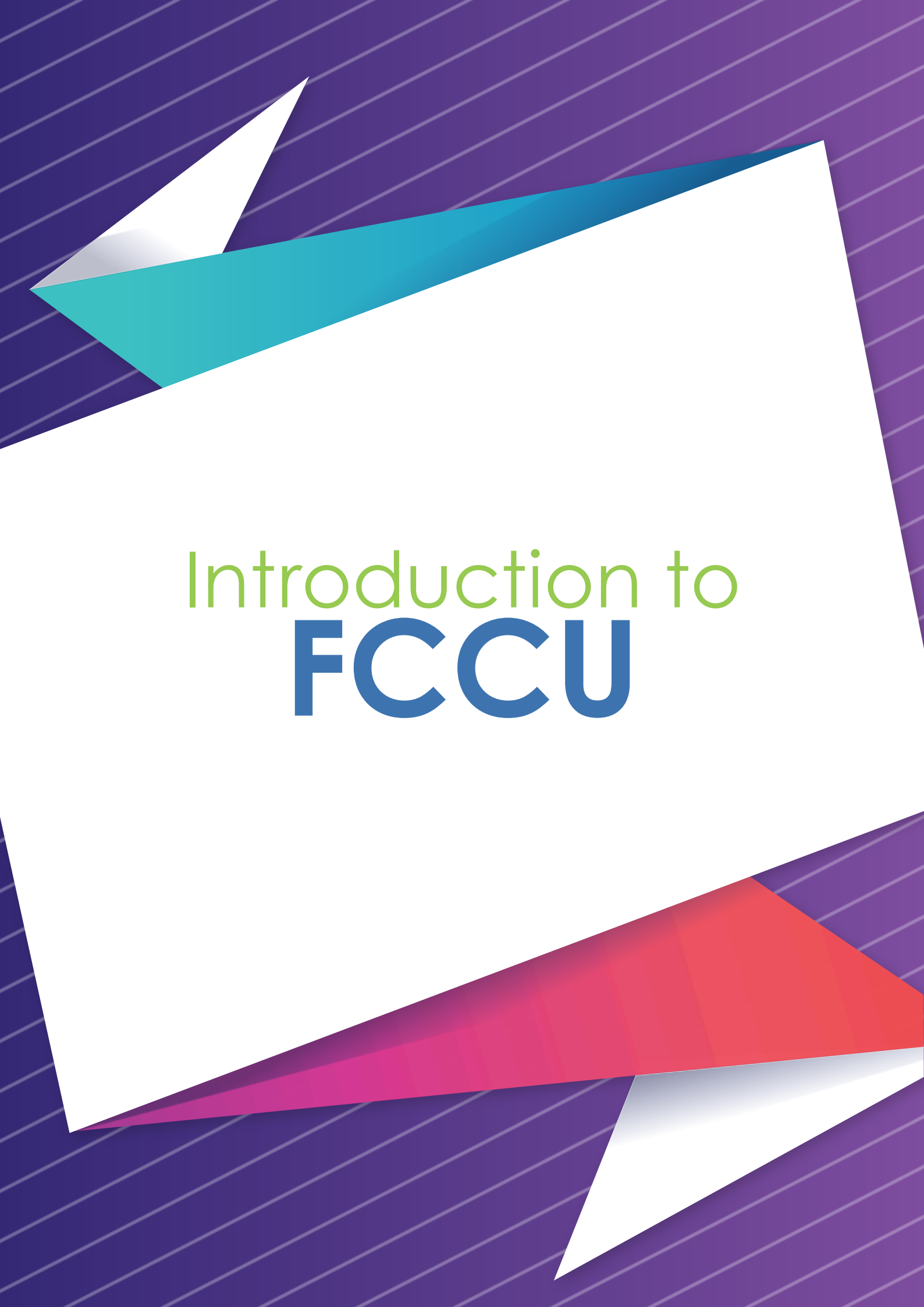
As you explore this catalog and consider the opportunities described within it, I encourage you to look beyond the courses and programs listed on these pages, even as you consider the questions you want to explore, the skills you want to develop, the communities you want to join, and the person you want to become.

For our part, we welcome you to the Forman community, even as we look forward to joining with you on this important journey of growth, learning, discovery, friendship, and purpose!

Dr. Jonathan S. Addleton

Rector

Forman Christian College (A Chartered University)



Introduction to **FCCU**

Introduction to FCCU

Forman Christian College (A Chartered University) has a rich history dating back to its founding in 1864 by Dr. Charles W. Forman, a Presbyterian missionary from the USA. Originally named Lahore Mission College, the institution was officially renamed in 1894 to Forman Christian College in honor of the founder. It started by awarding degrees through Calcutta University but later transitioned to degrees from the University of the Punjab. In 2004, FCCU was chartered as a university and began awarding its own degrees in 2009.

Throughout its early years, the college experienced remarkable growth in enrollment. From a modest start with 18 students in 1886, the college's student body expanded rapidly, reaching 1500 students by the time it was nationalized in 1972. Today, the university section alone enrolls more than 5,000 students.

The college's campus has seen significant development and relocation. The original location was Rang Mahal Mission High School inside the walled city. It moved to its present 108-acre campus along the scenic banks of Lahore Canal in 1940. Several major buildings including Ewing Hall (constructed in 1916) continue to serve the institution's educational mission.

Throughout its history, FCCU has been led by a distinguished lineup of educational leaders and teachers such as Dr. C. W. Forman, Nobel Laureate Dr. Arthur Compton, and many others. This collective leadership has contributed to the college's recognition as one of the finest educational institutions in South Asia. FCCU was the first college in the area in whose laboratories research work of Nobel Prize caliber took place. Dr. Arthur Compton received the Nobel Prize in 1932 for research conducted, in large part, at FCCU. In 1902, the college was the first in Punjab to admit women. Notable alumni of FCCU include two presidents and two prime ministers of Pakistan, as well as leaders in politics, law, and various other fields. The college's contributions extend to international diplomacy, security, and a wide range of other domains.

Service to the nation is a core part of FCCU's ethos. The college played a pivotal role during significant events, including the conversion of two hostels into United Christian Hospital for refugees during independence, and active involvement in relief efforts during natural disasters like the Kangra Valley earthquake (1905) and the Quetta earthquake (1935). Social service by students was popularized by faculty members like Prof. D. J. Fleming.

In 1972, the college was nationalized by the government but was later returned to private ownership on March 19, 2003. In March 2004, the government granted university status to FCCU. The university entered an exciting new phase in its history in September 2005, launching a four-year Bachelor's Degree Program aligned with world-class standards for accreditation. Since then, FCCU has continued to expand its academic offerings, now featuring over 20 postgraduate programs.

Forman Christian College (A Chartered University) embodies a legacy of excellence, service, and continuous growth, fostering a community of scholars, leaders, and changemakers.

Mission

The mission of Forman Christian College (A Chartered University) is to impart, create, and disseminate knowledge and to develop informed, ethical, and responsible citizens who are prepared and committed to learn, lead, and serve; persons who exemplify the FCCU motto, "By love serve one another".

Vision

Forman's vision is to be recognized as one of the best liberal arts institutions of higher learning in Pakistan and across South Asia.

Goals

Our institutional goals are as follows:

- **Empowered learners** with strong written, oral and quantitative skills that they can use to evaluate a constant flood of information. The idea is to create in them the ability to think independently and critically, solve problems and continue a lifetime of self-directed learning.
- **Informed learners** who understand global and cross-cultural relationships, value the philosophy and history underlying the nation of Pakistan and are fluent in both their native language and English.
- **Responsible learners** who understand the ethical consequences of actions and are well- groomed to be active citizens who accept their public duty and participate in the decision- making process of a democracy.

Our Commitments

Commitment to Excellence

Forman Christian College (A Chartered University) operates all of its programs in accordance with the highest standards of excellence in education. The educational programs are designed and implemented in accordance with world-class standards of accreditation. The University has begun the process of seeking accreditation with one of the six regional accrediting associations in the USA.

Commitment to Individual Development

FCCU is concerned with the development of the whole person, and therefore encourages the intellectual, spiritual, cultural, social, emotional and physical growth of each student. We seek to prepare students for the basic responsibilities of life, and especially for competent and humane leadership and service. The FCCU experience is designed to help students go beyond the limitations caused by ignorance, narrowness, conformity, self-centeredness and irresponsibility. Our goal is to help individuals achieve excellence in thought and conduct.

Commitment to Core Values

The faculty and staff of FCCU seek to live by, and to teach students, its core values. In a variety of different settings, students are asked to learn and live by the following values beginning with signing a 'Shared Commitment' document that highlights the practice of the core values on a regular basis.

1. **Integrity:** I will speak the truth and keep my commitments. I will take my responsibilities seriously and fulfill them to the best of my ability.
2. **Excellence:** I will be steadfast in my pursuit of excellence. I will set high standards in my intellectual life, personal behavior and interpersonal relationships. I will honor the traditions of the University and preserve the beauty of the campus.

3. **Respect for the Dignity of Each Human Being:** I will treat others with respect, kindness, generosity of heart and compassion. I will accept and tolerate differences. I will handle disagreements with candor and civility.
4. **Discipline and Accountability for my Actions:** I will uphold the policies of the University and follow the rules and regulations. I understand that behavior has consequences. This understanding is an essential component in the development of my self-discipline.
5. **Fairness and Justice:** I will be fair in all of my decisions and work towards justice for others.
6. **Service:** I will live by the motto “By love serve one another”, knowing that serving others is a way of life that will enrich the community and the nation in which I live.
7. **Community:** I will take the concerns of others in the University community to heart. Because we are bound together by common purpose, objectives and values, the welfare of all will be my concern.

Commitment of Faculty to Students

The faculty of FCCU is committed to student learning and to helping students succeed in their studies and be well prepared for a meaningful and productive life after University. Students will form a close personal relationship with one or more members of the faculty, and this close student-faculty contact has been one of the strengths of FCCU throughout its history. Faculty members provide assistance to students, as needed, outside of the classroom, and they do not charge tuition for this help. Indeed, their contract with the University prohibits faculty members from charging tuition for extra assistance.

Commitment to General Education

While FCCU is committed to helping students develop competence in a specific field, it is equally committed to general education. The general education program is designed to provide a foundation for lifelong learning by helping students to develop a love of learning. It prepares students for responsible citizenship by teaching them the lessons of history, by creating awareness of their cultural heritage, by helping them understand the causes of social and political unrest, and the conditions for stable governance and sustainable economic development. Through studies in the humanities, the general education program seeks to help students explore various perspectives on the central concerns of human existence. The general education program is designed to help students assume increased responsibility for their own growth, to master the skills that are necessary to understand and deal with a rapidly changing and increasingly complex world. The program requires students to take courses in each of the following areas of human knowledge: Humanities, Natural Sciences, Social and Behavioral Sciences, Mathematics and Computer Sciences.

Commitment to Career Preparation

Enriched with the enduring qualities of a liberal arts education, FCCU seeks to graduate students who are well-prepared for success in their careers. Through the major field of study selected by the student, he or she will receive a basic knowledge of a particular field in enough depth to be successful in entry level positions in a career and to advance successfully to increased levels of responsibility on the job.

However, it is impossible to predict what a person will need to know for success in the job twenty years from now, but we do know that in most jobs new knowledge will have to be mastered that does not even exist today. Therefore, it is more important to learn how to learn, how to think, how to solve problems, and how to communicate effectively rather than just focusing narrowly on the content of an academic discipline. The educational program is designed to help students develop these skills.

Commitment to Co-education

The Bachelor's Degree Programs of FCCU is co-educational. FCCU first admitted women in 1902, and it seeks to provide a learning environment in which both men and women can learn effectively and develop the character traits and personality that will enable them to succeed in later life. The core value of respect for the dignity of each human being is also an important consideration for creating a wholesome and positive atmosphere for learning for both men and women.

Commitment to Lifelong Learning

FCCU seeks to prepare students for a lifetime of self-directed learning. This will be essential for success in a rapidly changing and increasingly complex world. The faculty models this commitment by constantly learning about new knowledge in their academic discipline, and by participating in a variety of professional development programs presented to them by the University management to help them learn new approaches to teaching and learning.

Commitment to Providing an American-style Education

The proceeding commitments reflect the commitment to provide an American-style education. The American system of higher education is widely recognized as the very best in the world and we seek no less than the best for Pakistan. This commitment is an approach to education rather than a statement about the specific content of the curriculum. At FCCU, the role of the faculty and the students, the balance between the breadth and the depth of learning (General Education and a major field of study) and the structure of the program reflect best practices in American higher education.

Commitment to Equality of Opportunity

At FCCU, students, faculty and staff are free within the University from all forms of discrimination based upon gender, race, age, ethnicity, nationality, religion or physical disability. Decisions regarding employment and admission to the University are based upon merit. Grades in courses and graduation from the University are based upon the performance of the student in meeting courses and graduation requirements.

Financial Integrity

FCCU is a private, not-for-profit educational institution. All tuition and other fee income goes directly to the support of the educational program. Tuition and fees pay only a portion of the educational costs per student. Thanks to the support of donors, the balance of costs is paid from endowment and gift income from individuals, churches, corporations and foundations.



Our CAMPUS

Our Campus

Forman Christian College (A Chartered University) has an impressive and well-maintained campus with all the facilities needed to create an environment that is truly academic and conducive to purposeful learning. Centrally located in a beautiful residential area of Lahore, the campus sprawls over 108 acres along the east bank of the canal.

There are two purpose-built buildings for University students. Inaugurated in 2007, the Business and Social Sciences Building houses the Social Science Education and Humanities disciplines including the School of Management. The Armacost Science Building is a modern state-of-the-art science building for the Departments of Chemistry, Pharmacy, Environmental Sciences, Physics, Computer Science, Mathematics, Statistics, and the Kauser Abdulla Malik School of Life Sciences. It was inaugurated in February 2010.

The Ewing Memorial Library has been functioning since 1943 and now contains over 100,000 volumes. It has an automated Library Management System and online and electronic reference services. Students can use the web-based Information Portal to search the library catalog and log in to their accounts to view their activity information. The Digital Library (associated with the Ewing Memorial Library) in the Campus Center has computer and multimedia workstations, printers, study rooms, an information literacy classroom, and a research help desk.

The Jim Tebbe Campus Center is the first of its kind in Pakistan and adds an important new dimension to the university. Measuring more than 180,000 square feet of covered space, this modern state-of-the-art facility is air-conditioned and houses a digital library (known as Friends Library) with more than 400 study carrels, a business incubation center, art gallery, dining room, coffee shop, offices, multiple meeting rooms, large and small auditoriums, several lounges including student lounges, a female lounge, and a faculty lounge, smart bank, campus store and other facilities.

Sinclair Hall houses the largest auditorium of the University (seating 740 people). This is where major events including the annual play and Christmas pageant are held. The Ahmad Saeed Administration Building (N Block) houses administrative offices including the Rector, Vice Rector, and Registrar offices.

The main cafeteria offers a place for students to relax between classes. The Lucas Center houses a basketball court, a gymnasium, and table tennis facilities, along with offices of the Health and Physical Education Department. FCCU has a vast sports ground in the middle of the campus and a running track that goes around it. There is also a 25-meter swimming pool, a baseball field, a gym, a badminton court, four tennis courts and a central cricket pitch on campus.

Out of seven hostels on the campus, three cater exclusively to university students: Shirazi and West Hall for men and the Cheryl Burke Hope Tower for Women. Shirazi and West Halls offer lodging for around 140 men each, while Hope Tower, along with a separate hostel for female postgraduate students, accommodates approximately 400 women. Also, as a result of the FCCU/TSA partnership, accommodation for nearly 100 female undergraduate and graduate students is available at a newly renovated hostel not far from the Forman Campus. All these hostels feature common rooms, mess halls, and spacious outdoor recreational areas.



Student SUCCESS

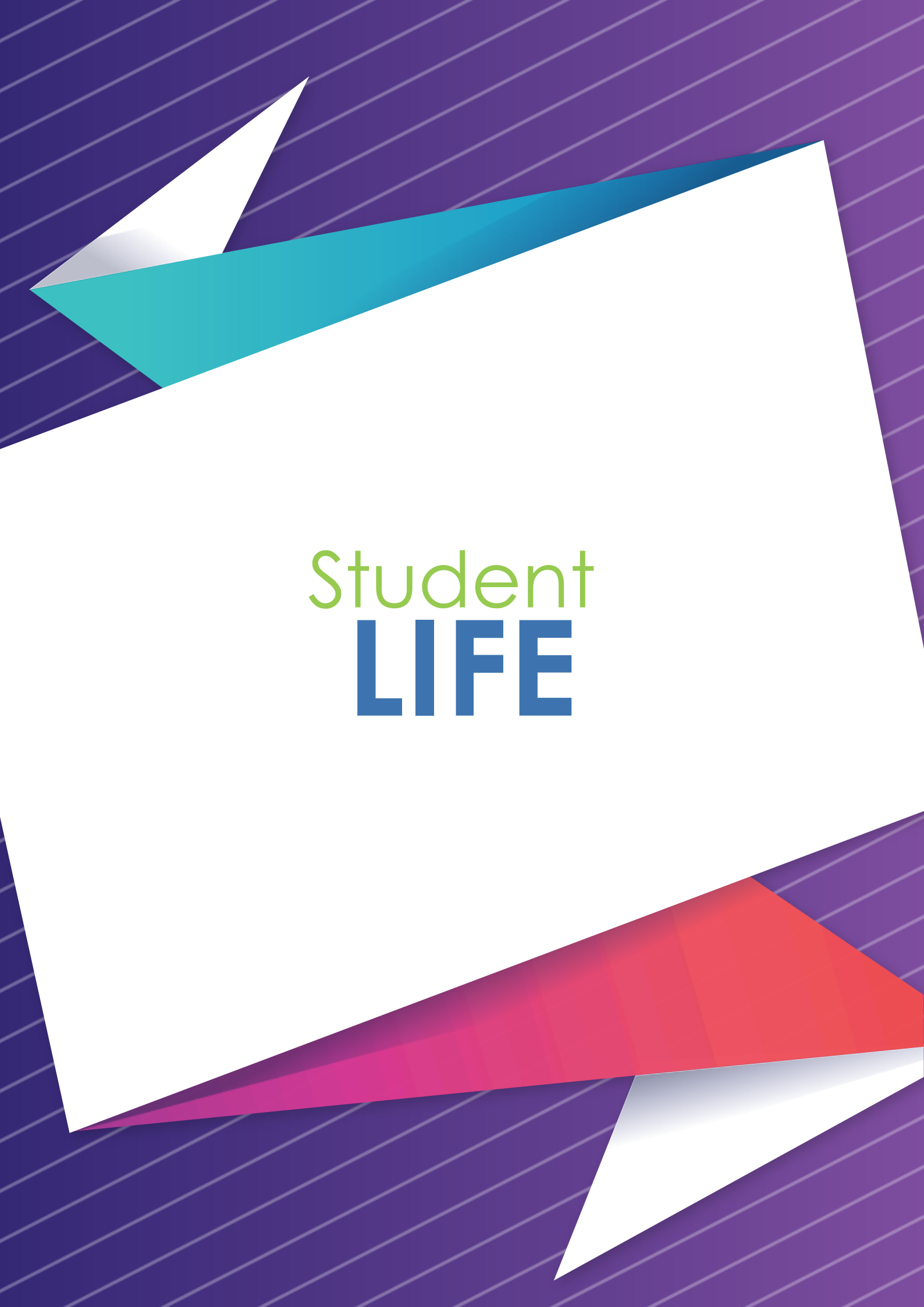
Student Success

Our Vision for Student Success:

At FCCU, we wholeheartedly believe that the success of our students is at the core of our mission. Our dedicated support system is designed to empower students in their pursuit of self-directed and lifelong learning, ensuring they not only achieve their academic and professional goals but also contribute meaningfully to their family, country, and the global community. We are devoted to fostering an environment where students lead with purpose, embracing responsibility for a sustainable and impactful future. Your journey of success is our shared mission, and we are here to support you every step of the way.

Student Success Definition:

At FCCU, we see student success as a journey towards holistic development – one that integrates academic excellence, personal growth, interpersonal relationships, and social responsibility. Our students embark on a transformative journey of self-directed and lifelong learning, empowering them to achieve their academic, personal, and professional goals. Our aim is to prepare discerning individuals who lead with purpose and responsibly serve their family, country, and global community for a sustainable future.



Student **LIFE**

Student Life

Learning is not confined to the classrooms and many of the most important lessons learned during the University years are learned through participation in co-curricular and sports programs. Forman Christian College (A Chartered University) Lahore offers a great variety of programs that provide opportunities for students to participate in activities that contribute to their learning and development. Participating in co-curricular activities helps students to understand the importance of critical thinking skills, time management, and academic and intellectual competence.

Residential Life

Residential life strives to provide a safe and supportive residential environment that complements and extends the educational experience of students. Residential facilities are available to students who come from cities outside Lahore. The Residential Life Office allots rooms to freshmen male students for two years (four semesters) and female students for four years (eight semesters) on a first-come, first-serve basis. The hostels are supervised by the Guardians and Assistant Guardians with the assistance of senior students, who serve as Residential Assistants to supervise activities and engage students in cocurricular activities.

Co-Curricular

FCCU is committed to providing holistic education. Classroom learning is supplemented by opportunities for students' intellectual and moral growth through carefully planned literary, academic, cultural, and recreational activities and programs. The Student Activities Office coordinates and promotes activities through student societies. Besides campus-wide societies, all academic departments also have their own student societies. Each society plans and conducts programs throughout the year that enrich the learning experiences of students and provide them with leadership opportunities.

The following societies are currently functioning:

- Armacost Psychological Society
- Benade Physics Society
- Dean Geographical Society
- Ewing English Society
- Forman Art Junction Society
- Forman Biology Society
- Forman Character Building Society
- Forman Computer Science Club
- Forman Debating Club: Forman Debating Wing
- Forman Debating Club: FORMUN
- Forman Dramatics Club
- Forman Earth Watch Society
- Forman Education Society
- Forman Entrepreneurial & Leadership Club
- Forman Islamic Society
- Forman Journalism Society
- Forman Mathematics Society

- Forman Music & Dance Society
- Forman Pharmaceutical Society
- Forman Philosophy Society
- Forman Photography Society
- Forman Political Science Society
- Forman Rotaract Club
- Forman Sociology Society
- Forman Sports Society
- Forman Statistics Society
- Forman Urdu Society
- Forman Women Empowerment & Leadership Club
- Forman Youth & Peace Development Corps
- Griswold History Society
- Lucas Economics Society: Forman Economics Wing
- Lucas Economics Society: Forman Finance Wing
- Speers Chemical Society
- TEDx FCCU

Religious Life

FCCU is committed to cultivating solid values and strengthening positive character and discipline in its students. For Muslim students, there are two mosques on campus, with Juma prayers offered at the main mosque. The university also convenes Dars-e-Quran classes along with symposiums and discussions, inviting eminent Muslim scholars to deliver talks and engage students in discussions on important religious, social and moral issues. For Christian students a weekly chapel service is offered on Friday. In addition to the weekly chapel program, the university provides regular Bible study groups and opportunities for volunteer service. No classes are scheduled on Friday during Juma or Chapel time.

Sports

The University Sports Office along with the Sports Society organizes, promotes, and conducts various games. The Sports Society oversees a very active intramural sports program with competition in athletics, basketball, cricket, football, baseball, hockey, table tennis, wrestling, lawn tennis and swimming. Players at all skill levels are welcome to participate, and often these competitions and matches are quite competitive. Selection for the University team and participating in inter-varsity competitions are a source of pride for any player.

Sports Facilities

FCCU's sports program is very comprehensive, encompassing both Sports facilities and infrastructure on the campus. Our facilities include eight lanes standard 400m grassy track, a short course Swimming pool (25m), Cricket, Football, and Field Hockey fields. It also has a well-equipped gymnasium, Table Tennis Hall, Badminton, Handball, Basketball, and Pickleball Courts. There are four grass tennis courts as well as one hard court. Students must use only the area marked for the sport they want to participate in (see the map at the end of the handbook). Students found using the wrong areas will be asked to leave.

Cricket Ground is a multiple use ground being utilized for Rugby and various Track and Field Events as well as for a variety of cricket matches which are being held regularly in FCCU. Cricket must be played with a hardball. Other type of balls are not allowed.

Cafeterias

The cafeteria provides a relaxed setting for informal interaction among students, faculty, and staff. There are several food options at the cafeteria.

On-Campus Health Services

Mercy Health Center is an on-campus facility focused on promoting wellness and providing medical care for our campus community. The medical team includes a full-time doctor as well as several fully-certified nurses. The Health Center has on-going relationships with several nearby hospitals and medical laboratories for cases that require specialized attention. Mercy Health Center provides the following services:

- Treatment of minor injuries and illnesses
- Confidential medical guidance and support
- Health guidance for students living with chronic illness/disability
- Referrals to outside medical care and community resources
- Seminars and small group classes on various healthy living topics

Campus Counseling Center

The Campus Counseling Center, located on the first floor of Mercy Health Center, is a facility to help students deal with concerns and difficulties that they may not want to discuss with family, friends or their teachers. It offers an emotional and physical space for the Forman community to share openly and honestly without the fear of judgment. Individual Counseling plans are devised to suit the unique needs of each client. The Campus Counseling Center provides the following services:

- Individual and confidential counseling
- Referrals to other professionals when needed
- Crisis intervention
- Seminars and workshops
- Access to books and articles that deal with mental health and emotional wellbeing
- Groups focusing on specific topics such as anxiety and stress management

Discipline

All students are expected to uphold standards of dignity, self-respect, honesty, and courtesy in their conduct. Strict adherence to disciplinary standards is mandatory. The Rector's decisions regarding disciplinary matters are final and legally binding on all students. Proctors are responsible for maintaining discipline, enforcing rules of good conduct, and taking necessary disciplinary actions when required.

Expectations of Conduct

1. **Rules and Regulations:** Students must observe all rules and regulations governing their studies, including both theoretical and practical components, as established from time to time.
2. **Attendance:** Students are required to attend all lectures, laboratory sessions, and academic activities for the courses in which they are enrolled.
3. **Academic Integrity:** Acts of dishonesty and cheating, particularly during examinations, are strictly prohibited and will result in punitive action if proven.
4. **Behavioral Standards:** Students must refrain from any behavior that poses a threat to fellow students, faculty, staff, or any other individuals associated with the University.

Disciplinary Consequences

Students involved in any of the following activities will face serious disciplinary consequences, which may include fines, parental undertakings, suspension, or expulsion from the University for a period of one semester to one year, depending on the severity of the offense:

- Causing, attempting to cause, or threatening physical injury to another person (e.g., fighting).
- Possession of dangerous objects that may cause harm or injury (e.g., metal rods, knives, firearms).
- Selling, possessing, or using alcohol, drugs, intoxicants, or controlled substances.
- Possessing or smoking tobacco or using any tobacco products.
- Possession or use of vapes.
- Engaging in hate violence or degrading/inflammatory behavior towards others, including hazing.
- Offering to sell or delivering a controlled substance or a substitute.
- Continued disruption of class or university activities, or repeated defiance of university authorities.
- Intimidation, bullying, or making threats towards university staff or students.

By adhering to these guidelines, students contribute to a safe and respectful learning environment for everyone at the University.

Dress Code

The purpose of the FCCU dress code is to ensure that our students are dressed in a dignified manner. This means that the clothing worn should be clean, neat, modest and reflective of the culture in which we are operating. The FCCU ID card must be visibly displayed at all times on campus.

Sexual Harassment Policy

Sexual harassment is unacceptable behavior at Forman Christian College (A Chartered University) and such behavior will be subject to disciplinary action. Harassment refers to behaviors that are intended to be offensive, threatening or disturbing to the recipient. To harass is to persistently annoy, attack, or bother someone.

Sexual harassment is defined as any unwelcome sexual advance, request for sexual favors, or other verbal or physical conduct of a sexual nature that is offensive, embarrassing, intimidating or humiliating. This includes:

- Instances when harassment has the purpose or effect of unreasonably interfering with an individual's work performance or creating an intimidating, hostile or offensive environment.
- Instances when submission to the harassment is made either explicitly or implicitly a term or condition of fair treatment.

Specific examples include, but are not limited to:

- Touching in an inappropriate way
- Staring or leering
- Requests for sex
- Subtle pressure for sexual activity or sexual innuendos
- Display of sexually explicit picture
- Repeated references to various parts of the body at inappropriate times
- Requests for dates when the other person has made it clear that she or he is not interested
- Hooting, whistling or other suggestive noises or gestures
- Suggestive comments or jokes
- Insults, name-calling or taunts based on a person's gender
- Derogatory graffiti referring to a person's character or making sexual implications • Sexually explicit emails, text messages, social media communication, etc.
- Spreading rumors about another person's sexual behavior
- Intrusive questions about a person's private life or body
- Any romantic or sexual behavior that you would consider to be inappropriate if directed at a member of your family

Sexual harassment does not refer to compliments or other behaviors that are considered to be socially appropriate.

There should be no relationships of a romantic or sexual nature between any faculty or staff member and a student. There is no exception to this. A student should not attempt to initiate such a relationship for any reason. There should be no attempt by a student to gain better grades or access to exams or assignments by encouraging or offering such relationships. Any pursuit of such relationships by a faculty or staff member should be immediately reported to the Student Discipline and Support Office or the Dean of Students.

All faculty and staff members are required to report instances of harassment if they are aware of any. Any faculty or staff member encouraging a student not to report such instances will be subject to disciplinary action.

FCCU has adopted the Code of Conduct from the Protection against the Harassment of Women Act of 2010.

The link to that full document can be found at www.aasha.org.pk.



Career Services and **INTERNSHIPS**

Career Services and Internships

The Career Services & Internships Office (CSIO) at FCCU supports students in building meaningful professional futures. We connect academic learning to real-world opportunity — equipping students with the skills and experience they need through internships, employer partnerships, and hands-on career preparation. CSIO is open to all enrolled students from their first year onwards and maintains ties with FCCU alumni as they grow in their careers.

Our Services

- Career Counselling and Guidance
- Career Readiness Series — resume writing, interview preparation, professional etiquette, and more
- Job Fairs and Recruitment Drives
- Mock Interviews
- Industrial Liaison & Visits
- Guest Speaker Sessions — bridging students with industry leaders and professionals
- Job Vacancy Announcements
- Graduates Booklet
- Graduate Employment & Employer Feedback Survey

Internships

CSIO coordinates internship placements across a wide range of industries, including credit-bearing internships that count toward degree requirements. In line with the **Higher Education Commission (HEC) policy**, internships are a mandatory component of undergraduate education, ensuring students gain practical, real-world experience before graduation. These are typically completed during the summer semester under the supervision of a faculty member.

Students receive support at every stage — from resume preparation and employer outreach to orientation sessions covering workplace expectations and professional conduct. CSIO stays in active contact with employers throughout the internship to ensure productive experience on both sides.

Credit Internship

A formal, faculty-supervised internship, supported by our industry partners, that contributes to a student's degree requirements. Students are required to complete the formal registration process through the university's student management system, with specific requirements outlined by their respective academic departments.

Non-Credit Internship

An optional internship that students may pursue independently for professional development. CSIO provides guidance and employer connections to support these placements. Please note that during the summer semester, priority is given to credit-bearing internship students.

The International
Education
OFFICE

International Education and Global Engagement Office

The International Education and Global Engagement Office (IEO) at Forman Christian College (A Chartered University) is committed to developing students' personal, academic and professional skills and enhancing learning experiences through international exposure and cross-cultural engagement. The office collaborates with partner institutes from different countries to facilitate admission opportunities, scholarships and semester exchange programs for FCCU students.

To enrich students' learning exposure globally, IEO actively organizes a diverse range of activities throughout the year.

Key initiatives include;

- Semester Exchange Program Sessions
- International Education Advising Fairs/ Expos
- Awareness Sessions on Admission and Scholarship Opportunities Abroad
- Leadership Development Programs and Joint Research Initiatives
- Seminars, Webinars and other Interactive Study-Abroad Preparedness Session

Writing and
Communication
CENTRE

Writing and Communication Center

The Writing and Communication Center (WCC) at FCCU supports undergraduate students in developing the communication skills necessary for academic success and lifelong learning. Recognizing that university-level communication places new and often demanding expectations on students, the Center provides a supportive environment where learners can strengthen their writing, speaking, reading, and critical thinking skills.

The WCC offers individualized, one-on-one consultations with trained tutors who assist students with academic assignments, essays, reports, presentations, personal statements, and other communication tasks. Students may schedule appointments or visit the Center during designated hours to receive guidance tailored to their specific needs.

In addition to individual consultations, the Center organizes workshops and seminars on academic writing, presentation skills, critical reading, research fundamentals, and professional communication. Through these services, the WCC seeks to empower students to become confident communicators, effective learners, and engaged members of the university community.



Fee STRUCTURE

Fee Structure

Fees are expected to increase each year depending on inflation and other costs:

- **Billing Cycle:** Billing occurs on a semester basis, with full-time and part-time status impacting the fee structure.
- **Full-Time Students:** Students taking between 12 to 18 credit hours are considered full-time and are charged the full semester fees. Pharmacy students will be charged on flat rate basis according to their roadmap.
- **Part-Time and Overload Charges:** Students taking 11 credits or less are considered part-time and are charged on a per-credit-hour basis. Overloads above 18 credits are charged on a per-credit-hour basis as well.
- **Payment Deadlines:** All tuition, hostel and other fees must be paid by the date specified by the University which are also published on the university website. Any outstanding balance will result in a business hold on the student's empower account, until dues are cleared. Late payments will result in late fee fines.
- **Hold Clearance and Add/Drop Registration:** Students can get their holds removed and re-register during add/drop. All courses on a student's schedule at the end of add/drop will be charged.
- **Withdrawal:** In case of withdrawal after the Add/Drop period, tuition fee will remain charged, and students must pay fee for withdrawn courses.

Additional Course-Related Charges

- **Extended Research:** The extended research fee will also be charged if the research period goes beyond the normal time.
- **Internship and Student Teaching Practice Fees:** Internship and Student Teaching Practice courses are charged at a different rate than standard courses. Detailed information on these rates can be found in the accounts section of the University website.
- **Applicable Taxes:** Students are responsible for any government and withholding taxes where applicable.

For further information, see the FCCU website.

Refund Policy (Tuition & General Security)

FCCU refund policies are strictly observed. No refunds are issued for students who drop to part-time status after registration. Tuition fee is refundable under the Higher Education Commission's National Level Refund Policy as follows:

- Application for a refund of Fee can be made by completing the appropriate form and submitting the proof of payment. Any refund will be paid to the student or his/her parents/guardians/ sponsoring body.

The refund amount of Tuition Fee for new admissions is subject to the following schedule:

Timelines	Fee Refund Policy
10th day of commencement of classes	No tuition fee will be charged.
Up to 15th day of commencement of classes	20% of the first installment of tuition fee will be charged.
Up to 20th day of commencement of classes	40% of the first installment of tuition fee will be charged.
Up to the 30th day of commencement of classes	50% of the first installment of tuition fee will be charged.
31st day onwards of commencement of classes	Tuition Fee will be charged as per installment plan.

- The date on which written notice of leaving the university is given to the admission office will be considered for calculating the timeline for refund of tuition fee. The timeline will be calculated continuously, covering both weekdays and weekends.
- A student expelled from the university due to disciplinary misconduct shall not be eligible for any refund, except for the refundable Security Fee.
- Excess Tuition Fee and other fees will not be refunded to the continuing students. Their fee will be adjusted against next semester's fees.

General Security

- Payable by all new students together with the first-semester tuition fee, this fee is refundable after deductions are made for the damage/loss (if any) caused by the student.
- The deposit is fully refundable after the student has completed his/her duration of studies or has withdrawn/transferred to another university.
- A security fee refund is subject to clearance of all outstanding dues.

Non-Refundable Fee

- **Admission Fee:** The fee paid as an admission fee is nonrefundable.
- **Registration/Application Fee (FEAT):** Any Fee, payable at the time of initial application is non-refundable under any circumstances.

Hostel Dues and Charges

- **Installment Structure:** Hostel dues are payable as per the installment plan.
- **Initial Payment Requirement:** The first installment of hostel dues must be settled before room allocation.
- **Mess Charges:** Compulsory mess charges for the semester are to be paid as per installment plan while extra mess (voluntary) meal charges are charged monthly.

- **Leaving the Hostel:** For leaving the hostel during the term, one month's notice should be given to the Residential Life Office. If no notice is given, one month's notice period will remain charged from the date of leaving.

Hostel Refund Policy

Hostel dues are charged in three categories: Hostel Registration, Occupancy Charges and Meal Charges.

Application Process: Refund applications should be submitted to the Accounts Office by the student through Residential Life Office. The refund process will be initiated after the clearance from the Residential Life Office.

For continuing students, hostel security and dues will be transferred/adjusted in tuition fee account. No cash refunds will be disbursed.

The Hostel Registration fee is non-refundable.

The Hostel refund policy is:

Hostel Security	Full refund, if all outstanding dues are cleared and the hostel is vacated.
Occupancy Charges and Compulsory Mess Charges	One month's notice should be given for leaving the hostel. Occupancy and compulsory mess will be charged on the pro- rata basis till the last day of the notice period.

Financial Appeals Committee

This committee is comprised of staff members from all related departments and is responsible for hearing appeals involving issues with financial impact on students.

- Appeals on account of financial hardship do not fall under the purview of this committee but will be forwarded to Financial Aid or other relevant offices.
- The decision of this committee regarding the appeals will be deemed final and binding on both parties.



Financial Aid and Merit **SCHOLARSHIPS**

Financial Aid and Merit Scholarships

Forman Christian College (A Chartered University) - FCCU provides merit-based scholarships to academically eligible university students and need-based financial aid to students with limited financial means, with the aim of making quality education accessible and attracting exceptional talent. In the ongoing academic year (2025-26), we have disbursed over Rs. 400 million to more than ~1450 students and have allocated a budget of Rs. 440 million for the next academic year (2026-27). Multiple scholarships are facilitated and made available across various categories as shared below:

Merit Scholarships for High performers in Cambridge exams

Merit Scholarships against program tuition fee are offered to Global and Pakistan distinction holders in A-level exam.

- 100% tuition fee scholarship for all candidates with Global distinction from Pakistan.
- 75% tuition fee scholarship for all candidates with Pakistan distinction.
- 50% tuition fee scholarship for all candidates with 3 A*.

Continuation Criteria: Students would be required to secure and maintain a CGPA of at least 3.75 throughout the course of their studies.

Merit Scholarships for HSSC exam position holders from across Pakistan

100% Merit Scholarships against program tuition fee are offered to HSSC position holders from any Board of Intermediate and Secondary Education across Pakistan.

Continuation Criteria: Students would be required to secure and maintain a CGPA of at least 3.75 throughout the course of their studies.

Merit Scholarships for High Achievers:

Merit Scholarships are offered to high achievers based on their T-Score, which is an aggregate of past academic performance and the Forman Entrance Admission Test (FEAT Accuplacer). The Merit Scholarship award is for program tuition fee only.

- 25% tuition fee scholarship for all candidates with T-Score 69.
- 50% tuition fee scholarship for all candidates with T-Score 70 and above.

Continuation Criteria: Students would be required to secure and maintain a CGPA of at least 3.5 throughout the course of their studies.

Rector and Vice Rector Merit Scholarships for Exceptional Students:

Prospective students with T-Score of 70 and above would be eligible to compete through a rigorous interview process for:

- Rector's Merit Scholarship - 100% Tuition Fee waiver for one prospective student.
- Vice Rector's Merit Scholarship - 75% Tuition Fee waiver for one prospective student.

Eligible students will be notified by the Financial Aid and Scholarships Office.

Continuation Criteria: Students would be required to secure and maintain a CGPA of at least 3.75 throughout the course of their studies.

Need-Based Financial Aid for Deserving Students

The University believes that deserving students should not give up on their dream of higher education due to financial constraints and dedicates a significant portion of their budget to support the financially disadvantaged students.

Provisional Assessment on Admission Offer:

Candidates with valid admission offers may apply for provisional assessment by visiting the Financial Aid & Scholarships Office with a Provisional Assessment form and necessary supporting documents.

Applying for Financial Aid after Admission:

Following admission acceptance, students who are interested in applying for financial aid should comply with the steps listed below:

- Step 1: From your Formanite email ID send email to financialaid@fccollege.edu.pk to request financial aid form access
- Step 2: You will receive online form access, username and password, and documents checklist.
- Step 3: Complete the online form and submit, get prints, attach all the required documents before submission in the Financial Aid & Scholarships Office Room #15 on/ before the prescribed deadline, as mentioned in the email, either in-person or through courier.

Work-Study Program

Work-Study Program (WSP) is intended to facilitate students by providing on-campus employment and professional development opportunities. Students may apply online, and the list of eligible students is shared by the Financial Aid & Scholarships Office with the hiring department for selection of the student(s). A stipend is provided to the beneficiary based on the number of hours worked, which is adjusted against the tuition fee.

All financial aid beneficiaries are encouraged to apply for this program to gain valuable work experience and develop key skills.

Sports-based Scholarships

FCCU offers tuition fee scholarships up to 50% to highly talented and competent players in different sports categories under individual games. The screening process to select a suitable candidate is performed by the Sports Scholarship Committee through trials and interviews. The admission criteria remain the same as for other regular candidates and the following terms and conditions apply:

- Scholarships will primarily be offered for individual games, which include athletics, boxing, bodybuilding, swimming, taekwondo, weightlifting, and team games on a case-to-case basis.

- The winner of the Board of Intermediate and Secondary Education (BISE) tournament(s) will be considered for a scholarship.
- The scholarship will be applied to the tuition fee only for one year and will be renewable based on annual performance.
- For scholarship renewal, students must bring at least one medal by participating in national games, provincial games, and Higher Education Commission (HEC) intervarsity competitions.
- Scholarship recipients must maintain a minimum CGPA of 2.0 and a clean disciplinary record.

FCCU Women Empowerment Scholarship Program

Forman Christian College (A Chartered University) offers up to ten (10) fully funded tuition and hostel fee scholarships (except meals) to meritorious and deserving female students joining our four-year undergraduate program.

The purpose of this scholarship is to prepare women for leadership while also attracting additional students from remote areas of Pakistan to Forman campus by inspiring and enabling women to pursue quality higher education. Interested female students from Khyber Pakhtunkhwa, Balochistan, Sindh, Gilgit-Baltistan; Chitral and Kalash valley and Kashmir are encouraged to apply with following terms and conditions:

- Successful completion of admission requirements with a T-score of 55 or above.
- Must have Domicile issued by the provincial government of Khyber Pakhtunkhwa, Balochistan, Sindh, Gilgit-Baltistan (Chitral and Kalash) or Kashmir.
- Must be first generation students and family have a demonstrated financial need.
- Candidates will mandatorily be required to participate in the work-study program from Year two onwards to be eligible for up to 20% of the scholarship award.
- Participation in outreach and promotional activities of FCCU through the Financial Aid and Admission offices.

Continuation Criteria: Students would be required to secure and maintain a CGPA of at least 3.0 throughout the course of their studies.

Zakat Fund:

The university believes that quality higher education should be accessible to all meritorious students, regardless of their ability to pay to attend an accredited university. The Zakat Fund (Certified) was founded in 2013 to support underprivileged Muslim students in Pakistan's higher education system. Since its inception, the Zakat Fund has provided Rs 29.8 million to about 480 Muslim students from poor families.

Support in Applying for Government and Other Scholarships:

The university also helps students apply for other external scholarships offered by the Punjab Educational Endowment Fund (PEEF), the Balochistan Educational Endowment Fund (BEEF), and the Higher Education Commission (HEC), as well as determining eligibility for designated scholarships and donor-specific categories of scholarships.

Punjab CM Honhaar Scholarships

Punjab Higher Education Commission has partnered with FCCU to provide 100% tuition fee scholarships to students. 50% funds shall be contributed by PHEC while 50% relief shall be provided by FCCU. Students shall be facilitated to apply for CM Honhaar scholarships as per the notified criteria as follows:

- Applicants must have domicile of Punjab province
- Monthly Family income of the applicant must be less than Rs. 500,000 for Private Sector Universities
- The applicant must be enrolled in the 1st semester/year of Selected Disciplines in the Selected Universities / Colleges at the time of application.
- Minimum Intermediate marks for students at Private Universities $\geq 70\%$ for Sciences and $\geq 65\%$ for Arts and Social Sciences
- Scholarships will be awarded on merit basis. Merit criteria will have Matric marks weightage 30% and Intermediate marks weightage 70%

The Punjab Educational Endowment Fund (PEEF)

The Punjab Educational Endowment Fund (PEEF) in partnership with FCCU offers twenty (20) partial tuition fee scholarships to talented and deserving enrolled undergraduate students fulfilling the prescribed criteria.

- Must have Punjab domicile.
- Scored at least 60% marks in last degree annual exams.
- Passed the examination as a regular student from:
 - Government Institution, or Continuing scholar of the respective institution
 - As a Private Candidate or
 - Private Institution
- Declared monthly income of parents (from all sources) is equal to or less than sixty thousand (Rs. 60,000/-).
 - Children of Govt. employees in BPS 1-4 are exempted from the condition of monthly income if salary is the only source of income.
 - Children of civilians martyred in terrorist attacks are exempted from the condition of monthly income.
- Secured admission in the institution as a full-time student in the same academic year as of passing the annual examination.
- Fulfill the criteria for award of financial aid at the respective Center of Excellence (COE) and that is FCCU.
- Not availing of any other educational scholarship during the respective academic year.

The Balochistan Educational Endowment Fund (BEEF)

The Balochistan Educational Endowment Fund (BEEF) in partnership with FCCU offers partially funded tuition fee scholarships to talented and deserving enrolled undergraduate students fulfilling the prescribed criteria.

- Local/Domiciled students of Balochistan enrolled in Pakistan based Public Sector and BEEF Private Panel Institutions situated outside Balochistan.
- Having Secured at least 60% marks or 3GPA/CGPA and their names are included in the merit list provided to BEEF by concerned institution.

- For Special Quota Scholarships on need based or special vulnerability (Orphan, Disables, Minority and Transgender) students must submit an application individually having all documents/ transcripts and required certificates attested from Concerned Institution.
- Students availing of any other scholarships are not eligible to apply.
- Students enrolled in BS, Masters & MS/MPhil must apply through the Focal Person/Financial Aid Office (FAO) or any other designated office of the concerned institution.
- No individual Scholarship Application Form will be accepted except a special quota.
- Students enrolled in BS 2 Years, Bridging, or Lateral Entry Programs are not eligible for BEEF Scholarships under any category.

Important Notes:

1. The deserving students may apply for need-based financial aid despite availing any other scholarship offered under various programs.
2. The total financial aid and scholarship awards shall not exceed the sum of fees charged to the student in a semester.
3. Financial aid and scholarship awards shall be awarded for the standard duration of the degree program only.

DEPARTMENT CONTACT DETAILS

Mailing Address:

Room # 15, Financial Aid Office

Ahmad Saeed Administration Building

Forman Christian College, Ferozepur Road, Lahore. 54600

Telephone: 042-99231581-8, Ext: 331

Email: infoscholarsips@fccollege.edu.pk, financialaid@fccollege.edu.pk



Academic Policies and **PROCEDURES**

Academic Policies and Procedures

Documents Release Policy

Application Process:

Formal Request: Students must apply in-person or via email to request release of documents at the Academic Service Office.

Collection Process:

Authorized Representatives: Those who cannot come in-person to collect the requested document can authorize any representatives. They have to authorize via email or provide a formal authorization letter from the student.

Identification:

A copy of the student's University Card or CNIC (in case of authorized person) is required.

University Holds:

If students have any hold (Business Office Hold, Proctorial Hold, etc.) on their account, documents will not be issued in any circumstance. Holds must be cleared for the issuance of the documents.

Academic Suspension:

If the student is suspended from University Rolls either Academically or for Disciplinary actions, the student has the right to receive official documents. If the access to campus is denied, the student can authorize someone else to receive the documents.

Medium of Instruction

At FCCU, the medium of instruction is English. Students entering the Bachelor's programs with deficiencies in English language proficiency will be required to enroll in pre-university English language courses. These courses are specifically designed to support students in transitioning effectively into the academic demands of the Bachelor's programs.

Class Attendance

Students are expected to attend all classes and laboratory sessions in the courses for which they are registered. Students who miss classes are far less likely to succeed in meeting the requirements of the course. The university's minimum accepted attendance is 67%, however individual teachers may set higher requirements. Each teacher outlines his or her expectations for class attendance in the course syllabus. Teachers are expected to keep accurate records of student attendance. If a student does not attend the percentage set by the instructor on the course syllabus of the class and laboratory sessions, he/she will not be permitted to take the final examination in the course.

Standards of Academic Progress

Forman Christian College (A Chartered University) upholds high standards for student performance, as we believe that excelling in coursework is essential preparation for future leadership opportunities. We encourage our students to strive for excellence and will monitor their academic progress to ensure they are aware of any

emerging concerns. Normal progress toward graduation is defined as completing five courses per semester (equivalent to 15 to 18 credits) with a minimum grade of C or higher.

Good Academic Standing, Probation, and Dismissal

At the end of each semester, the Academic Review Committee evaluates the progress of all students to determine their status in "Good Academic Standing." To maintain Good Academic Standing, students must meet the following requirements:

Classification	Semester Credit Hours Completed	Minimum CGPA Required for Good Standing
Freshmen	01-30	1.50
Sophomore	31-60	1.75
Junior	61-90	2.00
Senior	90 or More	2.00

Additionally, students are required to declare a major by the end of their fourth semester and must be enrolled in courses that align with the timely completion of their declared major. Students who fail a course have multiple Withdrawals, Incompletes, and/or receive "D" grades will also undergo review.

Academic Dismissal and Re-enrollment

Students are expected to maintain good academic standing throughout their academic career. Official notifications regarding academic standing will be issued to all full-time students whose performance does not meet the University's prescribed standards. After each semester, the Academic Advising Center, the Academic Review Committee and Associate Vice Rector for Academic Affairs will assess whether students experiencing academic difficulties will be placed on Academic Probation, suspended on Probation, or dismissed from the University. Academic Dismissal is subject to specific conditions (outlined below).

The designation "Academic Probation" will appear on a student's transcript if their academic performance falls below the required standards for Good Academic Standing. This serves as a formal alert to students that their academic performance is insufficient for graduation, and that, unless improvement is demonstrated in the following semester, they may face dismissal.

Students placed on Academic Probation must adhere to the following conditions:

- Enroll in no more than five courses during a regular semester.
- Attend all classes for each enrolled course.
- Meet with an Academic Advisor at least once every two weeks.

- Regularly visit the Academic Advising Center to discuss academic plans and progress with academic support officers or student success advisors.
- Comply with any additional requirements set by the Academic Review Committee.

Any student who fails to comply with the conditions of Academic Probation will be subject to review by the Academic Review Committee and Associate Vice Rector for Academic Affairs at any time and may be dismissed from the University.

The Committee may decide to drop a student whose academic performance consistently falls below the required minimum standards and who remains on probation for more than one semester. In such cases, the student may request re-enrollment after completing a full, regular semester. (Generally, students will not be allowed to re-enroll during Summer semester after being Dropped).

Students requesting re-enrollment should meet with an Academic Advising Centre and submit a written letter. Note that a student who is dropped for a second time will not be eligible to return to continue their degree program, effectively terminating the opportunity to complete their degree at FCCU.

Removal from Academic Probation

Probationary status will remain in effect until the student's CGPA meets the required threshold for Good Academic Standing.

Suspension of Co-Curricular and Sports Activities

A student placed on academic probation or facing dismissal for a second consecutive semester will be suspended from participating in co-curricular and sports activities. This suspension allows the student to dedicate their full attention to academic work. Additionally, no student with a CGPA below 2.0 is eligible to compete in any officially sanctioned sports event (whether FCCU or HEC) until back in good academic standing.

Academic Integrity Issues

The Academic Integrity Committee will review all cases involving breaches of academic integrity. This includes, but is not limited to, instances of forgery (such as students signing on behalf of advisors, faculty, or staff) and plagiarism. All cases of forgery will result in a fine and an automatic hearing by the designated committee.

Program and Majors

Students must fulfill the requirements of a major field of study to be eligible for graduation. The number of credit hours required for the major varies depending on the degree program/ Major chosen, but it will not be fewer than 52 credit hours. (Please refer to the departmental section for more information.)

FCCU offers 6 primary academic programs, featuring 24 distinct majors. Within these fields of study, specialized tracks are also offered for a few majors and programs. Detailed course requirements for each program and major can be found in the respective departmental sections of this catalog.

Program, Major and Specializations Offered

Programs

1. Bachelor of Science in Biotechnology
2. Bachelor of Business Administration with specializations in:
 - a) Accounting & Finance
 - b) Operations Management
 - c) Marketing and Sales
 - d) Human Resources
 - e) Business Analytics
3. BS Computer Science (without specialization and with specialization in:
 - a) Data Science
 - b) Artificial Intelligence
 - c) Software Engineering
 - d) Cyber Security
4. Doctor of Pharmacy (PharmD)
5. Bachelor of Science with major /specialization
6. Bachelor of Studies with major /specialization

Majors

Bachelor of Science

1. Biological Sciences
2. Microbiology
3. Chemistry
4. Data Analytics
5. Economics
6. Education
7. Linguistics
8. Environmental Sciences
9. Geography (with specialization in):
 - a) Human Geography
 - b) Physical Geography
10. History
11. Mass Communication (without specialization and with specializations in:
 - a) Print and Electronic Media
 - b) Advertising and Public Relations
12. Mathematics
13. Physics
14. Political Science
15. Psychology
16. Applied Psychology
 - a) Mental Health
 - b) Industrial and Organizational
 - c) Leadership and Management
 - d) Artificial Intelligence

- e) Forensic Psychology
- 17. Public Policy and Administration
- 18. Sociology
- 19. Statistics

Bachelor of Studies

- 20. English
- 21. Philosophy
- 22. Islamic Studies
- 23. Christian Studies
- 24. Urdu

Change of Program (Online Form)

Please note that a program change request will not be approved if the students' T-score was below the required merit for the program they are requesting to transfer into. The students' request will be reviewed by the following:

- The Chairperson of the department for the program the student wishes to enter.
- The Academic Dean of the new program.
- The decisions made by the Academic Dean and the department Chairperson are final.
- Once the change is approved, a copy of the approval will be forwarded to the Admission Office to matriculate according to the change of program request and approval.
- The form will be forwarded to the Academic Services Office to change the program.
- Once the program is changed on empower the request is forwarded to the Accounts office to adjust the fee structure.
- The new fee structure will take effect in the semester following the approval.
- After the fee structure has been changed, Finally, the student's advisor and major will be updated by the Academic Advising Center.
- If a change of program request is denied, the student is permitted to seek new admission into the desired program.
- For any further questions or clarifications, please contact the Academic Services Office.

Degree Completion Requirements

In alignment with the Higher Education Commission (HEC) of Pakistan's policy mandating eight regular semesters (fall and spring), Forman Christian College (FCCU) has established the following guidelines:

Students are required to complete 8 semesters (fall and spring) within 6 years to graduate. The student may request an extension of 1 year subject to the approval of the statutory body. Summer semester may be taken at the student's discretion, but these terms should only be used to ensure timely graduation or to address any academic deficiencies. They should not be used as an attempt to graduate early.

While FCCU will recognize degrees once all requirements are fulfilled, please note that the HEC will only attest degrees if the program includes a minimum of eight regular semesters.

Students may request a change of academic catalogue, subject to approval by the department or Head of Academic Services Office. Upon approval, the student's academic record is updated and reflects the new catalogue on the portal. Students must meet the requirements of the new catalogue, including any additional coursework or credits. This provides students with flexibility and opportunities to align their academic requirements with updated curriculum and program offerings.

Multiple Majors and Minors

Students are allowed to pursue more than one major. However, it is important to note that the Higher Education Commission (HEC) recognizes only the first major listed on the transcript. If a student intends to graduate with multiple majors or a combination of a major and a minor, all requirements must be completed according to the same catalog.

Students can only earn one major from any Academic Department. If a student completes a double major, the two majors must be from two different Academic Departments.

External review of credit bearing Capstone Project

Students taking the compulsory Capstone projects within their chosen field are required to complete the work under the supervision of a faculty member. Upon completion, the department chairperson will forward the project report, along with internal review, the Turnitin report, to the Controller of Examinations Office for external evaluation. The Controller of the Examinations Office will then notify the Academic Services Office of the grade awarded by the external examiner.

Bachelor of Studies and Bachelor of Science in 23 Majors

The following requirements must be fulfilled by all students to qualify for formal recommendation by the faculty for the Bachelor of Studies/Bachelor of Science Degree:

- The satisfactory completion of a minimum of 124 credit hours (Detailed course requirements for major can be found in the respective departmental sections of this catalog).
- A cumulative grade point average (CGPA) of 2.0 or better as well as a minimum grade point average (GPA) of 2.0 in the major.
- The completion of at least 12 upper-level courses labeled at the 300 or 400 level.
- The satisfactory completion of a major field of study from a list of those offered by the University and specifically to one catalog only. For the Bachelor of Studies/ Bachelor of Science core and elective courses must be completed as stated in the requirements of the selected major.
- The satisfactory completion of the General Education requirements of the University.
- The successful completion of the senior capstone courses in the major as required.

Bachelor of Business Administration

The following requirements must be fulfilled by all students in order to qualify for formal recommendation by the faculty for the Bachelor of Business Administration

- The satisfactory completion of a minimum of 124 credit hours with a cumulative grade point average (CGPA) of 2.0 or better as well as a minimum grade point average (GPA) of 2.0 in the major.
- The completion of at least 12 upper-level courses labeled at the 300 or 400 level.
- At least 75 credit hours (including core courses) of Business courses as specified in the catalog.

- The satisfactory completion of the General Education requirements of the University.
- The successful completion of the senior capstone courses in the major as required.

BS Computer Science

The following requirements must be fulfilled by all students to qualify for formal recommendation by the faculty for the BS in Computer Science:

- The satisfactory completion of at least 138 credit hours with cumulative grade point average (CGPA) of 2.0 or better as well as minimum a grade point average (GPA) of 2.0 in the major.
- The completion of at least 12 upper-level courses labeled at the 300 or 400 level.
- At least 96 credit hours of Computer Science courses as specified in the departmental section.
- The satisfactory completion of the General Education requirements of the University.

Bachelor of Science in Biotechnology

The following requirements must be fulfilled by all students to qualify for formal recommendation by the faculty for the Bachelor of Science in Biotechnology:

- The satisfactory completion of minimum 124 credit hours with a cumulative grade point average (CGPA) of 2.0 or better as well as a minimum grade point average (GPA) of 2.0 in the major.
- The completion of at least 12 upper-level courses labeled at the 300 or 400 level.
- At least 54 credit hours (including core courses) of Biotechnology and Biological Science courses as specified in the catalog.
- The satisfactory completion of the General Education requirements of the University. ● The successful completion of the senior capstone courses in the major as required.

Doctor of Pharmacy (PharmD)

The following requirements must be fulfilled by all students to qualify for formal recommendation by the faculty for the Doctor of Pharmacy (PharmD) (A five-year professional degree).

- The satisfactory completion of 208 credit hours with a cumulative grade point average (CGPA) of 2.0 or better as well as a grade of 'D' or better in each course, as per curriculum approved by the Pharmacy Council of Pakistan/Higher Education Commission of Pakistan.
- The satisfactory completion of the General Education requirements of the University Major GPA.
- The GPA calculated by tabulating ALL courses in the major taken by the student.

Requirements for Minor

An approved minor is outlined in the department's section of the catalog. To pursue a minor in a specific discipline, the student must complete the requirements for the minor as specified in the catalog.

Universal minimum CGPA of 2.00 for any undergraduate minor offered by FCCU to appear on the transcript. The CGPA for a minor is calculated based on the courses taken to fulfill the minor requirement. There will

be no minimum CGPA to declare a minor. If 2.00 is not met in all the courses taken for the minor, the courses will remain on the transcript but not the official minor.

Undergraduate Internship

Departments can offer credit-bearing internships but are not required to offer internships, nor require them of all students. Thus, it is up to the department to offer a required internship for all its students, to offer it as a core option, to offer it as a departmental elective, or not to offer one at all.

This policy replaces all previous ones, including the one adopted by the Academic Council on May 19, 2022. Although the 2022-23 (p. 29), 2023-24 (p. 20), and 2024-25 (p. 31) undergraduate catalogs stated that an internship was a mandatory graduation requirement, this is repudiated effective with the action of the Academic Council (anticipated in Spring 2025) and will be applied retroactively on these three catalogs and prospectively to the 2025-2026 catalog and henceforth. Thus, internships will only be required for students in majors that have an internship requirement. Any department is at liberty of amending its program offerings, including internships, through the ordinary channels for program modification.

Treating an internship like any other credit-bearing course so that it is a meaningful learning experience and meeting NECHE requirements.

General Education

Forman Christian College (A Chartered University) is a liberal arts university. The purpose of the General Education requirement is to introduce students to a broad range of intellectual pursuits, to provide sufficient breadth of knowledge to prepare them for their role as citizens, to equip them with communication and analytical skills, to help them to integrate knowledge that comes from different disciplines and to prepare them for a lifetime of self-directed learning.

Distribution Requirement

There are two components of the General Education requirement. First is the demonstration of competency as determined by performance on University assessment examinations given as final exams in the general education course related to each of the following five areas:

- Written Communication in Urdu (URDU 101)
- ENGL 106: Functional English
- ENGL 108: Expository Writing
- Quantitative skills (MATH course or FEAT Math's score at the time of admission)
- Information Technology (CSCS 100 or COMP 102)
- There are four categories of General Education. A total of 17 required courses fulfills General Education requirements. Further explanation is stated below.
 - Double dipping" will no longer be allowed. That is, a course can count towards General Education or the major, but not both, unless permitted by an Accreditation Council for a professional program. Courses for any minor can double dip from any major and from General Education.

- Any course from the departments of English, Foreign Languages, History, Mass Communication, Philosophy, Religious Studies, Urdu, or Visual and Performing Art that assesses Humanities-related general education learning outcomes
- Any course from the departments of Economics, Education, Geography, Political Science, Psychology, or Sociology that assesses Social Science-related general education learning outcomes
- Any lab course from the departments of Biology, Chemistry, Environmental Sciences, or Physics that assesses Natural Science-related general education learning outcomes

General Education	48-51 Credits
UNIV 100 Foundations of University Education All University Freshmen students must take UNIV 100 during their first semester or after they have finished the IEAP program. Transfer students with 60 or more credits are exempted from UNIV 100. See details for Transfer based exemption.	3
ENGL 106: Functional English	3
ENGL 108: Expository Writing	3
URDU 101: Communicative Urdu	3
ISLM 101 (required for all Muslim students) or CRST 152	2
ISLM 100 A & ISLM 100 B: Fehm-e-Qur'an (two 1-credit courses) or for non-Muslims additional 2-3 Religious Studies credits	2-3
UNIV 200: Critical Thinking, Ethics, and Artificial Intelligence (<i>Not required for Business students</i>)	3
Arts and Humanities (outside the major) Course must be selected English, Foreign Languages, History, Mass Communication, Philosophy, Religious Studies, Urdu, or Visual and Performing Art	3
PKST 101: Pakistan Studies	2
PKST 103: Ideology and Constitution of Pakistan	2
SOCL 104: Civics and Community Engagement	2

Social Sciences (outside the major) Courses from: Economics, Education, Geography, Political Science, Psychology, or Sociology	3
Natural Sciences (outside the major) 2- Science courses with labs from Biology, Chemistry, Environmental Sciences, or Physics	3 or 4 (2 or 3 + 1)
2 MATH courses as per FEAT score	
Below 75 on a score range of 20-120, or below 263 on a score range of 200-300 MATH 108: Quantitative Reasoning I (Basic) MATH 112: Quantitative Reasoning II (Basic)	3 3
75 or higher on a score range of 20-120, or 263 or higher on a score range of 200-300 MATH 109: Quantitative Reasoning I (Advanced) MATH 113: Quantitative Reasoning II (Advanced)	3 3
CSCS 100: Applications of Information and Communication Technologies	3 (2 +1)
BUSN 220 Entrepreneurship	2

Any current student (enrolled Spring 2026 or earlier) will be permitted to take Functional English to fulfill the WRCM 101 requirement and/or Expository Writing to fulfill WRCM 102. Taking these courses will also replace previously earned grades for WRCM.

Current students will be allowed a 1-credit relaxation for any of the following courses they have not taken yet for general education: ISLM 101 Islamic Education or CRST 152 Christian Ethics; PKST 101 Pakistan Studies; and natural science lab courses. These general education credits do not need to be replaced so long as the student earns 124 credits to graduate and at least 43 credits of general education in all 15 required courses.

Current students will be allowed to opt for the new curriculum by switching to the 2026-2027 catalog for all degree requirements, but only if this will not endanger them meeting degree tenure requirements.

Academic Credit

Credit towards a degree is awarded for satisfactory course completion, independent study or academic work certified by another accredited degree granting institution (covered in the Transfer Credit policy).

One credit is defined as one hour of classroom or direct faculty instruction and a minimum of two hours of out-of-class student work each week of the semester. A credit hour means teaching a theory course for 50 minutes each week throughout the semester. At least an equivalent amount of work is required for other academic activities including laboratory work, internships, practical, studio work, and other academic work leading to the award of credit hours. Thus, one credit hour in laboratory or practical work/project requires lab contact of a minimum of two hours per week throughout the semester and one hour per week of out-of-class work, for a total of three hours.

Ordinarily credit is earned by course completion. A normal full-time academic load is five courses per semester. A student may take a minimum of 12 semester credit hours or a maximum of 17 semester credit hours and still be considered a full-time student. Additional permission is required for overloads above 17 credit hours.

A student may take a minimum of 12 semester credit hours or a maximum of 17 semester credit hours and still be considered a full-time student (Students enrolled in Computer Science Program can register up to 18 credits in a regular semester as the BS in Computer Science degree requires the completion of 138 credits). Additional permission is required for overloads above 17 credit hours. If a student enrolls in fewer than 12 credit hours, he/she is considered to be a part-time student. Enrollment for more than 17 credit hours is very rare and requires the approval of the Academic Advising Center and payment of any additional fees, if applicable.

Complete URDU Exemption

Exemption for URDU 101 is granted by the Associate Vice Rector for Academic Affairs for the following types of academic backgrounds:

- Pakistani students who graduated from American or other foreign High School systems, either in Pakistan or in a foreign country.
- Foreign students who graduated from a High School system outside of Pakistan.
- Students who studied URDU as a second language/Easy Urdu in O level.
- An exemption of URDU 101 requires substitution of another course in the Humanities. It is the student's responsibility to have this course properly substituted by the Academic Services Office.

FEAT-Based Exemption

MATH 108 and Mathematics Competency: Students who achieve a score of 75 or higher on a score range of 20-120, or 263 or higher on a score range of 200-300 on the Forman Entrance Admissions Test (FEAT) have met the Mathematics Competency requirement and are exempt from studying Math 108. Exemption of Math 108 through FEAT requires that any course with a MATH prefix from the Science and Mathematics category course be studied to replace this general education requirement.

Transfer-based Exemption

UNIV 100 Students who transfer into FCCU with 60 or more credits are exempt from studying UNIV 100. This is given at the time of transfer and admission and is not credit based. The required credits must be earned through the study of an additional course in the humanities category.

Exemption Requiring Substitution

Exemptions made in the above manner require course substitution by appropriately replacing the exempted course for degree audit purposes. Replacement must be done in consultation with the academic advisor and notified to the Academic Services Office after grading in the replacement course appears on the transcript.

A Level-based Credit

FCCU will discontinue the policy of awarding credit for A-Level, Edexcel, and International Baccalaureate (IB) exams, effective FA24-Levels (or similar exams) can be used interchangeably with an associated course (see list below) to fulfill prerequisites for other FCCU courses. Only grades B and above will be used for meeting prerequisite requirements and only if the student wants to exercise this privilege. The student does not get credit for the exempted course and still needs to take courses to fulfill all credit hour requirements per their respective catalog. For example, MATH 201 Calculus II has a prerequisite of MATH 102. If a student earns a B or higher in the "Pure Mathematics 3" A- Level, then s/he has fulfilled the prerequisite for MATH 201 and can take MATH 201. The student does not get credit for MATH 102; the only benefit is that the prerequisite for MATH 201 has been met by the A Level exam score of B or higher.

Biological Science for BIOL 105

Chemistry for CHEM 100

Physics for PHYS 100

Computer Science for CSCS 100

Pure Mathematics 1 or 2 for MATH 101

Pure Mathematics 3 for MATH 102

Economics for ECON 100

Geography for GEOG 101

History for HIST 102

Psychology for PSYC 100

Islamic Studies for ISLM 101

Christian Studies for CRST 152

Sociology for SOCL 100 Urdu for URDU 103

Independent Study (Online Form)

Independent Study (IS) is a credited course meant for students to pursue their individual interest during a semester under the supervision of a faculty member. Such courses are private in spirit but directed and based on serious expectations. These courses are to be designed in the field of the student's academic needs to result in substantive outcomes, be they a project, a written paper, a written report or other essential requirements containing significant analysis and interpretation of the approved topic.

Instructions

Students wishing to register for an IS course must first make arrangements with a faculty member who holds expertise in the desired area. This faculty member will supervise the students' work. The student and the

supervisor should agree on the course title, plan of study, objectives and expectations, as well as on the nature of final requirements and evaluation criteria.

Please note that the Independent Study involves a rigorous academic experience equivalent to that of any other FCC undergraduate course and should only be approved under the following conditions:

- A student may be considered for special enrollment if they are in their final semester or facing a degree tenure limitation, and require a specific course to graduate, provided the academic year remains the same.
- Students may be granted exemption from the minimum contact hours requirement only in cases of a documented medical condition that prevents physical attendance.
- All other requests for remote/online study will not be approved.
- The required course is not being offered in the current or upcoming semester.
- There is no alternative course or offering that fulfills the same requirement.
- Courses that require laboratory work, field participation, and physical education courses are not eligible to be offered as Independent Study courses.

In order to earn credits, the student and their supervisor will consent on a drafted course outline along with a timeline or timetable. The course outline, like any other course outline, student learning outcomes, should include the academic instructions, detailing a periodic chart of activities based on readings, frequent writings relevant to the course design, teacher's comments on these writings, revisions, teacher's comments about notified improvement, etc. The course outline should also mention the number of tutorials and frequency of meetings between the students and the supervisor and must be submitted to the approving authorities along with this form.

Independent Study Course Outline Requirements

To earn academic credit for an Independent Study (IS) course, the student and supervising faculty members must jointly develop and agree upon a detailed course outline and timeline. The course outline must include:

- Clearly defined Student Learning Outcomes (SLOs).
- A weekly or periodic schedule of academic activities based on assigned readings.
- Frequent writing assignments aligned with the course objectives.
- The supervisor's written feedback on each submission and evidence of revisions made in response to feedback.
- Comments noting improvements in the students' work overtime.
- Additionally, the outline must specify the number of tutorials to be held and the frequency of meetings between the student and the supervisor.
- This completed course outline must be submitted along with the Independent Study form for review and approval by the designated academic authorities.

Time Commitment and Supervisor Meetings

As a general expectation and apart from the individual input of the student, **9 hours per week**, amounting to a total of **150 study hours per semester**. This includes time spent on **library research, reading, writing,**

and other project-based academic activities. In addition to independent work, the student must meet with the Supervisor at least once every fortnight (15 days), with a minimum of 8 weeks (times) during the semester.

Guidance and Responsibility for Independent Study Requests

- If faculty or department requires additional information regarding a student's eligibility for Independent Study (such as degree tenure issues or graduation requirements), they may contact the Advising Center for feedback and assessment.
- Students are expected to take full responsibility for planning their academic roadmap effectively by using student support services. This includes aligning their course selections aligning with the regular course offerings published by the department.
- Independent Study (IS) should not be used as a substitute for poor planning or failure to enroll in required courses when they are regularly offered. Requests for IS based solely on scheduling conflicts due to inadequate planning will not be approved.

Independent Study Completion and Registration Guidelines

All requirements finalized for the IS must be completed during the semester for which the student is registered. Failure to do so, may result in an "I" or Incomplete grade. The supervisor is responsible for evaluating the student's progress and submitting a final grade by the end of the semester.

To initiate an Independent Study, the student submits the completed IS Permission Form to the Academic Services Office no later than the end of drop/add period of the semester in which the "IS" is to be taken. Once approved, the course will be officially registered with an "IS" course code.

Transfer Credit

Credit is earned by migration or transfer from another degree-granting institution recognized by the Higher Education Commission or accredited in the USA or UK, up to a limit of 64 credit hours. A student transferring to FCCU from another institution should request a transcript of work done in the other institution to be sent to the Admissions Office. When the transcript has been evaluated by the Academic Services Office, the applicant is notified of credit acceptance by the Admissions Office.

Course Credit that is Acceptable for Transfer

1. An official request for credit transfer must be made through:
 - The Admissions Office, prior to new admission.
 - The Academic Services Office, for students participating in approved student exchange programs during their studies at FCCU.
2. An official transcript is required for all transfer credits to be reflected on the FCCU transcript.
3. Transfer courses must be from an accredited institution.
4. The courses must align with FCCU's degree requirements.
5. The grade for each transfer course must be "C" or better at the undergraduate level to be accepted by FCCU.
6. Applicants with a CGPA below 2.0 must await case approval following evaluation.

7. The Chairperson of each department determines how transfer credits fulfill the requirements for FCCU's major degree requirements.
8. The transfer of General Education requirements will be assessed by the Academic Services Office.

Course Credit that is Unacceptable for Transfer

Transfer of credit based on the following is unacceptable:

1. Applicants that have previously withdrawn officially from FCCU.
2. Applicants that have been previously released for academic, or disciplinary measures, including dismissals based on Academic Probation.
3. Unapproved study at another institution while an active student at FCCU.
4. Unapproved simultaneous enrollment at two institutions (one being FCCU) in the same semester.
5. Distance learning programs that have not been properly approved prior to beginning.
6. Transfer credit request based on unofficial or photocopied transcript.
7. Transfer credit request based on private conventional BA/BS or equivalent qualification while a regular student at FCCU.
8. Courses studied at non-accredited institutions.

Counting a Conventional Degree towards a 4-Year Bachelors

1. A petition for conversion must be submitted to the Admissions Services Office.
2. The transcript or result of the accredited university's conventional BA/BSc degree must be submitted with this petition.
3. The Head of Academic Services Office will make a merit-based decision on the petition and communicate this in writing to the student.
4. If the petition is approved, the standard transfer credit conditions of transfer will apply.
5. A maximum of 30 credit hours can be achieved through transfer at Forman at the time of admission.

Re-enrollment to the University

Students who have not been enrolled for three consecutive semesters will be inactive from the university. They must seek re-enrollment to the University to resume their studies by submitting a Request for re-enrollment to the Associate Vice Rector for Academic Affairs for final approval. It is the student's responsibility to submit a copy of the re-enrollment approval to:

1. The Accounts Office (for re-enrollment fee and tuition).
2. The Academic Services Office (for registration).

Termination of Study

A student at Forman can have his/her study terminated – enrolment annulled, on the following grounds:

- Completion of the Degree after obtaining accounts clearance (completion of degree audit and official listing in graduation booklet at commencement).
- Withdrawing from the University after obtaining clearance.
- Completion of the Tenure of the program six years (12 regular semesters).

- Removal from the rolls on disciplinary or administrative reasons (breach of academic integrity, etc.).
- Decision of Academic Review committee recommending dismissal.

Such students can reapply for enrolment anew, other than ones removed from rolls on discipline grounds.

Registration Policies

Unofficial Presence in Class

Students may attend class only if they are officially registered. Instructors confirm registration by reviewing the class roster. Students are required to attend all courses they are registered for in a given semester and must attend the correct section to receive a grade. Unofficial section changes are prohibited; students who switch sections without approval or never showed up in the class will receive an "NS" (No Show) designation.

Registration Time Frame

Registration dates are published in the academic calendar available on the official website of FCCU. All courses for which the student wishes to earn credit must be registered through student web services. The student is responsible for every course listed on his/her account schedule and can receive no credit for courses not listed there.

After registration, official changes in registration may be made only during official add/drop periods when access to registration is again available online. No course may be added after this deadline.

See '*Cancellation of Courses due to Low Enrolment*' for that time frame.

Overloading Courses in a Semester (Online Form)

Under normal academic guidelines, students register for 5 courses, or up to 17 credits, through the online system.

However, credit overloads may be permitted under the following conditions:

- Students with a CGPA of 3.00 or higher may request a credit overload through the Academic Advising Center.
- Senior students who have completed at least 90 credit hours may be allowed to take an overload, regardless of CGPA, if the additional course is required for timely graduation.
- Students with a CGPA below 3.00 may be considered for an overload on a case-by case basis, only if there is a compelling academic need to ensure timely completion of their degree.
 - In such cases, this policy applies only to students in semesters 7 and 8, as well as summer sessions following semester 7.
 - The minimum CGPA for overload eligibility is 2.50. Requests from students with a CGPA below 2.50 will not be accepted.

Computer Science students are allowed to register for up to 18 credits in a regular semester, in accordance with the degree structure. The Bachelor's in Computer Science requires the completion of 132 credit hours.

No Seat in Selected Course (Request Online for Seat Exception)

Due to high demand, seats in certain courses may fill quickly. However, students with a compelling need to take a specific course in the current term may request entry. (Approval is at the discretion of both the course instructor and the department chair.

When seeking a seat in a course that is already full:

1. Apply for Seat Exception online by going through the Online Seat Exception process guide, which is available on FCCU's website <https://www.fccollege.edu.pk/wpcontent/uploads/Online-Seat-Exception-Requests-Guide-for-Students.pdf> in Student's Tab and on Academic Services Office page Resources Tab.
2. The student must ensure that this course does not conflict in time with other courses registered online before submitting the request for approval to the instructor.
3. Any discrepancy must be dealt with in the applicable semester.

Prerequisite Waivers

Students must successfully complete all prerequisite courses before enrolling in a course that requires them. In exceptional cases, if a prerequisite has not been met, the student may be allowed to register only with the instructor's endorsement and written approval from the department chair. This documentation must be submitted to the Academic Services Office within the official registration period for processing.

Confirmation of Registration

Upon completion of the registration procedures as outlined in the registration post on the University website, the student's registration is confirmed on payment of the estimated bill provided on the student's web account. Payment must be made in entirety by the last date mentioned on the academic calendar.

Business Holds

Students whose dues remain in arrears will be put on business holds. All courses on a student's schedule at the end of add/drop will be charged. No courses will be deregistered, but a business hold will be placed on the account and no further registrations will be done until all fees are cleared. A fine will be charged for fees/dues paid after the stated deadline.

Cancellation of Courses due to Low Enrolment

1. After the close of add/drop, classes with low enrolment are likely to be closed, particularly lower-level elective courses.
2. For Spring and Fall, 10 students for all level courses.
3. For Summer (optional semester), low enrolment is 10 students for all courses. However, in optional semesters the Low enrolment sections are not dropped but shall continue if the faculty teaching their courses agree to keep them active.
4. The Chairperson will be informed of the closure of courses due to low enrolment
5. All instructors teaching courses with low enrolment must inform students of the likelihood of closure, prior to the end of add/drop.

6. The Chairperson can also make the decision to continue with such a course provided there is appropriate justification to do so.
7. If a registered course has been canceled, students of that course can visit the Academic Services Office to arrange a satisfactory substitute, preferably within the add/drop period or within 4 days of add/drop closure.

Grading Policies

Grades are final as given by an instructor unless a reason exists for change as stated below. All grades will be locked onto the transcript at the time of degree awarding and will not be changed subsequently. (See Grade Change Policy).

Grading Legend

The grading system for undergraduate students is as follows:

Transcript Updates

Grade	Point Value	Numerical Value	Meaning
A	4.00	93-100	Superior
A-	3.70	90-92	
B+	3.30	87-89	
B	3.00	83-86	Good
B-	2.70	80-82	
C+	2.30	77-79	
C	2.00	73-76	Satisfactory
C-	1.70	70-72	
D+	1.30	67-69	
D	1.00	60-66	
F	0.00	59 or below	Failing
NS	0.00	0.00	
W	-	-	Did not show up in Class
AW	-	-	Officially Withdrawn
AU	-	-	Administrative Withdrawal/Dismissal
I	-	-	Audit/Listener Status
T	-	-	Incomplete
			Transferred credit

Students are accountable for reviewing their transcripts after each semester's grading period and notifying the Academic Services Office of any inaccuracies, such as incorrect placements on Academic Probation or misapplied Repeat course credits.

Moreover, if a course is taken as a replacement for an exempted course, it must receive proper approval and be submitted to the Academic Services Office for substitution immediately following the grading period.

Courses that have dual designators (coding) are identified with an asterisk (*) to signify their status as cross-listed courses. It is imperative for students to choose the correct designator pertinent to their specific program, as these designations will not be altered on the transcript post-grading.

In cases where a student registers for a cross-listed course twice under different departmental codes (except when the initial registration resulted in a W, NS, or F grade), the course credits will only be counted once. Consequently, the second registration will be marked as "Academically Withdrawn" and assigned an AW grade.

Grade Change Policy

If an instructor of a student determines that a grade was issued incorrectly because of a clerical or procedural error (a calculation error or one in transcribing the grade), it can be corrected by submitting a grade change to the Academic Services Office by submitting a grade change request remotely via EMPOWER login.

The procedure is as follows:

1. The grade change ONLINE must be filled completely:
 - a) The reason for the change must be stated clearly.
 - b) If the instructor is no longer on faculty, the grade change can be processed by the department Chairperson with approval from the respective Dean of Faculty.
2. All grade changes carry a time limit.
 - a) Incompletes in regular courses and any other grade changes due to typing errors or miscalculations will continue to be accepted by the Academic Services Office up to 8 weeks of the following regular semester.
 - b) An Incomplete in Internship can remain unchanged for one semester.
 - c) Capstone Project should be awarded an "I" in the semester registered and must be completed and graded by the following semester and no later than one academic year of bachelors.
 - d) A late grade change can be submitted to the Academic Services Office after approval of the Dean of Faculty, but final approval will be granted by the Associate Vice Rector for Academic Affairs.

The grade change must be submitted by the instructor within the stated time frame.

Grade of 'I' (Incomplete)

1. A grade of "I" (Incomplete) indicates that, although a substantial proportion of the course requirements have been met, the student has not completed all course requirements by the end of the term.
2. In the judgment of the instructor the student:

- a) Has been in good academic standing.
 - b) Is facing an emergency beyond his/her control.
3. A student must submit the work required within six (6) weeks of the following semester. If the work is not completed the grade of Incomplete will be changed to F grade.
 4. An Incomplete should never be given to a student who has performed poorly during the entire semester and wants extra time to improve the grade.
 5. An Incomplete in Internship can remain unchanged for one semester.
 6. Capstone Project should be awarded an "I" in the semester registered and must be completed and graded by the following semester and no later than one academic year of bachelors.

Retaking a Course

- The first is where a student has received a letter grade of D or F in a course. In this situation he/she is allowed to retake that course. The better of two grades will be counted and be factored into the CGPA. If a student retakes a course for the purpose of earning a higher grade outside of the conditions above, the second enrollment will be recorded as "Academically Withdrawn" (AW) on the transcript.
- If repeating a course with a D grade has caused double credits to appear on a transcript, the student must apprise the Academic Services Office of this anomaly and get it corrected immediately to avoid unforeseen circumstances delaying graduation.

Policy on Multiple Final Exams in One Day

If a student has three or more final exams scheduled on the same day, s/he has the option to reschedule one or more of the exams beyond two. It is the responsibility of the student to contact his/her faculty member and make this arrangement. The new day and time must still fall during the days of final exams, not before or after the final exam window. This arrangement shall be communicated to the Department Chair and Dean for their approval, with onward notice given to the Academic Services Office.

Withdrawal Policies

Course Withdrawal (Online Form)

- Students are allowed to withdraw from a course until the end of the 10th week of the regular semester. Students who withdraw from a course by the withdrawal deadline will receive a grade of "W" from the respective instructor.
- If a student does not officially drop a course or withdraws from a course after the deadline, he/she will receive a grade earned or F or NS.
- In circumstances where Academic Policy has been breached or disciplinary action taken, the Associate Vice Rector for Academic Affairs may award an AW (Administrative withdrawal) to a student and withdraw the grade given for the applicable course.
- In extreme circumstances beyond the student's control, such as illness, accident or death of a parent, permission will be granted to withdraw after the withdrawal deadline. In extenuating circumstances "W" will be awarded by the Associate Vice Rector for Academic Affairs.
- Withdrawal from a course or courses does not affect a student's fee status. Students must pay fees for withdrawn courses.

Temporary Withdrawal/Leave of Absence (Discontinuing for one semester or year)

- Temporary Withdrawal indicates a student's decision to discontinue studies for one semester or one academic year.
- Students considering withdrawal should first consult with their Academic Advisor, followed by a meeting with the Academic Advising Center (AAC). They must complete the withdrawal form and provide supporting documentation for their request. The AAC will then submit the form, along with their recommendation, to the Vice Rector's Office.
- Students receiving financial aid are required to consult the Office of Financial Aid to understand how the withdrawal may affect their financial aid eligibility.
- Once the withdrawal decision is finalized, the Vice Rector's Office will forward the withdrawal form and supporting documents to the Academic Services Office for processing.

University Withdrawal (Online Form)

A University Withdrawal is defined as leaving the university permanently. If a student decides to leave the University, the following steps must be taken:

1. The University Withdrawal Clearance Form must be downloaded from the website.
2. The student is responsible for obtaining clearance from the Library, Computer Lab, Proctor Office, Accounts Office, and Science Laboratories.
3. The university ID card must be returned to the Accounts Office.
4. Once the completed form with all necessary clearances has been received, the Academic Services Office will issue a *Letter of Release*.
5. To collect the security deposit, the student must provide a copy of the Letter of Release to the Accounts Office.
6. If a student quits and fails to inform the university about the decision to discontinue at the university, he/she will receive failing grades for all registered courses of the respective semester. If the student does not cancel his/her registration prior to the drop deadline, he/she will be held financially responsible for applicable tuition fees.

Preliminary Degree Audit

Degree Audits show progress towards a degree being earned at the University. It is mandatory that the student meet regularly with his/her advisor to keep a check on progress and plan for courses towards degree completion. The authority in degree audit is the published catalog which must be followed strictly.

This is a commonly accepted way of counting courses and is done in other universities that have General Education requirements for graduation apart from the Major.

All senior students are advised to complete the degree audit checklist from <https://www.fccollege.edu.pk/degree-audit-checklists/> according to the catalog year, ensuring grades are accurately recorded and no sections are left blank. Have the advisor review and endorse the checklist, then submit it (with advisor's signature/comments) to degreechecklist@fccollege.edu.pk. Check the transcript for any academic policy violations such as double-counted repeated courses or cross-listed course duplications.

The Academic Advising Center will review submitted checklists and contact you if needed - note that checklists without advisor endorsement will not be accepted.

Official Degree Audit

After students have completed all degree requirements from their catalog, an independent degree audit is done by the Academic Services Office to determine compliance with Catalog requirements and eligibility for degree awarding.

All majors or minors added to the degree after the primary major has been declared must be fulfilled from one catalog to pass the independent degree audit.

The Academic Services Office will email and send SMS to students prior to commencement each year, that reflects degree completion status and eligibility for participation in commencement. It is the student's and advisor's responsibility to ensure that the requirements for graduation have been met specifically as stated in the published catalog.

Ineligibility for Commencement

If requirements have not been met, the student will need to report on the matter to the Academic Advisor and Head of Academic Advising, as well as to appraise his/her department Chairperson of the situation. The degree application will need to be canceled at the Academic Services Office and participation in commencement will be withdrawn.

Urgent Degree Application Process

Students may apply for an urgent degree at any time before commencement, provided all degree requirements, as outlined above, have been fully satisfied.

If a student applied for commencement but their degree application was withdrawn due to noncompliance, they may reapply for an urgent degree once all requirements are met at the time of application. Requests for an urgent degree must be submitted to the Academic Services Office, require additional payment, and take approximately three weeks to process.

Addressing Anomalies in Departmental Offerings and Catalog Requirements

If anomalies exist between semester offerings and catalog requirements, students, along with their advisors, must immediately apprise the department Chairperson of the need to facilitate them by offering a course that is listed on their catalog whether this is a core or a required elective.

One Course in Lieu of Another

If the department decides to change a core or required elective from a past catalog through deletion or dormancy or inactive, while the catalog is still active, the department Chairperson must:

- Acquiring approval of the Dean of Faculty regarding any exception being made in lieu of such a course, justifying its applicability to one or all students.
- The Dean will then formally communicate this approval to the Academic Services Office in writing.

Policy on Privacy

Forman Christian College guarantees both the privacy and the confidentiality of all students' educational records as well as each student's right to access those records. The official custodian of student records is the Head of Academic Services Office with authorized processing carried out by the Academic Services Office staff.

Access to student records is limited to the student but can, with the student's written authorization and accompanied with the student ID, be granted to parents or guardian, current instructors, counseling and administrative staff with legitimate interests, or any party designated by the student. In cases involving **serious mental health concerns, life-threatening situations, or behavior posing imminent risk to self or others**, the University reserves the right to share limited, relevant information with parents or guardians **without the student's prior consent**, in accordance with applicable laws and institutional policies.

Authorized officials of the government, accrediting agencies, as well as persons bearing a lawful judicial order or subpoena may also request access to student records by presenting proper documentation with a reason supporting such access.

A student or former student has the right to access his or her records. However, the University may deny access if the student has unpaid financial obligations to the University. Requests for access or copies of records must be made in writing to the Head of Academic Services, who will comply within seven business days. Following review, a student may request any portion of his or her record to be edited, provided that supporting documentation is produced or available.



Academic Support for **STUDENTS**

Academic Support for Students

Forman Experience – Academic Support for Incoming Freshmen

The Academic Advising Center organizes a comprehensive two-week orientation program for all incoming freshmen. Freshmen Program at FCCU consists of three core components: the UNIV 100 course, freshman seminars, and social events designed to support new students' academic and social transition to university life.

UNIV 100: Foundation of University Education

This course is especially designed for freshmen to help them make the best use of their years of university study at FCCU. UNIV 100 course facilitates student integration into university life; assists them to make informed choices about their four years at FCC & beyond and equips freshmen with necessary academic skills to support their success. Each student will become more aware of his or her strengths and weaknesses in learning by working on a variety of skills. By examining their values and developing learning strategies this course will help students become successful in FCCU's liberal arts program.

All entering students must take UNIV 100 during their first semester or after they have finished the IEAP program. University 100 is a mandatory course that must be completed in the first year at FCCU. A student cannot drop UNIV 100 during the drop period. The instructor of the UNIV 100 course will be the student's academic advisor for the first two years until the student declares a major. Transfer students with 60 or more credits are exempt from UNIV 100 and will be assigned an Advisor by the Head of Advising directly.

Academic Advising Center

The Academic Advising Center (AAC) serves as the heart of student-centered support at FCCU, complementing the faculty advising system by directly engaging with students to ensure their academic success and personal growth. Unlike the Faculty Advising Office, which focuses on training and equipping faculty advisors, the AAC is a hands-on resource where students can access immediate guidance for course selection, degree planning, and navigating university policies. Each student is paired with a faculty advisor who provides mentorship in academics, career planning, and holistic development, while the AAC's Student Academic Support Officers and Student Success Advisor step in to address challenges such as probation requirements, academic difficulties, or uncertainty about university procedures. The center acts as a safety net, ensuring that no student is left without support even arranging continuity through the Head of Advising if a faculty advisor is unavailable. By balancing structured guidance with the freedom for students to take responsibility for their progress, the AAC embodies FCCU's commitment to fostering resilience, accountability, and success.

Faculty Advising Office (FAO):

The Faculty Advising Office (FAO) leads the faculty advising system at FCCU, which is designed to enhance student learning, academic success, retention, and timely degree completion. The FAO provides continuous professional development to faculty advisors, equipping them with current policies, effective advising strategies, and best practices to support students throughout their academic journeys.

The office organizes university-wide advising initiatives, including Advising Weeks, Academic/Advising Alerts, and the Annual Advising Fair. These initiatives assist students with course registration, academic planning, major declaration, and timely interventions. In collaboration with academic departments and student support units, the FAO promotes the meaningful relationships between advisors and advisees. It advances

FCCU's mission of academic excellence, holistic student development, and institutional effectiveness. As Pakistan's first structured faculty advising system, it demonstrates FCCU's ongoing commitment to student success and quality education and assurance.

Campus Accessibility Committee - Academic Support for Student with Disabilities

The Accessibility Committee at FCCU is responsible for determining and arranging reasonable accommodation for students with disabilities. The committee provides information and guidance to students, parents, and faculty regarding available support services and best practices for inclusion, while also advising the institution on improving accessibility across admissions, examinations, and other academic affairs in compliance with disability legislation.

Student with disabilities seeking support and accommodations can contact: a.vicerector@fccollege.edu.pk.



AWARDS

Awards

Vice Rector's List

The Vice Rector's List is published following the Fall Semester and the Spring Semester and includes names of students who completed at least four courses with a grade point average of 3.75 or better. Students with incomplete grades at the time of publication are not eligible. This achievement appears on the student official transcript, and a certificate can also be issued if 'requested'.

Honors at Graduation

Latin Honors

FCCU awards diplomas with Latin Honors to qualified students in each graduating class. The criteria and designation for graduation with Latin Honors are:

Summa Cum Laude (Highest Honors)	3.90 or above CGPA
Magna Cum Laude (High Honors)	3.70 to 3.89 CGPA
Cum Laude (Honors)	3.50 to 3.69 CGPA

Migrated (transferred) students are eligible to graduate with Honors if they have completed more than half of their credit requirements for graduation at FCCU.

Scholar Athlete Trophy

This trophy is awarded to the students who fulfill the following criteria:

- Senior standing (transfer students MUST have completed at least half the required credits to graduate at FCCU).
- CGPA at the end of Junior year 3.00 or above.
- No academic probations.
- Involved in team sports/track and field/swimming.
- Participated in a minimum of nine intervarsity competitions over the three years.
- Won first or second place in a minimum of three intervarsity competitions.
- Participated in intramurals.
- Won intramurals.
- Cleared the fitness test.
- Participated and won a position in at least two events at the Annual Sports Gala for two years.
- Consistently, displays sportsmanship.
- Clear of any disciplinary actions.

Medals

"All medal awardees should have a clean proctorial record".

Prof. M.S. Bhatti

In honor of Professor M. S. Bhatti, a renowned former teacher. The medal is awarded to the student securing the highest CGPA amongst the university candidates in the Faculty of Humanities.

Sir. Mian M. Shafi

In honor of Sir Mian M. Shafi, a renowned alumnus. The medal is awarded to the student securing the highest CGPA amongst the university candidates in the Faculty of Social Sciences.

Sir. Shaikh Abdul Qadir

In honor of Sir Sheikh Abdul Qadir, a renowned alumnus. The medal is awarded to the student securing the highest CGPA amongst the university candidates in the Faculty of Education.

Swami Ram Teerth

Dedicated to a renowned former teacher and saint; it is awarded to the student securing the highest CGPA amongst the university candidates in the Faculty of Computer and Mathematical Sciences.

Dr. J.H. Orbison

In honor of Dr. J. H. Orbison, a renowned former teacher and founder of the Biological Science Department. The medal is awarded to the student securing the highest CGPA amongst the university candidates in the Faculty of Natural Sciences.

Dr. E.D. Lucas

Dedicated to a renowned former Principal and Economist; it is awarded to the student securing the highest CGPA amongst the university candidates in the Department of Economics.

Prof. E.J. Sinclair

In honor of Professor E. J. Sinclair, a renowned former Principal. The medal is awarded to the student who secures the highest CGPA amongst the graduating class in the university.

Khan Bahadur Muhammad Sanaullah

In honor of Khan Bahadur Muhammad Sanaullah, a renowned alumnus. The medal is awarded to the student securing the highest CGPA amongst the university candidates in the Faculty of Business and Management.

Professors'

Sponsored by the faculty of the School of Management, these medals are awarded to the students securing the top three CGPAs amongst the university candidates in Bachelor of Business Administration.

Unilever

Unilever is a globally recognized brand for fast moving consumer goods; this medal is awarded to the student securing the highest CGPA amongst the university candidates in Bachelor of Business Administration specialization in Marketing and Sales.

Shezan

Shezan is one of Pakistan's oldest brands in the food industry; this medal is awarded to the student securing the highest CGPA amongst the university candidates in Bachelor of Business Administration specialization in Operations Management.

Askari Investment

Askari Investment Management is the asset management company of Askari bank and is at the forefront of providing investment [products for individuals and institutions; this medal is awarded to the student securing the highest CGPA amongst the university candidates in Bachelor of Business Administration specialization in Accounting and Finance.

Dr. C. M. Hussain

Dedicated to the father of Dr Wasiq Hussain (ex-Dean of the Faculty of Information Technology & Mathematics). Dr CM Hussain was a renowned mathematician. This medal is awarded to the student securing the highest CGPA (at least 3.0) amongst the university candidates in the Department of Mathematics without any F or W grades.

Dr. Peter H. Armacost

In honor of Dr Peter H Armacost, whose decade of dedicated service to Forman Christian College (A Chartered University) as the Principal and Rector, transformed FCCU once again into a premier educational institution of Pakistan. This medal is awarded to the student securing the highest CGPA amongst the University candidates in the Department of Psychology.

Dr. Robert F. Tebbe

In honor of Dr Robert F Tebbe, who served the cause of education in Pakistan for almost four decades as Professor of Chemistry at Gordon College, Rawalpindi and Professor and Principal at Forman Christian College (A Chartered University). This medal is awarded to the student securing the highest CGPA amongst the university candidates in the Department of Chemistry.

Aziz and Shaheen Haque

Dedicated to Col (r) Aziz-ul-Haque, a former alumnus of Forman Christian College (A Chartered University), and Mrs Shaheen Haque, parents of Dr Saad ul Haq, an MD in USA. The medal is instituted to encourage a spirit of healthy competition and promotion of the discipline of Geography. This medal is awarded to the student securing the highest CGPA (3.7 or better) in Bachelor of Science amongst the University candidates in the Department of Geography.

A. Karim

Dedicated to the father of Zafar Iqbal Qureshi, former Dean, faculty of Business and Management Studies at FCCU from 2005 to 2010; this medal is awarded to the student securing the highest CGPA amongst the University candidates in Business strategy in MBA.

Shafqat

Dedicated to the father of Dr Saeed Shafqat, Director Center for Public Policy and Governance at Forman Christian College (A Chartered University); it is awarded to the student securing the highest CGPA amongst the graduating class amongst the University candidates in Master of Public Policy and Governance.

Dr. Myron Miller

In honor of Dr Myron Miller who was founder chairperson, Department of Philosophy at Forman Christian College (A Chartered University). This medal is awarded to the student securing the highest CGPA amongst the university candidates in the Department of Philosophy.

Chaudhry Ahmed Saeed

In Honor of Chaudhry Ahmed Saeed an alumnus, philanthropist and President of the FCC Alumni Association. This medal is awarded to the student securing the highest CGPA amongst the university candidates in Bachelor of Science in Biotechnology.

Roll of Honor (Co-Curricular)

Co-Curricular Roll of Honor is awarded to the students who demonstrate outstanding performance in a specified category of co-curricular activity on the recommendation of the Societies' Board.

Certificate of Merit (Co-Curricular)

Co-Curricular Certificate of Merit is awarded to the students who demonstrate excellent performance in a specified category of co-curricular activity on the recommendation of the Societies' Board.

Shanky Lal Sheets Sports Medal and College Color

This medal and the College Color is awarded to the students who demonstrate outstanding performance in a specified category of a sports activity on the recommendation of the Societies' Board.

Qamar Zamani/Dr. K.F. Yusuf Gold Medal

Dr. Saeed Shafqat, Professor and Founding Director, CPPG, would like to institute a Gold Medal in the name of his two lifetime benefactors, his mother Qamar Zamani and his teacher Dr. Kaniz Fatima Yusuf, the first Women Vice-Chancellor of Quaid-e-Azam University, Islamabad. *“This Gold Medal will be awarded annually to the Mphil. or PhD student who writes an outstanding, innovative, and evidence-driven Mphil. thesis or PhD dissertation in the discipline of Public Policy and secures a high CGPA in the batch (3.8 or above)”*.

Dr. Henrick John Malik (Professor Emeritus) Gold Medal

This Gold Medal was instituted by Dr. Shazma Azeem from the Department of Chemistry in honor of her uncle Dr. Henrick John Malik (Professor Emeritus), a renowned former professor and warden of Velte Hall at Forman Christian College (A Chartered University), Lahore from 1955 to 1960. His services as an educationist are also acknowledged by many other universities internationally. This Gold Medal will be awarded to the student securing the highest CGPA amongst the University candidates in the Department of Statistics.

Sarfraz Begum

In Honor of the mother of Dr. Mian Wajahat, Professor of Biological Sciences and Controller of Examinations. *This medal is awarded to the student securing the highest CGPA amongst the University Candidates majoring in Bachelor of Science School of Life Sciences.*

Zafar Hayat Kahlon Medal

Dedicated to Zafar Hayat Kahlon, a lawyer of High Court Lahore (1978-2011), political and social figure from Faisalabad, father of Dr. Laila Zafar Kahlon, Associate Professor of Physics at FCCU. He received a Law degree from Lahore Law College in 1975. Zafar Hayat Kahlon was a well-known Social and Political figure of Tehsil Samundri, Faisalabad and is also known for his contribution at Samundri Bar Association and for providing justice to the deserving population. This medal is awarded to the student securing the highest CGPA (3.500 or above) amongst the university candidates in Bachelor of Science in Physics.

Justice (R) Rao Muhammad Naeem Hashim Khan Gold Medal

In honor of the father of Dr. Rao Raza Hashim, Dean of Faculty of Business and Management. This Gold Medal is awarded to the Bachelor of Business Administration student with the Specialization of Human Resource Management who secures the highest CGPA in the batch (3.0 or above).

Scholar Athlete Trophy

This trophy is awarded to the students who fulfill the following criteria:

- a) Graduating (transfer students MUST have completed at least half the required credits to graduate)
- b) CGPA above 3.000
- c) No academic probations
- d) Involved in team sports/track and field/swimming
- e) Participated in a minimum of nine intervarsity competitions over the three years
- f) Won first or second place in a minimum of three intervarsity competitions
- g) Participated in intramurals
- h) Won intramurals
- i) Cleared the fitness test
- j) Participated and won a position in at least two events at the Annual Sports Gala for two years
- k) Consistently displays sportsmanship
- l) Clear of any disciplinary actions



Liberal **ARTS**

Liberal Arts

Liberal Arts Education

Forman Christian College (A Chartered University) is currently the only University in Pakistan offering a true liberal arts Bachelors' education, which provides the students with a good balance of depth and breadth of learning. This is done by ensuring that in addition to their selected major, all Bachelor's students take a minimum number of courses in Humanities, Natural Sciences, Mathematics and Information Technology and Social and Behavioral Sciences.

Depth of learning gives students the opportunity for career preparation; it teaches them to think critically, communicate effectively, solve problems and become lifelong learners, while breadth of learning provides the basis for students to understand modern issues as they develop, and to live a quality life. It develops in them a healthy curiosity about the world and above all to be good citizens of their countries and the world.

The 4-year Bachelor's program is designed to prepare graduates to succeed in a wide variety of occupations and to advance to positions with higher responsibilities. The program provides a world-class education at Pakistani prices. Our graduates are able to enter postgraduate programs abroad without taking further courses.

Majors/Areas of Concentration/Minors

A liberal arts education gives students the opportunity to choose from a wide variety of majors, minors and specializations in a major in consultation with their academic Advisor. Students can also choose to do double majors, minors and more than one specialization after discussion with the relevant department and their Advisor.

Students can only earn one major from any Academic Department. If a student completes a double major, the two majors must be from two different Academic Departments.

General Education

An important part of the program for each student is General Education. Students must take a total of 17 courses in the General Education program. This includes courses in each of the following areas of human knowledge: Humanities, Social and Behavioral Sciences and Science and Mathematics.



ADMINISTRATION

Administration

Vice Rector's Office

Dr. Douglas E. Trimble

Vice Rector for Academic Affairs

Dr. Gloria Calib

Associate Vice Rector for Academic Affairs

Dr. Hee Sook Song

Associate Vice Rector for Faculty Affairs

Vice Rector Office: vicerecortoffice@fccollege.edu.pk

Academic Advising Office

Dr. Anam Muzamill

Head of Academic Advising

advisingcenter@fccollege.edu.pk

Academic Services Office

Mr. Rohail Shahzad

Head of Academic Services

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Accounts Office

Mr. Furqan Seith Mall

Chief Financial Officer

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Mr. Muhammad Ahmad Awan

Manager Accounts and Finance

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Admissions Office

Ms. Zainab Hamayun Sanik

Head of Admissions

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Campus Counseling Center

Ms. Bandgi Zafar Administrator

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Career Services Office

Ms. Mariam Khalil
Head of Career Services
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Controller of Examinations

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Deans

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Robin Dass
General Manager Food and Beverages
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Student Affairs Office

Dr. Elizabeth Schwaiger
Dean of Student Affairs

Department of
PUBLIC POLICY

Bachelor of Science in Public Policy and Administration

Program Purpose

The Bachelor of Science (BS) in Public Policy and Administration at Centre for Public Policy and Governance (CPPG), Forman Christian College (A Chartered University) is designed to inform and equip students with the knowledge, skills, and ethical foundation necessary to comprehend and address complex societal challenges through effective policy making, policy analysis and reformed administrative practices. It aims to foster a new generation of leaders who can analyze, design, and implement evidence-based policies that promote equity, sustainability, and the public good, and administer them effectively in an increasingly interconnected, competitive, and challenging societal environment. By bridging policy and its administration, the program prepares graduates to contribute meaningfully to public service, non-governmental organizations, the private sector, and international institutions, ultimately positioning them as instruments of positive change in democratic processes and public administration.

Program Description

The Bachelor of Science in Public Policy and Administration is a four-year undergraduate program spread over eight semesters that provides comprehensive general, interdisciplinary and focused education. Drawing from political science, economics, sociology, data analytics, etc. Its curriculum emphasizes critical thinking, evidence-based decision making and practical administration of policies informed by field experience/ internships, and research capstone projects. During the studies students will explore topics such as policy formulation and analysis, public administration and governance, global policy issues, sustainable development and ethics. The program includes general education, foundation and core courses, electives, and experiential learning opportunities, with a total of 124 credit hours required for graduation. It is structured to accommodate diverse interests, allowing specialization in economic, sociology development policies, public administration and management skills. The Degree shall have the title of “Bachelor of Science Public Policy and Administration.

Program Vision

A rigorous interdisciplinary academic program in public policy and administration that informs and inspires potential leaders who through technology-enabled innovative solutions create an inclusive, resilient, and prosperous society.

Program Mission

To prepare informed leaders in public policy and administration, capable of objectively and ethically comprehending public policies and administrative structures and procedures, leaders; imbued with collaborative spirit, reinforced by experiential learning and focused on addressing and reforming policy and administrative issues.

Program Objectives

Upon completion of the Bachelor of Science in Public Policy and Administration, graduates will be able to:
Develop a sound understanding of public policy, public administration, good governance structures in Pakistan.

- i. Apply analytical, ethical, and evidence-based approaches to identify, evaluate, and address policy and administrative problems.
- ii. Use quantitative, qualitative, and data-informed methods for policy analysis, administrative diagnosis, and decision-making.
- iii. Design practical and context-sensitive, transparent and inclusive policy and administrative responses that improve delivery of public good. Through a choice of Elective courses
- iv. Communicate and collaborate effectively in professional, civic, and public-service settings through research, writing, presentation, and team-based work through experiential learning (Internship).
- v. Effectively deliver in Governmental organizations, INGOs, NGOs, and think tanks, and succeed in competitive examinations.

Program Learning Outcomes

Upon successful completion of the Bachelor of Science in Public Policy and Administration graduates through the eightfold learning outcomes will be able to:

- i. Analyze public policy problems, institutional contexts, stakeholder dynamics, and policy outcomes using established policy frameworks and theoretical approaches. Evaluate alternative policy options by considering the contributing political, administrative, and socio-economic factors.
- ii. Integrate ethical considerations into policy design and administrative practices, addressing issues of equity, inclusivity, and sustainable development.
- iii. Utilize technological tools to gather evidence, formulate policy and administrative recommendations, and communicate effective findings to varied audiences.
- iv. Apply qualitative and quantitative data tools to collect, analyze, and interpret evidence and inform policy and administrative decision-making. Translate data insights into actionable recommendations for diverse policy audiences.
- v. Work collaboratively to design innovative solutions and administrative reforms to public policy and administrative challenges while drawing on interdisciplinary perspectives and practical case analysis.
- vi. Demonstrate leadership and communicate policy and administrative ideas clearly and persuasively in written, oral, and digital formats for academic, professional, and public audiences.
- vii. Integrate theory, methods, and field evidence in a capstone research project that addresses a real policy or governance challenge.

Upon successful completion of minor in Public Policy and Administration graduates will be able to:

- i. Analyze public policy problems and administrative problems: evaluate alternative policy and administrative reforms by considering the contributing political, administrative, and socio-economic factors.
- ii. Apply knowledge of public policy and administration in multidisciplinary settings.
- iii. Secure employment in Governmental organizations, INGOs, NGOs, and think tanks and secure employment through competitive exams.

Intended Audience of Program

Students who aspire to a career in public services in governmental, non-governmental and nonprofit organizations, the private sector and international institutions and those desiring to compete in competitive exams for service in the government sector.

Requirements for BS Major Public Policy and Administration

Core Courses	Elective Courses	Total Credits	Required MCGPA
11 (27+6=33 credits)	7 (21 credits)	54	2.0

Core Courses	Credits
CPPG 101	3
CPPG 111	3
CPPG 121	3
CPPG 201	3
CPPG 211	3
CPPG 221	3
CPPG 301	3
CPPG 311	3
CPPG 321	3
CPPG 498	3
CPPG 499	3
Core Courses Total	33
Elective Courses	Credits
CPPG 302	3
CPPG 303	3
CPPG 312	3
CPPG 313	3
CPPG 322	3
CPPG 323	3
CPPG 401	3
CPPG 402	3
CPPG 403	3
CPPG 404	3
CPPG 411	3
CPPG 412	3
CPPG 413	3
CPPG 414	3
CPPG 421	3
CPPG 422	3
Elective Courses Total	21

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54

University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements of a Minor in BS Public Policy and Administration

Core Courses	Credits
CPPG 101	3
CPPG 111	3
CPPG 201	3
CPPG 211	3
Core Courses Total	12
Elective Courses	Credits
CPPG 302	3
CPPG 303	3
CPPG 311	3
CPPG 312	3
CPPG 313	3
CPPG 321	3
CPPG 322	3
CPPG 323	3
CPPG 401	3
CPPG 402	3
CPPG 403	3
CPPG 411	3
CPPG 412	3
CPPG 413	3
CPPG 414	3
CPPG 421	3
CPPG 422	3
Elective Courses Total	6
300-400 Level Courses	

CPPG 101: Introduction to Public Policy (3 credits)

The course introduces students to the theory and practice of policymaking, showing how public problems are identified, prioritized, and addressed within political and administrative systems. It covers core concepts, key actors and institutions, stages of the policy cycle (agenda-setting, formulation, implementation, evaluation), and basic analytical tools used in policy analysis. Using comparative cases and examples from Pakistan and the region, students learn to apply critical frameworks and simple research methods to assess policy choices and outcomes. By combining case studies and practical exercises, the course prepares students for advanced study and careers in public administration and policymaking.

CPPG 111: Introduction to Public Administration (3 credits)

Introduction to Public Administration familiarizes students with the structure, functions, and processes of government organizations and public service delivery. The course explores bureaucracy, public management, intergovernmental relations, budgeting and financial management, human resource systems, and accountability mechanisms, with attention to administrative reform and ethics. Through Pakistan and regional examples, students examine challenges in service delivery, corruption control, and decentralization while learning practical tools for organizational analysis and performance improvement.

CPPG 121: Introduction to Economics and Public Policy (3 credits)

This course introduces students to the economic foundations of public policy and the ways in which governments respond to market failures, distributional concerns, and development challenges. It covers core economic concepts such as scarcity, incentives, supply and demand, inflation, unemployment, public goods, externalities, taxation, and fiscal policy, and shows how these concepts shape policy choices in the public sector. Students also examine how public policies affect growth, equity, efficiency, and welfare, with reference to Pakistan's economic and governance context.

CPPG 201: Public Policy Theories and Analysis (3 credits)

Pre-Requisite: CPPG 101

This course covers the formulation of public policies; its stages; theory and practice of policy analysis, issues surrounding the delivery, implementation, and evaluation of public policies; and current debates and perspectives in public policy. The course trains students to apply these frameworks to analyze policy formulation, implementation, and change, blending normative debates with empirical methods. Using case studies from Pakistan and South Asia, students critically assess competing explanations for policy outcomes and learn to choose appropriate analytical tools. By the end, they will be equipped to diagnose policy problems, compare theoretical interpretations, and craft evidence-based recommendations.

CPPG 211: Public Administration in Pakistan (3 credits)

Pre-Requisite: CPPG 111

Public Administration in Pakistan examines the country's administrative institutions and their historical evolution from the colonial civil service to today's federal-provincial and local government structures. The course analyses key challenges—devolution and decentralization, administrative capacity, accountability, corruption, and public-sector reform—using Pakistan-specific case studies and policy evaluations. Students engage with constitutional and legal frameworks, reform initiatives, and practical tools for assessing service delivery and governance performance. By the end, they will be able to diagnose institutional bottlenecks and design evidence-based reforms suited to Pakistan's political and administrative context.

CPPG 221: Statistics, Economics & Accounting in Public Policy & Administration (3 credits)

Pre-Requisite: CPPG 121

This course introduces students to the quantitative and financial tools used in public policy and administration. It combines basic statistics, economic reasoning, and introductory accounting concepts to help students interpret data, analyze public problems, and understand how resources are allocated and managed in the public sector. The course emphasizes evidence-based decision-making, fiscal awareness, and the use of analytical tools in policy formulation, budgeting, and administrative evaluation. Students will work with applied examples from Pakistan's public sector to develop practical skills in policy analysis and public financial understanding.

CPPG 301: Public Policy Process, Advocacy and Implementation (3 credits)

Pre-Requisite: CPPG 201

Public Policy Process, Advocacy, and Implementation explores how policies move from ideas to concrete outcomes, focusing on agenda-setting, stakeholder engagement, advocacy strategies, and the mechanics of implementation and monitoring. The course examines advocacy tools, coalition-building, communications, and tactics for influencing policy within political and administrative constraints, alongside common implementation challenges such as capacity gaps, incentives, and coordination failures. Using practical case studies from Pakistan and the region, students practice designing advocacy plans, implementation roadmaps, and monitoring frameworks. By the end, they will be able to bridge analysis and action—crafting realistic strategies to advance policy objectives and track results.

CPPG 311: District Administration & Local Government (3 credits)

Pre-Requisite: CPPG 211

CPPG 321: Data Analytics for Public Policy and Administration (3 credits)

Pre-Requisite: CPPG 221

This course introduces students to the use of data for evidence-based decision-making in public policy and governance. The course covers data collection, cleaning, visualization, and basic analysis using spreadsheets and statistical software, along with interpretation and communication of findings for policy audiences. Students learn to work with real-world datasets from government and development sectors to explore trends, evaluate programs, and support administrative reforms. By the end, they will be able to transform raw data into actionable insights, improving the effectiveness and accountability of public institutions.

CPPG 302: Technology, AI and Public Policy (3 credits)

Pre-Requisite: CPPG 201

This course examines how digital technologies shape public governance, service delivery, and regulatory choices. The course covers digital transformation, e-governance, data governance and privacy, cybersecurity, platform regulation, AI and algorithmic accountability, and innovation policy. Using Pakistan and regional case studies, students assess opportunities and risks, and learn practical tools for policy design, impact assessment, and stakeholder engagement. By the end, they will be able to evaluate technology-driven interventions and craft balanced, evidence-based policy responses.

CPPG 303: Public Policy and Communication (3 credits)

Pre-Requisite: CPPG 201

This course examines the role of communication in public policy formulation, advocacy, implementation, and evaluation. It introduces students to the principles and practices of policy communication, including audience analysis, framing, message design, media engagement, stakeholder communication, and the preparation of policy briefs, memos, speeches, and visual materials. The course also explores how governments, advocacy groups, and public institutions communicate policy decisions to diverse audiences in politically and socially complex environments. Using Pakistan-based examples, students learn to communicate policy ideas clearly, ethically, and persuasively across different platforms and audiences.

CPPG 312:E-Governance & Public Administration (3 credits)

Pre-requisite: CPPG 211

E-Governance in Public Administration develops the practical skills and theoretical understanding needed to provide e-governance in public-sector organizations. The course covers management theories, public-sector managerial competencies, ethical and accountable decision-making, project and change management, and stakeholder engagement, with insight into tools for e-governance. By the end, they will be prepared to effectively manage e-governance mechanisms in Pakistan and possess the understanding to foster new e-governance tools.

CPPG 313: Organizational Theory and Public Administration (Pre Req 211)

Pre-requisite: CPPG 211

Organizational Theory in Public Administration within the context of Institutional theory examines major theories and concepts used to understand, diagnose, and reform public organizations. The course covers classical and Weberian models, human relations and behavioral perspectives, contingency and institutional theories, and newer approaches to networks and organizational culture. Using Pakistan and regional case studies, students learn practical tools for organizational diagnosis, design, performance measurement, and change management, and explore how structure, incentives, and environment shape service delivery. By the end, they will be able to apply theoretical frameworks to recommend evidence-based reforms that improve effectiveness, accountability, and public value.

CPPG 322: Public Financial Management (3 credits)

Pre-requisite: CPPG 121

Public Financial Management examines the institutions, processes, and instruments that govern how public money is raised, allocated, spent, and audited. The course covers the budget cycle, public expenditure management, tax and revenue policy, intergovernmental fiscal relations, debt management, procurement, audit and internal controls, and performance-based budgeting, along with practical tools like fiscal forecasting and basic budget analysis. Using Pakistan-specific examples (budget processes, provincial finances, and donor-funded projects), students practice preparing budget notes, tracking expenditures, and diagnosing fiscal bottlenecks. By the end, they will be able to analyze fiscal trade-offs and propose reforms to improve transparency, efficiency, and accountability in public finance.

CPPG 323: Cost-Benefit Analysis (3 credits)

Pre-requisite: CPPG 321

This course introduces students to the principles and practice of cost-benefit analysis as a tool for public policy and administrative decision-making. It covers the identification, measurement, and comparison of costs and benefits associated with public projects, programs, and regulatory interventions. Students learn how to evaluate policy alternatives using economic reasoning, basic discounting, opportunity cost, and social welfare considerations. The course emphasizes practical application to Pakistan's public sector through case studies in infrastructure, education, health, environment, and governance.

CPPG 401: Public Policy and Development (3 credits)

Pre-requisite: CPPG 211

This course examines the relationship between public policy and socio-economic development, with a focus on developing countries. It explores how governments design and implement policies to promote economic growth, reduce poverty, improve human development, and achieve sustainable development goals. The course introduces key development theories, policy frameworks, and institutional approaches, and analyzes the role of the state, markets, and international actors in shaping development outcomes. Using Pakistan-focused and

comparative case studies, students learn how public policy interventions influence development trajectories and address complex challenges such as inequality, governance, climate change, and service delivery.

CPPG 402: Public Policy and the Impact of the 18th Amendment in Pakistan (3 credits)

Pre-requisite: CPPG 201

This course examines the impact of the 18th Constitutional Amendment on public policy, governance, and intergovernmental relations in Pakistan. It analyzes the devolution of powers from the federal government to provinces, changes in legislative and administrative authority, and the implications for policy formulation, implementation, and service delivery. The course explores sectoral impacts in areas such as education, health, social policy, and local governance, as well as challenges related to coordination, capacity, fiscal federalism, and accountability. Through case studies and policy analysis, students assess how constitutional reforms reshape governance and influence policy outcomes in Pakistan.

CPPG 403: Global Policy Processes and International Influences on Policy Making in Pakistan (3 credits)

Pre-requisite: CPPG 201

This course examines how global policy processes and international actors influence public policy making in Pakistan. It explores the role of international organizations, development agencies, transnational networks, and global policy frameworks in shaping national policy agendas, reforms, and implementation strategies. Particular emphasis is on United Nations SDGs, World Bank and IMF. The course analyzes policy diffusion, conditionality, global governance, and the interaction between domestic institutions and international pressures. Using case studies from sectors such as health, education, environment, and economic policy, students assess how global influences affect policy choices, sovereignty, and development outcomes in Pakistan.

CPPG 404: Environmental Policy and Sustainable Development (3 credits)

Pre-requisite: CPPG 201

The course focuses on the policy approaches used to balance economic growth with environmental protection. Topics include climate change regulations, natural resource management, energy policies, and the social contexts through which environmental challenges are addressed, preparing students to handle sustainability issues. The course addresses one of the most pressing, complex, and high-priority areas in contemporary policy development.

CPPG 411: Governance Challenges in Pakistan (3 credits)

Pre-requisite: CPPG 211

This course examines the key governance challenges facing Pakistan, focusing on institutional capacity, policy implementation, accountability, and public service delivery. It analyzes how political, administrative, and socio-economic factors shape governance outcomes and explores persistent issues such as corruption, weak institutions, policy discontinuity, fiscal constraints, and center–provincial relations. The course also evaluates reform efforts in areas such as decentralization, civil service, e-governance, and regulatory systems. Through case studies and policy analysis, students develop a critical understanding of governance problems and identify practical strategies for improving state effectiveness and public sector performance in Pakistan.

CPPG 412: Federalism and Provincial Autonomy in South Asia (3 credits)

Pre-requisite: CPPG 201, CPPG 211

This course examines federalism and provincial autonomy in South Asia through a comparative public policy perspective. It explores how federal systems allocate authority, resources, and responsibilities across different levels of government, and how these arrangements shape policy outcomes, governance, and development. The course analyzes constitutional structures, intergovernmental relations, fiscal federalism, and decentralization in countries such as Pakistan, India, Bangladesh, and Nepal. It also examines challenges related to regional disparities, political dynamics, capacity, and policy coordination. Through comparative analysis, students develop a deeper understanding of how federal systems operate in diverse political and socio-economic contexts.

CPPG 413: Role of Non-Profit Management (3 credits)

The course explores the principles and practices essential for effectively leading and managing nonprofit organizations. Students will examine key topics such as strategic planning, fundraising, and resource development, board governance, financial management, and program evaluation, while addressing unique challenges and opportunities within nonprofit sectors. Through case studies, practical exercises, and discussions, the course equips students with the skills to navigate the complexities of nonprofit leadership, foster stakeholder engagement, and drive mission-driven impact in diverse organizational contexts. Designed for aspiring nonprofit professionals, this course emphasizes real-world applications and ethical decision-making to prepare students for leadership roles in the nonprofit world.

CPPG 414: Human Rights, Minorities and Gender in Pakistan (3 credits)

This course examines the legal, institutional, and policy frameworks that protect and promote human rights in contemporary states, with particular attention to minority communities and gender as a cross-cutting dimension. It explores how constitutional guarantees, international human-rights instruments, and public policies shape the lived experiences of religious and gender groups, and how governments design, implement, and evaluate policies to ensure equality, non-discrimination, and accountability. The course provides practical knowledge of applying a rights-based approach to public policy, thus translating legal norms into concrete programs, institutional safeguards, and measurable outcomes in diverse and plural societies such as Pakistan.

CPPG 421: Informal Economy (3 credits)

Pre-requisite: CPPG 121

This course examines the nature, dynamics, and policy implications of the informal economy, particularly in developing countries. It explores why informal economic activities persist, how they relate to formal institutions, and their role in employment, livelihoods, and economic development. The course analyzes key issues such as labor informality, regulation, taxation, social protection, urban informality, and gender dimensions. Using Pakistan-focused and comparative case studies, students assess how public policy can engage with, regulate, and support the informal sector to promote inclusive and sustainable development.

CPPG 422: Political Economy and Public Budgeting (3 credits)

This course examines how political and economic factors shape public budgeting and resource allocation in the public sector. It explores the political economy of budgeting, including the roles of institutions, incentives, stakeholders, and power dynamics in fiscal decision-making. The course covers budget processes, public expenditure management, taxation, and fiscal policy, with a focus on how political considerations influence efficiency, equity, and accountability. Using Pakistan-focused and comparative case studies, students analyze how budgets are formulated, negotiated, and implemented, and how they affect development outcomes and public service delivery.

CPPG 498: Internship (3 credits)

In order to graduate, every student shall have to complete an internship program of at least 9 continuous weeks duration at any time after completion of second semester. The internship can be on site or virtual, can be paid or unpaid, and the performance review will be part of the student academic record.

CPPG 499: Capstone Project (3 credits)

Capstone Project Course bridges academic theory with real-world governance, public administration, and policy analysis. It acts as a professional-grade portfolio piece enabling students to demonstrate their ability to evaluate public systems, analyze regulatory impacts, or design strategic governance interventions. It is non-substitutable, supervised course.

Department of
**VISUAL AND
PERFORMING
ARTS**

Introduction

The Department of Visual and Performing Arts is dedicated to redefining the role of art in society through a multidisciplinary approach. From prehistoric cave paintings to contemporary digital media, artistic expression responds to the needs, values and visions of its time. Our program situates art within this dynamic tradition, inviting students to explore how visual culture both shapes and is shaped by social, political and personal realities.

With an integrated curriculum of studio practice and critical theory, students develop creative excellence alongside analytical clarity.

Highlights include hands-on studio work, interdisciplinary projects and community-based learning—distinctive experiences that connect curriculum to real-world impact, employing context-specific strategies tailored to the liberal arts standards. We invite prospective students, to join a vibrant learning community where artistic practice and critical inquiry prepare graduates to be conscientious leaders and responsible global citizens

Student Learning Outcomes Visual and Performing Art Program

Outcome 1: Creative thinking

- Students will demonstrate a well-rounded liberal arts education including historical, social and philosophical awareness contextualized in visual culture and the humanities in general.
- Students will demonstrate innovation, imagination and visual problem solving.

Outcome 2: Skill and Technique Development

- Students will be able to demonstrate familiarity with the selected media in and through studio courses.

Outcome 3: Communication

- Students will be able to communicate their ideas visually and verbally.
- Students will be able to demonstrate fluency with formal vocabulary to articulate and engage with current debates as well as comment on the works of art and art practices.

Outcome 4: Thinking/making/learning

- Students will demonstrate self-learning behavior which they will carry outside of the classroom. Through making, thinking, experiencing in the class to make connections and learning from their social engagements, physical environment and visual culture.

Minor in Art History & Minor in Visual Arts

The Dept. is currently offering:

1. Minor in Art History and
2. Minor in Visual Arts

It invites students from all disciplines to see art as more than reflection—it is a powerful engine for empathy, critique and civic engagement. Through a rigorous foundation in visual literacy and art history, students

examine the ethics of place, power and historical legacy, expanding their “medium” to include conversation, policy and human relationships.

These programs challenge students to move beyond representation, engaging in collaborative creative processes and embedding visual arts within the civic fabric. Practical experiences—like community projects, oral history archives, and participatory performances—connect campus to neighborhood, embodying FCCU’s ideals of “Education for the Common Good.” Graduates emerge as artist-citizens, prepared for diverse careers and meaningful, socially engaged lives.

Minor in Visual Arts and Art History

Introduction

Art serves as the collective memory of a civilization, offering a visual record of human struggle, triumph and identity that shape our shared social consciousness. By analyzing these creative legacies, we recognize that art is not merely a reflection of society but a vital engine for empathy and cultural critique. This program challenges students to move beyond mere “representation” toward “engagement,” where the creative process is co-authored with the public. By embedding the visual arts within the civic fabric, we fulfill the liberal arts mission of fostering responsible citizenship.

Central to this approach is a rigorous foundation in visual literacy and art history, which allows students to grapple with the ethics of place, power dynamics and historical legacies. They learn that their “medium” extends beyond paint to include conversation, policy and human relationships. This pedagogical shift addresses an evolving professional landscape. Graduates are uniquely prepared for careers bridging the arts and public life, such as museum curation, urban planning and community advocacy.

Furthermore, this program applies the “Education for the Common Good” by transforming the campus from an “institution” into an active partner with its neighborhood. Through shared projects, oral history archives and participatory performances, the degree creates a reciprocal exchange of knowledge. Ultimately, this approach reimagines the academic environment as an active public square, preparing students for a meaningful life as artist-citizens.

- *These minors’ welcome students from any major that can benefit from the interdisciplinary nature of this program*
- *Core for one minor automatically becomes an elective for the other/ any other being offered or cross listed from other department in consultation with the department Advisor & Chair.*

Requirements for Minor in Visual Arts and Art History

Core Courses	Elective Courses	Total Credits	Required MCGPA
3(9 credits)	3(9 credits)	18	2.0

Core Courses for Minor in Visual Arts	Credits
ARTS 101	3

ARTS 111	3
ARTS 102	3
Core Courses Total	9

Core Courses for Minor in Art History	Credits
ARTS 101	3
ARTS 200	3
ARTS 201	3
Core Courses Total	9

Elective Courses	Credits
ARTS 103	3
ARTS 104	3
ARTS 105	3
ARTS 106	3
ARTS 121	3
ARTS 200	3
ARTS 203	3
ARTS 204	3
ARTS 210	3
ARTS 213	3
ARTS 300	3
ARTS 301	3
ARTS 303	3
MUSC 101	3
MUSC 250	3

Course Descriptions

Art Courses

ARTS 101: Introduction to Art (3 credits)

The course is an interesting mix of theory and practice. Where students will not only be introduced to the practice of art but will also dive into the process of art making and the mapping and documentation of the creative process. What is art? Can one define art? Why and how is it significant? Can one learn to read and

create art? These are the questions that will be explored during this course. Students will learn to look at and define art through new perspectives such as art as expression (individual and art), art depiction and celebration of beauty (art for the sake of art, art's function to beautify and embellish), art as the byproduct of fear (fear and creative process), art as resistance, Anything and everything is art (Marcel Duchamp), art as War (ISIS and Talin, Zizek), art is violence (Francis Bacon, Ann Bogart), art is a tool (socially engaged art, instrumentalization of art by development organizations and state institutions), art is memory (mythologies, culture, stories and documentation), art is labor of love (Artivism, social engagement, art for social change), everyone is an artist (Joseph Beuys, art that can't be art, art works not created in the name of art). The course will focus on the design, context and means of art and art making by exposing students to the basic principles and elements of art as well as introducing them to different media such as painting, drawing, sculpture, performance, text, video, photography and installation.

ARTS 102: Painting I (3 credits)

This is a studio course directed to teaching the fundamentals of painting. Students will learn to see color, shape, line and tone and to relate as well as use these elements together in a composition. The course will expose students to various media using traditional and experimental approaches in painting such as landscape, still life, figurative and non-objective. Students will understand the symbolic significance of certain colors and their use in translating ideas. Various methods and techniques of applying paints will also be explored. Students will be encouraged to find their own style of painting which is more comfortable and personal to them for materializing their ideas in the form of paintings.

ARTS 103: Sculpture I (3 credits)

This Introductory level course will teach students the basic elements, techniques and history of Sculpture. In order to learn the additive and subtractive or reductive processes students will explore various media such as wood constructions, clay modeling, plaster casting and stone carving thus will become familiar with the new method and technique specific to each material. Students will learn the handling and tools for each material and will explore the potential and limitations of the materials. The course focuses on the form, whether representational or non-representational, and its relationship with the space and viewer as well as its design.

ARTS 104: Photography I (3 credits)

The camera is by far the most radical invention that changed the course of history. Images now dominate our world, and we make sense of this world in and through images. This course is a combination of practice and theory that aims to help students learn the basics of photography such as understanding the camera and its functions, learning how to see, composition and translating ideas into visuals. This course will explore the role of cameras in the contemporary world and the concept of curation and archiving in visual culture. The main focus is on teaching students to dissect images in order to understand the power of an image and how and what makes a successful image successful. Students will be exposed to a variety of images from various genres in photography in order to understand the diversity of photography as a medium and also its agency as a medium by discussing canonical works as case studies through the course of time, thus being introduced to the history of photography as well.

ARTS 105: History of South Asian Art (3 credits)

This cross-cultural exploration while considering art and other creative practices as the medium of investigation traces the development of mind, knowledge and civilization in South Asia as informed by various religious streams. In South Asian culture, art is created by inscribing it with diverse content and

meaning; emotions, historical and mythological events or more often political or religious ideologies. Framed around two subthemes, the course aims to examine how politics of power and religion have inspired creative geniuses. Through a diverse range of case studies, students will analyze how art has been used to convey spiritual ideas, evoke emotions and create sacred spaces.

ARTS 106: Women, Art & Politics (3 credits)

This interdisciplinary course explores the intricate relationships between women, art and politics, examining how women's experiences, identities and perspectives have both shaped and been shaped by art and political movements. Beginning with the 19th-century roots of women artists in feminism, the course will delve into the impact of second-wave feminism on art from the 1960s to the present. Moving beyond a narrow Western perspective and with a special focus on South Asia, the course will adopt a global lens. Students will engage with works by key women artists across cultures, theories and methodologies, analyzing art through the intersecting categories of gender, race, culture, sexuality and class. In the broader scope, the course investigates how women have used art to resist, subvert and transform societal norms and power structures.

ARTS 111: Drawing I - Fundamentals (3 credits)

Drawing is the most essential tool for understanding the world as well as translating it in visual form. It remains the pivotal activity to the works of many artists. Drawing is the basis of it all, it is where exploration, development and refinement of ideas happen. This is a basic level course where students will learn mark making through various media in drawing as well as learn the elements and principles of image making such as line, shape, form, color, movement, perspective and composition. From drawing out of imagination and then gradually building on drawing from observation the course will encourage ways of looking such as various viewpoints and perspectives as well as the conscious use of all the elements and principles in drawing/designing a visual.

ARTS 121: Drawing II - Drawing Methods and Techniques (3 credits)

This course will teach advanced drawing methods and techniques that enhance observation as well as help students draw realistically. They will be taught drawing with measurements, starting with the objects/still life and proportions in relationship to space both negative and positive. Students will also experiment with different media in drawing. They will be encouraged to find their choice of medium and also their style of drawing through extensive practical exercises and projects.

ARTS160: Theatre I - Introduction (3 credits)

This course is process oriented and participatory. It focuses on the creative learning process by exposing students with various theater making methodologies and techniques through practical workshops and training. It addresses all the stages of theater production from gaining technical skills in acting to developing the script. Students learn to improvise and perform on stage and get the basic know-how in (production) design and artistic management. The course brings in acting techniques from diverse backgrounds, western (Boal, Stanislavsky, Brook) as well south Asian. Importantly, the course strengthens the practice with theoretical knowledge for the intellectual growth of the students. Majorly focusing on the historic trends and developments in theater tradition both global and local, the course also touches upon the theoretical perspectives of performance studies at the basic level.

ARTS 200: History of Art I - Prehistoric to Renaissance (3 credits)

We understand the history of humanity through art. From prehistoric depictions of wild beasts to contemporary abstraction, artists have addressed their time, place in history, and have expressed universal human truths for tens of thousands of years. This course chronologically traces human creativity from as early as 40,000 years ago up till the Renaissance, providing a systematic exploration of major civilizations and their culture, social structure, technology, iconography and architecture. The aim of the course is to acquaint students with visual imagery from prehistoric and ancient civilizations, spanning about 40,000 years. Students will develop skills in image recognition, vocabulary formation, and the use of visual evidence in argument and interpretation.

ARTS 201: History of Art II-Renaissance to Contemporary (3 credits)

The study of the history of art cultivates a deep understanding of humanity's fundamental need to create. It traces how artists across various time periods and regions responded to their environment, society, and religion, and thereby produced artworks of lasting resonance and insight into the cultural practices of the time. This course is like time travel that constantly moves back and forth through the windows of time in order to understand the canonical works of art and as to why, when and where they were created, and furthermore making connections to the present. It aims to instill in them the analytical approach needed to understand why and how things have shaped through the course of time. The course encourages students to compare periods, styles, materials and subjects etc. and to make connections. Within each chronologically set period, discussions will be thematic with particular attention to portraiture, religion, resistance, arts, technology, gender identities and the role of the artist in society. In addition, frequent connections will be made with contemporary art, with the aim that students will learn to not only stay linked to art being made today, but to also derive inspiration and relevance from the content studied in the course. The course will, through assignments, introduce students to not just writing about art, but also creative formalization of their ideas. More importantly, this course aims to make history a relevant tool for students to understand visual culture through connecting past and present.

ARTS 202: Theatre II: Improvisation and Devising (3 credits)

Prerequisite: ARTS 160

The course focuses on exploring the body as a tool for acting. The aim of this course is to have sufficient knowledge of the body and to be able to improvise a performance with and without a script. Students will warm up with yoga and theater exercises. They will be introduced to inhibition breaking and trust building exercises. The students will learn interpretative skills (devise a character in a situation and engage with other characters in that situation in order to create sense of reality) technical skills (voice, movement, speech etc.) and knowledge of the improvisation and devising process (to be able to improvise and devise scenes for both unscripted and scripted performance). The students will improvise solo and group performances in class in order to learn the above skills and will have discussions about the performances. The course will expose and engage students in Improvisation games to justify the character through performance as well as understanding and show commitment to imaginary circumstances in order to create convincing realities.

ARTS 203: Art and Architecture of the Muslim World (3 credits)

Prerequisite: ARTS 01 or any other Level one course or permission of the instructor

This course examines the art and architecture of the Muslim world as it developed through centuries of adoption, adaptation and innovation which emerged as Islam spread throughout Eurasia and Muslim communities came into contact with other Muslim and non-Muslim societies. Focusing on the early modern empires – Ottoman, Saffavid and Mughal and looking at a number of prominent sites, buildings and art objects

plurality of traditions and artistic practices of the Muslim world are explored. All along the way, buildings and art objects will serve as snapshots that open a window into the social, cultural, economic and political life of these Muslim societies.

ARTS 204: Contemporary Art in South Asia (3 credits)

Prerequisite: ARTS 01 or any other Level one course or permission of the instructor

This course introduces the student to contemporary art in South Asia with a focus on artistic movements as they unfolded during the 20th century and onwards. Doing so, it explores the history and legacy of South Asian art and its significance and relation to the contemporary international art world. Students will see the works of artists and different art movement which occurred, starting from the pre-partition (British Raj) and how it entered the modern era up to the contemporary times. Students will be encouraged to examine issues arising from historical and social histories, formation of identities and ideologies and will learn to analyze artwork within these contexts. Issues of expectations, creation and reception of South Asian art in the global markets is also explored.

ARTS 210: PAINTING II (3 credits)

Prerequisite: ARTS 111 or with permission of the instructor.

Building on foundational painting skills, this course explores still life and landscape subjects through direct observation. Students develop their ability to depict form, light, and color using diverse painting techniques. Through guided exercises and projects, the course encourages both technical refinement and individual artistic expression in observational painting.

ARTS 211: Drawing III- Advanced Drawing (3 credits)

Prerequisite: ARTS 111/ARTS 121

This course focuses on developing the skills to observe and draw the human figure, and may involve the study of nature, plants, animals and objects in their environment. It develops a better understanding of anatomical and muscular structure, proportion, and rendering of form. Students will learn the basic human anatomy, major joints to understand the form of the human body in various poses and the role and engagement of muscles and joints. The course will focus on the realistic translation of the human form composed in different spaces with relation to various objects and architectural structures serving as backdrop. Emphasis is laid on developing observation, memory, perceptual skills, and individual style or way of expression.

ARTS 213: Calligraphy (3 credits)

The course enables students to embark on a journey through the sacred art of Islamic calligraphy, where the beauty of the written word meets spiritual devotion. This combined theory and studio course introduces students to the timeless elegance of traditional khat styles, exploring their significance in Islamic art and culture. Students delve into the evolution of scripts over the centuries and practice Khata-Nastaliq, a popular South Asian style that's been a staple since the Mughal Empire. Through hands-on practice and research, students unlock the secrets of this revered art form and connect with its spiritual essence, learning the foundational techniques of shaping letters.

ARTS 300: Images of Terror: Violence in Visual Culture (3 credits)

The course focuses on performance and the discourse on the interplay of art and violence. It examines the inevitability of violence in our daily lives. We experience, talk, fear, consume and even produce violence unconsciously. We produce more violence even while trying to counter violence. Thus, the course creates a

debate around questions that link human aesthetics to the experience of horror and terror: How and when did violence become an integral part of our popular visual culture? Has the wave of terror changed our aesthetics? Do horror and violence together intrigue, thrill and entertain us? In order to answer these questions, the course looks at the discourse between art and violence, and their interplay. It looks into the aesthetics of intimidation, fear and resistance by going through art projects such as the post-9/11 works of Pakistani artist, Marina Abramovic' Rhythm, Tania Bruguera' Self Sabotage, ISIS and Taliban released videos and images, media coverage of bomb blasts, and political performances around the world after Arab Spring, Occupy Wall Street movement, hate graffiti and military parades, as well as Hollywood movies like Saw and Pakistani movies like *Maula Jutt* and *Waar*.

ARTS 301: Performance Art (3 credits)

The course will explore the body as a medium and also as a site of performance. Students will learn the potential of the body interacting with space and objects and the awareness of this interaction through practice across three key areas: text/theory, body/ movement and materials/space. Students will practice and create small pieces throughout the course. By incorporating theoretical knowledge of key issues in and around performance, methods of reflective practice and processes of documentation and criticism, students will be engaged in debates and will learn to theorize their creative process.

ARTS 302: Survey of Pakistani Art (3 credits)

Prerequisite: ARTS-101 or any other Level one course with the permission of the instructor.

This course explores the vibrant world of contemporary art in Pakistan, from its rich history to its significance in the global art scene. It takes the student on a journey through various art movements, starting from the pre-partition era to the present day, introducing and examining the works of influential Pakistani artists. Students delve into the complexities of identity, ideology and social histories that shape artistic expression, learn to analyze artworks within their cultural and historical contexts. Through discussions and critiques, they shall gain a deeper understanding of Pakistani art and its contribution to the global art world.

ARTS 303: Mark Making (3 credits)

The studio-based course offers a relevant introduction and insight into the process and practice of making art. It seeks to provide students with opportunities to work with a variety of traditional and non-traditional mediums, develop skills in observation and technique while learning to solve visual problems. The work done in the studio will be accompanied by lectures and readings that trace the historical significance of each project.

ARTS 400: The Social Turn: Art and Social Practice (3 credits)

Social turn was first used in 2006 to describe the recent return to socially engaged art that is collaborative, often participatory and involves people as the medium or material of the work. The course is an artistic exploration of socially engaged art through practice and theory. This course will include the key critical and disciplinary debates in order to understand art which involves people and communities in debates, collaboration, or social interaction. The students will understand the meaning of collaboration as well as the significance of participation of communities in making art and how it guides the creative process. They will study and explore the term Artivism as well as look at the development of art and its discourses critically. Students will also understand practice through engagement by collaborating with communities and designing research-based art projects. The primary aim of this course is to investigate the conception, process and materialization of art within a society and its impact on society. Students will be engaged in debates concerning ethics of public interventions and creative collaboration and in some cases its discontent.

Music Courses

MUSC 101: Music Practicum I - Vocal (3 credits)

This performance-based course focuses on developing vocal performance skills in learners. Students learn 12 notes and 10 thaats in two vocal performance I & II courses in two different semesters. The vocal performance I course focuses on the recognition and verbal reproduction of the twelve notes that constitute a scale/octave. The recognition includes the notes in the lower and upper octaves. Students first recognize the notes in the lower tetra chords and the upper tetra chords then learn to reproduce these notes verbally in a similar fashion. They get command over the notes by learning various combinations and permutations of the notes. They also learn to tune the tanpura, which is an essential component of the course. Students are not allowed to take the vocal performance I course along with the vocal performance II course in the same semester.

MUSC 201: Music Practicum II – Vocal

- Definition of Nad, Shurti, Sarwa, Alankar, Rag, Vadi Samvadi, Anuvadi, Vivadi, PakarTaan, Meend, Andolan, Gamak Kan, Alaap, Bol, Taan, Barhat, Dugan, Tigan, Chugan
- Forms of Classical Music Dhrupad, Khayal, Thumri, Hori, Lakshangeet, Sargam, Tarana
- Taals - Ektaal, Jhaptaal, Rupak Ragas - Bhairvai, Rageshveri, Kaushik Dhun
- Life Sketches - Amir Khusro and Sultan Hussain S Sharqi

MUSC 301: Music Practicum III –Vocal

- Thaat System
- Ten Thaats System
- Timetable of Ragas
- Process of evolving ragas from thaats Sudh, Chaya and Sankeeran ragas
- Life Sketches - Ustad Amir Khan, UstadBadr Ghulam Ali Khan
- Practical - Revision of previous ragas, Raga Desh, Bhopali, Bindrabani, Sarang
- Taals - Ektaal Slow, Rupak, Dhamar

MUSC 401: Music Practicum IV –Vocal

- Confluence of all the previous rafas and theory with an advanced approach.
- This section of the course would add some ragas such as, Puriya Dhanashri, Mianki, Todi, Marwa
- Life Sketches - Nazakat Ali Khan & Salamat Ali Khan, Amanat Ali Khan & Fateh Ali Khan, Roshan Ara Begum
- Taal - Practice and mature previous taals along with Ada Chautala, SulFakhtah

MUSC 210: Music Practicum II –Instrumental (Sitar/Flute/Other) (3 credits)

This hands-on performance-centered course familiarizes students to musical instruments like Sitar or Flute, its structure, tuning, sitting position, angle of approach, techniques and the performance of *iver Notes and of Raga Aimen*.

MUSC 220: Music Practicum III –Instrumental (Tabla/Other) (3 credits)

This hands-on course focuses on developing Tabla playing skills in students. They learn the Bols of Teen Tala along with the recitation of these with Talis, playing the Tala with their own hands.

MUSC 250: Music Appreciation (Introduction to Music, Theory, & Terminologies) (3 credits)

The course focuses on genres, terminologies, and forms of South Asian classical and semi classical music, and classifications, differences and improvisation of Ragas and Thatahs. The course exposes them to meanings, philosophies and role of music in history.

MUSC 400: Musicology (3 credits)

Prerequisite: MUSC 250

The Music Appreciation I course focuses on appreciation of the stylistic features of big six South Asian gharanas based on the following elements:

- Vocal Quality
- Use of Timber
- Use of Volume
- Treatment of Sawaras
- Treatment of Lai
- Selection of Ragas
- Style of Composition
- Method of Improvisation
- Treatment of Vilambitlai
- Treatment of Duratlai
- Types of Tanas
- Alankaras

Department of
CHEMISTRY

Introduction

The Department of Chemistry is one of the oldest at Forman Christian College (A Chartered University). Its rich tradition as a leading institution of chemical knowledge is reflected by eminent scholars of chemistry who served it. Special mention must be made of Prof Dr Carter Speers (1940-1951), who was Head of the Department and Professor of Technical Chemistry at the University of the Punjab; Dr RF Tebbe, an eminent Organic Chemist and learned scholar, who served the Department as Professor and later as Principal of the College for twelve years; and Prof Khairat M Ibne Rasa, an internationally known chemist, who was Professor of Organic Chemistry and Head of the Department.

Chemistry is an important discipline of scientific knowledge that plays a central role in our daily life. It is vital to our understanding of molecular transformations that are important to the production of food, medicines, fuel and countless other manufactured and extracted products. Chemistry helps to sustain the natural resource base for life and contributes to environmental protection and economic development. Design and synthesis of new organic and inorganic molecules, new methods of synthesis via microwave or ultrasonic irradiation, isolation and structural elucidation of novel natural products, development of biopolymers for targeted drug delivery, use of electroanalytical techniques for studying molecular interaction, computer aided drug design and discovery, and design and synthesis of catalysts for photocatalytic degradation of textile and other waste molecules are some of the areas of focus.

Career opportunities for Chemistry graduates are available in education, R&D organizations, research, and a wide range of industries such as pharmaceuticals, fertilizers, cement, leather, textile, paints and dyes, petroleum, sugar, soap and detergents, food and beverages, cosmetics and many more.

The Department of Chemistry at FCCU currently offers BS (4-year), MPhil (2-year) and PhD (3-year) programs. The main objective is to train students intellectually and professionally to work in a variety of capacities in their future life providing informed and skilled human resources for national and international needs. It offers a wide range of courses in analytical, inorganic, organic, physical, applied, biochemistry, and other interdisciplinary areas.

The Department of Chemistry has a rich research culture with extensive research facilities and excellent faculty, all being PhD and HEC approved PhD supervisors. The teaching/research equipment available in the Department includes GC-MS, HPLC, CHNSO Analyzer, Atomic Absorption Spectrophotometer, Flame Emission Spectrophotometer, TGA-DSC, FT-IR Spectrometer, Cyclic Voltammetry, 60 MHz Benchtop NMR (^1H , ^{13}C , DEPT, COSY), Magnetic Susceptibility Balance, Digital Polarimeter, Freeze Dryer, Plate Reader, among others.

The Department of Chemistry is part of the Faculty of Natural Sciences.

Bachelor of Science in Chemistry

The program prepares the students to demonstrate the ability to understand, analyze and address the real-world problems, from academia to industry, using Chemistry; and provides them with plenty of opportunities to excel both in the diverse job market, and in future learning.

Program Objectives

- i. To educate students in foundational concepts of major disciplines of chemistry (inorganic, organic, physical, analytical, applied and biochemistry).
- ii. To educate students the fundamental chemistry lab methods/techniques and skills to conduct experiments and communicate the findings.
- iii. To equip students with skills to solve real-world problems related to chemistry.
- iv. To educate students how to think critically about the questions/problems pertaining to the area of chemistry.
- v. To promote awareness among students about ethical aspects of chemistry such as safety, environmental issues and plagiarism policy.
- vi. To educate students about employment opportunities and entrepreneurship skills based on the knowledge of chemistry.
- vii. To train students with effective professional skills (communication & presentation) in the area of chemistry.

Program Learning Outcomes

- i. **Knowledge**
Demonstrate an understanding of major concepts, theoretical principles and experimental findings in chemistry.
- ii. **Lab Activities**
Perform various laboratory experiments successfully by applying fundamental chemistry methods and techniques.
- iii. **Real-World Application**
Apply chemistry concepts to address real-world problems through effective solutions.
- iv. **Critical Thinking**
Demonstrate critical thinking and analytical reasoning skills in the basic areas of chemistry
- v. **Ethical Values**
Demonstrate understanding of ethical issues, norms, and implications in the domain of chemistry related to research, safety and environmental issues.
- vi. **Career Opportunities**
Describe the range of career and higher education opportunities available to a chemistry graduate.
- vii. **Effective Communication**
Students will demonstrate adequate scientific communication and presentation skills in the domain of chemistry.

Requirements for BS Chemistry Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
11 (42 credits)	12 credits	54	2.0

Core Courses	Credits
CHEM 250	4
CHEM 261	4
CHEM 270	4
CHEM 311/ENVR 311	4

CHEM 320	4
CHEM 330	4
CHEM 350	4
CHEM 361	4
CHEM 370	4
CHEM 498	3
CHEM 499	3
Core courses Total	42
Elective Courses	Credits
Any course offered by the department out of which 3 courses of 400 level	12
Electives courses Total	12

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Recommendations

During their Freshman and Sophomore years, students are recommended to take the following courses which will prepare them for higher level courses: CHEM 150, CHEM 160, CHEM 170, CHEM 173, CHEM 250, CHEM 260, CHEM 261 and CHEM 262. Taking these courses at the initial stages will help them to complete the degree on time with sufficient number of courses in their desired specializations.

Recommended Elective Courses: In order to gain sufficient mastery of the subject, a student majoring in chemistry is advised to take as many elective courses as he/she can, giving special attention to his/her desired area of specialization, and selecting at least one course from each of the following categories.

Inorganic-Analytical Chemistry

- CHEM 413: Instrumental Methods of Analysis
- CHEM 450: Advanced Inorganic Chemistry
- CHEM 453: Chemical Applications of Group Theory
- CHEM 454: Inorganic Electronic Spectroscopy
- CHEM 455: Inorganic Reactions Mechanism

Organic-Biochemistry

- CHEM 260: Principles of Organic Chemistry
- CHEM 331: Principles of Biological Chemistry
- CHEM 462: Spectroscopy of Organic Compounds
- CHEM 464: Advanced Organic Chemistry
- CHEM 465: Natural Products and Medicinal Chemistry

Physical Chemistry

- CHEM 271: Quantum Chemistry
- CHEM 470: Polymer Chemistry
- CHEM 471: Advanced Physical Chemistry
- CHEM 473: Surface and Solid-State Chemistry

Requirements for the Minor in Chemistry

Minor in Chemistry is open to students having basic science background at Intermediate and A Level.

Core Courses	Elective Courses	Total Credits	Required Minor CGPA
3 (12 Credits)	6 Credits	18 credits	2.0

Core Courses	Credits
CHEM 150	4
CHEM 160/ENVR 160	4
CHEM 170	4
Core courses Total	12
Elective Courses	Credits
Any other courses of 200 level or above.	6
Electives courses Total	6

Course Descriptions

CHEM 100: Introduction to Chemistry (3 credits)

Open for those students who have not taken chemistry at Intermediate or A Level Chemistry as a basic science, matter and states of matter, elements and periodicity, atomic structure, concept of mole and elementary stoichiometric calculations, acids and bases, elementary redox reactions and electrochemical cells, organic functional groups and major classes of organic compounds and their importance, and environmental aspects of chemistry.

CHEM 150: Introduction to Inorganic Chemistry (4 credits)

Prerequisite: Intermediate or A Level Chemistry

Various theories of bonding include valence bond theory, molecular orbital theory, Werner's theory, crystal/ligand field theory, three center bonds, bonding theory of metals and intermetallic compounds, bonding in electron deficient compounds, hydrogen bonding, shapes of molecules (VSEPR model).

CHEM 160/ENVR 160: Introduction to Organic and Biochemistry (4 credits)

Prerequisite: Intermediate or A Level Chemistry or CHEM 100

Bonding and structure of organic compounds, study of hydrocarbons including additions to multiple bonds and substitution reactions of benzene, petroleum products, chemistry of food and its components including carbohydrates, proteins, lipids, nutrition, and caloric intake.

CHEM 161: Foundation of Chemistry Entrepreneurship (3 credits)

Chemical entrepreneurship is a rapidly growing field that is especially important to contribute to a knowledge-based economy. The objective of the course is to provide students with the skills necessary to start their own business based on chemistry knowledge. There are many business opportunities available for chemistry students that this course will focus on such as herbal products, cosmetics, essential oils and perfumes, skin care products and so on.

CHEM 170: Introductory Physical Chemistry (4 credits)

Prerequisite: Intermediate or A Level Chemistry

Physical states of matter: structure and physical properties of gases, liquids, and solids, Bragg's Law, intermolecular forces. Kinetic Theory of Gases: deviation from ideal behavior, Van der Waals equation, distribution of velocities, Boltzmann Distribution Law. Chemical Kinetics: rate of reaction, reaction velocity, rate laws, integrated rate law, half-life of reaction, determination of order of reaction. Chemical Thermodynamics: First Law of Thermodynamics and Thermochemistry. Quantum Chemistry: development of quantum theory, wave mechanics, Schrodinger equation, wave function.

CHEM 172: Chemometrics (3 credits)

Prerequisite: Intermediate or A Level Chemistry

Chemometrics is one of the important chemistry courses that undergraduate students are being taught worldwide. It mainly helps the students to get deep insight into chemistry while applying the various concepts of mathematics and statistics tools. A few basic concepts of mathematics starting from order of operations to graphs and few statistical tools specific to chemistry problems will be discussed in this course to understand lab experiments and instrumental analyses.

CHEM 173: Introductory Nuclear Chemistry (3 credits)

The course focuses on structure and properties of atomic nuclei and related phenomena; nuclear structure, nucleons, binding energy and decay processes; fission and fusion; isotopes and isotopic labeling; nuclear transmutation; applications in the field of medicine and energy; nuclear power reactors; future prospects.

CHEM 180: Introduction to Forensic Chemistry (2 credits)

Prerequisite: Science Background

Introduction to forensic chemistry, chemical analysis in forensic science, time of death; blood, DNA in forensic, fingerprinting; polymers and fibers; firearms, narcotics, toxicology; case studies.

CHEM 181: Introduction to Food Chemistry (2 credits)

Prerequisite: Science Background

Introduction to food science and technology, food components, food processing, commercially important fats and oils, manufacture of regular and calorie-reduced margarines, fat replacers, flavor enhancers, hydrolyzed vegetable protein, browning reactions, fluid milk and pasteurization, yogurt and cheese, vegetable pigments and flavoring compounds, and food preservatives.

CHEM 182: Fundamentals of Textile Chemistry (2 credits)

Prerequisite: Science Background

Chemistry of natural and synthetic fibers, mercerization, decorative fabrics, textile dyeing and printing, sizing, bleaching and pretreatment of fibers, fashion designing, textile raw materials and environmental issues of textile industry.

CHEM 250: Chemistry of Main Groups Elements (4 credits)

Prerequisite: Intermediate or A Level Chemistry

Structural characteristics, reactivities, simple compounds, coordination compounds, metal crowns, organometallic compounds of s and p block elements, noble gases and their compounds, interhalogens, pseudohalogens and polyhalides. Anomalies in periodicity, the use of d-orbitals by non-metals, reactivity and d-orbital participation, $p\pi-d\pi$ bonds, multicenter bonding in electron deficient molecules, three-center-two electron and three-center four electron bonds.

CHEM 260: Principles of Organic Chemistry (4 credits)

Prerequisite: Intermediate or A Level Chemistry

Basic concepts of organic chemistry like resonance, inductive effect, isomerism including stereochemistry, geometric isomerism, acids and bases, their relative strength and factors affecting acidity and basicity, significance of pH, pK_a and pK_b, chemistry of alcohols, phenols, thiols and ethers and their industrial applications.

CHEM 261: Organic Chemistry I (4 credits)

Prerequisite: Intermediate or A Level Chemistry

Reaction mechanisms with full emphasis include free radical, electrophilic and nucleophilic substitution, addition and elimination reactions; chemistry of alkyl halides, amines and organometallic compounds, catalytic reactions and their importance.

CHEM 262 Physical Organic Chemistry (3 credit)

Prerequisites: CHEM 150 or CHEM 160 or CHEM 170

The course attempts to explain physical aspects of organic molecules and their reactions. It includes an overview of the theories of chemical bonding, energy changes during chemical reactions and energy profiles, thermodynamic properties, concept of stability and spontaneity, equilibria and reversibility, transition state, energy of activation and Hammond postulate, solvation and solvation effects, and catalysis. It will also cover charge transfer complexes, quantitative structure-activity relationship and computational chemistry and its applications.

CHEM 270: Thermodynamics and Equilibrium (4 credits)

Prerequisite: Intermediate or A Level Chemistry

Chemical Thermodynamics: Second and third laws of thermodynamics, concept of entropy, Helmholtz and Gibbs Energy functions, Spontaneity and equilibrium, chemical potential, system of variable composition, interrelationship of thermodynamic functions. Phase equilibrium:

Claapeyron equation, solid-liquid, liquid-gas, solid-gas equilibria, Phase diagrams, Phase rule. Solutions: ideal and non-ideal solutions, Raoult's law, Colligative properties, Osmotic pressure, depression of freezing point, elevation of boiling point.

CHEM 271: Quantum Chemistry (3 credits)

Prerequisite: CHEM 150 or CHEM 170 or CHEM 270

Postulates of quantum mechanics, quantum mechanical operators, Schrodinger equation, wave functions and their properties. Quantum mechanical systems: Particle in a 1 – D box, Harmonic oscillator, Rigid rotator, Hydrogen atom, Variation principle, Huckel method.

CHEM 311/ ENVR 311: Fundamental Analytical Chemistry (4 credits)

Open to Sophomores and Above

Gravimetric and volumetric methods of analysis including buffers, complexometric titrations, redox titrations, non-aqueous titrations, Karl-Fischer titrations, UV/VIS spectroscopic analysis, IR Spectroscopy, treatment of measurement errors; usage and handling of standards, sampling, precision, accuracy, signal-to-noise ratio, limits of detection and quantitation, statistical evaluation of data; quality control and quality assurance.

CHEM 320: Industrial Chemistry (4 credits)

Open to Sophomores and Above

Efficiency and yield, common chemical industries with special reference to Pakistan including cement, surfactants, paper and pulp, glass and ceramics, leather, metallurgies of important metals, liquid crystals and inorganic polymers. Environmental industrial impacts and industrial environmental management.

CHEM 330: Biochemistry (4 credits)

Prerequisite: CHEM 160 or CHEM 261 or equivalent

Detailed structure and physiological function of primary metabolites including carbohydrates, proteins, lipids and nucleic acids, nature and role of enzymes and coenzymes, metallo-proteins and enzymes, mechanism of enzyme action, kinetics and regulation of enzymes and their industrial applications.

CHEM 331: Principles of Biological Chemistry (3 credits)

Prerequisite: CHEM 330

Interconnections between chemistry, biology and underlying chemical logic of biomolecules and metabolic pathways, genes and genomics.

CHEM 340/ENVR 340*: Environmental Chemistry Fundamentals (3 credits)

Prerequisite: CHEM 100 or Chemistry at Intermediate/A Level

Viewing the earth as five integrating systems i.e. the hydrosphere, the atmosphere, the geosphere, the astrosphere and the biosphere, the course deals with the chemical phenomenon involved in these systems and how human activities are affecting these. With a strong emphasis on the chemical aspects of environmental science, the course covers such topics as Water chemistry; oxidation/reduction; phase interaction; chemicals pollutants: origin, fate and hazards; atmospheric chemistry; chemicals and other air pollutants; photochemical smog; and hazardous waste chemistry.

CHEM 342: Green Chemistry Applications

Prerequisite: Open to Sophomores and Above

The course focuses on the environmental issues related to products and processes of chemical and related industries, and exploring their environmentally benign alternatives; green and sustainable chemistry, its principles and their applications in various chemical and pharmaceutical industries and research; new trends, environment friendly new materials, processes, and green methods in organic/ inorganic synthesis and green

solvents and catalysis; various chemical strategies to remove inorganic and organic pollutants from water, soil and air.

CHEM 350: Coordination Chemistry (4 credits)

Prerequisite: CHEM 150 or CHEM 250

Historical background of coordination compounds, nomenclature and stability, geometry of complexes having coordination number 2 to 9, explanation of optical and magnetic properties of coordination compounds, Jahn-Teller effect, isomerism and stereochemistry, stabilities of coordination compounds, characterization and applications of coordination compounds, metal-based drugs, organic reagents used in inorganic analyses.

CHEM 361: Organic Chemistry II (4 credits)

Prerequisite: CHEM 260 or CHEM 261

Study of carbonyl compounds including aldehydes, ketones, carboxylic acids, esters, amides, enolates and conjugate additions. Chemistry of vegetable oils and waxes.

CHEM 370: Kinetics and Mechanism (4 credits)

Prerequisite: CHEM 170 or CHEM 270

Chemical Kinetics: first and second order reactions, reaction mechanisms, unimolecular reactions, complex and chain reactions, theories of reaction rates, fast reactions, reaction in solutions. Electrochemistry: electrode potential, ion selective electrodes, electrochemical cells, measurement of electrode potential, electrical work, temperature dependence of cell potential, electrolysis, polarography, cyclic voltammetry, impedance. Surface chemistry: adsorption, chemisorption, heterogeneous catalysis.

CHEM 372: Chemistry and Energy (3 credits)

Prerequisites: CHEM 250 or CHEM 261 or CHEM 270

The course attempts to explain the diverse role of chemistry in solving energy problems. It covers the broad areas of electrochemistry, solar energy, biofuels, and chemical and biochemical processes. Fossil fuels and need for alternative energy sources, electrode potential and its measurement, galvanic cells, cell potential and free energy change, Nernst equation, various types of fuel cells and their thermodynamic and electrochemical aspects, nature of solar radiation and its use as source of energy, photovoltaic converters, dye-sensitized solar cells, biofuels, biomass as fuel, ethanol production, fermentation, and anaerobic digestion. Recent research and future prospects.

CHE3M 413: Instrumental Methods of Analysis (4 credits)

Prerequisite: CHEM 311 or CHEM 370

Atomic spectroscopic techniques: atomic absorption and emission techniques. Thermal analysis: TGA, DTA, DSC. Chromatography: introduction to separation techniques, solvent extraction, chromatography (paper, TLC, HPLC, GC, GPC) and electrophoresis. Hyphenated techniques: gas chromatography-mass spectrometry (GC-MS), liquid chromatography-mass spectrometry (LC-MS), MS-MS, LC-FTIR; inductively coupled plasma-mass spectrometry, matrix-assisted laser desorption/ionization-time of flight (MALDI-TOF) mass spectrometry, tandem mass spectrometry, ion trap mass spectrometry, other topics of interest. Nuclear techniques: neutron activation analysis, nuclear quadrupole resonance, isotope dilution method, isotope ratio mass spectrometry, Mössbauer spectroscopy, radioimmunoassay, x-ray techniques.

CHEM 421: Pharmaceutical Chemistry (3 credits)

Prerequisite: CHEM 260 or CHEM 261 or CHEM 330

Types and physicochemical properties of drugs and pharmacologically active products, structure and activity relationship, drug design, metal ions as information carriers, chemistry and mode of action of some common drugs.

CHEM 450: Advanced Inorganic Chemistry (4 credits)

Prerequisite: CHEM 250 or CHEM 351

Advanced inorganic chemistry topics including chemistry of carbonyl and nitrosyl compounds, lanthanides and actinides, and non-aqueous solvents. The course provides in-depth study of electronic structure, oxidation states, spectral properties, coordination chemistry, and separation of f-block elements, with applications in catalysis and nuclear chemistry. Molecular orbital treatment of π -acceptor ligands (CO and NO) and metal–ligand bonding. Synthesis, structure, bonding, and applications of metal carbonyls and nitrosyls, including the EAN rule and mononuclear and polynuclear carbonyls. Structural characterization of carbonyl and nitrosyl complexes by IR spectroscopy, including linear and bent M–NO bonding modes.

CHEM 453: Chemical Applications of Group Theory (3 credits)

Prerequisite: CHEM 250 or CHEM 350

Symmetry; symmetry elements and operations; point groups; matrices and transformation matrices, group representations, character tables; reducible and irreducible representations; applications of group theory to bonding theories, infrared and Raman spectroscopy, crystal field theory, optical activity.

CHEM 454: Inorganic Electronic Spectroscopy (3 credits)

Prerequisite: CHEM 250 or CHEM 350

Electronic spectroscopy of coordination compounds, Russell-Sanders coupling scheme, derivation of term symbols of for p1-p6 and d1-d10 systems, pigeonholes diagram, magnetism, magnetic susceptibility, magnetic moments, Faraday's and Gouy's methods, orbital contribution to magnetic moment, effect of temperature on magnetic properties of complexes.

CHEM 455: Inorganic Reactions Mechanism (4 credits)

Prerequisite: CHEM 250 or CHEM 350

Classification of reaction mechanisms, rate laws, steady state approximation, inert and labile complexes, substitution reactions, octahedral complexes, acid hydrolysis, acid catalyzed reaction, base hydrolysis, attack on ligands, steric effects of inert ligands, square planar complexes, nucleophilic reactivity, trans-effect, cis-effect, effect of leaving group, mechanism of substitution, and racemization reactions, reactions in non-aqueous inorganic solvents, classification of solvents, types of reactions in solvents, effect of physical and chemical properties of solvent, detailed study of liquid NH₃, H₂SO₄, HF, SO₂, BrF₃ supercritical fluid (water and CO₂), reactions in molten salts and ionic liquids.

CHEM 462: Spectroscopy of Organic Compounds (4 credits)

Prerequisite: CHEM 260 or CHEM 261 or CHEM 361

Study of IR, Mass and NMR spectroscopy as tools of structure elucidation, functional groups identification by IR spectroscopy, fragmentation pattern of main classes of organic compounds, 1-D proton and C-13 NMR spectroscopy and fundamental 2-D NMR techniques.

CHEM 464: Advanced Organic Chemistry (4 credits)

Prerequisite: CHEM 260 or CHEM 261 or CHEM 361

Study of reactive intermediates, pericyclic reactions, rearrangement reactions and oxidation reduction reactions, retrosynthesis and disconnection approach, design and synthesis of organic compounds of industrial importance.

CHEM 465: Natural Products and Medicinal Chemistry (4 credits)

Prerequisite: CHEM 330 or CHEM 361 or equivalent

Introduction to natural products and their medicinal importance, biosynthesis of terpenoids, alkaloids, flavonoids and steroids, total and partial synthesis of some representative natural products, chemistry of perfumes and aromatherapy, drug discovery.

CHEM 470: Polymer Chemistry (3 credits)

Prerequisite: CHEM 261 or CHEM 270

Introduction to polymers, step-growth, chain polymerization, copolymerization, kinetics of polymerization. Physical aspects of polymers: molecular weight, distribution, averages, and methods of determination, characterization of polymers.

CHEM 471 Advanced Physical Chemistry (3 credits)

Prerequisite: CHEM 271 or CHEM 370

Angular momentum, approximate methods, perturbation theory, multi-electron atoms, rotational, vibrational and rotational-vibrational spectroscopy, symmetry and group theory, molecular orbital calculations, recent developments in physical chemistry, computational chemistry.

CHEM 473: Surface and Solid-State Chemistry (4 credits)

Prerequisite: CHEM 270 or CHEM 370

Crystal structures, unit cells and Miller indices, X-ray diffraction, adsorption and desorption, Langmuir and BET isotherms, surface reactions and reactivity, ultrathin films and interfaces, techniques for the study of surfaces.

CHEM 498: Internship (3 credits)

Students intending to major in Chemistry

Registered students will take off-campus Industrial Training for 6-8 weeks in a reputed organization. At the end of the training, the student will submit a report on his/her internship. The internship will be evaluated by the Department.

CHEM 499: Capstone Project (3 credits)

Open to Junior year students intending to major in Chemistry

The project must be completed in one semester. At the end of the semester, the student will submit a project report. The report will be evaluated by the Department.

Department of
**COMPUTER
SCIENCE**

Introduction

The Department of Computer Science is part of the Faculty of Computer and Mathematical Sciences. It offers a rigorous undergraduate program for capable and motivated students who are excited by the influence of computing in our lives and who want to tackle the challenges of tomorrow. Computer Science is a fast-developing field, and our program aims to equip students with the fundamental knowledge that enables them to keep abreast of the latest developments and to contribute.

The Department offers the following degree program:

- **BS Computer Science**

BS Computer Science

The 4-year *BS* degree program at Forman Christian College (A Chartered University) is duly accredited by NCEAC (National Computing Education Accreditation Council) of HEC. The program provides a solid understanding of the theory and techniques of Computer Science (CS) so that upon graduation the students can enter various industries or pursue graduate studies. The curriculum offers a holistic view of discipline. It emphasizes both the state-of-the-art and the essential skills of modeling, abstraction and problem solving that open up a wide range of industries to the students. Our courses are a mix of various areas that reflect the excitement and joy of computing, e.g. Mathematics, Electronics, Algorithms, Programming, Theoretical CS, Artificial Intelligence, Application Development, Software Engineering, Networks, Operating Systems, Database, Computer Vision, Human-Computer Interaction and more.

Vision - Department of Computer Science

To be a leading Computer Science department in the region recognized for excellence in academics, research, service, and entrepreneurship.

Mission - Department of Computer Science

Creating an enabling educational environment that prepares students in Computer Science and related disciplines for global impact through high-quality programs, collaborative research, innovation, and service-orientation ensuring success of a diverse student, faculty, and alumni community.

Program Mission - BS Computer Science

The mission of the BS Computer Science program is to prepare graduates for professional careers, higher education, and research in Computer Science by providing a strong foundation in core concepts and practices and enabling them to apply their knowledge responsibly to address real-world, societal, and entrepreneurial challenges

Program Objectives

- Core Knowledge:** Graduates will possess core computing knowledge and skills to establish themselves in professional roles.
- Professionalism:** Graduates will fulfill their professional responsibilities taking due account of the ethical dimension of their conduct.

- iii. **Leadership:** Graduates will be effective as members/leaders of inclusive and multidisciplinary teams, communicating effectively, and engaging with society for creative problem solving.
- iv. **Continuous Improvement:** Graduates will be acquiring new knowledge professionally or academically.

Program Learning Outcomes

- i. **Academic Education**
Completion of an accredited program of study designed to prepare graduates as computing professionals.
- ii. **Knowledge for Solving Computing Problems**
Apply knowledge of computing fundamentals, knowledge of computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
- iii. **Problem Analysis**
Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
- iv. **Design/Development of Solutions**
Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
- v. **Modern Tool Usage**
Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
- vi. **Individual and Teamwork**
Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.
- vii. **Communication**
Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
- viii. **Computing Professionalism and Society**
Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
- ix. **Ethics**
Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.
- x. **Lifelong Learning**
Recognize the need and have the ability to engage in independent learning for continual development as a computing professional.

Total Degree Requirements

Requirement Category	Total Credits
General Education	42

Computer Science courses Requirements	96
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	138

General Education for BS Computer Science

General Education	42 Cr.
UNIV 100: Foundations of University Education	3
ENGL 106: Functional English	3
ENGL 108: Expository Writing	3
URDU 101: Communicative Urdu	3
ISLM 101 (required for all Muslim students) or CRST 152	2
ISLM 100 A & ISLM 100 B: Fehm-e-Qur'an (two 1-cr. courses) or 2-credit Religious Studies course for non-Muslims	2
Arts and Humanities (outside the major)	3
PKST 101: Pakistan Studies	2
PKST 103: Ideology and Constitution of Pakistan	2
SOCL 104: Civics and Community Engagement	2
Social Sciences (outside the major)	3
Natural Sciences (outside the major)	3 (2 + 1)
MATH 109: Quantitative Reasoning I (Advanced)	3
MATH 113: Quantitative Reasoning II (Advanced)	3
CSCS 101: Applications of Information and Communication Technologies (Advanced)	3 (2 + 1)
BUSN 220: Entrepreneurship	2

Requirements for BS Computer Science

BS Computer Science degree requires the completion of 138 credits in all as specified in the tables below.

Core Courses	14 (48 credits)
Professional certification COMP 496	1(3 credits)
Internship COMP 498	1(3 credits)
Capstone Project COMP 499 A& B	1(6 credits)
Interdisciplinary / Allied Courses	4 (12credits)
Elective Courses OR Specialization Elective Courses	8(24 credits)
Total Computer Science Credits	96

Important Note for Pre Medical background Students: *Applicants with Intermediate **Pre-medical** (subject combination of Physics, Chemistry and Biology) or equivalent 12 years education are eligible to apply for bachelor's in computer science. Such candidates, if admitted, are required to study additional two Mathematics courses, worth 6 credit hours, hours (MATH 101 and Math 102), during the first year of studies.*

Credit Hours per Course:

Though most of the courses are 3 credit hours, there are courses with a different number of credit hours as well. These are noted in the course details below.

Core and Inter-Disciplinary Studies (IDS) courses are shown in the following tables. IDS courses are also compulsory to take, but the term 'Supporting' is used to indicate that they are from other disciplines that support Computer Science curriculum:

Generic BS Computer Science Program

Students enrolled in the **BS Computer Science** program may choose elective courses from any specialization cluster without committing to a specific specialization. In such cases, **no specialization will be awarded, and no specialization title will appear on the student's degree or transcript.**

Specialization:

Students must complete the required **24 credit hours (eight elective courses)** by selecting **at least seven courses** from their chosen specialization cluster. The **remaining course** may be selected from **any specialization cluster** including the student's chosen specialization cluster.

Specialization clusters for BS Computer Science

- Data Science
- Artificial Intelligence
- Software Engineering
- Cyber Security

Core Courses	Credits
COMP 102	4
COMP 111	4
COMP 200	4
COMP 206	4
COMP 213	4
COMP 220	3
COMP 221	3
COMP 300	3
COMP 301	4

COMP 302	3
COMP 303	3
COMP 311	3
COMP 360	3
COMP 402	3
Core Courses Total	48
COMP 496	3
COMP 498	3
COMP 499	6
Total Credits	60

Interdisciplinary / Allied Courses <i>(these are also compulsory)</i>	Credits
MATH 111	3
STAT 115	3
CSCS 202	3
COMP 113	3
Allied Courses Total	12

Specialization Elective Courses	Credits
Data Science	
CSCS 364	3
CSCS 363	3
CSCS 366	3
CSCS 368	3
CSCS 455	3
CSCS 456	3
CSCS 457	3
CSCS 460	3
Specialization Elective Courses Total	24

Specialization Elective Courses	
Artificial Intelligence	Credits
CSCS 363	3
CSCS 454	3
CSCS 455	3
CSCS 456	3
CSCS 457	3
CSCS 460	3
CSCS 469	3
CSCS 471	3
Specialization Elective Courses Total	24

Transfer to BS Computer Science Degree

Any student who wishes to transfer from any other degree program to BS Computer Science must:

- i. Have the required T-Score at the time of admission.
- ii. Have a good previous academic and achievement record relevant to Computer Science.
- iii. Minimum 3.0 CGPA at the time of request to transfer.
- iv. Attain a grade of B in any Computer Science course(s) given to the student to gauge the student's ability with the permission of the Chairperson.
- v. The decision is also subject to the availability of seats in the department.
- vi. The Chairperson of the Computer Science Department reserves the right to accept or reject any application.

Students from other Majors

- i. Students from other majors can only take the following course from the computing domain: CSCS 100 Applications of Information and Communication Technologies to fulfil their general education requirement.
- ii. For registration in a course other than CSCS 100, students from other majors must take written permission from the Chairperson of the Computer Science Department. Such students will only be accommodated during the add/drop period if seats are available after registration by BS Computer Science students.

Course Descriptions

General Education Course (For Non-CS Students)

CSCS 100: Applications for Information and Communication Technologies (3 credits)

Computer Science students can take this elective course only in their Freshman year, and not afterwards, as part of their General Education requirements only.

This course provides a comprehensive introduction to Information and Communication Technologies (ICT) for freshman undergraduate students. The course covers fundamental ICT concepts, productivity tools, and practical applications across various domains including education, healthcare, finance, and e-commerce. Students will develop essential digital literacy skills, understand ethical considerations in technology use, and gain hands-on experience with contemporary ICT tools and platforms. The course emphasizes both theoretical understanding and practical application through a combination of lectures and laboratory sessions.

General Education Course (For CS Students)

CSCS 101: Advanced Applications for Information and Communication Technologies (3 credits) For CS Major students

This course provides a comprehensive introduction to Computer Science, covering both fundamental concepts and practical skills. Students will explore essential computing concepts including number systems, computer architecture, operating systems, and networking at a survey level. The course emphasizes hands-on proficiency with essential productivity tools (Microsoft Office Suite) and professional development tools (Git, LaTeX, and prompt engineering for AI). Students will develop computational thinking skills and gain exposure to cybersecurity and professional ethics. The emphasis is on achievable learning objectives with hands-on lab work each week, preparing students for specialized courses in programming, data structures, and other advanced CS topics.

Core Courses

COMP 102: Programming Fundamentals (4 credits) Lab Course

Prerequisite: None

Basic skills of problem solving and programming, problem analysis, algorithm design, program development and testing, structured design techniques, object-oriented thought process and basic tools.

COMP 111: Advanced Programming (4 credits) Lab Course

Prerequisite: COMP 102

The course includes the concepts and practical applications of classes, inheritance, class hierarchy, polymorphism, basic data structures, basic searching and sorting techniques.

COMP 200: Data Structures and Algorithms (4 credits) Lab Course

Prerequisites: COMP 111, COMP 113

More sophisticated data structures and algorithms required to manipulate them, selection or construction of suitable data structures for a wide range of problems, analysis of the efficiency of chosen solutions, standard problems such as sorting and searching, time and space complexity of computer programs.

COMP 206: Digital Logic Design (4 credits) Lab Course

Prerequisite: MATH 101 or A-Level Mathematics or Intermediate Mathematics

This course provides a foundation in design and analysis of the operation of digital gates, understanding basic operation of computer hardware, design and implement combinational and sequential logic circuits. It includes concepts of Boolean algebra, Karnaugh maps, flip-flops, registers, and counters along with various

logic families and comparison of their behavior and characteristics. It equips students with a sound knowledge of digital devices.

COMP 213: Database Systems (4 credits) Lab Course

Prerequisite: COMP 111

This course provides a foundational understanding of database concepts and their role in managing and organizing data effectively. Students will learn to design, implement, and query databases to support information systems and data-driven applications. Topics include data modeling, relational database design, SQL, normalization, and basic database management principles, with an emphasis on practical, hands-on use of database systems.

COMP 220: Software Engineering (3 credits)

Prerequisite: COMP 111

Basics of Software Engineering, the terminologies involved, and various principles, methods, tools and techniques used to produce quality software, two fundamental approaches of software engineering: structural and object-oriented. Various techniques used for requirements engineering, system/software design, implementation, and testing, fundamental issues of software measurement and project management.

COMP 221: Information Security (3 credits)

Prerequisite: NONE

This course introduces the fundamental concepts of information security for students with no prior technical background. Topics include the CIA triad (confidentiality, integrity, availability), common threats such as malware and phishing, password security, multi-factor authentication, backups, privacy, safe browsing, and basic incident response. Students will learn practical, real-world skills to protect their personal data and understand security challenges in everyday computing. The focus is on principles, best practices, and behavioral awareness rather than technical implementation.

COMP 300: Computer Organization and Architecture (3 credits) Lab Course

Prerequisites: COMP 206

This course introduces the basic concepts of computer organization and architecture, with an emphasis on the MIPS architecture and assembly language. Students learn how instructions are executed, how data is represented and stored, and how key components such as the CPU, memory, and input/output systems work together. The course combines fundamental theory with hands-on assembly programming to develop an understanding of low-level system design and performance.

COMP 301: Operating Systems (4 credits) Lab Course

Prerequisites: COMP 200, COMP 300

This course provides a comprehensive exploration of modern operating systems and their role in managing computer resources efficiently and securely. Students will master fundamental concepts including process management, CPU scheduling algorithms, inter-process communication, synchronization mechanisms, and deadlock prevention. The course covers memory management techniques such as paging, segmentation, and virtual memory, along with file system architectures, storage management, and I/O systems. Through analysis of real-world examples and implementation of simplified OS components, students will gain deep insights into how operating systems coordinate hardware and software, optimize resource utilization, and maintain

system security and reliability. By the end of the course, students will be able to explain fundamental OS principles, evaluate design alternatives, and analyze system behavior.

COMP 302: Theory of Automata (3 credits)

Prerequisite: COMP 200

Mathematical models of computation, definition and properties of formal languages and grammars, finite automata, regular languages and regular expressions, pushdown automata and context free languages, pumping lemmas and normal forms, Turing machines, Church's Thesis, Halting Problem and undecidability, overview of the theory of computational complexity.

COMP 303: Design and Analysis of Algorithms (3 credits)

Prerequisite: COMP 200

Basic notions of the design of algorithms and the underlying data structures, analysis of complexity of algorithms, number-theoretic and graph algorithms, basic algorithmic paradigms e.g. divide-and-conquer, dynamic programming, greedy algorithms, and introduction to computability theory.

COMP 311: Computer Networks (3 credits) Lab Course

Prerequisite: COMP 301

Engineering concepts underlying computer communication, including analog and digital transmission, circuit switching and packet switching, logical network structure and operation including network layers, network models (OSI, TCP/IP) and protocol standards, understanding of modern network concepts.

COMP 360: Introduction to Artificial Intelligence (3 credits) Lab Course

Prerequisite: COMP 200

This course introduces principles and practices of Artificial Intelligence, elements of intelligent behavior, techniques of knowledge representations, optimal solutions and complexities with heuristics, production systems and expert systems, introduction to machine learning, languages and their usage for implementation of intelligent behavior.

COMP 402: Cloud Computing (3 credits)

Prerequisite: COMP 311

Cloud computing services are being widely adopted by a variety of organizations from different domains. Cloud computing is the delivery of computing as a service over a network (usually the internet), where distributed resources are rented, rather than owned, as a utility by the end user. This course gives students an overview of the field of Cloud Computing. This includes a thorough understanding of cloud enabling technologies, primary building blocks of cloud computing, and hands-on experience by utilizing public cloud infrastructures (e.g., Google Cloud Platform, Amazon AWS, Microsoft Azure, etc.). The major topics covered in this course include fundamentals of cloud computing, cloud delivery models (IaaS, PaaS, SaaS), virtualization, containerization, Kubernetes, cloud computing mechanisms and architecture, storage, and cloud security.

COMP 496: Professional Certification (3 credits)

Prerequisite: Student should complete in Junior/Senior Year

This course enables BS Computer Science students to complete one or more internationally recognized professional certifications aligned with current industry standards and market needs. The course supports

guided selection of certifications in relevant computing domains such as software development, data science, cybersecurity, cloud computing, and emerging technologies, as recommended by industry and national bodies.

COMP 498: Internship

Prerequisites: COMP220

This course provides mandatory field-based learning experience for BS Computer Science students through structured engagement with industry, organizations, or approved university-led professional activities. Students are placed in real or simulated professional environments where they apply theoretical knowledge to practical computing tasks under the joint supervision of an industry mentor and a faculty supervisor.

COMP 499: Capstone Project (6 credits, 2 semesters, 3 credits each semester)

Prerequisites: COMP 213, COMP 220, Senior standing COMP 499A is the prerequisite of COMP 499 B

Requires students to research, conceive, plan and develop a real and substantial project related to computer science over the course of two semesters. It provides an opportunity to students to realize their acquired professional competence in the form of a demonstrable software product or other tangible result. The students must also submit a written report as per guidelines and make an oral presentation.

Interdisciplinary / Allied Courses

These courses are also compulsory and belong to disciplines that support Computer Science.

Similar Courses: Some courses in the supporting category are similar to courses from other departments. These are indicated below. Computer Science majors must strictly study the course as specified in their catalog and **cannot** study a similar course at any time.

COMP 113: Discrete Mathematics (3 credits)

Prerequisite: MATH 101 or A Level Mathematics or Intermediate Mathematics Similar to MATH 303

Foundations of discrete mathematics as they apply to Computer Science, understanding and appreciation of the finite nature inherent in most Computer Science problems and structures through study of logic, set theory, functions, recursive relations, combinatorial reasoning, iterative procedures, predicate calculus, tree and graph structures.

MATH 111: Calculus and Analytic Geometry (3 credits)

Prerequisite: MATH 101 or A Level Mathematics or Intermediate Mathematics This course is open only to Computer Science majors unless it is cross-listed. Similar to MATH 102, MATH 20.

The focus of this course is the study of the inverse relationship that exists between differential and integral calculus. Students will learn the fundamentals of calculus along with its application to interesting problems in science and engineering. Topics include limits, derivative, problem solving using differentiation, Fundamental Theorem of Calculus, integrations techniques, computation of areas and volumes by slicing, volumes of revolution and surface areas of revolution, infinite series and convergence tests and Taylor's series. Demonstration of various ideas will use mathematical software package(s).

STAT 115: Probability and Statistics (3 credits)

Prerequisite: MATH 101 or A Level Mathematics or Intermediate Mathematics This course is open only to Computer Science majors unless it is cross-listed. Similar to STAT 100, STAT 101, STAT 102, MATH 105, MATH 107

This course will introduce fundamental concepts of statistics and probability to Computer Science students. It includes a firm understanding of concepts like descriptive statistics, measures of central tendency, dispersion, moments etc. Different ways and laws of probability, random variable, conditional probability, independence Bayes' Theorem, some standard discrete and continuous probability distributions, concept of regression and correlation, along with exposure to statistical package(s).

CSCS 202: Computational Linear Algebra (3 credits)

Prerequisite: MATH 111 Similar to MATH 103, MATH 209

The course teaches concepts that are crucial to many areas of Computer Science, including computer graphics, image processing, computer vision and quantum computation. Topics include matrices, vectors, vector spaces, linear independence, linear transformations, eigenvalues & eigenvectors, solution of system of linear systems and singular value decomposition. Students will use ideas from linear algebra and apply them to data to perform tasks such as: 2D graphics transformations, image transformations and error-correcting codes. Software packages such as Mathematics or MATLAB will be used to implement various algorithms.

Computing Electives:

CSCS 290: Introduction to Java Programming (3 credits)

Prerequisite: COMP 111

The main purpose of the course is to equip students with knowledge of Java language programming methods. In this course students will acquire procedural programming: software development technique that imposes a hierarchical structure on the design of the programs. Moreover, students will learn the principles of object-oriented programming (OOP): programming techniques based on objects.

CSCS 291 Introduction to C/C++ (3 credits)

Prerequisite: None

The course provides a solid foundation in programming principles and techniques using the versatile and powerful programming language C/C++. The students learn various concepts of programming, including variables, data types, control structures, functions, arrays, pointers, classes, and object-oriented programming (OOP) principles through a combination of lectures, hands-on coding lab exercises, and assignments. In addition to exploring the syntax and semantics of the C++ language, this course will also emphasize good programming practices, code readability, problem-solving strategies, and algorithmic thinking. By the end of the semester, students will have developed the skills necessary to design, implement, test, and debug C++ programs, enabling them to tackle real-world programming challenges with confidence.

CSCS 306 Embedded Systems (3 credits) Lab Course

Prerequisite: COMP 111, COMP 206

An embedded system is centered around an embedded processor, usually a microcontroller, performing control related activities with minimum electronic component count. This course is focused on how to program and interface an embedded processor. Students will learn how to acquire data from analog as well as digital sensors, perform processing and take real time decisions to implement switching of motors/actuators.

CSCS 313: Software Requirements Engineering (3 credits)

Prerequisite: COMP 220

Role of requirements engineering within the software life cycle, comparison, contrast and evaluation of structured, object-oriented, data-oriented and formal approaches to requirements analysis, gathering necessary requirements from a customer to develop specifications and software.

CSCS 321: Cyber Security (3 credits)

Prerequisite: COMP 200

This course is designed to teach students the principles of cybersecurity and how they correlate to security practitioners and cyber adversaries. The course integrates networking and system management principles with a focus on system security through defensive strategies. Students will learn how to think like an attacker, how modern cyber-attacks and defenses work in practice, and how to evaluate risks and security systems. They will learn by working on hands-on lab settings and participating in real-world attack and defense scenarios while developing strong critical thinking skills. After completing this course, students will have the necessary core skills and understanding of typical risks and weaknesses as well as how to design a safe system using fundamental concepts and methodologies.

CSCS 322: Applied Cryptography (3 credits)

Prerequisite: COMP 221, COMP 113

This course provides a rigorous and practical introduction to the principles, algorithms, and protocols of modern cryptography. Students will develop a thorough understanding of both classical and contemporary cryptographic techniques, including symmetric and asymmetric encryption, cryptographic hash functions, digital signatures, and key management. The course bridges mathematical theory with real-world application, covering widely deployed protocols such as TLS/SSL, PKI, and Kerberos, as well as emerging topics including elliptic curve cryptography, zero-knowledge proofs, blockchain cryptography, and post-quantum cryptography. Hands-on lab sessions reinforce theoretical concepts through implementation using industry-standard libraries and tools such as OpenSSL and PyCryptodome.

CSCS 323: Vulnerability Assessment and Penetration Testing (3 credits)

Prerequisite: COMP 221

This course provides a comprehensive introduction to the methodologies, tools, and techniques used in vulnerability assessment and penetration testing (VAPT). Students will learn the legal and ethical boundaries of ethical hacking, the phases of a penetration test (reconnaissance, scanning, exploitation, post-exploitation, and reporting), and how to identify, exploit, and remediate vulnerabilities in web applications, and systems. The course emphasizes hands-on practice in controlled lab environments.

CSCS 342: Web Application Development (3 credits) Lab course

Prerequisites: COMP 213, COMP 220

Concepts, methods, technologies, and techniques of developing web applications that collect, organize and expose information resources, web application architectures, design methods and technologies, interface design, usability of web applications, security, accessibility, testing metrics, operation, deployment and maintenance of web applications, current and future web technologies.

CSCS 351: Software Quality Assurance (3 credits)

Prerequisite: COMP 220

Quality assurance and verification, avoidance of errors and other quality problems, inspections and reviews, testing, verification and validation techniques, process assurance versus product assurance, quality process

standards, product and process assurance, problem analysis and reporting, statistical approaches to quality control.

CSCS 352: Object-Oriented Analysis and Design (3 credits)

Prerequisite: COMP 220

Exploitation of the rich object-oriented modeling provided by Unified Modeling Language (UML), adaptation to changing requirements with iterative techniques and component-based design, design solutions optimized for modern object-oriented languages and platforms, application of proven design patterns, design heuristics, anti-patterns and refactoring techniques to refine analysis and design models, construction of unit and system tests to verify implemented designs.

CSCS 360: Kernel and Binary Exploitation (3 credits)

Prerequisite: COMP 221, COMP 301, C Programming (proficiency required)

This course provides an advanced, hands-on introduction to low-level software exploitation, focusing on memory corruption vulnerabilities in user-space binaries and operating system kernels. Students will learn how to analyze, exploit, and mitigate classic and modern vulnerability classes including stack overflows, heap overflows, use-after-free, and race conditions. The course covers the x86/x64 architecture, assembly language, debuggers, fuzzers, and kernel-level exploitation techniques (e.g., privilege escalation, rootkits). Emphasis is placed on writing custom exploits, bypassing common protections (ASLR, NX/DEP, Stack Canaries), and understanding defensive mechanisms in Windows and Linux kernels within a controlled lab environment.

CSCS 363: Generative AI (3 credits)

Generative AI is a fast-growing and highly influential area in computer science, transforming fields like healthcare, education, and software development. This course is designed to bridge the gap between basic AI concepts and modern generative models, helping students understand how these systems work and apply them in practice. It supports the department's AI specialization track and prepares students for capstone projects, internships, and careers in AI-driven industries. Additionally, the course aligns with HEC guidelines as part of the newly introduced AI specialization in the BS Computer Science program.

CSCS 364: Information Retrieval (3 credits)

Prerequisite: COMP 200

Information Retrieval is the process of responding to the user queries by retrieving the information from text-based information on a specific topic (i.e., text in-text out). In recent years, these concepts have also extended to image-based information retrieval (i.e., image in-image out). Web search is a specific application of IR techniques to the largest corpus of text i.e., the web, and it is a use case where many people interact with IR systems most frequently. This course emphasizes efficient text indexing, Boolean and vector-space retrieval models, positional encodings, IR approaches for the web, document clustering for reducing search space, and ranked retrieval based on queries.

CSCS 365: Introduction to Digital Image Processing (3 credits)

Prerequisite: COMP 200 and CSCS 202

This course introduces the students to the basics of digital images, their structure and formulation. Algorithms for image manipulation and characterization in spatial domains are included. Later formal treatment of images in frequency domain, including filtering and edge detection are described.

CSCS 366: Introduction to Natural Language Processing (3 credits)

Prerequisites: COMP 200 and STAT 115

The course is focused on different approaches to building systems for semantic understanding of the natural language. It not only covers the theoretical concepts but also aims to provide hands-on natural language processing tools like POS-tagger, sentence parser, named- entity recognizer, etc.

CSCS 367: Blockchain Technologies and Applications (3 credits)

Prerequisites: COMP 213, COMP 303 and COMP 311

This course will familiarize students with basic concepts of Blockchain technology. This course enables students to explain basic components of a Blockchain (transaction, block, block header, and the chain), its operations (verification, validation, and consensus model) underlying algorithms, and essentials of trust. Content includes the hashing and cryptography foundations indispensable to Blockchain programming. The business uses of Blockchain in various industries like Finance and Supply Chain will be discussed.

CSCS 368: Fundamentals of Data Visualization (3 credits)

Prerequisites: COMP 200 and STAT 115

Data Visualization is a process of obtaining detailed insights hidden in the data. It is a necessary component in the pipeline of any data science project. This course teaches skills specifically in terms of how to effectively present the data and findings. Further, this course provides hands-on skills using a selection of tools and language libraries for data exploration and visualization.

CSCS 370: Secure Software Design and Development (3 credits)

Prerequisite: COMP 220, COMP 221

This course provides a comprehensive foundation in Secure Software Design and Development. The course equips students with the knowledge and practical skills to design, build, verify, and maintain software that is resilient against modern cyber threats. The course integrates foundational security principles with industry-leading frameworks and the latest international guidelines, including the UK NCSC Software Security Code of Practice (2025), OWASP SAMM, and OWASP ASVS.

CSCS 371: Wireless and Mobile Security (3 credits)

Prerequisite: COMP 221, Basic networking concepts (TCP/IP, OSI model), Familiarity with Linux command line

This course provides a comprehensive introduction to the security assessment of wireless networks and mobile applications. Students will learn the unique threat models, attack surfaces, and penetration testing methodologies for Wi-Fi (802.11), Bluetooth, and RFID/NFC systems (40% of course), followed by in-depth coverage of Android and iOS security architectures, static/dynamic analysis, and exploit development (60% of course). Emphasis is placed on hands-on practice using deliberately vulnerable mobile applications such as AndroGoat, Pentesting Dive Beta, Damn Vulnerable iOS App (DVIA), and OWASP MSTG crackmes. Students will also learn to write penetration testing reports tailored to wireless and mobile ecosystems.

CSCS 372: Cloud Security (3 credits)

Prerequisite: COMP 221, Basic networking concepts (TCP/IP, OSI model), Familiarity with Linux command line

This course provides a comprehensive foundation in cloud security, focusing on the principles, risk-based strategies essential for safeguarding information in any cloud environment, and governance frameworks. The curriculum examines the shared responsibility model, data protection standards, identity and access management architectures, network defense concepts, and incident response as universal building blocks.

Students will learn to evaluate security trade-offs across public, private, and hybrid deployment models, and to align technical controls with business, legal, and regulatory requirements.

CSCS 403/MATH 403*: Graph Theory (3 credits)

Prerequisite: COMP 113/MATH 303

Graphs, subgraphs, isomorphism, trees, connectivity, Euler and Hamiltonian properties, matching, vertex and edge colorings and planarity.

CSCS 405: Software Projects Management (3 credits)

Prerequisite: CSCS 220

This course describes the basic principles and informs about the application of activities involved in Software Projects Management. It provides an opportunity to develop the ability to plan and manage software development projects successfully and maximize the return from each stage.

CSCS 406: Human-Computer Interaction (3 credits) Lab Course

Prerequisite: COMP 220

Exploration of the differences in information processing by humans and machines using insights from psychology and cognitive science, design of human-computer interfaces and systems involving both human and computer components.

CSCS 410: Digital Product Management

Prerequisite: COMP 220

The Digital Product Management course is introduced to bridge the gap between software engineering and strategic product leadership, preparing CS students for industry roles beyond traditional development. Endorsed by the department's Industrial Advisory Board (CS- IAB), which also provided the initial course outline, it aligns with the growing demand for tech focused product managers in companies like Google, Amazon, and Microsoft. Unlike general business courses, this curriculum is software-centric, covering agile methodologies, UX research, AI-driven analytics, and SaaS business models. The course supports CS career diversification, equipping students to lead AI products, startups, and software development teams, ensuring their competitiveness in the digital economy.

CSCS 430: Embedded Systems and IoT Security

Prerequisite: COMP 221

This course provides a comprehensive introduction to security challenges, vulnerabilities, and mitigation techniques in embedded systems and the Internet of Things (IoT). Students will learn embedded hardware platforms (Arduino, Raspberry Pi, ESP8266/ESP32), client-server communication protocols (MQTT, CoAP, HTTP/HTTPS), hardware security interfaces (UART, JTAG), firmware reverse engineering, and side-channel attacks. The course emphasizes hands-on practice with vulnerable IoT applications (IoT Goat, Damn Vulnerable IoT Device, HomePwn) in controlled lab environments. Upon completion, students will be prepared to assess, exploit, and secure IoT deployments in real-world scenarios.

CSCS 440: Systems Programming (3 credits) Lab course

Prerequisites: COMP 301

Internal operation of system software including assemblers, loaders, macro-processors, interpreters and inter-process communication.

CSCS 441: DevOps: Principles and Practices (3 credits) Lab course

Prerequisites: COMP 311

This course introduces the fundamental principles, culture, and practices of DevOps, emphasizing the integration of software development and IT operations to enable faster and more reliable software delivery. It covers key concepts such as continuous integration and continuous delivery (CI/CD), automation, infrastructure as code, containerization, and cloud-based deployment. Students will gain hands-on experience with industry-standard tools to design, build, test, and deploy scalable applications in modern development environments. The course also highlights collaboration, communication, and teamwork as essential elements of DevOps culture, preparing students to work effectively in cross-functional teams.

CSCS 442: Internet of Things and Its Applications (3 credits) Lab course

Prerequisites: COMP 311

This course provides an overview of the Internet of Things (IoT), focusing on how interconnected devices collect, exchange, and analyze data in modern digital ecosystems. Key topics include IoT architecture, sensors and actuators, communication protocols, cloud integration, and real-world applications such as smart homes, healthcare, and smart cities. Students will gain hands-on experience with basic IoT systems and platforms. The course is suitable for undergraduate students in computing and related disciplines interested in emerging technologies.

CSCS 443: Cross-Platform Application Development (3 credits) Lab course

Prerequisites: COMP 200 and basics of JavaScript, Html and CSS

This course provides a comprehensive introduction to the principles and practices of cross-platform application development. It covers key topics such as cross-platform architectures and frameworks, application design and development, and the use of client-side and server-side APIs, tools, and libraries. Students will also learn to apply responsible engineering practices, including developing secure, accessible, and user-centered applications. The course is well-suited for students interested in building modern applications that run seamlessly across multiple platforms.

CSCS 450: Technology Management (3 credits)

Prerequisite: COMP 220

Introduction to technology strategy, corporate strategy, technology transfer, technology strategy development, product development strategy and innovation process.

CSCS 451: Introduction to Robotics (3 credits)

Prerequisite: COMP 200

Robotics, as a field of study, is undergoing massive research and development to improve the dynamic, perceptual, and behavioral aspects of robots. Students will learn the basics to model, simulate and design various robotic systems (with an emphasis on robotic manipulators). The course will cover introduction to different types of robots and their applications. It will also reflect upon coordinate systems, kinematic principles, locomotion, and robot perception techniques (which will include robotic sensors, data acquisition, use of computer vision techniques in robotics, image processing applied to robotic perception, and feature extraction from various sensor data).

CSCS 453: Computer Graphics (3 credits)

Prerequisites: COMP 200, CSCS 202

Graphics hardware including display devices, applications of computer graphics, development of graphics software, interactive graphics programming, Raster scan, conversion algorithms for line segments, circles, ellipses and general curves, 2D transformations, windowing and clipping including panning and zooming, line clipping and area clipping algorithms, region filling algorithms, 3D objects display techniques, 3D representations, 3D transformations, curve and surface design and representations, rendering, shading and animation.

CSCS 454: Reinforcement Learning (3 credits)

Prerequisite: COMP 360

The course will introduce students to the fundamentals of reinforcement learning (RL). We will start by introducing RL problem formulation, its core challenges, and a survey of consolidated approaches from literature, including dynamic programming, value-function learning and policy learning. We will then cover model-based RL and exploration strategies. Finally, the course will discuss more recent reinforcement learning models that combine RL with deep learning techniques. The course will leverage a combination of theoretical and applicative lectures.

CSCS 455: Data Mining and Data Warehousing (3 credits)

Prerequisite: COMP 213, STAT 115

Database concepts, different data models, data storage and retrieval techniques and database design techniques, data warehousing and data mining, emerging database technologies and applications.

CSCS 456: Introduction to Deep Learning (3 credits)

Prerequisite: COMP 200 and STAT 115

Deep Learning is a hierarchical learning methodology based on artificial neural networks which are algorithms inspired by the structure and function of the brain. It has applications in wide range of industries these days such as face- recognizers working at massive scales, robotics, speech translation, text analysis, improving customer experience, autonomous vehicles etc.

CSCS 457: Computer Vision (3 credits)

Prerequisites: COMP 200, CSCS 202

Introduction of theory and applications of computer vision and current problems, techniques and applications, computer vision systems, interaction of different components in a complete system, writing programs to solve computer vision problems through the use of several programming assignments and examples.

CSCS 458: Big Data (3 credits)

Prerequisites: COMP 213

The proliferation of unstructured data in quantities impedes the use of traditional statistical approaches. To examine such data, new techniques are required. In order to be responsive, new algorithms are required to deal with different techniques. New data storage and retrieval mechanisms are required. Many of the algorithms come from well- known big data owners like Google (search, AdWords), Amazon (recommended books), and Facebook (social network analysis). As more players enter the competition, new methods will emerge to meet their needs. The course objective is to develop understanding about the core concept of Big Data, why Big Data requires a different programming paradigm and mindset, and what are the various programming approaches used, and what type of data can be processed.

CSCS 460: Machine Learning (3 credits)

Prerequisites: COMP 200, CSCS 202

Supervised learning, classification, regression, bias-variance analysis, maximum-margin classification, kernel methods, evaluation, validation, unsupervised learning, clustering, dimensionality reduction, outlier/anomaly detection, sequential data modeling, deep learning, model maintenance.

CSCS 461: Principles of Programming Languages (3 credits)

Prerequisite: COMP 303

Theory and practice of programming language translation, languages, grammar and parsing, lexical, syntactic and semantic analysis, compile-time error handling, organization of programming languages including language processors, syntax data types and sequence control, storage management, comparison of language features from the functional, imperative, logical and object-oriented paradigms.

CSCS 462: Social Engineering and Insider Threats (3 credits)

Prerequisite: COMP 303

The aim of this course is to teach students the methods of social engineering and to equip them with the capability to recognize and identify insider threats. After taking this course, the students will be able to develop and implement responsive security measures and preventive security policies and regulations to counter social engineering and insider threats.

CSCS 465: Information Security Standards and Regulations (3 credits)

Prerequisite: COMP 303

This course aims to provide basic and advanced concepts relevant to information security standards / frameworks. At the end of the course, the students should be able to analyze, evaluate, initiate, implement, control, and improve an information security management system using ISO 27001 framework. Moreover, the students will have understanding about the other frameworks to standard / framework within their organization.

CSCS 466: Wireless Networks (3 credits)

Prerequisite: COMP 311

Techniques in design and operation of first, second and third generation wireless networks: cellular systems, medium access techniques, radio propagation models, error control techniques, handoff, power control, common air protocols, radio resource and network management.

CSCS 467: Network Security (3 credits)

Prerequisite: COMP 221

This course provides a comprehensive study of the principles, mechanisms, and practices used to secure computer networks against modern threats. Key topics include applied cryptography and PKI, network protocol vulnerabilities and defenses across the TCP/IP stack, firewall architectures, intrusion detection and prevention systems, VPN technologies, wireless security, and cloud/zero-trust network models. Students develop hands-on proficiency through a dedicated weekly laboratory using industry-standard tools such as Wireshark, Nmap, Snort/Suricata, and Metasploit, culminating in a team-based capstone security audit of a simulated enterprise network.

CSCS 468: Mobile Application Development (3 credits) Lab Course

Prerequisites: COMP 200

Programming of applications for mobile phones and mobile devices such as tablets in a popular mobile device platform and programming language.

CSCS 469: Production AI Systems with MLOps (3 credits)

Prerequisite: COMP 360

This course teaches students how to take a machine learning model from a Jupyter notebook to a live, monitored, and self-healing production system. Students will build real pipelines — not toy examples — using the same tools and practices employed in industry today. Starting from data ingestion and validation, students will progress through feature engineering, experiment tracking, model packaging, serving, continuous integration and delivery, and automated monitoring and retraining. Every concept is grounded in a single capstone project running from Week 9 to Week 15, requiring students to debug real problems on real data rather than following pre-packaged instructional examples. By the end of the course, students will have a fully functioning MLOps pipeline deployed using cloud-compatible, self-hosted open-source infrastructure, a monitored and automatically retraining model, and a production codebase demonstrable in a professional setting. The course employs an entirely free and open-source technology stack — Python, Docker, Apache Airflow, DVCs, MLflow, FastAPI, Evidently, Prometheus, Grafana, GitHub Actions, and Kubernetes — selected for industry relevance, pedagogical transparency, and zero financial barrier to access.

CSCS 470: Introduction to Game Development (3 credits) Lab Course

Prerequisites: CSCS 220

The course will cover a selection of topics in the design and development of 2D and 3D video games. Fundamental issues, implementation techniques, platforms, and programming languages will be discussed. Introduction to core game engine, basic graphics, Graphical User Interface (GUI), networking, Artificial Intelligence will be provided.

CSCS 471: Cybersecurity Governance, Risk, and Compliance (3 credits)

Prerequisite: COMP 221

This course provides a comprehensive understanding of Information Assurance (IA), with a strong emphasis on security governance, risk management, regulatory compliance, international standards, and the emerging domain of AI security management. The course progressively builds foundational and advanced knowledge grounded in the five core principles of Information Assurance (Confidentiality, Integrity, Availability, Authentication, and Non-Repudiation).

CSCS 472: Malware Analysis and Reverse Engineering (3 credits)

Prerequisites: COMP 221

This course provides a comprehensive introduction to the methodologies, tools, and techniques used in malware analysis and reverse engineering. Students will learn to analyze malicious software using both static and dynamic analysis approaches, disassemble and decompile binary executables, understand malware behavior, and develop indicators of compromise (IOCs). The course covers legal and ethical boundaries of malware research, and examines common malware families including ransomware, remote access trojans (RATs), rootkits, and botnets. Emphasis is placed on hands-on practice in isolated, controlled lab environments using industry-standard tools such as Ghidra, IDA Pro, x64dbg, and REMnux/FlareVM.

CSCS 473: Agentic AI Systems (3 credits)

Prerequisite: CSCS 365

This course explores the design, development, and evaluation of agentic AI systems capable of autonomous reasoning, planning, and tool use. Students will study core concepts such as LLM-based agents, planning strategies, memory architecture, tool integration, and multi-agent collaboration. The course emphasizes building real-world agentic systems and critically evaluating their ethical, societal, and operational implications.

CSCS 474: Digital Forensics and Incident Response (3 credits)

Prerequisites: COMP 221

This course provides a comprehensive introduction to digital forensics and incident response (DFIR), covering the methodologies and tools used to investigate cybercrimes and respond to security breaches. Students will learn the incident response lifecycle, evidence acquisition and preservation, disk and memory forensics, network traffic analysis, and forensic artifact examination on Windows and Linux systems. The course emphasizes hands-on practice in lab environments, legal and ethical considerations under Pakistani cyber laws (PECA), and professional reporting. Upon completion, students will be prepared to handle real-world security incidents and assist in forensic investigations.

CSCS 475: Technology Entrepreneurship (3 credits)

Prerequisites: COMP 220

Develop an understanding of different concepts of entrepreneurship and their application in technology enterprises for better entrepreneurial competence and ultimate success. The course will cover case studies from the tech domain. The students will get practical experience of entrepreneurship by completing a course project in Technology Entrepreneurship.

CSCS 482: Generative Artificial Intelligence Security (GenAI Security) (3 credits)

Prerequisite: Introduction to artificial Intelligence (COMP 360)

The swift embrace of Generative Artificial Intelligence (GenAI) systems that include the adaption of Large Language Models (LLMs) has posed a significant security risk to individuals, organizations and governments across the world. Therefore, it is vital to equip our students with necessary knowledge, skills and tools to mitigate AI threats and defend the GenAI systems.

CSCS 483: Cybersecurity for Artificial Intelligence Systems (3 credits)

Prerequisites: COMP 221, COMP 360

This course provides a comprehensive examination of security principles for generative and agentic artificial intelligence (AI) systems. The students will explore secure AI architecture design, vulnerability classes unique to Large Language Models (LLMs) and autonomous agents, examine adversarial techniques, and evaluate secure communication protocols for multi-agent coordination. The course further addresses the security hardening of AI infrastructure and concludes with the governance frameworks and compliance obligations.

CSCS 295/495: Special Topics in Computing / Themes (1-3 credits)

Prerequisites: As appropriate for the material

These courses allow the presentation of new or emerging areas of study in the computing domain not present in the current catalog.

COMP 452: Computer Architecture (3 credits)

Prerequisites: COMP 301

This course provides an understanding of design issues of computer systems from the perspective of performance measures and cost-performance tradeoffs. The course covers the fundamentals of modern processor design. Topics include instruction set design, RISC vs. CISC architectures, memory management, caches, memory hierarchies, interrupts, I/O structures, pipelining, parallelism, and multiprocessor systems.

*** Cross-listed Courses:**

Courses with two designators (coding) are marked with an asterisk (*) to identify them as cross-listed courses. Students must select the correct designator for their applicable program and requirement. Students may earn credit in only one of the two cross-listed courses.

Department of
ECONOMICS

Introduction

The Department of Economics at Forman Christian College (A Chartered University) has been offering the undergraduate program since 1915. The Department has evolved over time, bringing it up to par with contemporary needs by building resources, introducing a wide range of courses, upgrading curricula, and adding new scholars. Currently, there are more than 300 baccalaureate students majoring in Economics, and it is rated as one of the most popular degrees at FCCU. The Department of Economics is part of the Faculty of Social Sciences.

Vision

The vision of the Department of Economics is to develop responsible, capable, and ethical leaders in the field of Economics who can apply existing ideas and develop new ones to serve humanity in general and Pakistani society in particular.

Mission

The Department is working with the mission of creating a learning environment in the field of Economics, endowed with the best physical and human resources required for producing confident, competent, and credible professionals and sending them to seek success and serve society. The Department strives to produce and provide an atmosphere:

- Conductive to independent, critical, and creative thinking.
- Promoting competitive as well as collaborative attitudes of seeking knowledge.
- Cultivating an ability of self-motivated lifelong attitude of independent learning.

Bachelor of Science in Economics

The BS Economics program provides a thorough understanding of economic theories of global economic issues with a particular focus on Pakistan's economy. It enables the students to perform quantitative and qualitative analyses to present the findings effectively. The degree not only aims to enhance the student's personal and professional decision-making capacities but also empowers them to contribute to policy formulation and participate in the economic development of Pakistan.

Program Objectives

- i. To develop in our students an understanding of core principles and concepts of economics.
- ii. To develop in our students critical and analytical skills necessary to analyze socio economic issues faced by economic agents.
- iii. To encourage our students to articulate economic ideas and policies for the welfare of society.
- iv. To instill in our students ethical and moral values are necessary in professional and personal lives.
- v. To develop in our students' desire of learning and ability to explore and avail career and educational opportunities in the field of economics and related disciplines.

Program Learning Outcomes

- i. **Subject Knowledge**
Demonstrate knowledge of theories and concepts of economics for application in practical life.
- ii. **Policy Formulation Skills**
Formulate and communicate economic ideas and policies for the welfare of society
- iii. **Quantitative and Empirical Skills**

Apply statistical tools, software, and empirical techniques to collect, analyze, interpret data, test hypotheses, and investigate socio-economic issues.

iv. Research Skills

Demonstrate knowledge of the research process, survey design, ethical consideration and communicating research results in a meaningful way.

v. Ethical and Moral Values

Recognize ethical and regulatory practices in economic activities, research process and report writing.

vi. Career Skills

Explore and avail career and educational opportunities in the field of economics and related disciplines.

Requirements for BS Economics Major

- A minimum CGPA of 2.0 is required to declare Economics as a major.

Core Courses	Elective Courses	Total Credits	Required MCGPA
10 (30 Credits)	8 (24 credits)	54	2.0

Core Courses	Credits
ECON 101	3
ECON 102	3
ECON 103	3
ECON 201	3
ECON 202	3
ECON 203	3
ECON 300	3
ECON 302	3
ECON 498	3
ECON 499	3
Core Courses Total	30

Elective Courses choose from the list. (A minimum of six elective courses of 300 or 400 levels is required for majoring in Economics.)	Credits
ECON 206	3
ECON 210	3
ECON 303/ ENVR 306	3
ECON 304	3
ECON 305	3
ECON 306	3
ECON 307	3
ECON 309	3
ECON 311	3

ECON 312	3
ECON 313	3
ECON 315	3
ECON 320	3
ECON 322	3
ECON 323	3
ECON 325	3
ECON 326	3
ECON 355	3
ECON 395	3
ECON 400	3
ECON 402	3
ECON 403	3
ECON 404	3
ECON 406	3
ECON 411	3
ECON 413	3
ECON 417	3
ECON 418	3
ECON 422	3
ECON 490	6
Elective Courses Total	24

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Course Descriptions

ECON 100: Principles of Economics (3 credits)

This course is counted towards General Education credits.

The course cannot be registered simultaneously or after any of the following: ECON 101, ECON 102, ECON 103 BUSN and BUSN 225

Prerequisite: None

The aim of this course is to introduce students to the basic concepts and principles of microeconomics and macroeconomics. Main contents of the course include market forces of demand and supply, cost of

production, market structures, concepts of national income, inflation, unemployment, international trade and development economics.

ECON 101: Principles of Microeconomics (3 credits)

Prerequisite: None

The course is designed to build the foundation for major concepts of microeconomics. Students will learn how consumers and producers take rational decisions in their daily economic affairs. The course includes principles of demand and supply and interactions of market forces, optimization of output and pricing strategies under perfect and imperfect market structures.

ECON 102: Principles of Macroeconomics (3 credits)

Prerequisite: None

This course describes key macroeconomic concepts, theories, and policies; and explains how an economy operates in a country. Main contents of the course include national income accounting, productivity, standard of living, growth and public policy issues, introduction to consumption, saving and capital formation, inflation, unemployment, monetary system, quantity theory of money and introduction to monetary and fiscal policies.

ECON 103: Mathematics for Economists (3 credits)

Prerequisite: None

This course is designed to teach students basic tools of Mathematical Economics and their application to economic analyses. With the help of mathematical tools, it will be learnt how firms precisely maximize their profit, revenue and minimize cost. The main contents include real number system, set theory, comparative static analysis, linear models and matrix algebra, tools of algebra and calculus, application of calculus in economics, optimization of one and multivariable functions, optimization with constraints, and economic application of optimization.

ECON 201: Intermediate Microeconomics (3 credits)

Prerequisite: ECON 101, ECON 103

This course is a continuation of Principles of Microeconomics. By using mathematical models, it analyzes consumer and producer theories, firm behavior under perfect and imperfect market structures, and market inputs in perfect and imperfect competitive environments.

ECON 202: Intermediate Macroeconomics (3 credits)

Prerequisite: ECON 102, ECON 103

This course is a continuation of Principles of Macroeconomics, it covers Classical and Keynesian economic theories, derivation of AD and AS models and their implication for stabilization policies, short-term and long-term inflation-unemployment relationship, consumption and investment theories, economic growth, growth accounting and convergence.

ECON 203: Statistics for Economists (3 credits)

Prerequisite: None

This course is designed to introduce students with basic concepts of Statistics and their application in Economics. The course focuses on measures of central tendency and dispersion. Basic concepts of probability and its distributions, normal, chi-square, t and F distributions and their applications. Moreover, students will

be taught to develop and test hypotheses and apply regression and correlation analyses to analyze relationships between economic variables.

ECON 206: Banking and Finance (3 credits)

Prerequisite: ECON 102

This Course is designed to familiarize students with basic concepts and principles of banking and financial institutions. It will also provide a comparison between conventional and interest free financial institutions, their underlying principles, modes of financing and contractual mechanisms. The basic contents of the course include functions and role of money, credit creation, financial institutions and markets, and functions of commercial and central banks in economic development. For Islamic or interest free banking, the course covers norms of Islamic Finance such as, prohibition of *Riba*, *Ghrarar* and the need for cooperation. The course will also cover the Islamic modes of financing such as *Mudarabha*, *Musharakah*, and *Murabaha*; their contracts, types, basic rules and contractual mechanisms.

ECON 210: Population Economics (3 Credits)

Prerequisite: None

This course is designed to introduce students with basic concepts of demography and theories of population economics. The main contents of the course will include historical trends in fertility & mortality, demographic transition, determinants of changes in population composition and their socio-economic impact, theories of population and economic development, role of youth, consequences of aging, issues of education, health & labor market, and concept of demographic dividend. This course also aims to develop basic skills to use census and other population data sources to calculate, analyze and interpret different demographic indicators and indices.

ECON 300: Fundamentals of Econometrics (3 credits)

Prerequisite: ECON 201, ECON 202, ECON 203

The course will equip the students with the fundamental concepts and techniques of econometrics to conduct an econometric project and to understand the way in which economic theory and the real world are connected. The main topics are single-equation regression models, the method of ordinary least squares, dummy variable regression models; relaxing the assumption of classical model: multicollinearity, heteroscedasticity and autocorrelation.

ECON 302: Research Methods and Computer Applications (3 credits)

Prerequisite: ECON 300

This course is designed to learn about different methods and methodologies of research used in economics and how to apply them. The course will cover the topics of data collection methods, research design, sampling, report writing and the use of econometric softwares.

ECON 303/ ENVR 306 *: Environmental Economics (3 credits)

Prerequisite: ECON 101

This course is an Introduction to the economic and ecological principles essential for a clear understanding of contemporary environmental and natural resource management issues. The course covers economic, legal and ecological perspectives to better understand the causes and solutions to market failure and environmental degradation. Main contents of the course include economic efficiency and market failure, property rights, externality, measuring social welfare and welfare improvements, demand for environmental goods,

environmental valuation methodologies, environmental cost-benefit analysis, environmental policy and environmental regulations, common pool resource management and depletable resource management.

ECON 304: Managerial Economics (3 credits)

Prerequisite: ECON 201, ECON 203

The purpose of this course is to provide a basic understanding of economic theory and analytical tools that can be used in decision-making problems. It covers demand analysis, estimation and forecasting, production and cost theories and their estimation, decision making under risk and uncertainty, managerial theories and models of the firm.

ECON 305: Public Sector Economics (3 credits)

Prerequisite: ECON 201, ECON 202

This course covers the theory of public sector economics, that is, the role of the government in the functioning of the economic system. The course focuses on the topics, such as role of public sector, theories of public goods, externalities, distribution equity and economic welfare, concepts used in taxation, types of taxes and their shifting & incidence, resource mobilization, public expenditure evaluation, public debt, external debt modeling, budget deficit and cost benefit analysis of development finance.

ECON 306: History of Economic Thought (3 credits)

Prerequisite: ECON 202 and CGPA ≥ 3.0

This course intends to survey the evolution of economic ideas and main schools of thought that played an important role in the development of history of economic thought. The course will discuss the selected authors and their challenges in historical and institutional perspective. The course includes the evolution of Economics in ancient Greek and Roman medieval times; doctrine of Ibne Khaldun, Mercantilists, and Physiocrats. Marxists, Marginalists, Classical; Keynesian, Monetarists, and Institutionalists' contribution to economic theory will also be discussed.

ECON 307: International Trade Theory and Policy (3 credits)

Prerequisite: ECON 201

This course is designed to introduce basic principles of international trade and policies to improve trade balance. Evaluation of international trade, Mercantilists' views on trade, theories like absolute & comparative advantage, standard theory of international trade, neo- classical trade theory, offer curve and terms of trade, factor endowments and Heckscher- Ohlin model, factor price equalization and Stolper-Samuelson theorems, HO Model and new trade theories, trade based on economies of scale, product differentiation, technological gap and product cycle models, impact of economic growth on international trade, impact of international trade policies on nation's welfare like tariff and its effects, quota and its effects, nontariff trade barriers, new protectionism, economic integration, customs unions, and free trade areas.

ECON 309: Econometric Methods (3 credits)

Prerequisite: ECON 300

This course covers advanced econometrics techniques at undergraduate level. It includes nonlinear regression models and approaches to estimate them; qualitative response regression models: LPM, logit and probit; panel data regression models: fixed effect approach and random effect approach; simultaneous-equation models: simultaneous equation bias, identification problem and approaches of their estimation (ILS, 2SLS).

ECON 311: Development and Growth Economics (3 credits)

Prerequisite: ECON 202

This course focuses on theory and the history of economic growth and development. It covers the role of economic institutions in development, structures and characteristics of developing/developed countries, growth models of Harrod-Domer, Solow-Swan, Kaldor and Joan Robinson. Moreover, issues of poverty, inequality, population, urbanization, education, health, agriculture and rural development are also part of this course.

ECON 312: Behavioral Economics (3 credits)

Prerequisite: ECON 101

This course aims to introduce the students to the foundational theories and real-world applications of Behavioral Economics. During this course, students will learn how people make decisions in real life, rather than assuming that all individuals are perfectly rational. They will examine how emotions, habits, social pressures, limited information, psychological biases, and heuristics influence choices related to spending, saving, borrowing, health, education, voting, and technology use. The course introduces concepts such as bounded rationality, loss aversion, nudging, consumer behavior, financial decision-making, online shopping behavior, and the impact of social media on economic choices.

ECON 313: Monetary Theory (3 credits)

Prerequisite: ECON 202

This course helps students to comprehend the monetary aspects of an economy by focusing on the determinants of demand and supply of money and role of financial institutions. The main topics are the nature of monetary economics, money supply process and definition of monetary aggregates, theories of money demand, microeconomic determinants of demand for money, testing the demand for money, monetary transmission mechanism, price surprises, central banking and the money supply.

ECON 315: Advanced Mathematical Economics (3 credits)

Prerequisite: ECON 103

This advanced course in mathematical economics helps students who want to pursue higher degrees in Economics. The main contents include homogeneous and homothetic functions, application of linear homogeneity, Cobb-Douglas production functions, dynamic and integral calculus, definite and indefinite integral, proper and improper integral, economic application of integration, dynamic economic models, solving differential and difference equations of various types, plotting phase diagram and trace phase trajectories, second order differential and difference equations, dynamic economic models of first order and second order, linear programming with varying techniques, nonlinear programming, Kuhn-Tucker conditions, application of linear and nonlinear programming, and optimal control theory.

ECON 320 Economics of Investment and Finance (3 Credits)

Prerequisite: ECON 203 OR Stat 101/Math 107.

This course introduces the theory and practice of investment and finance in economics. The course will cover the framework, tools and techniques for making financial and investment decisions. Specifically, it covers the topics related to time value of money, stock and bond valuation, capital budgeting, modern portfolio theory and risk-return relationship. At the end of the course, students should have a clear understanding of financial decision-making practices using excel.

ECON 322 Data Analysis for Economists (3 credits)

Prerequisite: ECON 203

Data analysis for economists is a course designed to equip students with the essential computational skills and tools necessary to excel in the field of Economics. In an increasingly data-driven world, the ability to harness the power of computing is essential for economists to conduct robust analyses, make informed decisions, and communicate their findings effectively. This course bridges the gap between economic theory and computational proficiency, empowering students to navigate complex economic problems using modern computational techniques. The course will cover fundamental software applications, and data analysis techniques to enhance students' quantitative and analytical skills in the field of Economics.

ECON 323 Digital Economics (3 credits)

Prerequisite: ECON 101 OR ECON 102

This course focuses on the analysis of markets, valuation of factors of production, changing nature of work and consumption patterns, big data and data industry, and monetary theory during the age of digital Economics, especially in the presence of both crypto and platform-based currencies as well as rising growth and advancement of fintech. It will also probe the scenarios of the creative economy within the context of prospective economic growth where each economy must compete in a globalized world for the sustainability of economic growth.

ECON 325 Islamic Economics (3 credits)

Prerequisite: ECON 100 or ECON 101 or ECON 102

This course is designed to introduce students to the basic principles and foundations of Islamic Economics. It will provide an alternative perspective to conventional economic systems and offer solutions to the core economic issues faced by the world today. Students will explore the ethical and moral dimensions of economic activities, wealth distribution, sustainability, and social justice. The main topics include Sharia-compliant economic and financial systems, the prohibition of interest and uncertainty, the distribution of wealth, Islamic financial instruments, their mechanisms, and ethical and moral practices in business and Economics. Additionally, the course covers monetary and fiscal policies, rules of taxation, and the role of government in market regulations.

ECON 326: Economics of Artificial Intelligence (3 credits)

Prerequisite: ECON 101

This course explores the economic impact of AI by integrating economic theory with technological advancements. It examines AI's influence on supply, demand, and decision-making, including rational, bounded, and behavioral economics. This course explores the areas of information asymmetry, game theory, and multi-agent modelling for complex economic systems. Students will analyze AI's social effects on inequality, workforce composition, and economic output. Finally, the course covers AI's role in shaping market structure, competitiveness, and industry transformation.

ECON 355: Urban Economics and Real Estate (3 credits)

Prerequisite: ECON 101

This course introduces students to the key concepts of urban economics and examines how economic forces influence the structure, growth, and development of cities. It explores the interactions among households, firms, and governments within urban areas and analyzes their effects on land use, housing markets, transportation systems, and urban public policy. The course also investigates the emergence and expansion of

cities, the functioning of urban markets, and the geographical concentration of economic activities. In addition, it provides students with analytical insights into major urban challenges, including housing affordability, traffic congestion, environmental quality, local public finance, and regional disparities.

ECON 395 Micro Data Analysis (3 Credits)

Prerequisite: ECON 203 OR STAT 101 /MATH 107 OR instructor's permission

This course has a broader scope and will be available to all FCCU students and students of other universities as per FCCU policy.

This course aims to introduce the process of evidence generation through data. The learners will be able to make informed decisions using data and to communicate their results effectively. The focus of the course is the practical application of micro-econometric methods based on cross-sectional and panel data of individuals, households, firms, regions, etc. The course covers data management in STATA, like; ordering variables and observations, groups and subsets of data, comparison, and missing values. changing data values, data validity, data cleaning, outliers, combining and merging data files. The course also includes various econometric models like; count data, negative binomial distribution, and logit and probit modeling.

ECON 400: Issues in Pakistan Economy (3 credits)

Prerequisite: ECON 202

This course is designed to introduce leading issues of Pakistan's economy. The main feature of the course is that it covers economic issues with reference to the political and social environment of Pakistan. The strategy of the course will be class discussions on the selected readings. The main contents of the course include poverty and income distribution, state of inflation and unemployment, growth and development issues of major sectors of Pakistan's economy and role of fiscal and monetary policies in Pakistan. The course also covers the issues related to education, population, health, corruption, balance of payment, budget deficits and governance in Pakistan.

ECON 402: Project Planning and Appraisal (3 credits)

Prerequisite: ECON 201

This course is designed to equip the students of economics with capabilities required for preparation and execution of projects with high success. It covers project planning, its process, appraisal and evaluation with special reference to Pakistan. It includes technical, social, financial and economic analysis of projects; discounted measures of project worth, i.e., BC ratio, IRR, NPV etc.; techniques used in time and resource management like CPM, PERT and WBS, project monitoring and sensitivity analysis. Moreover, critical analysis of mega projects related to health, education, transport, irrigation, electricity and environmental preservation in Pakistan would be the part of this course.

ECON 403: Time Series Econometrics (3 credits)

Prerequisite: ECON 300

This specialized course focuses on time series data analysis. Main contents include times series techniques in econometrics: dynamic econometric models, autoregressive and distributed lag models, ARIMA modeling and forecasting, measuring the volatility in financial time series (ARCH and GARCH models), vector auto-regression (VAR), co- integration and error correction modeling.

ECON 404: Corporate Finance (3 credits)

Prerequisite: ECON 320

This course is designed to introduce students to financial instruments, and the process and analysis of investment in financial markets. Both the theory and practice of investment decisions, and the diverse securities used as investment tools to implement those decisions, will be addressed in this course. The modern tools of investment such as options and futures will also be introduced in this course.

ECON 406: Introduction to Game Theory (3 credits)

Prerequisite: ECON 201

This course applies microeconomics concepts within the context of firms/industries with focus on analytical tools to understand and consequently predict behavior in multi-person decision settings. Main contents include definitions and rules of games, games with perfect certainty, symmetric and complete information, mixed and continuous strategies, dynamic games with symmetric and asymmetric information, moral hazard and adverse selection, principal agent models, cooperative and non-cooperative games.

ECON 411: Development Policy (3 credits)

Prerequisite: ECON 311

The course will cover the process, tools and basic principles of policy formation in countries like Pakistan. The main aim of the course is to develop critical skills in students to analyze past development policies with the objective of learning lessons for the future. The course will discuss in detail future forces/drivers of development in a rapidly changing global world. The main contents of the course include issues and policies related to poverty, population, health, debt crises, monetary and fiscal management, agriculture and industrial development. A practical approach of the use of data analysis and application of economic and statistical concepts will be used to enhance the capabilities of students to formulate the right kind of policies for the social and economic development of a country like Pakistan.

ECON 413: Monetary Policy (3 credits)

Prerequisite ECON 313

An advancement after learning monetary theory, this course will deepen the use of monetary theory in policy design by covering these contents: Introduction to monetary policy, policy goals and instruments, monetary policy and macroeconomic models, monetary policy coordination, Tobin tax, market constraints and policy analysis in Keynesian models, monetary policy rules, price targeting vs. inflation targeting, optimal policy with endogenous contracts.

ECON 417: International Finance (3 credits)

Prerequisite: ECON 202

This course develops a specialized taste for financial matters among nations of the world through understanding and analysis of following concepts: balance of payments, foreign exchange markets, exchange rates, purchasing power parity theory, monetary approach to the BOP and exchange rates, elasticity approach to BOP, asset market model of exchange rates, macroeconomics of open economy and international monetary system, adjustments in BOP under fixed and flexible exchange rate systems, various adjustment policies to remove internal and external balances, impact of different policies (fiscal and monetary) on open economy through aggregate demand and aggregate supply curves under different exchange rate systems, evaluation of European Monetary and International Monetary systems.

ECON 418: Economics of Financial Technologies (3 credits)

Prerequisite: ECON 101

This course explores the economic foundations and technological innovations driving Financial Technologies (FinTech). It integrates financial theory with emerging technologies such as blockchain, cryptocurrencies, artificial intelligence, machine learning, big data analytics, and digital platforms. The course examines how FinTech reshapes payments, banking, lending, asset management, and financial intermediation. Students analyze the economic implications of digital disruption, platform-based business models, decentralized finance, regulatory challenges, cybersecurity risks, and ethical considerations.

ECON 421: General Equilibrium and Welfare Economics (3 credits)

Prerequisite: ECON 202 and CGPA ≥ 3.0

The course discusses the nature of general equilibrium analysis. Main topics include Walrasian model, general equilibrium and disequilibrium models, economic efficiency and Pareto principle, social welfare function and Arrow's Impossibility theorem. Moreover, income distribution and compensation principles, imperfect competition and efficiency, market based and Pareto solutions to public goods and externalities, market failure and theory of second best will be discussed in detail.

ECON 422: Macroeconomics Analysis (3 credits)

Prerequisite: ECON 202

This advanced level course in macroeconomics provides a rigorous framework for understanding modern macroeconomics. It covers long run economic growth, short run output and employment determination, areas of consensus and controversy among various schools, fiscal & monetary policies implications under classical, real business cycle and Keynesian school of thoughts; Monetary and fiscal policies policy in the open economy; Optimal monetary policy, competing strategies: targeting monetary aggregates or interest rates, debt sustainability, government debt and Ricardian equivalence, dynamic consumption theories and investment Theories.

ECON X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

ECON 490: Research Project and Paper Writing (6 credits Spread over two semesters)

Prerequisite: ECON 302 AND CGPA > 3.3

This is a research study course and will be registered as 490a and 490b in two consecutive semesters of final year. The project must be approved by the department before its commencement. Students will conduct a research project under the prearranged supervision of a faculty member of the department on a topic of economic significance. The departmental committee will conduct a viva-voce examination of the students. The student will submit 3 hard copies as per undergraduate research guidelines available with the advisors.

ECON 498: Internship (3 Credits)

Credit will be granted for a full-time internship of at least six weeks during the summer semester. Internship credit is not retroactive and must be pre-arranged with the Head of the Department. The students must provide verification of the completion of internship on the institution's letterhead from the head of the institution, not later than one week after such completion. The grading for internship will be done by the Chairperson based on the written report/viva voce by the student.

ECON 499: Capstone Project (3 credits)

Prerequisite: ECON 302

This course serves as the culminating integrative experience for Economics majors. It requires students to apply economic theory and analytical methods to real-world problems faced by government departments, NGOs, and both public and private organizations. Core topics include research design, economic modeling, policy evaluation, data analysis, and professional economic communication.

Department of
EDUCATION

Introduction

FCCU's Department of Education was revamped in 2003 when the college returned to private ownership. It is gaining a reputation for being one of the finest Education Departments in the country. It offers a 4-year BS degree in Elementary and Secondary Education.

Mission

Our mission is to send out into the field teachers who are fully prepared to meet the challenges of teaching in a constantly changing 21st century.

Bachelor of Science Education

The BS degree follows a liberal arts education framework. This provides every student with the opportunity to explore the breadth of knowledge by taking courses in varied disciplines to fulfill the general education requirements and be better prepared to face the world. In addition, it provides every student with the opportunity to explore in depth the field of education as their area of concentration.

Graduates from this program are well prepared to face the challenges of the real classroom and its impact on young lives as they teach. A unique feature of this program is a full semester of teaching practice during the final year which provides the student-teacher with real-world classroom experience. The student-teacher works in close association with the cooperating teacher at the school and the advisor from FCC. It helps the prospective teacher visualize his/her career. Our education graduates make a difference: they TEACH!

Program Objectives

- i. To develop among students a comprehensive understanding of the field of education.
- ii. To enable students to identify complex problems prevailing in school education and conduct research studies to understand concerned phenomena and improve practices.
- iii. To help students integrate ideas, disciplines, perspectives, and contexts to understand complex educational phenomena and improve professional practices.
- iv. To enable students to identify areas of personal interest and motivate them to commit to professional excellence, life-long learning, teaching as a career, and educational research.

Program Learning Outcomes

- i. **Demonstrating Conceptual Knowledge:**
Demonstrate concepts, theories and models of the field of education that may include but not limited to, teaching, learning, assessment, educational technology, educational research, counselling and educational administration.
- ii. **Practicing Research Skills:**
Conduct research projects to understand complex educational phenomena or solve educational problems.
- iii. **Applying appropriate pedagogical approaches:**
Apply appropriate pedagogical approaches and strategies to cater to the learning needs of all students in mixed ability classrooms.
- iv. **Integration of Information and Communication Technology (ICT):**
Demonstrate integration of Information and Communication Technology (ICT) in their teaching and assessment strategies and other academic activities using innovative ideas.
- v. **Devising self-regulated learning plans:**

Devise self-regulated learning plans to keep them abreast with advancements in the field of education and to make efforts to fill the gaps in educational phenomena.

vi. **Career path and advancement:**

Identify opportunities for career advancement and further education related to the field of education.

Total Degree Requirements

Requirement Category	Total Credits
General Education and Interdisciplinary	45
Major Requirements & Internship and Capstone Project	81+3+3=87
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	132

General Education and Interdisciplinary Courses

S#	Course Title	Credits
1	UNIV 100: Foundations of University Education	3
2	UNIV 201: Critical thinking, Reflective Practice, Ethics and AI (in Education)	3
3	Social Sciences: EDUC 160: School Community and Teacher OR EDUC 130: Peace Education OR EDUC 140: Comparative Education OR Any course from Health and Physical Education. OR Any other course from Social Sciences (outside the major)	3
4	ENGL 106: Functional English	3
5	ENGL 108: Expository Writing	3
6	Arts and Humanities: URDU 101 Communicative Urdu	3
7	ISLM 101 or CRST 152	2
8	ISLM 100 A & ISLM 100 B: Fehm-e-Qur'an (two 1-cr. courses) or 2-credit Religious Studies course (for non-Muslims)	2
9	PKST 101: Pakistan Studies	2
10	PKST 103: Ideology and Constitution of Pakistan	2
11	SOCL 104: Civics and Community Engagement	2
12	Natural Sciences -I (outside the major)	3 (2+1)
13	Natural Sciences -II (outside the major)	3 (2+1)

13	As per Feat Score MATH 108: Quantitative Reasoning I (Basic) OR MATH 109: Quantitative Reasoning I (Advanced)	3
14	As per Feat Score MATH112: Quantitative Reasoning II (Basic) OR MATH 113: Quantitative Reasoning II (Advanced)	3
15	CSCS 100: Applications of Information and Communication Technologies	3 (2 +1)
16	BUSN 220: Entrepreneurship	2

Requirements for BS Education Major

Requirement for Education	Credits
Professional Courses (Core and Supporting courses)	12+24 =36
EDUC 497 A, B,C Field Experience Courses (Core Courses)	9
EDUC 498 Internship (Core Course)	3
EDUC 499 Capstone Project (Core Course)	3
Specialization Courses (Elective Courses)	18
Pedagogical Courses (Core Courses)	18
Required Major CGPA	2.0 CGPA
Total Credits	87

Core Courses	Credits
EDUC 110	3
EDUC 120/ PSYC 175	3
EDUC 200	3
EDUC 250	3
EDUC 497A	3
EDUC 497B	3
EDUC 497C	3
EDUC 498	3
EDUC 499	3
Core Courses Total	27

Elective Supporting Courses	Credits
Any Elective Supporting course offered by the department	
EDUC 210	3
EDUC 215	3
EDUC 220	3
EDUC 230	3
EDUC 242	3
EDUC 260	3
EDUC 270	3
EDUC 280	3
EDUC 290	3
EDUC 316	3
EDUC 340	3
EDUC 366	3
Supporting courses Total	24

Electives – Specialization Courses	Credits
EDUC 302	3
EDUC 336	3
EDUC 346	3
EDUC 352	3
EDUC 386	3
EDUC 396	3
Electives – Specialization Courses Total	18

Any Elective-Specialization course offered by the department.

Pedagogy Courses	Credits
EDUC 322	3
EDUC 325	3
EDUC 356	3
EDUC 360	3

EDUC 376	3
EDUC 380	3
EDUC 390	3
Pedagogy Courses Total	18

Any other Pedagogy course offered by the department

Requirements for Minor in Education

Minor in Education (Curriculum Studies) [EDUC 120, 200, 250 + 3 courses from Specialization course list]

Course Descriptions

General Education and Interdisciplinary Courses

UNIV 201: Critical thinking Reflective Practice, Ethics and AI (in Education) (3 credits)

This foundational course introduces Education students to critical thinking, reflective practice, professional ethics, and the emerging role of Artificial Intelligence in education. Students develop essential skills for analyzing educational issues, engaging in reflective teaching, making ethical decisions, and responsibly using AI tools while preparing to become thoughtful and ethical educators.

EDUC 130: Peace Education (3 credits)

This course develops students' ability to critically examine the concepts of peace, conflict, and nonviolence through global and local perspectives. Students analyze key thinkers, evaluate real-world cases, and design classroom practices that promote harmony, empathy, and social responsibility in diverse school communities.

EDUC 140: Comparative Education (3 credits)

Comparative Education provides a systematic analysis of global education systems, focusing on their similarities, differences, and influencing political, social, cultural, and economic factors. It emphasizes comparative study of selected countries and Pakistan to critically evaluate policies on access, equity, quality, curriculum, assessment, and educational reform.

EDUC 160: School, Community and Teacher (3 credits)

This course explores the vital connection between schools and communities. Students examine the teacher's role in community development, family collaboration, and inclusive practices. Topics include cultural factors influencing schooling, stakeholder engagement, and utilizing community resources to enhance student learning while understanding teachers' ethical responsibilities toward families and the broader community.

Professional Courses (Core Courses)

EDUC 110: Foundations of Education (3 credits)

Perspectives on economic, cultural, political, ideological, philosophical, aesthetic and psychological foundations of education, history of education in Pakistan.

EDUC 120/PSYC 175 *: Educational Psychology (3 credits)

Principles of Psychology as applied to the educational process, characteristics of the individual learner, the teacher, the classroom, methods and other relevant factors in the learning process, various stages of growth and development, brief introduction to psychological measurements and creativity in children.

EDUC 200: Instructional Methods and Strategies (3 credits)

Prerequisite EDUC 120

This course introduces foundational teaching methods and their applications in diverse classrooms. It covers direct instruction, active learning, cooperative strategies, and technology integration to engage students and promote critical thinking. Designed for aspiring educators, it emphasizes practical lesson planning and adaptation for mixed-ability groups through hands-on activities.

EDUC 250: Classroom Management (3 credits)

Prerequisite EDUC 120

This course provides foundational strategies for creating positive, inclusive learning environments. It covers proactive management, behavior interventions, relationship-building, and culturally responsive practices to address diverse student needs. Aimed at aspiring educators, it emphasizes ethical decision-making and self-regulation to foster equitable classrooms through practical simulations.

Field Experience (Core Courses)

EDUC 497A: Practicum – I: School Observation (3 credits)

Prerequisite: EDUC 120, EDUC 200

The Practicum-I: School Observation course introduces students to the school and classroom environment through guided observations and participation in school activities. Students observe classroom settings, teaching practices, learner engagement, and classroom management while assisting cooperating teachers in academic and co-curricular activities. The course also includes weekly reflective sessions at the university to help students discuss and critically reflect on their field experiences and professional learning.

EDUC 497B: Practicum –II: Student Teaching Experience (3 credits)

Prerequisite: EDUC 497A, three courses from the Electives – Pedagogy Courses

An extended guided field-based course with real-life experience in the classroom. All aspects of the teaching-learning process will be looked at from a practical perspective.

EDUC 497C: Practicum –III: Student Teaching Experience (3 credits)

Prerequisite: EDUC 497B; Six courses from the Electives – Pedagogy Courses

Practicum-III is an independent field-based course that provides students with extended teaching experience in real classroom settings. Students independently plan, teach, and assess lessons while developing classroom

management, reflective practice, and professional teaching competencies. The course concludes with a teaching portfolio documenting lesson plans, assessments, and professional growth.

EDUC 498: Internship (3 credits)

Prerequisite: EDUC 120, EDUC 200

This internship course provides BS Education majors with experiential learning opportunities in NGOs, educational foundations, and community organizations. Students engage in program coordination, outreach, administrative support, and community-based educational initiatives to understand the broader educational ecosystem beyond classroom teaching.

EDUC 499: Capstone Project (3 credits)

Prerequisite: EDUC 497B, EDUC 497C

The Capstone Project provides students with an opportunity to identify an educational issue or challenge during their school observations in Practicum I and investigate it using a systematic and scientific approach to explore its possible causes and propose practical solutions.

Professional Courses (Elective: Supporting Courses)

EDUC 210: Education Policies of Pakistan (3 credits)

Various educational policies of Pakistan, analyses of the successes and failures of each with identification of probable causes, analysis of policies at the preschool, elementary, secondary, higher education and teacher education levels.

EDUC 215: Learning Theories (3 credits)

Prerequisite EDUC 120

This course introduces major learning theories and their implications for teaching. Students critically compare behaviorist, cognitive, constructivist, and social learning perspectives, analyze classroom scenarios through multiple lenses, and design theory informed instructional strategies. Emphasis is placed on reflective application and developing a foundational, theoretically grounded approach to teaching.

EDUC 220: Introduction to Research Methods in Education (3 credits)

Prerequisite EDUC 120

Introduction to Research Methods introduces students to the fundamentals of research. It covers research design, question formulation, data collection, and basic analysis using qualitative and quantitative approaches. The course also highlights ethical considerations, enabling students to understand, evaluate, and conduct simple research studies effectively.

EDUC 230: Introduction to Curriculum Development, Planning and Evaluation (3 credits)

Prerequisite EDUC 120

This course introduces students to the principles and practices of curriculum design and evaluation in education. Students will explore major curriculum models, learn how to conduct needs assessments, evaluate curriculum alignment with learning objectives, and design effective instructional units.

EDUC 242: School Management (3 credits)

Prerequisite EDUC 120

This course equips students with school management skills in the Pakistani context. It explores administrative functions, leadership, human and financial resource management, curriculum planning, school culture, and community engagement. Students examine real-life scenarios and apply theory through practical tools and strategic planning.

EDUC 260: Inclusive Education (3 credits)

Prerequisite: EDUC 120 or PSYC 100

Introduction to the teaching of exceptional children either exceptionally intelligent or with difficulties, strategies for use in a regular classroom setting, methods of identifying disabilities and giftedness, ways of finding available resources to facilitate the learning process.

EDUC 270: Contemporary Literacies (3 credits)

Prerequisite EDUC 120

This course explores evolving forms of literacy in the 21st century, including digital, media, visual, cultural, and information literacies. Students critically engage with contemporary texts and technologies to understand, evaluate, and produce content across contexts, preparing them to be reflective educators and informed global citizens in a digitally mediated world.

EDUC 280: Classroom Assessments (3 credits)

Prerequisite EDUC 120

This course introduces essential principles and techniques of classroom assessment for diverse learners. It covers formative and summative methods, test construction, data interpretation, feedback strategies, and alignment with learning objectives. Education majors develop inclusive assessment tools and explore technology integration to support equitable instruction and student growth.

EDUC 290: Introduction to Guidance and Counseling (3 credits)

Prerequisite EDUC 120

The course will equip students with the knowledge and skills to support students' holistic growth and development during their school years and design appropriate programs and intervention strategies to support students become aware of and dealing with different situations affecting their academic performance and personal lives.

EDUC 316: Education for Sustainable Development (3 credits)

Prerequisite: EDUC 120

This course assists students to explore Education for Sustainable Development (ESD) principles, emphasizing sustainable thinking, ethical responsibility, and transformative learning. Students examine environmental, social, and economic sustainability in educational contexts. The course fosters critical thinking and community engagement to drive meaningful, sustainable changes through theory and practical application.

EDUC 340: Early Childhood Education (3 credits)

Prerequisite: EDUC 120

Areas of early childhood development include theories of development, discipline and guidance, and instructional methodologies for preschool children.

EDUC 366: Development of Individualized Education Plan (3 credits)

Prerequisite: EDUC 120

This course examines the development of Individualized Educational Plans (IEPs) for students with diverse learning needs in inclusive settings. It covers assessment of strengths and needs, writing measurable goals, selecting accommodations and services, multidisciplinary collaboration, and legal/ethical compliance. Upper-level Education majors engage in case studies and IEP simulations.

Elective: Specialization Courses

EDUC 302: Instructional Material Development (3 credits)

Prerequisite EDUC 230

This course focuses on principles and techniques for developing effective instructional materials. Key topics include needs analysis, media selection, design of print/digital resources, and evaluation. It emphasizes creating engaging, accessible materials for diverse learners.

EDUC 336: Models of Curriculum (3 credits)

Prerequisite: EDUC 230

This course will introduce theoretical and practical approaches to curriculum design, including objective-based, cyclical, inductive, and competency-based models. It develops students' skills in curriculum planning, alignment with standards, and application of key models such as Tyler, Taba, Wheeler, and CIPP to improve teaching and learning.

EDUC 346: National Curriculum Studies (3 credits)

Prerequisite: EDUC 120

This course provides a comprehensive overview of Pakistan's national curriculum frameworks, including the Single National Curriculum (SNC), provincial adaptations, governance, reform history, and current challenges.

EDUC 352: Localization and Contextualization of Curriculum (3 credits)

Prerequisite EDUC 120

This course explores how curriculum can be localized to meet the cultural, linguistic, and contextual needs of learners. Students analyze contextual factors and develop context responsive adaptations for curriculum design.

EDUC 386: Curriculum Planning, Design and Implementation (3 credits)

Prerequisite: EDUC 230

This course enables students to examine the principles and processes of curriculum planning, design, and implementation in educational settings. Key topics include needs assessment, objective setting, content selection, instructional strategies, and alignment with standards. It is suitable for students pursuing careers in teaching or curriculum development, emphasizing practical application through design projects.

EDUC 396: Curriculum Evaluation and Assessment (3 credits)

Prerequisite: EDUC 230

This course will enable students to examine the process of curriculum evaluation and assessment that includes examining the effectiveness, quality, implementation, and outcomes of a curriculum to determine whether educational goals are being achieved and how the curriculum can be improved.

Pedagogy Courses

EDUC 322: Teaching Islamiat (3 credits)

Prerequisite: EDUC 120, ISLAM 101 and Fehm-e-Quran I & II

This course prepares students to teach Islamiat effectively at the elementary level through learner-centered pedagogies, reflective teaching, lesson planning, and activity-based learning. It emphasizes moral development, character building, peaceful coexistence, and practical teaching skills suitable for the Pakistani context.

EDUC 325: Teaching Religious Studies (3 credits)

Prerequisite: EDUC 120, CRST 152 (For Christian Students) Or EDUC 130: Peace Education (For non-Muslims other than Christians) Religious Studies (For Non-Muslim Students)

The course will prepare students to support school students in reflecting on their religious experiences and religious identity and learn to interact and engage sensitively with their peers from different groups, religious or non-religious

EDUC 356: Teaching Languages (3 credits)

Prerequisite: EDUC 120, Functional English (GE) and Expository Writing (GE)

This course provides a comprehensive introduction to theory and practice of teaching languages in school settings. Students will explore theoretical, psychological, and socio-cultural perspectives on language learning and examine a range of instructional and assessment approaches. The course emphasizes the practical application of these concepts for effective classroom teaching.

EDUC 360: Teaching Science (3 credits)

Prerequisite: EDUC 120, Natural Science I and Natural Sciences II

This course explores curriculum concepts, teaching methods, and instructional materials for elementary science education. Emphasis is placed on the discovery and constructivist approaches, enabling prospective teachers to design engaging, student-centered science lessons that promote inquiry, critical thinking, and conceptual understanding

EDUC 376: Teaching Computer Science and Artificial Intelligence (3 credits)

Prerequisite: EDUC 120, UNIV 201 Critical Thinking, Reflective Practice, Ethics and AI (in Education), Applications of Information and Communication Technologies

This course will equip students to effectively teach Computer Science and AI at the school level. It focuses on pedagogical approaches, curriculum design, instructional strategies, assessment methods, and integrating AI tools. Emphasis is placed on making CS and AI accessible, engaging, and relevant for elementary school students.

EDUC 380: Teaching Mathematics (3 credits)

Prerequisite: EDUC 120, Quantitative Reasoning I & Quantitative Reasoning II

This course prepares students to effectively teach mathematics. It focuses on the key components of elementary mathematics, organizing content, instructional methodologies, and the development of appropriate teaching aids and support materials to enhance student learning.

EDUC 390: Teaching Social Studies (3 credits)

Prerequisite: EDUC 120, Pakistan Studies, Ideology and Constitution of Pakistan

This course provides conceptual frameworks and practical insights into the effective teaching of History and Geography at the elementary level. It equips prospective teachers with appropriate methodologies, resources, and strategies to make social studies meaningful, engaging, and relevant for young learners.

EDUC X95: Themes in Education (1-3 credits)

Sections: A-E of 1 credit F-J of 2 credits K-Z of 3 credits

The course explores themes in education keeping the current trends and practices in mind. It is more exploratory and experiential learning than simply listening to the instructor. Field visits and surveys will be conducted.

Department of
ENGLISH

Introduction

Established in 1864, Forman Christian College stands among the foundational institutions in Pakistan to offer English as a formal field of study. The Department of English possesses a distinguished lineage of academic excellence and professional preeminence in both English Language and Literature. Over the decades, it has served as the academic home to renowned luminaries—including Dr. J.C.R. Ewing, Prof. J.B. Weir, Prof. Rev. H.D. Griswold, his son Dr. Griswold, Dr. F. M. Velte, Dr. E. J. Sinclair, Dr. S. L. Sheets, Dr. F.H. Ritz, Prof. Eric Cyprian, Prof. Z. Bede, Prof. Talat Mehmood, Prof. Gillani Kamran, Prof. Dr. Kamal ud Din, Prof. Dr. Waseem Anwar, Prof. Dr. Fatima Syeda and many others—whose dedicated service has shaped the intellectual legacy of the department.

The department of English remains peerless in its structural milestones. It has conducted undergraduate as well as postgraduate classes since the inception of the department in 1901 and holds the distinction of pioneering co-education by welcoming female students to its cohorts. Today, highly qualified faculty members, with local and international PhD degrees, advance this tradition through rigorous undergraduate and postgraduate programs. Equipped with premier research facilities, our scholars engage in advanced, comprehensive research that addresses contemporary literary contexts. Our curriculum offers diverse and specialized courses reflecting recent trends in literary studies, including Critical Thinking and Reading, Introduction to Digital Literary Studies, Trauma Literature and Theory, Partition Fiction, Media and Literature, Contemporary Political Context in Pakistani Literature, Science Fiction, Research Paradigms, Postmodern Fiction, Russian Literature, Children's Literature, Ecocriticism, and many other exciting courses, thereby cultivating profound critical and theoretical sensibility in our students.

Beyond the curriculum, the department also offers a lively academic atmosphere where students and faculty participate in many literary and scholarly events. These activities are enjoyed not just by the Forman community but also by attendees and participants from other universities. The scholars engage in collaborative projects with other national and international universities to present their research on different forms such as oral presentations, research projects, poster presentations, panel discussions, digital exhibits etc. Other key events organized by the department include international conferences, joint research ventures within and without FCCU, and the Forman Literary Festival, which is organized in partnership with the Ewing English Society. There are also regular forums such as the Graduate Seminar Series, The Writer's Lounge, and the Faculty Research Forum for which experts, writers and scholars are invited on regular basis. Our biannual publication, The English Dispatch, continuously documents these milestones. Furthermore, the department has an active social media presence on different digital forums. These digital platforms help the Department of English to extend its legacy of academic excellence by sharing scholarly achievements, updates, and research contributions with the wider world. The department profile may be viewed on its official webpage <https://www.fccollege.edu.pk/department-of-english/>

Historically, the graduates of the English department FCCU have attained immense distinction across the world as educators, civil servants, jurists, diplomats, authors, and entrepreneurs who continue to contribute meaningfully to national and international intellectual communities.

BS Degree Programs

The department offers two majors at undergraduate level i.e. BS English and BS Linguistics. Both degree programs are designed to provide strong literary & linguistics foundations and advanced critical training. The purpose is to cultivate analytical reading, academic writing, research literacy, and communication skills, while postgraduate degrees emphasize research methodology, theoretical engagement, and original scholarship. This structure enables students to move confidently into doctoral research, university teaching, policy-related careers, and professional leadership roles. In addition to these degree programs, the department offers courses in Functional English and Expository Writing. These courses prepare the students to excel in their written and oral expressions required not only for all the other courses during their degree completion period, but also for their life beyond university.

Bachelor of Studies in English

The Bachelor program trains the students for lifelong effective communication in English. It teaches them to connect literature with history, theology, culture, and civilization and interpret them in the perspective of liberal education. The main objectives are to broaden the vision of students, to enlighten their minds, and to give them deep insight into literature. This program opens various options for English majors and prepares them for pursuing postgraduate research in language and literature within and outside Pakistan. They also have the option of joining various careers such as newscasting, teaching, civil services, law, journalism, editing, creative writing and publishing.

Mission of BS English Program

The mission of BS English is to educate and train students for long life effective communication in English by connecting English literature with history, theology, culture and civilization and interpret them in the perspective of liberal arts education to prepare them for careers such as teaching, creative writing, civil services, journalism, newscasting, editing and publishing.

Objectives BS English Program

- i. To cultivate intellectual growth and critical awareness through the study, interpretation, and appreciation of literary texts from diverse cultural and historical contexts.
- ii. To develop a comprehensive understanding of major literary genres, movements, and traditions, enabling the students to engage meaningfully with a wide range of literary works.
- iii. To broaden students' perspectives and enhance their analytical and critical thinking abilities through the evaluation of literary texts and theoretical approaches.
- iv. To equip students with effective research, academic writing, presentation, and publication skills, enabling them to produce well-documented and scholarly research in the field of literary studies.

Program Learning Objectives (PLOs) for BS English Program

- i. **Critical Reading:**
The students as active readers will express their appreciation for ambiguity and interpret multiple perspectives.
- ii. **Appraisal of Literary Texts:**
The students will appraise the diversity of literary and social voices within – and sometimes marginalized by – major traditions of literature in prose.
- iii. **Application of Theory:**
The students will read works of literary, rhetorical, and cultural criticism, and apply ideas from these texts in their own reading and writing.

iv. Interpreting Poetry:

The students will appraise, interpret and analyze various forms of poetry in different eras

v. Employing Skills:

The students will identify topics, formulate questions for productive inquiry, use appropriate methods and sources for research, evaluate critically the sources they find, and employ their chosen sources effectively in their own writing.

vi. Practicing Core Values:

Students will demonstrate integrity by writing plagiarism-free academic papers and will also abide by the FCCU core values.

Values

Students will demonstrate integrity by writing plagiarism-free academic papers and will also abide by the FCCU core values.

Requirements for BS English Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
9 (27 credits)	9 (27 credits)	54	2.0

Core courses	Credits
ENGL 201	3
LING 217	3
ENGL 301	3
ENGL 307	3
ENGL 315	3
ENGL 331	3
ENGL 403	3
ENGL 498	3
ENGL 499	3
Core courses Total	27
Elective Courses	Credits
The remaining courses will be chosen depending on personal interests and career goals.	27
Electives courses Total	27

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22

Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Note: **Linguistics courses can also be taken by the students majoring in English.*

Requirements for Minor in English

Core Courses	Elective Courses	Total Credits	Required Minor CGPA
4 (12 credits)	2 (6 credits)	18	2.0

Core courses	Credits
ENGL 201	3
ENGL 301	3
ENGL 307	3
ENGL 315	3
Core courses Total	12
Elective Courses	Credits
The remaining courses will be chosen depending on personal interests and career goals.	6
Electives courses Total	6

- A minor in English is open to students from all disciplines that have a minimum CGPA of 2.50.
- The two elective courses are supposed to be 300 and 400 level courses.

Bachelor of Science in Linguistics

The study of linguistics draws on methods and knowledge from a wide range of disciplines. Pursuing a major in linguistics allows students to study language scientifically and at the same time exposes students to related disciplines, all while developing insight into one of the most intriguing aspects of human knowledge and behavior. Majoring in linguistics means that students will learn about many components of human language, including the physical properties and structure of sounds (phonetics and phonology), words (morphology), sentences (syntax), and meaning (semantics). It involves looking at how languages change over time (historical linguistics); how they vary from situation to situation, group to group, and place to place (sociolinguistics and dialectology); how people use language in context (pragmatics); and how people acquire or learn language (language acquisition). Students will learn to make insightful observations, formulate and test clear hypotheses, develop arguments and draw conclusions, and communicate their findings to a wider community. A Linguistics major is therefore well equipped for a variety of jobs.

Mission of BS Linguistics Program

The mission of BS Linguistics is to educate and train students to study and interpret different aspects of language scientifically at micro and macro levels by exposing them to a wide variety of disciplines such as psychology, sociology, history, education, anthropology, philosophy, computer science, and to prepare them for careers in teaching, translation, creative writing, civil services, journalism, newscasting, editing and publishing.

Objectives BS Linguistics Program

- i. To develop a comprehensive understanding of linguistic inquiry.
- ii. To train the students to identify, relate and emphasize the importance, role, and function of language in other disciplines and vice versa.
- iii. To introduce students to the theoretical and practical aspects of the English language.
- iv. To equip students with research skills in favor of writing, presenting and publishing well documented research.

Program Learning Objective (PLOs) for BS Linguistics Program

- i. **Application & Analysis of Language:**
The students will apply and analyze the theoretical concepts and empirical findings of modern linguistics, its relationship to other disciplines and describe the synchronic and diachronic changes that English has been going through.
- ii. **Appraisal & Application of Speech Sounds:**
The students will appraise the production, perception and transmission of speech sounds and their meaningful combinations in various varieties of English.
- iii. **Interpretation of Linguistic Structures:**
The students will identify, analyze and interpret the internal structure of words, phrases and sentences and the rules that govern them
- iv. **Demonstration of Inter & Intracultural Communication:**
The students will describe and differentiate meanings in real and contextual settings through various theoretical approaches of meaning formation and demonstrate insight into inter and intracultural communication.
- v. **Interdisciplinary Linguistics:**
The students will theoretically analyze and critique the role of psychology, sociology, power, dominance and social inequalities in acquiring, learning and using language effectively.
- vi. **Employing Linguistic Skills:**
The students will identify topics, formulate questions for productive inquiry, use appropriate methods and sources for research, evaluate critically the sources they find, and employ their chosen sources effectively in their own writing.
- vii. **Practicing Core Values:**
The students will demonstrate integrity by writing plagiarism-free academic papers and will also abide by the FCCU core values.

Requirements for BS Linguistics Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
11(33 credits)	7 (21 credits)	54	2.0

Core courses	Credits
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LING 207	3
LING 217	3
LING 301	3
LING 302	3
LING 304	3
LING 319	3
LING 410	3
LING 413	3
LING 495	3
LING 498	3
LING 499	3
Core courses Total	33
Elective Courses	Credits
The remaining courses will be chosen depending on personal interests and career goals.	21
Electives courses Total	21

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements Minor in Linguistics

Core Courses	Elective Courses	Total Credits	Required Minor CGPA
4 (12 credits)	2 (6 credits)	18	2.0

Students will choose 6 courses from the given list.

Minor courses	Credits
LING 217	3
LING 301	3
LING 302	3
LING 304	3
LING 319	3

LING 410	3
LING 413	3
LING 414	3
Minor courses Total	18

Course Descriptions

ENGL 106: Functional English (3 credits)

This course is designed to develop students' functional proficiency in English through the integrated teaching of reading, listening, writing, and speaking, with systematic support for grammar, vocabulary development, and academic and real-life communication practices. The course emphasizes meaningful communication in academic, professional, and everyday contexts while fostering critical comprehension, clarity of expression, and intercultural awareness. Students will engage with authentic texts, participate in discussions and presentations, and produce a range of academic and practical written genres. The course aims to cultivate confident, ethical, and effective communicators who can navigate academic environments and real-world communication situations.

ENGL 108: Expository Writing (3 credits)

Prerequisites: ENGL 106

In the second course of the Writing and Communication Program, students will build on what they learned in the first course while learning how to research and incorporate academic sources. In the first of two major papers and presentations, students will analyze and report the history and status quo of a current controversy in Pakistan or the world. The second paper and presentation build on the first with the addition of the students' view on and possible solution for the controversy. For the second major presentation students will analyze their audience beforehand in order to craft and deliver a maximally persuasive speech in the PechaKucha style.

ENGL 110: Contemporary Short Story (3 credits)

Prerequisites: ENGL 108

This course is an introduction to major short story writers in English and American literature; emphasis on familiarity with composition, technique, style and thought process; understanding the elements of a short story that make it different from other forms of prose fiction.

ENGL 175: Critical Thinking and Reading (3 Credits)

Prerequisites: ENGL 108

This foundational course serves as an introduction to a wide range of methods for literary, critical and textual reading. This course aims to provide the students with a set of interpretive tools which they can use to analyze and critique texts in English classes and beyond. This course has a special emphasis on developing close reading and critical analysis of the text through close reading. Emphasis will be on speed reading, previewing, skimming, scanning, brainstorming and developing extensive reading habits in the students.

ENGL 201: Introduction to English Literature (3 credits)

Prerequisites: ENGL 108

This course revolves around the basic questions on the nature and function of literature, interpretation, discussion and evaluation of literary texts through a diverse and rich variety of selections from English and American literature.

Note: This course is a prerequisite for all 300-400 level literature courses.

ENGL 204: English Prose (3 Credits)

Prerequisites: ENGL 108

This course will give an overview of English non-fiction prose. It is a selective survey of representative works of (non-fiction) prose in order to create a critical understanding of what distinguishes one form of writing from another; the impact of the times and movements that bring about change in a writer's form, thought, and style. In fact, students will be exploring the development of philosophical and critical thoughts of the English nation in this course.

ENGL 207/MCOM 207*: Media and Literature (3 credits)

Prerequisites: ENGL 108

This course centers around the role of media as a means of understanding literature; importance of language and literature with reference to specific social and cultural issues.

ENGL 209: Introduction to Children's Literature (3 credits)

Pre-requisite: ENGL 108

This course offers a comprehensive introduction to the study of Children's Literature, with a focus on cultivating critical perspectives on texts primarily created for young audiences. Students will engage with a wide array of literary forms, from nursery rhymes and picture books to myths, legends, and other narratives that children encounter in their formative years. Through critical examination, students will explore the historical, cultural, and social contexts of these works, gaining a deeper understanding of the impact of children's literature on both individuals and society.

ENGL 214: 18th Century Novel (3 credits)

Prerequisites: ENGL 108

This course provides an overview of prominent English novelists; literary criticism; appreciation of the themes of different English novels both in their historic milieu and from the perspective of current language philosophies.

ENGL 220: Partition Fiction (3 credits)

Prerequisites: ENGL 108

This course aims to introduce students specifically to short stories and novella about Partition. These fictional works record the trauma of this historic moment witnessing the mass migrations, brutality and psychological trauma of the division. The authors studied during this phase will include Manto, Khushwant Singh, Bapsi Sidhwa, Joginder Paul, Jahanara Shahnawaz. This will include stories written in English and translated into English from other languages.

ENGL 250: Creative Writing (3 Credits)

Prerequisites: ENGL 108 In addition, students will have to present their writing portfolios to the instructor before they are allowed to register for this course.

This course is taught in a workshop format. Students will bring original works of poetry and fiction to be critiqued by the workshop facilitator and fellow peers. The works will be produced and revised based on the critical analysis of relevant poetry and fiction texts, discussions, and lectures by guest speakers on literary craft. As part of the course requirements, the students will attend a certain number of literary events and activities. They will be assessed for attendance and participation in these events.

ENGL 270: City & Story (3 credits)

Prerequisites: ENGL 108

What is a city? What do cities mean? Who creates, maintains and owns a city? How does a city shape an individual, a people, a country? In a global context today, are cities sites of despair, turmoil or resistance? This course employs an interdisciplinary approach and aims to familiarize students with major cities of the ancient, medieval, and modern Indian subcontinent through the lens of literature (storytelling). Each study of a selected city is paired with a literary text and scholarly work in order to impart a richer understanding of the subcontinent upon the student. This course follows a chronological order and some of the cities explored here are Kandahar, Delhi, Lahore, Peshawar, Dhaka etc.

ENGL 300: An Introduction to Russian Literature (3 credits)

Pre-requisites: ENGL 201

Russian Literature is believed to have influenced World Literature in terms of philosophical thoughts, emotions and most importantly on Russian identity and Nationalism. This course will focus on writers whose works were signified as voices of the land, produced from the late 18th century to 19th century. It will examine the main socio-political and cultural trends in Russia that influenced writers such as Pushkin, Gogol, Chekov, Dostoevsky and Tolstoy. Russian Romanticism (patriotism/nationalism), Sentimentalism, Cosmism and Realism as trends will be emphasized. This course includes all genres of literature i.e. Prose (novel & short story), drama and poetry and all texts will be studied in English. In this course students will explore the historical significance of Russian literature and trace its development into the nineteenth century.

ENGL 301: The 19th Century English Novel (3 credits)

Prerequisites: ENGL 201

This course introduces and explores the characteristics of Victorian culture, literature and history, exploring ambition, enlightenment, social, moral and economic issues of the 19th century.

ENGL 303: Science Fiction (3 credits)

Prerequisites: ENGL 108

This elective course has been specially designed for the students of English Literature, Computer Sciences, and Natural Sciences to reflect upon the entanglements of science, technology, and humanity in the contemporary world. It is aimed at informing and educating students about the dynamics of the genre of Science Fiction by critically analyzing seminal novels written in the 19th, 20th, and 21st centuries. During the classes, students will discuss the repercussions of the increasing enmeshment of humans with technology and its possible futures.

ENGL 307: Classical Drama (3 credits)

Prerequisites: ENGL 201

This course brings a critical awareness of Drama as a genre or a tradition with focus on its origin in Greece and onwards. The course aims to review and analyze the nature and function of ancient Greek and Roman

drama in its theatrical, historical and social context. The texts will be selected from the major dramatists such as Aeschylus, Sophocles, Euripides, Aristophanes, Seneca and others from classical antiquity whose works are preserved. Through all of this, our goal will be to reconstruct as fully as possible the evolution of ancient play-making. The main objective will be to integrate, as broadly as possible, the picture we receive of classical drama into that of ancient history, society and thought.

ENGL 308: Elizabethan Drama (3 Credits)

Prerequisites: ENGL 201

This course explores English drama from the Elizabethan to Jacobean Age, with attention to such playwrights as Lyly, Kyd, Marlowe, Ben Jonson, and Shakespeare. Plays will be selected from these playwrights to create critical awareness in the students regarding the questions of genre, tradition, innovation and elements of universality in these Elizabethan plays.

ENGL 310: British Romantic Poetry (3 Credits)

Prerequisites: ENGL 201

This course especially covers a critical survey of the Romantic Movement and prominent romantic poets. It will describe the distinctive position that the English Romantic poets celebrate in this movement. This course may include selected poems by romantic poets such as William Blake, William Wordsworth, S.T. Coleridge, John Keats, P.B. Shelley and others.

ENGL 315: Poetry I (3 credits)

Prerequisite: ENGL 201

In this course, students will explore the history of English poetry, looking at its changing forms in the Elizabethan, Jacobean, Restoration, Neoclassical, Reflective, Pre-Romantic, Romantic and Victorian ages.

ENGL 317: Existential Philosophy and Literature

Prerequisite: ENGL 201

The most distinctive focus of existentialism (or existential philosophy) is the nature and meaning of human existence. Why do we exist? What is our purpose in life? What is the ultimate meaning of life or is there even one to think about? These questions are intriguing as well as mind-boggling to say the least. These questions sometimes lead us to what is infamously known as “Existential Crisis”. But why does this crisis arise in the first place, is an overarching idea waiting to be dissected on an individual as well as collective level. While many avenues exist within Existentialism.

ENGL 318: The Art of Truth: Creative Nonfiction (3 credits)

Prerequisite: ENGL 201

This course will familiarize students with contemporary examples of creative nonfiction, also called the fourth genre or narrative nonfiction. Students will read different sub-genres of CNF like personal essays, literary journalism, creative cultural criticism, nature and travel writing etc. They will also try their hand at writing in these sub-genres.

ENGL 322: American Prose and Fiction (3 credits)

Prerequisites: ENGL 201

This course is designed to conduct an in-depth survey and exploration of leading developments in fiction and non-fiction prose in the United States in various ages.

ENGL 323: American Poetry (3 Credits)

Prerequisites: ENGL 201

The course offers a general survey of American poetry representing various poets such as Whitman, Emily Dickinson, Robert Frost, Sylvia Plath, Adrienne Rich others.

ENGL 325: American Drama (3 credits)

Prerequisites: ENGL 201

This course offers a selective study of the cross section of American playwrights. The idea is to develop a taste and understanding for American Drama in particular. This course aims to include selected plays from the works of playwrights such as Arthur Miller, Tennessee Williams, Eugene O'Neill, Edward Albee, Sam Shepherd, August Wilson, Thornton Wilder and others.

ENGL 328: Trauma Literature and Theory (3 credits)

Prerequisites: ENGL 201

The purpose of this course is to establish a foundational understanding of Trauma Literature & Theory in the students. To achieve this purpose, the students will be introduced to Trauma Literary and theoretical texts in English featuring geographical and historical contexts. The texts to be studied in this course are selected from the works of Jonathan Safran Foer, Mohsin Hamid, Don DeLillo, Kamila Shamsie, Khaled Hosseini, Nadeem Aslam, Sorayya Khan, Basharat Peer, Dan Fesperman and others. All these writers, in their works, have depicted the various aspects of trauma including its traces from the Holocaust, WW II, Indian Partition 1947, Hiroshima/Nagasaki atomic explosions, the separation of the East & West Pakistan, Kashmir Conflict and 9/11 attacks followed by terrorism, militancy and War and Terror.

ENGL 331: Research Methods for English Literature and Language (3 credits)

Prerequisites: ENGL 201

This Course focuses on the techniques of writing professional abstracts or proposals; thesis statement and contention; finding appropriate topics for research papers; preparation and annotation of bibliographies; writing reviews from critical and analytical perspectives; using theoretical framework, getting into the process of writing the first or later drafts; editing research according to academic requirements. For those students who are interested in pursuing a thesis, this course creates a foundation for them for preparing and submitting a proposal for a research paper, which is the final requirement for ENGL499: Thesis.

ENGL 350: Advanced Creative Writing (3 credits)

Prerequisites: ENGL 250

This course is developed for students who wish to improve their writing of fiction, literary nonfiction, and poetry in a creative, professional studio arts workshop atmosphere. Please note that this course must be capped at 15 students approximately.

ENGL 400: Re-Contextualizing Shakespearean Drama (3 credits)

Prerequisites: ENGL 201, ENGL 308

This course offers our students the chance to review and critically approach selected Shakespearean plays. The course focuses on understanding the socio-political context of the plays and their relevance to the local context and influences on local literature. The course will include selected well acknowledged Shakespearean plays (both tragedy and comedy).

ENGL 401: Forms of Fiction, Nonfiction, or Poetry

Prerequisites: ENGL 201

This course is designed for students who wish to learn more about the narrative structures and forms that underpin different literary genres. It is open to students interested in developing both creative writing and critical skills. Our major focus will be on two literary traditions from two different geographic regions: Latin America and Africa. Primary studies will focus on writing techniques with a special focus on learning the techniques the writers adopt to develop character, plot, voice, and structure. Also, varying literary histories and traditions will be presented and discussed.

ENGL 403: Contemporary Literary Criticism (3 credits)

Prerequisites: ENGL 201

This course explores the latest trends in literary criticism and theory to trace influence and interaction of contemporary literary criticism within the diverse range of literary genres.

ENGL 404: Poetry II - Contemporary Poetry in English (3 credits)

Prerequisites: ENGL 201

In this course students will study Modern poetry, Postmodern poetry, and Poetry written in the 21st century. The course will not be confined only to English poetry written by American and British poets. It will cover the poets of other English-speaking countries like Canada, Australia, and New Zealand etc. It will also include poetry in English by poets from non- English-speaking nations and translated works in English.

ENGL 405: Modern & Post-Modern Drama (3 credits)

Prerequisites: ENGL 201

This course will be taught as a series of seminars exploring the shift from modernist to postmodernist theater from the early 20th to the 21st century. It examines several plays that address important issues of modernity and postmodernity, both in their form and content. This course will explore the contribution and response of drama to social and cultural debates around the role of art, gender and sexuality, the family, the state and the nation. The students will study a different play each week, give a short presentation in one of the seminars and engage fully in the discussion of all the texts. The activities include study of plays by such dramatists as Beckett, Chekov, Ibsen, Bond, Miller, Williams, Pinter, Soyinka with background readings from other dramatic literatures.

ENGL 406: Literature, Gender, Sexuality (3 Credits)

Prerequisites: ENGL 201

“Literature, Gender, Sexuality” aims to practice reading some renowned and popular literary works from gender diversity perspective (such as feminist, queer and trans studies). Following any acknowledged yet debatable definition/ description of gender (man- masculine, woman feminine, transgender-androgynous, etc.) and sexuality (hetero, homo, bi, trans, etc.), the course highlights historical and socio-cultural constructs along with discursive [mis]conceptions that tend to multiply marginalization and increase psychological complexities. In a way, the course foregrounds literary representation of complex and at times tabooed gender identifications that are commonly blocked by dominant mindsets in a society. Introductory in spirit, critical readings selected for this course enhance debate around the aforementioned controversy related to literary representation of gender and sexuality.

ENGL 407: Pakistani Literature: Contemporary Political Context (3 credits)

Prerequisites: ENGL 220

This course will focus on the Pakistani Literature produced in the past three decades. The overlapping themes and concerns related to diaspora, hybridity culture, and gender will be discussed. Emerging political issues such as the Kashmir crisis, post 9/11 responses from Pakistan, neo-colonialism within the independent states, and marginalization of minorities will be explored. Some of the authors discussed in this course will include Basharat Peer, Wajahat Ali, Mohammad Hanif, Mohsin Hamid and Kamila Shamsie and others.

ENGL 408: Modern and Postmodern Fiction (3 credits)

Prerequisites: ENGL 201

This course explores the development of avant-garde British Modern and Postmodern fiction writers. Selected novels written by Joseph Conrad, James Joyce, Virginia Woolf, Iris Murdoch, and Don DeLillo will be discussed in the classes. Novels will be selected from the larger oeuvre of these authors to create a critical awareness in the students regarding the questions of genre, tradition, innovation and elements of universality in these texts.

ENGL 409 Transcultural Literature (3 credits)

Course Prerequisite: ENGL 201

In an era of globalization and mass migration, Transcultural Literature crosses political borders to highlight the interconnectedness of human experience. It challenges hegemonic discourse and encourages the students to explore culturally diverse perspectives via textual analysis. With its special focus on empathy, the course challenges individualistic, self-centered, and oppressive metanarrative of success and existence. It aims to connect everyday experiences with global implications and universal humanistic appeal via literature. In this course, the students will be familiarized with a range of transcultural writers and will get an opportunity to explore transculturality in regional literature as well. Including a range of genres, eras, and culture, this course navigates complexities in ideas like home, migration, identity, freedom, and belongingness. The goal is to have a scholarly understanding of transcultural literature alongside highlighting its humanistic appeal.

ENGL 410: Introduction to Digital Literary Studies (3 credits)

Prerequisites: ENGL 201

Literary Studies have evolved with the advent of digital technology and electronic gadgets. This has helped the literary scholarship to discover newer ways of approaching Literature. This course aims to familiarize the students with the emergent contours of literary studies in the wake of the digital world and the ways in which literary studies are progressing alongside the advancement of technology. Secondly, the purpose of this course is to develop a digitally informed practice that may lead to an effective application of different digital tools by the students of Literature. The students will be dealing with the digitized forms of Literature such as digital literary texts, video adaptations of literary texts, digitized Folk Wisdom, and digital Literary Archives etc. This will encourage the students to use digital tools to produce new interpretations and hitherto untapped perspectives in Literary Studies. Thus, at baccalaureate level, this course will provide the students with a foundational understanding of Digital Literary Studies.

ENGL 430: Postcolonial Literature & Theories (3 credits)

Prerequisites: ENGL 201

The focus of this course is the analysis of literature produced in the former colonies of the European Empire including Africa, South Asia, Australia and Canada. The literature produced in these countries reflects on

theorizing the issues related to power and domination, native/settler tussle, identity issues, conceptualizing diaspora, mimicry, ambivalence and hybridity. Literary analysis of fiction and poetry from postcolonial nations within the broader theoretical framework offered by postcolonial theorists.

ENGL 440/PHIL440*: Literature and Philosophy (3 credits)

Prerequisites: ENGL 201

The focus of this course is to trace the connections between two disciplines: Literature and Philosophy; students survey how writers make use of philosophical ideas, themes, vocabulary and language; study of how literary texts can focus on important philosophical issues; emphasis on both Philosophy in literature and Philosophy as Literature.

ENGL 445: Literature & Environment (3 credits)

Prerequisites: ENGL 201

Ecocriticism asks how the literary arts—one of the richest arenas for the practice of human imagination—does, has, or could shape environmental thought and action. We read critical environmental theory, literature and poetry, and examine selected films to pry open new and urgent questions about the past and present, to build alternative visions for the future. Grounded in the research and writing methods of literary studies, this course also asks participants to be global citizens and polymaths – to think across national borders and disciplinary boundaries – to open up earthy and alternative ways of interpreting the ecological crisis that are arguably relevant to students in any study program. Practically speaking, the course is broken into two parts “Ecocritical Theory” and “Ecocritical Reading”. In the first half of the course we will read and investigate a variety of contemporary ecological theoretical writings, including *Ecocriticism: A History* by Pippa Marland, *Literature and Environment* by Lawrence Buell, Ursula K. Heise, and Karen Thornber, *Writing the Anthropocene: An Introduction* by Tobias Boes and Kate Marshall; students will also be introduced to important conceptual terms such as “the Anthropocene” and learn strategies for critiquing the distinction between nature and culture. The second part of the course encompasses the reading of selected eco-conscious literature from around the world.

ENGL 496: Research Seminar Series (3 credits)

Pre-requisites: ENGL 201, ENGL 331

The course intends to prepare students for conducting research independently and being able to learn from and participate in academic discussions. Weekly seminars led by faculty, varying in topical content and scope according to faculty interest. A graduate student tutor will be assigned to each research seminar to assist in writing the weekly review, every phase of the research and writing process. The faculty member will also be responsible for preparing the students for writing one academic paper as a final assignment at the end of this course. The assessment of this course will be based on attending 90% of the seminars, preparing one page brief for each seminar (graded), preparing a set of questions and writing a one-term paper based on the experience of each seminar attended and presenting it in front of the entire group. The research seminar culminates in the writing of a major research paper by each student enrolled in. The faculty instructor will grade students' research papers. The seminars organized each semester will revolve around a single theme related to the literary, socio-political and critical debates surrounding one topic.

ENGL 498: Internship (3 credits)

Minimum 2.75 CGPA required, and it's recommended that students opt for the internship in their junior year or must have completed 70 credit hours (General Education Literature courses).

ENGL 498 is an Internship Program offered by the department of English during the Summer semester. The objective is to expose students to the job market. This course will provide them the opportunity to enhance their English language and academic skills at a broader scale. The course also helps them understand how they can utilize their degree by joining various professional domains.

ENGL 499: Capstone Project (3 credits)

Mandatory final Assessment for English Literature Majors Prerequisites: ENGL 331, ENGL 403

The students majoring in Literature will present a well-thought thesis, using primary and secondary sources to develop a critical, well-thought and well-argued, theoretically apt 11, 000word research paper (following the latest MLA style bibliography) under the supervision of the assigned instructor. The assessment will also include a viva assessed by an external examiner. The department will conduct an Internal Examination/entry test to select the students who can opt for English as Major. This examination will be conducted once every semester. The students who fail this exam will have the option of doing English Minor or re- appearing in the departmental exam for achieving the required assessment standard of the department.

ENGL X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

LING 207: History of English Language (3 credits)

Prerequisites: ENGL 108

Built around the notion of 'all languages change over time', this course surveys the development of English language from its Germanic origins in the Anglo-Saxon period till today; emphasis is on tracking some of the linguistic changes English has undergone from its origin, including changes in words, spellings, sentence structure, meaning, and sound system; major periods in the history of English language from old English to Middle English and Early Modern English to Modern Standard English; keeping to the central notion of language variation and change, the present English accents and dialects, 'World Englishes', will also be briefly considered and their implications discussed for Future English.

LING 217: Introduction to Linguistics (3 credits)

Prerequisites: WRCM 102

Theoretical concepts and empirical findings of modern linguistics on a non-technical level; highlights of the connection between linguistics and other disciplines; study of language, understanding of related disciplines.

LING 301: Phonetics and Phonology (3 credits)

Prerequisites: LING 217

Awareness of the sound system of languages with particular emphasis on English; speech- sounds and how they differ; pattern-formation, changes in sounds according to context, stress and intonation affecting meaning; introduction to the system of phonetic transcription.

LING 302: Morphology and Syntax (3 credits)

Prerequisites: LING 217

This course covers the basic concepts of morphology and syntax. Though an overview of other approaches is provided, this course takes a functional approach in the analysis of the structure of words, phrases, and

sentences. Morpho-syntactic structures, such as affixation, cliticization, simple and complex constructions, and information structure, are examined from a wide variety of languages to provide students with the skills they will need for the linguistic description and analysis of any language.

LING 304: Semantics (3 credits)

Prerequisites: LING 217

This course is a study of word meaning in human languages, especially English, and introduces students to the subtlety and richness of the cognitive capacities and explains how a cognitive system computes the meanings of complex linguistic expressions in an infinite variety of thoughts and experiences. Various theoretical approaches of meaning formation within semantics, including an analysis of the relation between words in a language system, words and concepts/objects, parts of words and sentences, approaches to context and inference, modality, componential analysis, reference, sense, mental images/representations, prototypes, dictionary definitions and lexical relations will be discussed in this course.

LING 306: Introduction to Digital Humanities for Linguistics (3 credits)

Prerequisites: LING 217

This course introduces the interdisciplinary field of Digital Humanities methods in linguistics. The course will explore the theories, methods, and tools that underpin the study of language and linguistics using digital technologies. It will also provide students with practical skills in digital analysis, as well as critical thinking and interpretation of results.

LING 308: Translation Studies (3 credits)

Prerequisites: LING 217

The course introduces some of the major concepts in translation theory and focuses on their application to translation practice. It deals with the issues of equivalence, formal properties of texts as objects for analysis at linguistic, semantic, discourse, and pragmatic levels, and emphasizes the importance of adopting a functional approach to translation practice, and a descriptive and sociological approach to translation research. Students will be provided with a comprehensive overview of the discipline of translation studies, making them aware of both the diversity of possible approaches to translation and the relationships between these approaches.

LING 309: Teaching English as a Second Language (3 credits)

Prerequisites: LING 217

Teaching reading and writing to second language students; research and theory; connections to first language/literacy models, techniques, materials and tasks that facilitate the acquisition of second language literacy.

LING 315: World Englishes (3 credits)

Prerequisites: LING 217

This course provides in-depth analysis of various varieties of English. The similarities and differences of multiple linguistics elements such as phonetic, phonological, semantic, syntactic, pragmatic and social among various varieties of English besides unparalleled growth of English as a language form the crux of this course.

LING 319: Pragmatics (3 credits)

Prerequisites: LING 217

Pragmatics helps us gain insight into inter-and-intra-cultural communication. This course studies conversations and different genres through the theories of pragmatics (strategies, routines, politeness, turn taking patterns, speech acts and motivations). The intended meaning will be broken down into speaker meaning, contextual meaning, extra-linguistic communication and relative distance. This course also helps with individual cultural nuances, experiences, and worldviews.

LING 320: Historical Linguistics (3 credits)

Prerequisites: LING 217

Historical Linguistics is a diachronic exploration of language change in terms of phonology, morphology, syntax, semantics and discourse. This course will focus on the theoretical frameworks which address why and how languages change within themselves and cross linguistically in the aforementioned areas of any language and the sociolinguistic influences which cause creole, convergence and language death.

LING 333: Bilingualism and Multilingualism (3 credits)

Prerequisites: LING 217

Bilingualism or multilingualism is an undeniable socio-psychological phenomenon which is accepted in every society. This course aims at introducing the role of socio-psychological aspects in being bilingual/multilingual and exploring theories, processes and issues involved in bilingualism and multilingualism. The concept of bilingualism and multilingualism will be viewed at three levels: individual, community and state. The students will learn about the process of code switching and mixing language in oral communication at individual level, the intervening factors which make a group give up its ancestral language and shift to a new one at community level and the language planning and policies at government level which support or hinder bi/multilingualism.

LING 340: Language and Gender (3 credits)

Prerequisites: LING 217

This course explores the relationship between language and society, with particular reference to the differences in linguistic behavior between women and men, from a cross-cultural perspective. Topics include a historical overview of linguistic stereotyping and discrimination, a sociolinguistic analysis of sex differentiation and conventions of politeness. Sex differences reflected in discourse and their social consequences are also examined. In addition, the course will consider issues concerning language structure and ideology, including sexism in English and the relative success of gender-based language reform efforts.

LING 343: Language Variation and Change (3 credits)

Prerequisites: LING 217

This course combines aspects of historical linguistics and sociolinguistics. The social factors driving language change as well as the typological constraints limiting language change are discussed and exemplified through real-world case studies. Other topics include dialects, pidgins and creoles, language maintenance, and quantitative analysis techniques.

LING 353: Second Language Acquisition (3 credits)

Prerequisites: LING 217

The course introduces students to the principal findings, concepts, and models in the field of first and second language acquisition. The pivotal question addressed is: how (healthy) children acquire grammar (including structure of sounds, words, and sentences) of their native language in a relatively short period of time apparently without extensive external assistance; emphasizing primarily on phonetics, phonology,

morphology, and syntax, the course surveys and examines theories that have been proposed to explain the observed developmental phenomena in these domains. Issues related to second language acquisition, particularly the ways it differs from first language acquisition, shall also be briefly discussed.

LING 370: Language, Culture and Identity (3 credits)

Prerequisites: LING 217

The course aims at developing an understanding of what constitutes identity and how it is related to or shaped by language and culture. The course also seeks to conceptualize views of language, literacy and cultural practices in different contexts. Some topics that will be discussed include types of identity (religious, ethnic, linguistic, cultural, and national), the issue of identity in multicultural societies, identity crisis, language attitudes, ethnic conflicts, linguistic conflicts, linguistic inequality, linguistic imposition, cross-cultural communication and culture shock.

LING 379: Language Planning and Policy (3 credits)

Prerequisites: LING 217

This course explores the main theories and issues of language planning and policy, a field that continues to grow in importance due to migration, greater recognition of minorities, and technological advances. It examines the ideologies behind language planning and policy as well as their social, political, educational, and historical consequences. This course also delves into how the spread of English has affected the language policy of various countries around the world. In addition, students will receive insight into language maintenance and revitalization. While examples from around the world are discussed in this course, special focus is given to the policies and planning related to the languages of Pakistan.

LING 410: Critical Discourse Analysis (3 credits)

Prerequisites: LING 217

This course aims to determine critical discourse analysis, its aims, special methods, and theoretical foundation. CDA mainly focuses on the study of power, dominance and social inequality using theoretical and empirical approaches in texts and non-verbal communications, while providing information as to how they are constructed through social relations, identity, knowledge and power in any written and spoken text in communities, cultures and literature. This makes it a highly trans-disciplinary course which helps to critically analyze both talk and text.

LING 413: Sociolinguistics (3 credits)

Prerequisites: LING 217

This course addresses the interdependence of language and culture in various sociological contexts and covers basic sociolinguistic theoretical models and methodologies including social network theory and ethnography. It also focuses on various aspects of human behavior and sociocultural interaction that affect language use, and explores diversity in language in relation to age, gender, region, social class, ethnicity and national origin. An exploration of attitudes and ideologies about the language varieties and choices with an emphasis on speech community, dialect and vernacular will be of particular importance to understanding this relationship in diverse cultures.

LING 414: Psycholinguistics (3 credits)

Prerequisites: LING 217

This course focuses on the psychological components of the English language. It covers areas such as lexical ambiguity, language comprehension, language production, language acquisition and methodology to understand and help change human behavioral patterns. Students will learn about the nature of language, how it is used, how it illuminates understanding of the properties of the mind and brain and has an influence on human language.

LING 415: Forensic Linguistics (3 credits)

Prerequisites: LING 217

This course prepares students to examine the complex relationship between language and law through a multidisciplinary investigation of spoken and written texts, linguistic behavior, forensic phonetics and legal issues/scenarios as they relate to detective investigations, criminal or civil disputes. The course content will be composed of theories, techniques and observations which are necessary to analyze linguistic evidence and understand characteristics of legal language based on a range of topically relevant areas, including dispute of meaning and utterance, speaker/author profiling, authorship analysis and attribution, voice/text comparison, plagiarism detection, forgery, language/behavioral crimes, and courtroom discourse analysis.

LING 422: Corpus Linguistics (3 credits)

Prerequisites: LING 217

This course aims to provide a general introduction to corpus-based language study by analyzing large digital collections of text (corpora). Students will explore the use and manipulation of language in society by analyzing and transcribing electronic texts used in conjunction with computer tools to uncover, quantify and exploit the patterns of linguistic variation by mainly looking at concordances, collocations and frequency lists within various genres

LING 429: Assessment in Second Language Acquisition (3 credits)

Prerequisites: LING 217

This course explores the theories and practices of language assessment and testing. Students will examine, present and evaluate classroom activities and create various assessment tools in different contexts to understand the theoretical and practical implications of the construction, interpretation and utilization of tests of second/foreign languages.

LING 441: Language Documentation (3 credits)

Prerequisites: LING 217

This course covers the basic principles of language and culture documentation, including issues such as how to collect data, choose proper equipment and use linguistic software. In this course, students learn how to properly plan, archive, and manage language data. This course also explores the different ethical and ideological issues related to documentation of endangered languages.

LING 442: Statistics for Linguistics (3 credits)

Prerequisites: LING 217

As usage-based approaches to linguistic analysis spread, it is increasingly important for linguists to be aware of the principles of statistics and be able to statistically model language data. This course provides students with the basic principles of descriptive and inferential statistics and focuses on the methods necessary for quantitative linguistic analysis, such as ANOVA, correlation analysis, linear and logistic regression,

conditional inference trees, and principal components analysis. These methods are demonstrated with real linguistic data and the programming language R.

LING 450: Stylistics (3 credits)

Prerequisites: LING 217

This course helps in equipping students to critically appreciate literary and non-literary texts using literary stylistics. By studying the language of the text, the course helps students to determine the communicative strategies, linguistic style and the purpose of the writer. The course will inform how the language used in texts differs from everyday language in purpose and function by studying the patterns of lexis, phonetic and metrical organization.

LING 495: Research Methods in Linguistics (3 credits)

Prerequisites: LING 217

This course prepares students for their final thesis. It covers methods of linguistic analysis, appropriate research questions in linguistics, as well as data-collection techniques. In addition, students learn how to create literature reviews, how to use a linguistic theoretical framework in their data description, and how to organize and draft a linguistic research paper. This course also addresses ethical concerns in linguistic research as well as IRB processes.

LING 498: Internship (3 credits)

Prerequisites: Senior year students who have passed three core courses of linguistics Students are advised to opt for the internship in their Senior year or must have completed 90 credit hours (General Education + 3 Linguistics core courses). LING 498 is offered in summer. It introduces students to multiple job opportunities and experiences. The internship will assist students in applying their newly gained knowledge of linguistics in several professional domains.

LING 499: Capstone Project (3 credits)

Mandatory final Assessment for Linguistics Majors Prerequisites for Linguistics majors: LING 217, LING 495

Presenting a well-thought thesis, using primary and secondary sources, synthesizing and analyzing these sources to develop a critical, well-thought, well-argued and theoretically apt 8,000-10,000 words research paper (following the latest APA style of referencing and bibliography) under the supervision of the instructor.

Department of
**ENVIRONMENTAL
SCIENCE**

Introduction

Environmental Science is an interdisciplinary subject that draws upon knowledge from the Biological, Physical, Earth and Social Sciences. Study of the subject gives students the opportunity to find out how we, through our use of water, land, and energy resources, are affecting our surroundings and the world at large. Knowledge is also gained about the actions that could be taken to reduce, control and, in some cases, reverse the damage caused to the environment. A degree in Environmental Sciences from FCCU will not only make the graduate a very well-informed person about some of the greatest challenges faced by humanity today but also point to several avenues for higher studies or gainful employment, both locally, nationally, and internationally, as worldwide demand for such professionals is growing. The Department of Environmental Sciences is part of the Faculty of Natural Sciences.

BS in Environmental Sciences is based on core courses that will strengthen the student's understanding of the broad subject of Environmental Sciences. Students can then broaden their knowledge of a specific area by studying several courses out of a large selection of elective courses on offer. At the end of six semesters, students meeting the minimum requirements can either opt for an internship or a research project while others must take two additional courses from the electives.

The core courses, offered by the department and as cross-listed courses by other departments at FCCU, begin for example with an Introduction to the subject of Environmental Sciences and the Sources of Energy, after which the student is exposed to the Earth's Physical Realm and the Statistical Techniques used for data collection and analysis. The remainder of the courses cover Environmental Analysis, Ecology and Evolution, and Environmental Impact Assessment. The final selection of courses includes Environmental Chemistry, Microbiology, and Toxicology. The elective courses are similarly available in the Departments of Chemistry, Economics, Environmental Sciences, Geography, and Physics.

The Department of Environmental Sciences has local and foreign trained faculty who are well qualified and highly experienced in teaching and researching the broad subjects making up Environmental Sciences. Their aim is to make studying Environmental Sciences at FCCU a good, rewarding, and meaningful experience.

Vision

The vision of the BS Environmental Sciences program is to contribute to the improvement of the environment globally but with an emphasis on Pakistan and to sustainable development through the efforts of its graduates.

Mission

The mission of the BS Environmental Sciences program is to prepare students for employment in environment-related areas and/or to pursue advanced degrees in environmental sciences related fields by imparting knowledge and developing understanding and relevant practical techniques and skills of environmental sciences.

Program Objectives

- i. To instill in our graduates an understanding of environmental (air, water, soil), chemical, waste management, energy and mineral resources, ecological and sustainability concepts.
- ii. Inculcate in our students an understanding of the institutional framework regarding the formulation and implementation of environmental laws and regulations.

- iii. Develop an understanding of the tools used for environmental risk assessment and evaluation of the environmental impacts.
- iv. To develop our students' critical evidence-based thinking and communication skills.
- v. Instill in our students the knowledge and skills in an ethical manner that reflects the values of the discipline and the Core Values of FCCU.
- vi. To prepare students for careers, citizenship, and environmental stewardship through experimental curricular and co-curricular opportunities.

Program Learning Outcomes

After completing the major in Environmental Sciences, students will be able to:

- i. **Subject knowledge**
Demonstrate knowledge to characterize and analyze human impacts on the environment.
- ii. **Methodology and skills**
Design and evaluate strategies, methods, and technologies, to sustainably manage environmental systems and to remediate or restore degraded environments.
- iii. **Subject knowledge of environmental laws and regulations**
Appraise and implement environmental laws and regulations when addressing environmental problems.
- iv. **Knowledge application on risk and impact assessment**
Apply the scientific tools and techniques needed for environmental risk and impact assessment
- v. **Subject knowledge communication**
Communicate complex environmental information to both technical and non-technical audiences through written reports and oral and visual presentations.
- vi. **Higher order critical thinking**
Demonstrate and apply critical thinking and problem-solving skills, allowing for the interdisciplinary nature of environmental sciences, to address environmental issues.
- vii. **Values**
Apply the ethical, cross-cultural, and historical context of environmental issues and the links between human and natural systems.
- viii. **Knowledge application**
Apply the acquired knowledge within an interdisciplinary context in solving environmental issues in different professional and research context.
- ix. **Future**
Identify/ describe/list opportunities for higher studies and career advancement in the field of environmental sciences.

Requirements for BS Environmental Sciences Major

Core Courses	Elective Courses	Total Credits	Required MCGPA
12 (36 credits)	18 credits	54	2.0

Core Courses	Credit
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ENVR 101	3
ENVR 151/PHYS 151	3
ENVR 210/GEOG 210	3
STAT 212/ENVR 212/BIOL 212/ MBIO 212	3
ENVR 250	3
ENVR 323/BIOL 323	3
ENVR 340/CHEM 340	3
ENVR 345	3
ENVR411/BIOL411	3
ENVR413	3
ENVR 498	3
ENVR 499	3
Core courses Total	36
Electives courses	Credit
ENVR 160/CHEM 160	4
ENVR 240/GEOG 240	3
ENVR 252	3
ENVR 253	3
ENVR 255/PHYS 255	3
ENVR 250	3
ENVR 260	3
ENVR 262	3
ENVR 270	3
ENVR 305	3
ENVR 306/ECON 303	3
ENVR 308	3
ENVR 309	3
ENVR 311/CHEM 311	4
ENVR 314/GEOG 314	4
ENVR 402	3
ENVR 403	3
ENVR 404	3
ENVR 405/BIOL 404	3
ENVR 406	3
ENVR 410	3
ENVR 414	3
ENVR 415	3
ENVR 416	3
ENVR 417/GEOG 416	3
ENVR 418	3
ENVR 419	3
ENVR 442/CHEM 442	4
ENVR 474/GEOG 474	4

Electives courses Total	18
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Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for Minor in Environmental Sciences

Core Courses	Elective Courses	Total Credits	Required MCGPA
3 (9 credits)	9 credits	18	2.0

Core Courses	Credit
ENVR101	3
ENVR323/BIOL323	3
ENVR340/CHEM340	3
Core courses Total	9
Electives courses Total	Credit
Students must select any 300/400 level courses offered by the department, as below:	
ENVR 305	3
ENVR 306/ECON303	3
ENVR 308	3
ENVR 309	3
ENVR 311/CHEM311	4
ENVR 314/GEOG314	3
ENVR 351/PHYS351	3
ENVR 402	3
ENVR 403	3
ENVR 404	3
ENVR 405/BIOL404	3
ENVR 406	3
ENVR 411/ ENVR 411	3
ENVR 414	3
ENVR 415	3
ENVR 416	3
ENVR 417/GEOG416	3

ENVR 418	3
ENVR 419	3
ENVR442/CHEM442	4
ENVR474/GEOG474	4
ENVR498	3
ENVR499	3
Electives courses Total	9

Requirements for Minor in Climate Change

Core Courses	Elective Courses	Total Credits	Required MCGPA
3 (9 credits)	9 credits	18	2.0

Core Courses	Credit
ENVR101	3
ENVR250	3
ENVR406	3
Core courses Total	9

Course code	Credit hours
ENVR 252	3
ENVR 253	3
ENVR 260	3
ENVR 309	3
ENVR 414	3
ENVR 419	3
STAT 206	3
BIOT 411	3
ENVR 306 /ECON 303	3
ECON 402	3
GEOG 201	3
GEOG 202	3
ENVR 240/ GEOG 240	3
GEOG 326	3
SOCL 170	3
SOCL 390	3
PSYC 330	3
Electives courses Total	9

ENVR X95: Themes (1-3 credits) Sections:

A-E of 1 credit F-J of 2 credits K-Z of 3 credits

*** Cross listed Courses:**

Courses with two designators (coding) are marked with a * to identify them as cross-listed courses. Students must select the correct designator for their applicable program to be counted towards the degree.

Courses that are cross-listed with lab courses in BIOL CHEM and PHYS will be counted towards the fulfillment of the Gen Ed. requirements.

Course Descriptions

ENVR 101: Introduction to Environmental Sciences (3 credits)

This foundational course introduces students to the core concepts, principles, and interdisciplinary nature of Environmental Sciences. It examines major global and national environmental challenges, including population growth, biodiversity loss, energy transitions, pollution, climate change, waste management, agriculture, and urbanization, while analyzing their scientific, socio-economic, and policy dimensions. Emphasis is placed on sustainability frameworks, environmental management strategies, and alignment with the Sustainable Development Goals (SDGs).

ENVR 151/ PHYS 151*: Introduction to Sources of Energy and Environment (3 credits)

Not recommended for first semester Freshmen

Conventional energy resources, fossil fuels including petroleum, natural gas, coal and tar sands, the promise and problems of nuclear energy, alternative energy sources, wind, solar, biogas, tidal etc., energy conservation, environmental pollution and its global effects.

ENVR 160/ CHEM 160*: Introduction to Organic and Biochemistry (4 credits)

Prerequisite: Intermediate or A-Level Chemistry

Bonding and structure of organic compounds, study of hydrocarbons including additions to multiple bonds and substitution reactions of benzene, petroleum products, chemistry of food and its components including carbohydrates, proteins, lipids, nutrition and caloric intake.

ENVR 210/ GEOG 210*: Earth's Physical Realms (3 credits)

Spatial and functional dynamics of major physical phenomena relating to the planet Earth – its evolution, interior state, atmosphere, lithosphere, hydrosphere and ecosphere, physical phenomena and related cycles and man-environment interactions.

ENVR 212/ BIOL 212/STAT 212*: Research Methods and Statistics for Natural Sciences (3 credits)

Research and its types, overview of research study designs, planning and implementation, ethics in research, selection of research topics, hypothesis formation, effective literature review and citation using software tools, Descriptive and Inferential Statistical methods, data collection and analysis tools, hypothesis testing, ANOVA, design of experiments and its significance. Training on software (e.g. SPSS, Mendeley, Qualtrics).

ENVR 250: Introduction to Sustainability and Climate Change (3 credits)

This course introduces sustainability and climate change as interconnected challenges shaping environmental, social, and economic systems at local, national, and global scales. It covers the foundations of sustainable

development, climate science basics, drivers and impacts of climate change, climate risk and vulnerability, mitigation and adaptation strategies, and major international frameworks and policy responses. Students will apply sustainability thinking through case-based analysis, basic foot printing and scenario evaluation, and communication of solutions for real-world contexts. This course is appropriate for Environmental Sciences students and allied disciplines seeking an applied introduction to sustainability and climate-informed decision-making.

ENVR 240/ GEOG 240*: Global Environmental Issues (3 credits)

Earth's ecosystems, major issues relating to human use and misuse of environmental resources and possible courses of action for their conservation.

ENVR 252: Environmental Pollution (3 credits)

This course deals with major problems related to pollution of air, soil and water resources. It covers processes responsible for the occurrence and release of pollutants in the environment, dispersion mechanisms, the hazards associated with different types of pollutants, problems of accumulation of toxic substances, and procedures for the reduction of emissions and remediation of contaminated environments.

ENVR 253: Energy and Environmental Stewardship (3 credits)

This course explores the relationship between energy systems and environmental impacts. Students gain an in-depth understanding of the problems associated with conventional energy sources including air pollution, climate change, and ecological degradation while introducing sustainable and green energy alternatives. Students gain understanding of energy transitions, energy policies, and the scientific principles linking energy production to environmental stewardship. Accessible to students from Environmental Sciences and other majors, the course emphasizes awareness, critical thinking, and practical insight into the emerging clean energy future.

ENVR 255/ PHYS 255*: Introduction to Meteorology (3 credits)

Prerequisite: PHYS 102 or PHYS 103 or PHYS 151

Study of the physical processes of condensation, precipitation, radiation and radiative transfer, solar radiation, atmospheric motion measuring properties of the atmosphere, ionosphere and magnetosphere, Earth's magnetic field and charge density movement in the atmosphere.

ENVR 260: Public Health and Environment (3 credits)

This course examines the linkages between environmental systems and public health, focusing on how environmental exposures shape disease patterns, wellbeing, and health equity. It covers key environmental determinants of health, including air, water, sanitation, waste, chemicals, food, noise, and climate, along with fundamental concepts of exposure pathways and environmental health risk assessment. Through case-based analysis and applied assignments, students develop skills to interpret health data and propose evidence-based prevention and control strategies relevant to environmental health policy and practice.

ENVR 262*: Soil and Environment (3 credits)

This course examines soil as a dynamic environmental system and a critical component of ecosystem functioning and environmental management. Students explore the physical, chemical, and biological properties of soil, soil formation processes, nutrient cycling, and interactions between soil and environmental quality. Emphasis is placed on soil degradation, contamination, sustainable soil management, and the role of

soil in climate regulation and ecosystem services. The laboratory component develops practical skills in soil analysis, interpretation of soil properties, and environmental assessment relevant to sustainable land management.

ENVR 270*: Environmental Monitoring (3 credits)

This course develops essential sampling and analytical skills required for professional environmental monitoring across air, water, soil, and microbiological systems. Students will perform quantitative analysis of various physical, chemical, and biological parameters while gaining proficiency in standard protocols and procedures used in contemporary research. The curriculum covers diverse sampling methods and analytical frameworks for determining both established and emerging contaminant levels. Additionally, the course integrates a spatial modeling component, introducing students to spatial analysis techniques that are increasingly critical for natural resource management and informed decision-making processes.

ENVR 305: Population Dynamics and Urbanization (3 credits)

Relationship of population, resources and sustainability, factors affecting human population size, population age structure, consequences of increase or decrease in population, population projections, population growth models and methods of controlling population, population growth in relation to resource utilization and sustainability, history and future trends of population in Pakistan integrated with the economic realities.

ENVR 306 /ECON 303*: Environmental Economics (3 credits)

Prerequisite: ECON 101

This course is an Introduction to the economic and ecological principles essential for a clear understanding of contemporary environmental and natural resource management issues. The course covers economic, legal, and ecological perspectives to better understand the causes and solutions to market failure and environmental degradation. Main contents of the course include economic efficiency and market failure, property rights, externality, measuring social welfare and welfare improvements, demand for environmental goods, environmental valuation methodologies, environmental cost-benefit analysis, environmental policy and environmental regulations, common pool resource management and depletable resource management.

ENVR 308: Environmental Laws and Policies (3 credits)

It covers federal laws and acts concerning environmental quality standards and the use of resources, legal procedures for enforcing laws, and problems concerning enforcement. This course offers a review of the National Conservation Strategy, Agenda 21, and the Pakistan Environmental Protection Act 1997.

ENVR 309: Environmental Modeling (3 credits)

Models are essential tools in environmental science for interpreting data, understanding system behavior, and exploring environmental change. This course emphasizes structure-based and conceptually transparent modeling, rather than black-box simulations. Students practice constructing and interpreting simple models using environmental datasets, with attention to model assumptions and limitations. The course develops practical modeling literacy applicable to terrestrial and climate-related environmental systems.

ENVR 311/ CHEM 311*: Fundamental Analytical Chemistry (4 credits)

Open to Juniors and Seniors

Gravimetric and volumetric methods of analysis including buffers, complexometric titrations, redox titrations, non-aqueous titrations, Karl-Fischer titrations, UV/VIS spectroscopic analysis, IR Spectroscopy, treatment of

measurement errors; usage and handling of standards, sampling, precision, accuracy, signal-to-noise ratio, limits of detection and quantitation, statistical evaluation of data; quality control and quality assurance.

ENVR 314/ GEOG 314*: Hydrosphere Resources (4 credits)

Prerequisite: GEOG 101 or GEOG 210

Origin, structure and shape of the ocean basins, composition, thermodynamics, circulation and oscillations of marine water, maritime resources and their use by humans, water resources on land, their origin, distribution, availability and quality, use and conservation as a resource for mankind.

ENVR 323/ BIOL 323*: Applied Ecology (3 credits)

This course introduces students to the applications of ecological principles for understanding, managing, and restoring natural and human-influenced ecosystems. It covers Core topics like ecosystem structure and function, population and community dynamics, species interactions, biodiversity conservation, and ecological responses to environmental stressors. The course emphasizes practical applications such as ecological assessment, conservation planning, and sustainable resource management. It is designed for environmental science students and incorporates field-based observations and problem-solving activities focused on real-world environmental challenges.

ENVR 340/ CHEM 340*: Environmental Chemistry Fundamentals (3 credits)

Prerequisite: CHEM 100 or Chemistry at Intermediate/A Level

This course explores the earth as five integrated systems, the hydrosphere, atmosphere, geosphere, astrosphere, and biosphere, focusing on the chemical phenomena within these systems and the impact of human activities. It provides a foundational understanding of the sources, reactions, transport, and effects of chemical species in air, water, and soil. Students will examine critical topics such as aquatic chemistry, atmospheric stratification, photochemical smog, and the chemistry of hazardous wastes. The integrated laboratory component emphasizes hands-on training in environmental analytical techniques, including sample collection, preparation, and the use of classical and instrumental methods to measure physicochemical variables in environmental media.

ENVR 345*: Strategic and Environmental Impact Assessment (3 credits)

This course introduces students to Environmental Impact Assessment (EIA) and Strategic Environmental Assessment (SEA) as core tools for integrating environmental considerations into decision-making at both policy/plan/program and project levels. It covers the principles, legal and procedural frameworks, screening and scoping, baseline studies, impact prediction and evaluation, mitigation and alternatives analysis, stakeholder engagement, and environmental management and monitoring. Students develop practical skills through case-based learning and structured assessment tasks aligned with professional EA documentation and review practices. The course is suitable for Environmental Sciences students and related disciplines involved in planning, development, and sustainability-focused decision-making.

ENVR 402: Solid Waste Management (3 credits)

Sources, classification, generations, onsite handling and storage, collection, transfer recycling and disposal techniques of municipal solid waste (MSW), land filling, thermal conservation, composting, concept of integrated solid waste management, existing practices and their hazards, economic evaluation of the systems, hospital waste management.

ENVR 403: Occupational, Health, Safety and Environment (3 credits)

This course introduces the principles and practices of occupational Health, Safety, and Environmental management (HSE) in workplaces, with emphasis on preventing injuries, illnesses, and environmental harm. It covers hazard identification, risk assessment and control, safety management systems, legal and ethical responsibilities, incident investigation, emergency preparedness, and environmental protection measure relevant to industrial and service-sector settings. Students will apply HSE tools through case-based analysis, task risk assessments, and development of practical control measures and documentation. This course is appropriate for Environmental Sciences students and allied disciplines seeking professional competence in workplace safety and environmental compliance.

ENVR 404: Air and Noise Pollution (3 credits)

Prerequisite: ENVR 252

In this course the physics, human and environmental health impacts are considered covering the physics of the atmosphere, the pollutants themselves and their physics, detailed human health impacts as well as the serious consequences of air pollution on the environment. Various pollutants are covered in more detail as part of the various cycles that exist in the atmosphere.

We spend a lot of our time in-doors, hence indoor air pollution is an integral part of this course. Quantification of pollutants is critical in our understanding and management of the problem hence monitoring systems and strategies and control will be studied. Lastly, noise pollution is considered. Aspects of the impact and control of noise pollution make up the bulk of the material covered.

ENVR 405/ BIOL 404 *: Conservation Biology (3 credits)

The objective of this course is to familiarize the students with different forms of biodiversity, threats to biodiversity and an overview of different strategies for its conservation.

ENVR 406: Climate Change Mitigation and Adaptation (3 credits)

Pre-requisites: ENVR101/ENVR 250 /ENVR252

This course provides an in-depth examination of climate change science, its drivers, observed impacts, and projected future risks. Students evaluate adaptation and mitigation strategies at global, regional, and local scales, integrating scientific, socio-economic, and policy perspectives. The course emphasizes evidence-based approaches to addressing climate risks, including renewable energy transitions, carbon management, resilience planning, and international climate governance.

ENVR 410: Pollutant Movement in Soil

This course examines the nature, monitoring, and estimation of pollutants within the soil environment, providing an overview of contaminant sources and their transport through soil and water systems. Students will investigate the environmental fate and impact of various contaminants to effectively assess risks to both human and ecosystem receptors. The curriculum covers fundamental principles including reaction kinetics, mixing processes, and chemical equilibrium, while exploring pollutant transport across diverse spatial and temporal scales to inform the assessment and remediation of soil contamination.

ENVR 411/BIOL 411 *: Environmental Microbiology (3 credits)

Open to all programs under Biological Sciences

This course provides a comprehensive introduction to the role of microorganisms in environmental systems and their significance in ecological processes, environmental quality, and applied environmental management. Students examine microbial diversity, growth dynamics, and interactions within terrestrial, aquatic, and extreme environments, with emphasis on nutrient cycling, biodegradation, and bioremediation. The course integrates theoretical understanding with laboratory-based training to develop practical skills in microbial analysis, environmental sampling, and basic microbiological techniques relevant to environmental monitoring, agriculture, water quality management, and industry applications.

ENVR 413: Environmental Toxicology (3 credits)

The course will introduce the concepts of adverse effects of environmental chemicals, from natural and anthropogenic sources, on human and other living organisms. It is focused on providing knowledge related to dose response relationships, acute and chronic effects on organ system, their containment and control strategies; Toxicokinetics; Immunological considerations in toxicology; Toxicant testing techniques: *In vitro* and *in vivo*; Risk assessment and management.

ENVR 414: Sustainable Development in a Changing Climate (3 credits)

In this course, students will be introduced to the core principles of sustainable development, exploring the interconnections between economic growth, environmental stewardship, and social equity, while examining the complex relationship between sustainable development and climate change. The course examines how global climate challenges affect environmental, social, and economic sustainability, with a focus on achieving the Sustainable Development Goals. Through case studies and practical applications, the course equips students with the skills to design and implement climate-smart solutions for sustainable development in a rapidly changing world. The course is suitable for students of Environmental Sciences and related disciplines who want to contribute to sustainable solutions in areas such as renewable energy, environmental management, climate policy, conservation, or sustainable business practices

ENVR 415: Environmental Management Systems (3 credits)

This course will deliver understanding about the Environmental Management Systems (EMS) and how EMS leads to environmental benefits to industries. Introduction of ISO 14000 series of standards and their role in environmental management; Certification of EMS; Principles of Cleaner production, tools of sustainable consumption and production (eco label, eco-design, cleaner technologies, etc.), Corporate social responsibility (CSR).

ENVR 416: Wastewater treatment (4 credits)

Pre-requisites: CHEM100/ENVR340

This course provides an in-depth introduction to the principles and practices of wastewater management, focusing on the generation, characteristics, and treatment of wastewater. Students examine major sources of wastewater and their physical, chemical, and biological characteristics, followed by the design and operation of collection systems and treatment processes, with an emphasis on biological treatment of domestic wastewater. Through integrated laboratory activities, students gain hands-on experience in wastewater sampling and quantitative analysis and develop practical skills essential for professional work in environmental and water-related fields.

ENVR 417/ GEOG 416*: Natural Hazards and Management Issues (3 credits)

Prerequisite: ENVR 240/GEOG 240

The course analyzes the natural phenomenon causing hazards, related issues and problems; means and techniques of ascertaining their distribution, impact on human life and possibilities of recurrences; options for disaster preparation and loss mitigation.

ENVR 418: Water Treatment (3 credits)

Pre-requisite: CHEM100/ENVR340

This course provides a thorough understanding of the key concepts and principles of water treatment and management, emphasizing the unit operations used for surface and groundwater treatment (physical and chemical processes), their effects on water quality, and their roles within a complete treatment train. Case studies illustrate how water utilities integrate multiple processes in real facilities. The course develops both theoretical foundations and real-world application of each unit process.

ENVR 419: Environment Social and Governance (3 credits)

Pre-requisite: ENVR101/ENVR252/ENVR250

This course provides an in-depth understanding to students about Environmental, Social, and Governance (ESG) principles and their significance in shaping sustainable development and responsible business behavior. Students study climate-related disclosures, biodiversity risks, environmental justice, and governance mechanisms guiding sustainable decision-making across public and private sectors. The course prepares Environmental Sciences students for roles in environmental management, regulation, sustainability reporting, and research.

ENVR 442/ CHEM 442*: Green Chemistry (4 credits)

Open to Juniors and Seniors

Green chemistry, principles, evaluating materials, feed stocks and starting materials, types of reactions in chemical transformation, evaluation of methods to design safer chemicals, green chemistry and future trends.

ENVR 474/ GEOG 474*: Geographical Information Science (4 credits)

Prerequisite: GEOG 201 or GEOG 202 or GEOG 203

Principles of geographical information science, functions of geographic information systems, relationship between GIS and remote sensing.

ENVR 498: Internship (3 credits)

Prerequisite: Open to Junior and Senior students majoring in Environmental Sciences

This course provides supervised professional experience in environmental sciences through structured placement in government agencies, industry, research organizations, or non-governmental institutions. Students apply disciplinary knowledge in real-world settings, engaging in activities such as environmental monitoring, compliance support, sustainability initiatives, data collection, or technical reporting. The internship requires a minimum of 240 hours of professional engagement and emphasizes professional ethics, workplace communication, and applied problem-solving. Assessment is based on supervisor evaluation, reflective documentation, and a final report and/or presentation demonstrating learning outcomes and professional competency.

ENVR 499: Capstone Project (3 credits)

Prerequisite: Open to Senior Students majoring in Environmental Sciences

This compulsory capstone course serves as the culminating academic and professional experience of the BS Environmental Sciences program. It requires students to integrate scientific knowledge, analytical tools, legal frameworks, sustainability principles, and interdisciplinary problem-solving approaches to address a real-world environmental issue. Students design and implement an independent or supervised applied research project, policy analysis, environmental assessment, GIS-based investigation, environmental management plan, or sustainability strategy. The course emphasizes structured research design, ethical compliance, data analysis, professional reporting, and public presentation. The capstone project demonstrates measurable attainment of Program Learning Outcomes and prepares students for professional practice, graduate study, consulting, regulatory work, and research roles.

Department of
**GLOBAL
LANGUAGES AND
CULTURES**

Introduction

Foreign language education at FCCU began in Fall 2018, laying the foundation for what is now the Department of Global Languages and Cultures (DGLC). DGLC promotes multilingualism, intercultural understanding, and global engagement through credit-based courses in French, German, Chinese, Korean, and Spanish, as well as academic degree programs, beginning with the French Minor.

Program Objectives

FCCU offers foreign language courses at two levels in French, German, Chinese and Korean. In the basic level courses, students learn to understand and use familiar and everyday expressions and very simple sentences which relate to their needs. In the higher-level course, they learn to communicate at a more complex level. Communication skills are taught through the four competencies of reading, writing, listening and speaking. The courses are taken by students who are interested in different languages as part of their general education requirements.

Mission

The mission of the Department of Global Languages and Cultures is to foster cultural understanding and international engagement through the discovery, interpretation, translation, and sharing of linguistic and cultural knowledge. The department is committed to preparing students to participate thoughtfully, ethically, and effectively in an interconnected global society.

French Minor

The French minor program, offered in collaboration with the French Embassy is designed for students from all academic disciplines and accommodates both beginners and those with prior exposure to French. Upon completing the minor, students will be able to communicate effectively in everyday and academic contexts, demonstrate awareness of Francophone cultures, and apply basic foreign-language teaching methodologies, enabling them to teach French at introductory levels or continue toward professional teacher training.

Requirements for French Minor

Core Courses	Elective Courses	Total Credits	Required MCGPA
18 credits	6 credits	24	2.0

Core Courses	Credits
FREN 101	3
FREN 102	3
FREN 201	3
FREN 202	3
FREN 301	3
FREN 302	3
Core Courses Total	18

Elective Courses	Credits
FREN 210	3

FREN 310	3
Elective Courses Total	6

Course Descriptions

French Minor Core Courses

FREN 101: French I (3 credits)

This course provides an introduction to the French language and its various aspects. Students will learn basic language skills, including listening, speaking, reading, and writing. Culture studies will include the history and geography of France with a focus on Paris, and also French speaking countries all over the world. Each lesson introduces new vocabulary and grammar concepts through listening comprehension, speaking, and writing activities. Students learn to talk about themselves and others, describe their surroundings and use numbers for dates and time. Regular verbs are introduced in the present tense. Simple grammatical structures are practiced in innovative and interesting ways with a variety of learning styles in mind. This course includes a broad range of lessons and activities that offer a variety of modalities for ultimate student engagement and content retention. Students develop an ability to communicate in real-life situations by acquiring reading, writing, listening, and speaking skills.

FREN 102: French II (3 credits)

Prerequisite: FREN 101

This course focuses on teaching how to communicate in a simple manner if the person they are speaking to speaks slowly and clearly and is willing to help. Communication skills will be taught through the four competences, i.e. writing, reading, listening and speaking. At the end of the course students will be able to talk about their tastes and interests, express preferences and desires and justify their choices, accept and refuse invitations, use the near future tenses, and talk about their activities in the past.

FREN 201: Intermediate French I (Oral Comprehension and Production) (3 credits)

Prerequisite: FREN 102

This course develops students' ability to understand and produce spoken French at the intermediate level. Emphasis is placed on listening comprehension, spoken interaction, pronunciation, and communicative fluency in everyday, academic, and cultural contexts. Through structured practice, role-plays, discussions, and audiovisual materials, students enhance their confidence in expressing opinions, narrating experiences, and participating in conversations in French.

FREN 202: Intermediate French II (Written Comprehension and Production) (3 credits)

Prerequisite: FREN 102

This course develops students' ability to read, understand, and produce written French at the intermediate level. Emphasis is placed on reading comprehension, structured writing, vocabulary development, and grammatical accuracy. Students engage with a variety of texts and practice writing coherent paragraphs, short essays, and functional texts related to everyday, academic, and cultural contexts.

FREN 301: French for Intercultural Communication - (B1–B1+ Level) (3 credits)

Prerequisite: FREN 201 and FREN 202

This course offers improved proficiency in French after the intermediate level through integrated practice in listening, speaking, reading, and writing. Emphasis is placed on fluency, accuracy, and complexity of expression in academic, professional, and cultural contexts. Students engage with authentic texts and media while refining grammatical control and communicative competence at the DELF B1–B1+ level.

FREN 302: Teaching French as a Foreign Language (3 credits)

Prerequisite: FREN 201 and FREN 202

This course explores the theory and practice of teaching French as a Foreign Language or Français Langue Étrangère (FLE) with a focus on developing reading, writing, speaking, and listening skills for elementary and lower secondary learners. Students examine language development, literacy instruction, lesson planning, classroom management, and assessment strategies in the French language classroom. The course integrates pedagogical theory with practical application through activity design, reflective practice, and lesson plan execution.

Electives for French

FREN 210: French grammar and language structure (3 credits)

Prerequisite: FREN 102

This course provides a systematic study of French grammar and language structure at the elementary to intermediate level. It focuses on the formation and use of grammatical structures, sentence construction, and linguistic accuracy in both written and spoken communication. Students develop analytical awareness of how French functions as a structured language while strengthening their ability to use grammar effectively in real-life and academic contexts.

FREN 310: French Culture and Civilization (3 credits)

This course introduces students to the historical, social, cultural, and intellectual foundations of French and Francophone civilization. Taught in English, the course explores key developments in French history, art, philosophy and everyday life. Students engage with cultural texts, visual media, and contemporary issues to develop an informed understanding of France and the wider Francophone world.

Other Credit Based Language Courses

GRMN 101: German I (3 credits)

This basic level course focuses on understanding and using simple everyday expressions which relate to satisfying concrete needs such as information about yourself and your family; daily routines and punctuality; hobbies and professions; shopping and directions in a new city; time with friends and celebrations. This course also presents essential vocabulary and grammar, and develops pronunciation, listening, reading, and writing skills necessary for basic communication and comprehension. At the end of the course students will be able to communicate everyday survival tasks including asking questions, counting numbers, giving directions, expressing needs, and making requests in a simple manner.

GRMN 102: German II (3 credits)

Prerequisite: GRMN 101

This course teaches students how to communicate in a simple manner if the person they are speaking to speaks slowly and clearly. Communication skills are taught through the four competences, i.e. writing, reading,

listening and speaking. At the end of the course, students will be able to talk about their daily work routine, health, home, studies and professions, clothes and holidays. Moreover, they will be able to write emails, text messages and blogs in the present and perfect structure combination. Additionally, students are prepared for the A1 Level of Common European Framework (CEFR).

CHIN 101: Chinese I (3 credits)

This basic level course teaches students how to recognize the (Mandarin) Chinese language in Pinyin and also the basic character set and its pronunciation. The course introduces students to Chinese culture to gain knowledge and understanding of the Chinese-speaking world. At the end of the course, students will be able to communicate at basic everyday survival tasks including self-introduction, asking questions, counting numbers, giving directions, expressing needs, and making requests in a simple manner.

CHIN 102: Chinese II (3 credits)

Prerequisite: CHIN 101

This course teaches students to communicate about basic tasks requiring a simple and direct exchange of information on familiar and routine matters. It enables students to describe in simple terms aspects of their background, immediate environment and matters in areas of immediate need e.g. basic personal and family information, shopping, etc. At the end of the course students will be able to understand the main ideas and supporting ideas when people talk about familiar topics and will know how to carry on a conversation on a variety of topics that are familiar to them. They will also understand short, simple readings on familiar topics and be able to write a series of simple sentences on topics of interest.

KORN 101: Korean I (3 credits)

This basic level course teaches students to read and write the Korean alphabet and learn its pronunciation. It also introduces Korean culture to gain knowledge and understanding of Korea. At the end of the course students will be able to communicate with native Korean speakers at basic everyday survival tasks such as introducing themselves and expressing their needs, asking questions and making requests in a simple manner, counting numbers, and giving directions.

KORN 102: Korean II (3 credits)

Prerequisite: KORN 101

This course teaches students to communicate about basic tasks requiring a simple and direct exchange of information on familiar and routine matters. It also teaches students how to describe in simple terms aspects of their background, immediate environment and matters in areas of immediate need. Students will learn more about Korean culture to gain further knowledge and understanding of Korea. At the end of the course, students will be able to understand the main ideas and supporting ideas when people talk about familiar topics and will know how to carry on a conversation on a variety of topics that are familiar to them. They will also understand short, simple readings on familiar topics and be able to write a series of simple sentences on topics of interest.

SPAN 101: Spanish I (3 credits)

Prerequisites: None

Spanish is the second most spoken native language in the world after Mandarin. It is a global language with approximately 600 million native and non-native speakers. It is the official language of 20 countries and one of the six official languages of the United Nations. As a result, it can be extremely useful for students whether in pursuing further studies abroad or in employment. A survey conducted in April 2024 by the QEC in FCCU

found that of the 324 undergraduates who responded, the largest number (181) were interested in learning Spanish. Their primary motivation was to learn about the cultural heritage of the language and the countries it is spoken in.

SPAN 102: Spanish II (3 credits)

Prerequisites: SPAN 101

Spanish 102 builds upon Spanish 101 to further develop students' foundational communication skills in Spanish. The course introduces new grammatical structures while strengthening command of the present tense. Students practice applying Spanish in practical contexts, including common daily interactions, travel scenarios, and discussions about personal plans and experiences. Designed for learners continuing their Spanish learning journey, this course emphasizes fluency, cultural awareness, and confidence in real-life communication.

Department of
GEOGRAPHY

Bachelor of Science in Geography

Introduction

The Department of Geography has the distinction of being pioneers in the subject in this part of the sub-continent, with glorious traditions and 100-year distinctive history. Geography is a systematic discipline that studies spatial and temporal interactions between people and their environment and explores the means to synthesize them for the benefit of society.

Vision

The vision of the Department of Geography is to provide its students with educational experiences of the finest quality, exciting a new, global generation of interdisciplinary thinkers to find ground-breaking outcomes to the interconnected environmental, social, and technological challenges facing our planet. We are committed to outstanding teaching and scholarly activities within the framework of academic freedom, the diversity of people and ideas, a spirit of inclusiveness, and a global perspective.

Mission

The Department's mission is to groom students as contributing persons to humanity and as inspiring leaders in the nation-building process besides focusing on intellectual acumen to take on labyrinthine challenges of today and the future. Its vision is to acknowledge the innovative and versatile community that can be looked upon to foster scholastic partnerships in seizing emerging horizons of knowledge.

Learning Objectives

Department embarks on the following objectives:

- Analyze and synthesize in-depth knowledge of different concepts and processes of physical, human and regional geography.
- Demonstrate competency in the use of fundamental geographic tools/techniques for data collection, display and analysis.
- Productive individuals in the service of humanity and develop themselves to become inspiring leaders in the progress and uplift of the country.
- Design an independent research project with competence to build a spatiotemporal profile of the phenomena under investigation that helps in reaching recommendations for development.
- Employ knowledge and skills that help in advanced study in the discipline for job placement.

The Department of Geography is part of the Faculty of Social Sciences.

Courses Recommended for General Education Requirement:

Any open course in lower or upper division. Options for other upper division course(s) may be considered by the chair on recommendation by the instructor.

Academic Programs

The Department of Geography is offering two Specializations under its umbrella:

- Human Geography Specialization
- Physical Geography Specialization

Human Geography Specialization

Program Learning Outcomes

- Nature and Society**
Able to articulate the theories, philosophies, and concepts in the field of geography, including fundamental themes of spatial patterns and processes of the interrelationship between people and places, and the interactions between nature and society.
- Human Spatial Patterns**
Understand global and regional patterns of cultural, political and economic institutions, and their effects on preservation, use and exploitation of natural resources and landscapes.
- Geospatial Technologies**
Apply mapping and geospatial technologies to analyze geographic data in geographic research and analysis.
- Research Skills**
Capable to clearly and effectively communicate geographic knowledge and research in oral, written, and visual form.
- Problem Solution**
Recognize and evaluate how geographic concepts apply in the workplace and in everyday life to solve real-world problems.

Student Learning Outcomes:

- Able to articulate the theories, philosophies, and concepts in the field of geography, including fundamental themes of spatial patterns and processes of the interrelationship between people and places, and the interactions between nature and society.
- Understand global and regional patterns of cultural, political and economic institutions, and their effects on the preservation, use, and exploitation of natural resources and landscapes.
- Apply mapping and geospatial technologies to analyze geographic data in geographic research and analysis.
- Capable to clearly and effectively communicate geographic knowledge and research in oral, written, and visual form.
- Recognize and evaluate how geographic concepts apply in the workplace and in everyday life to solve real-world problems.

Requirements for Human Geography Specialization

Core Courses	Elective Courses	Total Credits	Required MCGPA
9 (28 credits)	8 (24 credits)	52	2.0

Electives must include:

- At least Five (5) courses at the 300/400 level in Human Geography

- At least three (3) courses relating to GIS or Geospatial Analysis

Core Courses	Credits
GEOG 101	3
GEOG 201	3
GEOG 274	3
GEOG 301	3
GEOG 311	3
GEOG 313	3
GEOG 471	4
GEOG 498	3
GEOG 499	3
Core Courses Total	28

Elective Courses Human Geography	Credits
GEOG 322	3
GEOG 323	3
GEOG 324	3
GEOG 325	3
GEOG 326	3
GEOG 421	3
GEOG 422	3
GEOG 426	3
Total Required	15

Elective Courses GIS or Geospatial Analysis	Credits
GEOG 371	3
GEOG 372	3
GEOG 374	3
GEOG 474	3
Total Required	9

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	52

University Electives	21-24
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Physical Geography Specialization

Program Objectives

- i. Understand the physical principles and processes governing the circulation and characteristics of the atmosphere and climates on Earth.
- ii. To develop specific skills related to work techniques, particularly those related to the analysis, process and representation of geographical information and fieldwork.
- iii. Students will learn geographic theory and its use in understanding real world processes. **4. Determine the directional and locational systems employed on the surface of the Earth.**
- iv. Be able to use and analyze maps.

Program Learning Outcomes

- i. **Human Environment Interactions**
Understanding of physical geographic processes, the global distribution of landforms and ecosystems, and the role of the physical environment on human populations.
- ii. **Environmental Support System**
Able to explain the basic operations of environmental systems (biosphere, lithosphere, hydrosphere, atmosphere).
- iii. **Geospatial Technologies**
Apply mapping and geospatial technologies to analyze geographic data in geographic research and analysis.
- iv. **Research Skills**
Capable to clearly and effectively communicate geographic knowledge and research in oral, written, and visual form.
- v. **Problem Solution**
Recognize and evaluate how geographic concepts apply in the workplace and in everyday life to solve real-world problems.

Student Learning Outcomes

- Understanding of physical geographic processes, the global distribution of landforms and ecosystems, and the role of the physical environment on human populations.
- Able to explain the basic operations of environmental systems (biosphere, lithosphere, hydrosphere, atmosphere).
- Apply mapping and geospatial technologies to analyze geographic data in geographic research and analysis.
- Capable to clearly and effectively communicate geographic knowledge and research in oral, written, and visual form.
- Recognize and evaluate how geographic concepts apply in the workplace and in everyday life to solve real-world problems.

Requirements for Physical Geography Specialization

Core Courses	Elective Courses	Total Credits	Required MCGPA
9 (28 credits)	8(24 credits)	53	2.0

Electives must include:

- At least Five (5) courses at the 300/400 level in Physical Geography
- At least three (3) courses relating to GIS or Geospatial Analysis

Core Courses	Credits
GEOG 101	3
GEOG 201	3
GEOG 274	3
GEOG 301	3
GEOG 311	3
GEOG 313	3
GEOG 471	4
GEOG 498	3
GEOG 499	3
Core Courses Total	28

Elective Courses Physical Geography	Credits
GEOG 312	4
GEOG 314	4
GEOG 396	2
GEOG 411	3
GEOG 416	3
Total Required	16

Elective Courses GIS or Geospatial Analysis	Credits
GEOG 371	3
GEOG 372	3
GEOG 374	3
GEOG 474	3
Total Required	9

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	52
University Electives	21-24
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for Minor in Geography

Core Courses	Elective Courses	Total Credits	Required MCGPA
4 (credits)	28 (credits)	18	2.0

Core Courses	Credits
GEOG 210/ENVR 210	3
GEOG 220	3
GEOG 270	3
Core Courses Total	9
Elective Courses	Credits
minimum of two electives at the 300/400 level courses.	9
Electives courses Total	9

Course Categories				
General	Physical Geography	Human Geography	Regional Geography	Geographical Techniques and Methods
101	210	220	133	201
191	240	221	232	202
192	311	222	233	203
301	312	322		270
495	313	323		274
	314	324		371

	396	325		372
	411	326		374
	416	421		471
		422		474
		426		498
				499

Course Descriptions

GEOG 101: Fundamentals of Geography (3 credits) General

Builds perspective about geography as a discipline; familiarizes its thematic domains and fundamental concepts.

GEOG 133: Geographical Profile of Pakistan (3 credits) Regional Geography

It relates to major features of the physical environment, resources, culture, communications and trade of Pakistan. Analyses the major problems confronted by Pakistan relating to cross cultural relationships, socio-economic viability, environmental conservation, resource sustainability and development.

GEOG 191: ArcGis I (2 credits)

Familiarizes the essential skills needed to navigate and operate ArcGIS at a basic level using hands-on practices. Includes how to utilize GIS notions, approaches and procedures in conjunction with problem solving techniques to accomplish assigned real world examples.

GEOG 192: Remote Sensing Data Acquisition and Processing (2 credits)

A set of comprehensive practical hands-on approaches to navigate, search, acquire, integrate and process Remote Sensing data including Satellite Imageries, Digital Elevation Models (DEMs) and Environmental parameters datasets.

GEOG 201: Introduction to Geospatial Information Science and Technology (3 Credits) (2+2) Geographical Techniques and Methods

The course brings the use of geotechnologies, such as Global Positioning System (GPS), Remote sensing (RS), and Geographic Information Systems (GIS) to think about problems spatially without being encumbered by overlay complex software.

GEOG 202: Fundamentals of Remote Sensing (3 Credits) (2+2) Geographical Techniques and Methods

The course describes basic concepts in Remote Sensing and discusses how Remote Sensing tools may be used to study the Earth's environments and solve real world problems.

GEOG 203: Spatial Databases 3 Credits (2+2) Geographical Techniques and Methods Geospatial database design, implementation, management, and access course in organizing spatial and non-spatial data

in geospatially enabled enterprise Database Management Systems (DBMS). Students will gain practical database experience utilizing commercial database management system software and geographic information systems software to design and use spatial and non-spatial data.

GEOG 210/ENVR 210: Earth's Physical Realms (3 credits) Physical Geography

This course expresses the spatial and functional dynamics of major physical phenomena relating to the planet Earth - its evolution, interior state, atmosphere, lithosphere, hydrosphere and ecosphere. It further explores physical phenomena and related cycles, and man environment interactions.

GEOG 220: Human Domains of Geography (3 credits) Human Geography

The course focuses on concepts relating to the spatial and systematic organization of economic, cultural, political, demographic and occupancy milieu, arising out of human use of the earth's environment. It also deals with the importance of human attitudes and values in resource use and shaping of the patterns.

GEOG 221: Geography of Tourism (3 credits) Human Geography

This course focuses on physical and cultural factors affecting the location & relative importance of recreational areas and tourist attractions. Spatial analysis of tourist flows, modes of transportation, effects on regional economies, and impacts on environments.

GEOG 222: Globalization – An Introduction (3 credits) Human Geography

This course aims to focus on evolution and dynamics of globalization and its impact on spatiotemporal patterns of human culture.

GEOG 232: World Regional Geography (3 credits) Regional Geography

The emphasis is placed on the location, spatial distribution and interaction of human activities and resource patterns in a global context.

GEOG 233: Geography of South Asia (3 credits) Regional Geography

The course examines South Asia's Physiography, climate, settlement, population, historical geography, economic activities, and cultural landscape. It will also give an understanding of the major geographical patterns, processes, issues and problems of the region.

GEOG 240/ENVR 240: Global Environmental Issues (3 credits) Physical Geography

This course describes the earth's ecosystems; major issues relating to the human use and misuse of environmental resources, and possible courses of action for their conservation.

GEOG 270: Maps and their Interpretation (3 credits) Geographical Techniques and Methods

The course builds up the capability to understand & interpret different types of maps and their applications.

GEOG 274: Fundamentals of Cartography and Field Surveying (3 credits) Geographical Techniques and Methods

Lab Course

The course develops basic skills of map making, their use and contemplation techniques. Also, it aims to train students in the collection and processing of field data. It also includes basic training in field surveying for map making.

GEOG 301: Workshop on Geographical Thought and Concepts (3 credits) General

The course analyzes the current philosophical themes in geography, as well as the systematic doctrines and concepts that overwhelm the mainstreams of the discipline. Implications of the current strides on the cognitive domains are assessed and their impact on the future course of geographical avenues is envisaged.

GEOG 311 Principles of Atmospheric and Hydrospheric Dynamics (3 credits) Physical Geography

Prerequisite: GEOG 101 or 210

The course examines elements and forces generating weather phenomena, their dynamics and impact, climatic system and their classification. Features and dynamics of marine masses are also discussed; and a generalized appraisal of the water resources on land is reviewed.

GEOG 312: Meteorology and Climatology (4 credits) Physical Geography

Prerequisite: GEOG 101 or 210

The course focuses on the elements and forces producing lower atmosphere phenomena; the dynamics of heat flows, air movements, pressure changes, mass density, volume relationships and vaporization, release of moisture as applied to the changing state of the atmosphere, and production of disturbances and storms. It also examines the evolution of climatic systems, their spatial and temporal transformations and impact.

GEOG 313: Geodynamics and Geomorphology (3 credits) Physical Geography

Prerequisite: GEOG 101 or 210

The course provides comprehension about restructuring, isostatic balancing, geotectonics and modulation; processes, agencies and cycles of landscape sculpturing and evolution, including their temporal and spatial variations.

GEOG 314/ENVR 314 *: Hydrosphere Resources (4 credits) Physical Geography

Prerequisite: GEOG 101 or 210

The course examines origin, structure and shape of the ocean basins; composition, thermodynamics, circulation and oscillations of marine water; maritime resources and their use by humans; water resources on land, their origin, distribution, availability and quality; use and conservation as a resource for mankind.

GEOG 322: Economic Geography (3 credits) Human Geography

Focuses on principles governing multivariate interactions underlying the evolution and distribution of various economic activities and functions, the role of temporal and spatial variables responsible for changes in the economic systems and regions.

GEOG 323: Population Geography (3 credits) Human Geography

Deals with spatial, temporal and structural aspects of population characteristics including growth, distribution, density, composition and migration; the relationship of demographic variables to cultural, economic and environmental factors.

GEOG 324: Settlement Patterns and Processes (3 credits) Human Geography

Human settlement patterns: location, evolution, size, spacing, shapes and functional systems produced by interactive multivariate processes, forms and structures. Also includes problems relating to growth, congestion and evolution of ghettos.

GEOG 325: Political Geography (3 credits) Human Geography

The course emphasizes comparative study of global political regions and related systems. Varied approaches are explored such as power analysis, genetic analysis, functional analysis, thematic analysis and ethnic analysis of political units.

GEOG 326: Urban Environmental Issues (3 credits) Human Geography

The course deals with the dynamics of urban environment degradation and analysis of rectification mechanisms and policies. Focus upon current environmental issues of urban centers in Pakistan.

GEOG 371: Digital Cartography (3 credits) Geographical Techniques and Methods

Prerequisite: GEOG 201

Knowledge of computer software applications is preferred.

The course completely focuses on skill development in computerized map making and interpretation / analysis.

GEOG 372: Web GIS (4 Credits) Geographical Techniques and Methods

Prerequisite: GEOG 203

Knowledge of computer software applications is preferred.

The primary purpose of this course will be to examine such web-based technologies, and to help students develop the knowledge and skills necessary to plan, design, develop and publish a web-based GIS solution. Students will learn how GIS on the internet differs from the desktop experience, and how to adequately prepare spatial information for the web.

GEOG 374: Aerial and Satellite Imaging (3 credits) Geographical Techniques and Methods

Prerequisite: GEOG 202

Knowledge of computer software applications is preferred.

It includes the elements and interpretation processes pertaining to aerial photographs, remote sensing of earth resources and occupancy patterns; global positioning system [GPS]; geographic information science & systems [GIS]; digital image processing [DIP].

GEOG 396: Physical Geography Seminar (2 credits) Physical Geography

Prerequisite: GEOG 101 or 210

It comprises a departmental seminar investigating a selected field of Physical Geography. (Topic announced by the chair, prior to registration).

GEOG 411: Sustainable Management of Natural Resources (3 credits) Physical Geography

Prerequisite: GEOG 240 or instructor permission

It analyzes the parameters and principles governing sustainability of the Earth's resources. International and regional efforts to achieve sustainability are also focused.

GEOG 416/ENVR 417: Natural Hazards and Management Issues (3 credits) Physical Geography

Prerequisite: GEOG 240 or instructor permission

The course analyzes the natural phenomena causing hazards, related issues and problems; means and techniques of ascertaining their distribution, impact on human life and possibilities of recurrences; options for disaster preparation and loss mitigation.

GEOG 421: Cultural Geography (3 credits) Human Geography

Prerequisite: GEOG 220 or instructor permission

It deals with the patterns and processes of the world cultural realms such as language, religion, social traits and ethnicity serving as foci for an in-depth understanding of the world and its people and cross-cultural interactions.

GEOG 422: Spatial Planning for Economic Development (3 credits) Human Geography

Prerequisite: GEOG 322 or instructor permission

Theories of location and systematic modeling in describing nodes and hierarchy of economic clusters in terms of multivariate functional analysis and synthesis examined at evolving developmental modes.

GEOG 426: Spatio-Temporal Dynamics of Global Power Politics (3 credits) Human Geography

Prerequisite: GEOG 325 or instructor permission

The course provides an in-depth appraisal of the global power-politics patterns through times, their dynamics, evolution, exigencies, impact and fate.

GEOG 471: Qualitative and Quantitative Techniques in Geography (4 credits) Geographical Techniques and Methods

Prerequisite: Basic knowledge of computer software applications or instructor consent

The course provides the information on qualitative methodologies and quantitative techniques used by geographers in analysis and synthesis of systematic spatial phenomena. Application of statistical methods and thematic models for geographical analysis including the use of computer software and hands on experience.

GEOG 474/ ENVR 474 *: Geographical Information Science (4 credits) Geographical Techniques and Methods

Prerequisite: GEOG 201 or GEOG 202 or GEOG 203

The course relates to principles of Geographical Information Science, functions of geographic information systems, and relationship between GIS and remote sensing.

GEOG X95: Themes (1-3 credits) General Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

GEOG 498: Internship (3 credits)

Prerequisite: Open to the junior/seniors 90 completed credit hours with at least 9 upper division credit hours from Geography/GIS courses

An internship in a related industry, working with a private or public sector organization for six to eight weeks during summers. After the successful completion of training a student will have to submit a comprehensive report about the tasks done for his/ her evaluation.

GEOG 499: Capstone Project (3 credits)

The Capstone Project is a culminating academic experience that enables students to integrate and apply the knowledge, theoretical perspectives, and methodological skills acquired throughout the undergraduate

geography program. Typically undertaken after the fourth semester, students work under the supervision of a faculty member to design and complete a research-based or applied project addressing a contemporary geographical issue. The project requires the preparation of a written report and formal presentation demonstrating analytical, research, and problem-solving competencies developed during the program. .

Department of
**HEALTH AND
PHYSICAL
EDUCATION**

Introduction

The Health and Physical Education Department at FCC provides a range of sports activities to students for the total development of their personality. It focuses on the following areas, which contribute and are indispensable for growth and strength of sports in the university:

- Sports culture
- Coaching
- Officiating
- Competition opportunities and organization
- Facilities and equipment
- Scientific support

FCC has a very comprehensive sports program. Its sports facilities include an 8-lane standard 400m grass track and a short course; 25m swimming pool, five tennis courts and a playing field for cricket, football, hockey and handball. The Lucas Center has a well- equipped gymnasium, table tennis hall, basketball and badminton courts and offices for sports faculty and staff.

Intramural competitions in the above-mentioned sports take place all year round. Deserving sportsmen are awarded Medals of Distinction, College Colors and Certificate of Merit.

Vision

Sports is one of the important mediums through which we contribute towards the physical and mental development of individuals and the total development of the personality of a student.

The Health and Physical Education Department provides several different sports activities to ensure the physical, social, emotional wellbeing and development of students. As students, they will personify physical strength and learn the value of healthy activity and habits in their lives.

Mission

The Health and Physical Education Department's first commitment is to provide an opportunity for students to fully develop their sports potential. By way of practice, training and competition, the department strives to develop in each student:

- Good Sportsmanship.
- Offering a diverse range of sports.
- Promoting Sports culture among the students.
- Promoting an environment of healthy competition.
- To make them responsible and good citizens.
- Providing praise-worthy sports facilities from several sports fields.

Learning Objectives

- i. Demonstrate during play the rules and regulations by which various games/sports are played.
- ii. Demonstrate skills in sports and athletics.

- iii. Apply the techniques and skills needed to pursue positions in both school-based and commercial settings.
- iv. Demonstrate ethical sportsmanship while competing in individual and team sports.

Course Descriptions

Entry into any course requires students to be physically fit.

HPED 101: Cricket (1 credit)

Techniques of running, batting, bowling and fielding, working together both in offense or defense.

HPED 102: Football (1 credit)

Techniques of running, passing, kicking, tackling, blocking, heading and dribbling, coordination for offensive and defensive teamwork.

HPED 103: Hockey (1 credit)

Techniques of passing, dribbling, dodging, pushing with agility, speed and endurance.

HPED 104 (i): Swimming (1 credit)

Basics of freestyle swimming.

HPED 104 (ii): Swimming (1 credit)

Basics of the backstroke in swimming.

HPED 104 (iii): Swimming (1 credit)

Basics of breaststroke in swimming.

HPED 105: Volleyball (1 credit)

Techniques for passing, blocking and smashing.

HPED 106: Physical Exercises (1 credit)

General physical exercises, benefits of physical activities.

HPED 107 (i): Athletics (1 credit)

Techniques for sprints and javelin throw.

HPED 108: Table Tennis (1 credit)

Techniques to play table tennis for recreation or competition.

HPED 109: Badminton (1 credit)

Techniques to play badminton for recreation or competition.

HPED 110: Basketball (1 credit)

Skills of ball holding, dribbling, passing and shooting.

HPED 111: Tennis (1 credit)

Fundamental skills of playing tennis for recreation or competition.

HPED 112: Handball (1 credit)

Skills and knowledge of the game: ball holding, handling, dribbling, passing and shooting.

HPED 150: Advanced First Aid (4 credits)

The purpose of this course is to help students identify and eliminate potentially hazardous conditions in their environment, recognize emergencies, and make appropriate decisions on how to provide first aid with minimal or no equipment. Topics included are electric shock, fainting, drug overdose, accidents, bandaging techniques, CPR, burn injuries, natural disasters, hypothermia, hyperthermia, bleeding, crush injuries, bites, and poisons.

Learning Objectives/By the end of the course students will be able to:

- Provide immediate attention to a sick or injured person at the scene.
- Understand how and when to administer first aid.
- To minimize injury and future disability until the arrival of medical assistance.

HPED 151: SPORTS INJURIES (2 credits)

This course helps identify common sports injuries, prevention strategies, and treatment options. Topics include sprains, strains, contusions, fractures, dislocations, overuse injury, and skin injuries. This course is helpful for athletes, coaches, parents, and other active individuals. The course will cover athletic injury assessment, remedial exercises, diagnosis, treatment, and prevention of sports injuries.

Learning Objectives/By the end of the course students will be able to:

- The types of injuries are now so complex, and the sportsman so competitive, that he/ she must return to competition as soon as possible.
- This course will teach athletes Injury Assessment, Remedial Exercises, Diagnosis, Treatment and Prevention of Sports Injuries.
- This course is helpful for parents, coaches, athletes and anyone who is active and gets or worries about injuries.

Department of
**HISTORY AND
PAKISTAN STUDIES**

Bachelor of Science in History

Introduction

The Department of History and Pakistan Studies is one of the oldest departments in FCCU. It has produced many luminaries who have been associated with it as members of the learned faculty, such as Professor Griswold, Professor Brush and Dr ZH Zaidi. The Department, through its courses, aims to develop in the students an analytical, critical, scientific, argumentative and methodical thinking for assessing the past occurrences in order to draw conclusions which help to broaden their mental horizons, enrich their vision and enhance their knowledge. The Department of History is part of the Faculty of Social Sciences.

Learning Objectives

- i. To provide a wide range of knowledge of historical concepts, theories, and trends across. South Asian and world history.
- ii. To Develop the knack of communication, diversity, and inclusivity.
- iii. To Integrate and apply the knowledge of historical concepts, theories, facts, and trends into inter disciplinary mode and employment.
- iv. To encourage students to have a life-long love and engagement with historical learning.

Program Learning Outcomes

- i. **Academic Experience**
Graduates of this program will demonstrate advanced critical thinking and analytical reasoning by constructing well-supported arguments, engaging effectively with diverse audiences, and applying disciplinary knowledge through scholarly research, written expression, and informed communication.
- ii. **Critical Thinking & Historical Analysis**
Students will critically evaluate historical narratives, sources, and interpretations by applying evidence-based reasoning and comparative analysis to understand continuity, change, and causation.
- iii. **Knowledge and Understanding**
Students will demonstrate effective written and oral communication skills to convey historical knowledge and historical understanding.
- iv. **Diversity, Inclusiveness**
Students will develop skills to understand diverse cultures and religions and learn, plurality, inclusiveness and values of peaceful co-existence

Requirements for BS History Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
8 (24 credits)	12 credits	54	2.0

Core Courses	Credits
HIST 101	3
HIST 102	3
HIST 201	3
HIST 301	3

HIST 400	3
HIST 496	3
HIST 498	3
HIST 499	3
Core courses Total	24

Elective Courses	Credits
Survey Introduction (2)	6
HIST 103	3
HIST 104	3
HIST 105	3
OR any other Survey course offered by the department	
Mughal / Pre-Mughal South Asia (2)	6
HIST 250	3
HIST 255	3
HIST 302	3
HIST 305	3
HIST 308	3
Post-Mughal South Asia (1)	3
HIST 308	3
HIST 310	3
HIST 311	3
HIST 313	3
HIST 315	3
HIST 410	3
US/British/European History (1)	3
HIST 202	3
HIST 204	3
HIST 207	3
HIST 320	3
General Electives (4)	12
HIST 209	3
HIST 306	3
HIST 303	3
HIST 312	3
And any other Elective course offered by the department	
Electives courses Total	30

Total Degree Requirements

Requirement Category	Total Credits
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General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for the Minor in History

Core Courses	Elective Courses	Total Credits	Required Minor CGPA
3 (9 Credits)	9 Credits	18 credits	2.0

Core Courses	Credits
HIST 101	3
HIST 102	3
HIST 201	3
Core courses Total	9

Elective Courses	Credits
Post-Mughal South Asia (1)	3
HIST 308	3
HIST 310	3
HIST 311	3
HIST 313	3
HIST 315	3
HIST 410	3
Non-South Asia (1)	3
HIST 202	3
HIST 204	3
HIST 207	3
HIST 306	3
HIST 312	3
HIST 320	3
Electives	3
HIST 303	3
HIST 306	3
OR any other Elective course offered by the department	
Electives courses Total	9

Course Descriptions

PKST 101: Pakistan Studies (2 credits)

This course examines the historical evolution, ideological foundations, constitutional development, socio-economic structure, and contemporary challenges of Pakistan. It explores key themes including the Pakistan Movement, constitutional history, governance structures, economic planning, cultural diversity, and foreign policy orientation. The course develops students' critical understanding of Pakistan's political trajectory, regional engagements, and global positioning. It is designed as a compulsory foundational course for undergraduate students.

PKST 103: Ideology and Constitution of Pakistan (2 credits)

This course provides a foundational study of the ideology and constitutional development of Pakistan. It examines the historical evolution of the ideological foundations of Pakistan, the Two-Nation Theory, and the contributions of key leaders in the Pakistan Movement. The course also analyzes the structure, evolution, and major provisions of the Constitution of Pakistan (1973), including fundamental rights, principles of policy, and federal governance. It aims to develop informed and responsible citizenship through constitutional literacy and civic awareness.

HIST 101: Survey of South Asia (3 credits)

The course gives students a basic understanding of the general history of South Asia from the Indus Valley Civilization to modern times. The study of this course, which will be a prerequisite for all courses on South Asia in the department, will allow the students to become familiar with the broad periods of South Asian history, so that an in-depth analysis of historical periods and themes can be undertaken in upper-level courses.

HIST 102: Survey of Western Civilization (3 credits)

This course is a general survey of Western Civilization from Ancient Greece and Rome to the French Revolution in 1789. The aim of the course is to familiarize students with the broad periods and themes which have shaped Western Civilization, including an understanding of the scholarly debate on the issues. This course will be the basic prerequisite course for all upper level courses in the West.

HIST 103: Introduction to Archaeology (3 credits)

This is an introduction to the methods, goals, and theoretical concepts of archaeology. Archaeology analyzes the physical remains of the past, from million-year-old fossilized remains of our earliest human ancestors to 20th century buildings in South Asia, in pursuit of a broad and comprehensive understanding of human culture. Archaeology offers a unique perspective on human history and culture and helps us understand not only where and when people lived on the earth, but also why and how they have lived, examining the changes and causes of changes that have occurred in human cultures over time, seeking patterns and explanations of patterns to explain everything from how and when people first came to inhabit the South Asia, to the origins of agriculture and complex societies. Unlike history, which relies primarily upon written records and documents to interpret great lives and events, archaeology allows us to delve far back into the time before written languages existed and to glimpse the lives of everyday people through analysis of things they made and left behind.

HIST 104: Introduction to History (3 credits)

This introductory course familiarizes students with the foundations of the historical discipline, including nature, scope, value, and methods of history. It introduces history as an intellectual practice, examining how history is written, interpreted, and debated across civilizations and historiographical traditions. The course develops students' understanding of historical thinking, source criticism, and analytical writing while situating historical knowledge within interdisciplinary and contemporary contexts.

HIST 105: Survey of Cultural Heritage of Pakistan (3 credits)

This course introduces students to the cultural heritage of Pakistan by examining major monuments, historical sites, traditions, and preservation challenges across different historical periods. Students explore both tangible heritage sites, including archaeological sites, religious monuments, forts, museums, and colonial architecture—and intangible heritage such as festivals, rituals, crafts, and oral traditions. The course also examines national identity, heritage preservation policies, and development-related challenges in Pakistan. Through selected field visits, discussions, and written assignments, students develop a practical and informed understanding of Pakistan's diverse cultural legacy.

HIST 201: Research Methods (3 credits)

This course teaches steps in the research process and the tools and techniques of data collection. It is oriented towards Social Sciences research and will not include historiography, which, being a specialized field, will be taken up in the higher-level course.

HIST 202: Survey of Modern Europe (3 credits)

This course is a general survey of Europe from the French Revolution, and will concentrate on the major events, such as revolutions in Europe, the unification of Italy and Germany, the high noon of imperialism, the Russian Revolution, the two world wars and the Cold War. The course also discusses the key personalities and themes which shaped these developments.

HIST 204: History of the United States (3 credits)

This course is a survey of the United States since the American Revolution, concentrating on issues like the early development of the nation, the civil war, reconstruction, westward expansion, international relations, participation in the two world wars, and the emergence of the US as the superpower.

HIST 207: British History: From the Restoration to the Present (3 credits)

This course introduces students to basic themes, ideas, and influences in British History, from the restoration of the monarchy in 1660 to the present. The course will examine how Britain was transformed both at home and in its international role through the two world wars, imperial decline, Cold War, and in the present post-Cold War era in the twenty-first century. The purpose will be to situate the development of Britain internally and internationally up to the present times.

HIST 209: History of Early Muslims: 570–661 CE (3 credits)

This course examines the historical emergence of Islam within the broader context of Late Antique Arabia and the Near East. It explores the social, economic, religious, and political environment of pre-Islamic Arabia, the life and mission of the Prophet Muhammad (PBUH), the formation of the early Muslim community, and the development of early Islamic institutions. Emphasis is placed on historical sources, historiographical debates, and modern scholarly interpretations of early Islam. The course introduces students to critical historical inquiry while engaging with religious traditions and historical memory.

HIST 250: History of the Delhi Sultanate (3 credits)

This course studies the Sultanate period in Delhi (1206-1526), its origins, main features, and causes of its decline. The course will enable the students to analyze the development of art, culture, and architecture of the age, along with the evolution of state and society under the rule of the Sultans.

HIST 255: The Indus Valley Civilization (3 credits)

This course focuses on the inception, life, and eventual decline of the Indus Valley Civilization. The course will utilize current and early research and debates on the subject as well as visits to relevant sites to understand the period.

HIST 301/PHIL304*: Philosophy of History (3 credits)

This course studies in-depth the ideas behind the study of history, the different schools of thought in history, and the underlying theories. Special attention will be given to the development of South Asian and Muslim philosophy of history.

HIST 302: Mughal Rule in India (1526-1857) (3 credits)

Prerequisite: HIST 101

This course is a thorough study of the Mughal Empire from its inception to its decline and fall. Topics covered will include the condition of India before the Mughals, the rule of the great Mughals, their social, economic, religious, and political policies, and the development of art and culture during the period. The reasons for the fall of the empire as well as the rise of regional kingdoms will also be examined.

HIST 303: International Relations and Diplomacy (3 credits)

Prerequisites: HIST 102 or PLSC 101

This course focuses on the international system from the start of the First World War. The study begins with an examination of the key concepts in international relations, theory and practice, followed by an in-depth analysis of the relations between the great powers till the end of the Cold War.

HIST 305: Ancient History of the Subcontinent (3 credits)

This course will focus on a hugely diverse and dynamic span of history of ancient India. This will provide a systematic framework of the socio-political map of the ancient subcontinent starting from the Indus valley and culminating in the Gupta period. This will include the fundamental themes of polity, economy, philosophy and society. The times of glory, invasion, anarchy, factions and frictions will be discussed, highlighting Indus Valley Civilization, Epic Age, Buddhism and Class Structures. Culture and spiritual leanings throughout this span of time will be the focus of historical inquiry, exhibited in political ambition and philosophical thinking, art and literature that gave depth to one of the earliest periods of human history.

HIST 306: Islamic History (3 credits)

This survey course on the Caliphate, Umayyad and Abbasid periods examines their central and provincial administrations, expansion and conquests, religious policy, fiscal policy, foreign relations, the development of art and culture, and the causes of their downfall.

HIST 308: History of Lahore (3 credits)

Prerequisite: HIST 101

This course traces the history of Lahore from its origin to the present day. Topics like the importance of Lahore in different ages, its contribution to the development of art, architecture and culture, and its urban and rural dimensions are studied.

HIST 310: Issues in Pakistan's History (3 credits)

Prerequisites: HIST 101, HIST 201

This junior year seminar course focuses on particular issues in Pakistani history such as, but not limited to, the bureaucracy, army, judiciary, parliament, Kashmir issue, provincial autonomy, East Pakistan debacle, ethnicity and human rights.

HIST 311: India since Independence (3 credits)

Prerequisite: HIST 101

This course introduces students to themes in Modern India from independence to the beginning of the liberalization period in 1991. This course will provide students' knowledge of themes like nationalism, the Nehruvian system, domestic politics in India and the rise of Hindutva.

HIST 312/ PLSC 310 *: Modern Muslim World (3 credits)

Prerequisites: PLSC 101

This course explores the post- Second World War history of the Middle East with a special emphasis on the rise of Arab nationalism, Muslim solidarity, Israeli-Palestinian issue, the creation of modern states, and developments in state, society, religion and economy of the region.

HIST 313: British Rule in India (1757-1947) (3 credits)

Prerequisite: HIST 101

This course focuses on the political rise of the East India Company in 1757, the expansion of British influence through treaties and conquests, the Revolt of 1857 and the coming of the Raj, its policies, and its effect on the society, politics, economy and religious life of India.

HIST 315: Freedom and the Nationalist Movement in the Subcontinent (1857-1947) (3 credits)

Prerequisite: HIST 101

This course charts the start of the nationalist movement in South Asia with a focus on the key events and personalities which shaped society, politics and economy of the region during the period. The rise of Muslim separatism and the creation of Pakistan will also be studied.

HIST 320: Reformation and Renaissance (3 credits)

Prerequisite: HIST 102

This course focuses on Europe from the advent of the Protestant Reformation to the Catholic Counter Reformation, religious wars, the rise of the nation state, and the Enlightenment period. Special attention will be given to the renaissance in art, science, literature and culture during the period, up till the French Revolution.

HIST 400: Research Methodology (3 credits)

Prerequisite: HIST 201

This higher-level course shall build on the 200-level course on Research Methods to familiarize students with qualitative research and to introduce them to Historiography, its major traditions and debates. It will also focus on the use of sources in historical research.

HIST 410: Colonial Punjab, 1848-1947 (3 credits)

Prerequisites: HIST 101, HIST 201

This thematic course examines Punjab's history during colonial rule with a focus on several themes which shaped the period. The focus will be on the geography of the Punjab in historical perspective, the support of

local elites for colonial rule, the growth of the agrarian economy, and the political, social, and economic effects of these developments on the province.

HIST X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

HIST 496: Senior Seminar in History (3 credits)

Prerequisites: HIST 101, HIST 201

This seminar course comprises extensive readings on a selected topic (varying each semester) under the leadership of a faculty member who will facilitate in-depth analysis and discussion. The course will prepare students for independent research through a series of short papers and a final term paper on the course topic.

HIST 498: Internship (3 credits)

The Internship course (HIST-498) provides students with an opportunity to apply historical knowledge, research skills, and analytical thinking in real-world institutional settings. Students may work with archives, museums, cultural organizations, research institutes, NGOs, media organizations, or educational institutions. The internship emphasizes professional conduct, ethical responsibility, and reflective learning while helping students explore potential career paths related to history and heritage.

HIST 499: Capstone Project (3 credits)

Prerequisites: HIST 400, HIST 496

In this course the student will articulate a question to be investigated, draw up a plan of research, glean information from primary and secondary sources, analyze the material, and present his or her arguments and conclusion with 10,000 words count paper. The paper should integrate the methods learned in prior classes about historical analysis and research. Students will conduct their research under the guidance of a member of the History faculty or any other person approved by the Head of the Department.

Kauser Abdullah Malik
School of
LIFE SCIENCES

Bachelor of Science in Biological Sciences

Mission

The aim of the BS in Biological Sciences is to provide a comprehensive learning experience in Biological Sciences to produce lifelong learners for higher education and careers in plant, animal, environmental and health-related sciences, while maintaining the tradition of the liberal arts education. The program concentrates on a strong theoretical background and responsible lab and field work experience to empower students for employment and higher education in life sciences.

Program Objectives

- i. To give a thorough and comprehensive knowledge of fundamental concepts in biological sciences.
- ii. To enable students to interpret scientific literature.
- iii. To develop and enhance creative thinking skills in order to ethically solve real life problems.
- iv. To train the students to conduct and analyze scientific experiments.
- v. To produce competent and valuable future biologists for the betterment of society.

Program Learning Outcomes

- i. **Knowledge/concepts**
Students will be able to apply and formulate the key concepts effectively by demonstrating a clear understanding of theoretical principles, adapting them to practical situations, and synthesizing new ideas based on these foundations.
- ii. **Techniques**
Students will be able to implement and administer a variety of techniques acquired during their coursework in BS Biological Sciences. They will apply these methods effectively in their research projects, demonstrating proficiency in experimental design, data collection, and analysis. This will include hands-on skills in laboratory practices, fieldwork methodologies, and computational tools relevant to biological research.
- iii. **Scientific Literature**
Students will be able to critically evaluate scientific literature by identifying key research questions, methodologies, and findings in peer-reviewed articles. They will assess the validity and reliability of the data, analyze the strengths and limitations of experimental designs, and interpret results in the context of existing knowledge.
- iv. **Critical Thinking**
Students will solve various theoretical exercises that require advanced critical thinking, problem-solving, and the application of biological concepts to complex scenarios. They will engage in analytical reasoning by exploring multiple approaches to each problem, demonstrating a clear understanding of the underlying scientific principles.
- v. **Research**
Students will be able to differentiate and identify local microbial, plant, and animal species by applying standard laboratory procedures and field-based methodologies. They will demonstrate proficiency in techniques such as microscopy, culturing, staining, and morphological identification for microbial species. For plants and animals, students will utilize taxonomic keys, specimen dissection and anatomical comparisons to accurately classify and distinguish species. Additionally, they will follow appropriate protocols for sample collection, labeling, and preservation of specimen to ensure reliability and reproducibility.

vi. **Career**

Students will demonstrate the application of biological tools and technologies used in both academic research and industry settings. They will gain hands-on experience with biological techniques. Emphasis will be placed on understanding the theoretical principles behind each tool, as well as mastering proper usage, troubleshooting, and data interpretation. Students will also develop transferable skills, including teamwork, project management, and scientific communication, preparing them to integrate biological tools effectively into real-world academic research and industrial practices.

Requirements for BS Biological Sciences Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
10 (35 credits)	19 credits	54	2.0

Core Courses	Credits
BIOL 201/MBIO 201	3
BIOL 203/MBIO 203	3
BIOL 221	4
BIOL 222	4
BIOL 313/MBIO 313	4
BIOL 315/MBIO 315	4
BIOL 323/ENVR 323 /MBIO 323	3
BIOL 403	3
BIOL 498	3
BIOL 499	3
Core courses Total	34
Elective Courses	Credits
Any 200 or above level courses offered by the department	3 / 4 each
Elective courses Total	20

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Students majoring in Biological Sciences must take 8 credit hours of Chemistry courses (of any level, 100 to 400) in consultation with the faculty advisor.

Requirements for Minor in Biological Sciences

A minor in Biological Sciences is open to students in all disciplines with minimum CGPA of 2.5.

Core Courses	Elective Courses	Total Credits	Required Minor CGPA
4 (14 credits)	4 credits	18	2.0

Core Courses	Credits
BIOL 201/MBIO 201	3
BIOL 203/MBIO 203	3
BIOL 221	4
BIOL 222	4
Core courses Total	14
Elective Courses	Credits
Any 200 or above level courses offered by the department	4
Elective courses Total	4

Bachelor of Science in Microbiology

Mission

The goal of the BS in Microbiology program is to develop a dynamic community of aspiring microbiologists committed to utilizing the potential of microorganisms for the betterment of society. Through rigorous scientific training, ethical practice, and a culture of excellence, the program provides students with a comprehensive education that fosters curiosity, strengthens critical thinking, and cultivates a deeper understanding and appreciation of the microbial world.

Learning Objectives

- The graduate students will be able to demonstrate a solid comprehension of the fundamental concepts of microbiology, including the diversity, structure, function, and classification of microorganisms.
- They will acquire lab techniques skills, the roles of microorganisms in causing diseases, their modes of transmission, and the principles of disease prevention and control, agriculture, environment and industry.
- Additionally, students will be equipped to apply microbiological knowledge to real- world scenarios, such as diagnosing infections, developing vaccines, and improving food safety.
- Graduates will effectively communicate scientific concepts and findings to both expert and non-expert audiences through written reports, presentations, and discussions. They will be able to recognize the ethical considerations involved in microbiological research, including biosafety, biosecurity, and responsible conduct of research.
- Qualities of a good scientist like critical thinking, evaluating scientific literature, assessing experimental validity, and generating hypotheses, planning experiments, performing and analyzing data, interpreting as well as publishing, will be developed in students.

Requirement for BS Microbiology Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
11 (36 credits)	18 credits	54	2.0

Core Courses	Credits
MBIO 201/BIOL 201	3
MBIO 202	3
MBIO 203/BIOL 203	3
MBIO 204	3
MBIO 211/BIOT 211	3
MBIO 313/BIOL 313	4
MBIO 315/BIOL 315/BIOT 315	4
MBIO 316/BIOT 316	3
MBIO 319/BIOL 319	4
MBIO 498	3
MBIO 499	3
Total	36
Elective Courses	Credits
Any 200 or above level courses offered by the department	3 / 4 each
Total	18

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Bachelor of Science in Biotechnology

School of Life Sciences offers a program which is learner-centered, and job oriented in different specializations of Biotechnology. It will establish an environment of collaborative and reflective learning based on modern concepts of Biotechnology.

Mission

The mission of BS Biotechnology program is to produce competent biotechnologists with excellent knowledge who are capable to apply practical skills in different areas of Biotechnology for employment and higher education through providing curriculum based on liberal arts education to ensure greater breadth of knowledge along with in-depth knowledge of Biotechnology offering an opportunity to learn and experience various aspects of Biotechnology under one roof.

Learning Objectives

- i. To impart fundamental knowledge and skills in biotechnology.
- ii. To train the students to conduct and interpret scientific experiments.
- iii. To emphasize on scientific communication skills.
- iv. To use biotechnology knowledge for the betterment of society.
- v. To produce competent future biotechnologists.

Program Learning Outcomes

- i. **Knowledge**
Explain the key concepts and principles of biotechnology
- ii. **Application**
Apply the basic techniques used in biotechnology
- iii. **Analysis**
Be able to analyse and interpret the experimental data
- iv. **Critical thinking**
Demonstrate the use of various biotechnological mechanisms for human welfare
- v. **Scope**
Develop technical skills by hands-on training to use biotechnology tools in academia and industry

Requirements for BS Biotechnology Program

Core Courses	Elective Courses	Total Credits	Required Major CGPA
15 (51 credits)	9 credits	60	2.0

Core Courses	Credits
BIOL 201/MBIO 201	3
BIOL 203/MBIO 203	3
BIOL 313/MBIO 313	4
BIOL 315/MBIO 315	4
BIOT 201	3
BIOT 202/MBIO 212	4
BIOT 301/MBIO 301	3
BIOT 302	4
BIOT 313/MBIO 313	4
BIOT 314	3
BIOT 408/MBIO 408	4
BIOT 411	3

BIOT 412/MBIO 412	3
BIOT 498	3
BIOT 499	3
Core courses Total	51
Elective Courses	Credits
Any 200 or above level courses offered by the department	3 / 4 each
Elective Courses Total	9

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	60
University Electives	13-16
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

During Freshman and Sophomore years, students are recommended to take the following courses which will prepare them for higher level courses.

- Students who did not study Biology at higher secondary school or A-level or equivalent have to take BIOL 100; those who studied Biology at these levels, should not take BIOL 100, rather opt for BIOL 102 and/or BIOL, 105.
- Those who have not studied Chemistry at higher secondary school or A-Level or equivalent must take CHEM 100.
- Those who have not studied Physics at higher secondary school or A-Level or equivalent must take PHYS 100.

Course Descriptions

BIOL 100: Introductory Biology (3 credits)

Only for students who have not studied Biology in higher secondary school or A level or equivalent

The course includes basic concepts of biology with the cell as a building block, its function, reproduction, genetics, inheritance, principles of living systems, classification of plants and animals. The course is designed to provide non-science students with an overview of modern biology and to elucidate its importance in everyday life. How human beings depend upon other organisms, which fundamental processes are essential for living organisms? How does digestion and circulation take place? How DNA controls heredity characters? This course will answer all such questions.

BIOL 102: Introductory Plant Biology (4 credits)

Only for students who have studied Biology in higher secondary school/A -Level or equivalent This course covers the structure-function relationship of plants, basic principles of genetics and molecular genetics and Biotechnology as well as its use in modifying plants. Ecosystems, environmental issues and the relevance of flowering plants in human life are also examined.

BIOL 105: General Zoology (4 credits)

Only for students who have studied Biology in higher secondary school/A- Level or equivalent Classification, structure and function of all major animal groups including protozoa, porifera and metazoa (invertebrates and chordates).

BIOL 201 / MBIO 201: Cell Biology (3 credits)

Introduction to the course, applications of cell biology, research in cell biology, evolution of cell, comparison of plant and animal cells, biological molecules (carbohydrates), structure and function of plant cell wall, its comparison with prokaryotic cell wall, biological molecules (lipids and protein), ultra structure of cell: plasma membrane, movement across cell membrane and its significance, structure and function of chloroplast and mitochondria – comparison, the cytoplasm: endoplasmic reticulum, golgi complex, lysosomes and vacuoles. Peroxisomes and glyoxysomes, the cytoskeleton, significance of cytoskeleton, the nucleus: structure and function of nucleus, nuclear material, cell cycle, cell division and its control, apoptosis – programmed cell death, comparison to necrosis, introduction to cellular communications, types of cellular secretions, cellular communication – properties of receptors, cell signaling pathways, G coupled protein, tyrosine kinase and adhesion receptors.

MBIO 202: Microbial Taxonomy (3 credits)

This course will cover the structure of microbial communities, including the roles of microbes within an ecosystem, and the impact of environment on the community, the participation of diverse microorganisms in maintaining elemental cycles, physiological and molecular diversity within a community of microbes, methods and approaches used to study relationships and evolution (phylogeny) of microbes, particularly Bacteria and Archaea, and current classification of microbe groups, explore taxonomic strategies and approaches used to name microorganisms, and criteria used to define bacterial species and subspecific divisions within species, principles and methods behind studying and identifying cultured and uncultured microorganisms

BIOL 203 / MBIO 203: General Genetics (3 credits)

Introduction, Mendelian genetics, incomplete dominance, codominance, over dominance, multiple alleles, blood group system, gene interaction, lethality, environmental factors affecting phenotypes, sex determination mechanisms, sex linked inheritance, dosage compensation, nondisjunction phenomenon, linkage and crossing over, cytoplasmic inheritance, quantitative inheritance, variation in chromosome number, variation in chromosome structure, population genetics, problems related to the theoretical course.

MBIO 204: General Immunology (3 credits)

This course is designed for undergraduate students who are interested in developing a basic understanding of our immune system. It provides an overview of the immune system and functions of the organs involved. This course covers the basics of what goes into a functional immune system, how it has evolved to respond to our environment, and how it changes with time. Additionally, this course will give students an appreciation of how our immune system adapts to growing threats and how scientists have used this understanding to lead

immunology centered disease treatments. For students who are interested in further studies, it prepares them for a more advanced molecular immunology course.

BIOL 207: Applied Botany (3 credits)

This course will cover the study of plants from an economic point of view. The improvement of plants for better yield of their products and the strategies for the domestication and preservation of economically important plants. Plants as a source of food; beverage, herbs and spices; medicinal plants, psychoactive plants, poisonous and allergy plants, fibers, dyes, tannins, hydrogel, latexes and resins, wood cork and bamboo. Cultural and molecular approaches for the improvement of economic products, domestication and preservation of economically important plants.

BIOL 208: Applied Zoology (3 credits)

Pests of public health importance, parasites of human and animals, mites and ticks and their control, household insects, insect pests of cash crops, industrial aspects including apiculture (honey bee culture), lac culture, sericulture (silk-worm culture), aquaculture, snake venom, poultry industry, economic importance of mammals, leather industry, dairy industry, wool industry, pharmaceuticals from animals, bioactive substances from animals, pearl culture, wildlife and its management with values, conservational approaches.

BIOL 212/STAT 212/ENVR 212 / MBIO 212: Research Design and Biostatistics (3 credits)

Stages of research, selection of research topic, hypothesis formation, selecting the right materials and methods for the research being performed, experimental design and its significance, controls, data collection and analysis, hypothesis testing. Training on statistical software (e.g. SPSS, MINITAB).

BIOL 221: Plant Form and Function (4 credits)

This course is an introduction to plant structure and function. It deals with a general survey of plant kingdom and evolutionary trends in plants, morphology, anatomy, and physiology of plants. The aim of the course is to emphasize how structural botany is integrated with other modern fields. It provides knowledge and understanding in plant science for future employment in Biotechnology, agriculture, horticulture, and environmental areas.

BIOL 222: Animal Form and Function (4 credits)

Comparison of animals with one another: similarities and differences among the major phyla of animal kingdom, challenges in animal form and function, external and internal variations in organs and systems; integumentary, reproductive diversity among animal groups, digestive, sensory, cardiac systems and adaptations that enable them to live successfully in their respective environments.

BIOL 306: Integrated Pest Management (4 credits)

Prerequisite: BIOL 222

People, plants and pests, dynamics of pest populations, intensive agriculture, pest problems, concepts of Integrated Pest Management (IPM), cultural control, host plant resistance, parasitoids and predators, microbial control, botanical pest control, synthetic organic insecticides biotechnology approaches, bio-rational and other innovative approaches, IPM achievements, potential and challenges.

BIOL 313 / MBIO 313: Biochemistry (4 credits)

Prerequisite for non-science students: CHEM 160

This course is designed to provide a solid understanding of the organic structure of living systems. The topics include chemistry, structure and specific roles of: carbohydrates, lipids, amino acids, proteins and nucleic acids. General characteristics and properties of enzymes including enzyme kinetics will also be covered.

BIOL 315 / MBIO 315: Fundamentals of Microbiology (4 credits)

This course deals with History of microbiology, study of microbial life including bacteria, fungi, virus, protozoa, algae; comparison of prokaryotic and eukaryotic cells; pure culture techniques, microscopy, bacterial morphology, bacterial growth in different environments and on different media, control of microbes by using different methods, sterilization, BSL labs; bacterial genetics, mutation and its types, mutagens, transcription and translation in eukaryotes and prokaryotes; antimicrobial resistance; mechanisms microbes use for their protection; antimicrobial drugs and their mode of action; microbial products, applications of microbes in the food industry, biotechnology, environmental science, wastewater treatment, biofuels.

BIOL 319 / MBIO 319: Mycology (4 credits)

This course will deal with the study of structural specialization, diversity, and economic importance of fungi.

BIOL 323/ENVR 323 / MBIO 323: Applied Ecology (3 credits)

Introduction to ecology, climate and habitats, light, temperature, and water; Populations, distribution, species interaction, competition; Communities, their nature, species diversity, ecosystems, flow of energy, nutrient cycle, and food webs; Succession and stability in communities. Evolution and society, Darwinian natural selection, genetic variation and natural selection, mutation and migration, speciation; Interrelationships between ecology and evolution, evolution as an ongoing process.

BIOL 325: Human Physiology (3 credits)

This course is designed to provide students with an understanding of the physiology of the human system. Course content will include cell physiology and membrane permeability; gastrointestinal system; muscle physiology; cardiovascular system; respiratory system; excretory system; nervous system; sensory organs; endocrine system; homeostasis; reproductive system.

BIOL 327: The Molecular Biology of Cancer (3 credits)

Pre-requisites: BIOT-313: Molecular biology

This course will review the field of cancer research in depth. The molecular mechanisms and factors that invoke a normal cell to develop into a cancerous cell and the mechanisms that help cells avoid and fight cancer will be methodically surveyed. These include DNA damage and repair mechanisms, the regulation of gene expression in eukaryotes, tumor causing viruses, proto-oncogenes, oncogenes and tumor suppressor genes/proteins in cancer development, cellular and genetic basis of carcinogenesis, tumor growth and metastasis, genetic alterations that result in development of cancer molecular mechanisms and the biological principles that are a result of the above described alterations and the methodologies relevant to measure the extent and examine different cancer types.

BIOL 328: Taxonomy and Dendrology (3 credits)

In this course students will learn about the biology of woody plants (trees and shrubs), and their identification of morphological, ecological, and molecular aspects. It will introduce them to many economically important woody plants which are widely distributed in areas of Punjab. DNA barcoding techniques, various molecular techniques used in identification and preservation of important trees along with different strategies including

conservation of local flora. It also includes the role of important trees in phytoremediation. Students will also be introduced to the science of dendrochronology and dendroecology.

BIOL 329: Applied Endocrinology (4 credits)

This course is designed to provide students an understanding of the hormonal control and regulation of the animal bodies. Applied aspects of endocrinology will be discussed. Course contents will include an overview of endocrine systems in animals and humans, classical and non-classical endocrinology, hormone biosynthesis and metabolism, mechanisms of hormone action, interactions among hormones and environmental interactions with hormonal systems. Clinical disorders associated with endocrine glands and hormones will also be covered. Clinical case studies and endocrine-related technologies will also be included with emphasis on diagnosis and treatment.

BIOL 330: Radiobiology (4 credits)

Prerequisite: BIOL 201 or BIOL 203

This course will deal with the basic concepts of radiation, its sources, nature, types and properties and effects. Emphasis will be laid on the role of radiobiology in the health sciences and health care industry in terms of diagnosis and injury. Besides this, the course will also elaborate on the factors that affect the dose response elements processes by which radiation is absorbed, the biologic effects of radiation, including cell killing, carcinogenesis and mutation, radio-sensitivity, fractionated radiation, cell survival curves, concepts of linear energy transfer, radioprotection and applications of radiation in daily life.

BIOL 331/MBIO 331: Soil Microbiology (3 credits)

Prerequisites: BIOL 201

Soil Microbiology is a critical field for understanding the essential roles microorganisms play in nutrient cycling, organic matter decomposition, and plant- microbe interactions, which are foundational to ecosystem stability and agricultural productivity. In the face of global challenges like soil degradation, climate change, and environmental pollution, the study of soil microbiology offers sustainable solutions through innovations in bioremediation, biofertilizers, and carbon sequestration. This course equips students with both theoretical knowledge and practical skills to address real-world environmental and agricultural issues, making it a vital addition to the Biological Sciences curriculum and preparing students for careers in research, microbiology, biotechnology, and ecosystem management.

BIOL 332/MBIO 332: Phycology (3 credits)

Prerequisites: BIOL 201

This course aims to provide students with a comprehensive understanding of algae, covering their classification, morphology, ecological roles, and economic significance. This course highlights the significance of Phycology as a field that intersects with environmental science, industry, and public health, making it a crucial area of study for understanding and managing natural resources and ecosystems.

BIOL 403: Plant Physiology (3 credits)

Prerequisite for non-science students: BIOL 221

Application of physical and biological principles to the understanding of plant processes involved in assimilation, metabolism and regulation of growth and development.

BIOL 404/ ENVR 405 *: Conservation Biology (3 credits)

The objective of this course is to familiarize the students with different forms of biodiversity, threats to biodiversity and an overview of different strategies for its conservation.

BIOL 408: Applied Entomology (4 credits)

Prerequisite: BIOL 222

This course will cover all applied aspects of entomology in a broader perspective and will focus on; insects in the service of man, insects as enemies of man, causes of success of insects, collection and preservation of insects, classification and considerations for life histories, agricultural entomology, veterinary entomology, medical entomology, arthropods borne diseases including dengue, industrial entomology, forest entomology, forensic entomology and nutritional entomology.

BIOL 411/ ENVR 411 / MBIO 411: Environmental Microbiology (3 credits)

This course will provide an awareness and understanding to the students about the role of microorganisms in the environment. After completion of this course, students will be able to understand the significance, role and application of microorganisms in the environment.

BIOL 412 Neurobiology (3 Credits)

Prerequisite: BIOL 201

Introduction, basic anatomy of nervous system, cellular organization of brain, neuronal morphology, synapses and spines, different neuronal types: inhibitory and excitatory neurons, neuromodulatory neurons, neuroglial cells, astrocytes, cerebral vasculature. Modes of neuronal communication, catecholamines, serotonin, GABA neurotransmitters, biosynthesis, roles and release. Classical and non-classical neurotransmitters, excitatory and inhibitory amino acid transmitters, unconventional transmitters. Neurotransmitters receptors: ionotropic receptors, nAcCR, 5HT3, GABA receptors, glycinergic and glutamate receptors. NMDA and non-NMDA receptors, G-protein coupled receptors, adrenergic, purinergic and dopamine receptors. Quantal release of neurotransmitters: excitation - secretion coupling, short term synaptic plasticity.

BIOL 413 Developmental & Regenerative Biology (4 credits)

Prerequisite: BIOL 201 and BIOT 313

Focused on animals and human beings, this course provides current knowledge of the processes involved and phases of development and differentiation from embryo to adulthood emphasis on molecular mechanisms using examples from current studies. The course also includes the complex process of development of behavior, the underlying principles of regeneration in invertebrates and vertebrates and medical implications.

BIOL 414: Plant Growth Regulation (4 credits)

Prerequisite: BIOL 221

This course focuses on recent developments related to plant growth regulation; it covers hormones, senescence, abscission, phytochromes, photomorphogenesis, phototropic and gravitropic responses with emphasis on molecular mechanism of hormones and signal transduction.

BIOL 415 / MBIO 415: Food Microbiology and Safety (4 credits)

Prerequisite: BIOL 319 OR BIOL 201 OR BIOL 315

Food microbiology and microbial lab safety, overview of microbial cellular structures, identification of microbial food contaminants, cell growth and kinetics of probiotics, microbial metabolism and application of metabolites in food system, probiotics and their influence on gut microflora, microbial aspects of food

preservation, food safety and quality, Principle and applications of HACCP in food chain, quality assurance (QA) and total quality management (TQM) as combined approach towards food safety.

BIOL 416 / MBIO 416: Medical Microbiology (3 credits)

Prerequisite: BIOL 201 OR BIOL 315

Introduction to Medical Microbiology, commensal and pathogenic microbial flora in humans, sterilization, disinfection and antisepsis, general principles of laboratory diagnosis, innate and antigen specific immune responses to infectious agents, antimicrobial vaccines, classification of microbes (bacteria/viruses/fungi/parasites), structure and replication of microbes, mechanisms of pathogenesis and laboratory diagnosis of infectious agents, brief introduction to antimicrobial agents, intestinal and urogenital protozoa, blood and tissue protozoa, nematodes, cestodes, arthropods.

BIOL 469: Plant Pathology (4 credits)

Prerequisite: BIOL 221 OR BIOT 202 OR BIOL 102

This course deals with basic principles of plant pathology, disease symptoms, groups of plant pathogens and plant diseases; diseases of economically important plants; methods of plant disease control.

BIOL 473 / MBIO 473: Industrial Microbiology (3 credits)

Prerequisite: BIOL 315 OR BIOL 201 OR BIOL 319

This subject is designed to extend the student's understanding and appreciation of the attributes of microorganisms and the applications of modern techniques in the applied areas of industrial and environmental microbiology. It will also develop an advanced understanding of the applications of microorganisms in the industrial production of foods and other useful products.

BIOL X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

Taught by multiple faculty members including guests and foreign faculty. Advanced students in life sciences explore exciting topics in modern biology. Covering several topics over a semester expands the breadth and depth of student learning. The format will be review and journal articles. Topic possibilities are numerous and may include biological rhythms, epigenetics in development, inheritance, and disease, emerging diseases, and co- evolution of species.

BIOL 495: Human Genetics (2 Credits)

Pre-requisites: BIOT-313

It will cover different inheritance patterns and molecular genetics in humans. Students will be taught to identify and characterize the genetic basis, diversity of the normal human traits and genes involved in human diseases. The course can be modified so that current issues relevant to human and medical genetics are included.

BIOL 495: RNA Structure, Function and Metabolism (2 Credits)

Pre-requisites: BIOT-313

Introduction to RNA Biology, RNase P and RNase MRP RNA in tRNA and rRNA processing, snoRNAs, signal recognition particles (SRP) and telomerase RNA, trans splicing, RNA processing and its regulation: alternative splicing and alternative polyadenylation, RNA splicing couples to NMD, mRNA export from the nucleus to the cytoplasm, mRNA quality control, RNA modifications, The power of 5'UTR and 3'UTR on translational regulation, Internal ribosomal entry sites, Diverse roles of nuclear non coding RNAs in eukaryotic gene expression, Role of RNAi splicing.

BIOL 498: Internship (3 credits)

Pre-requisites: Completion of 6th semester

The internship is designed to provide students with hands-on experience in the fields of biotechnology and biological sciences. The internship aims to bridge the gap between academic knowledge and practical application, offering students an opportunity to apply theoretical concepts in real-world settings. During the internship, students will work with professionals in industries such as pharmaceuticals, agriculture, environmental sciences, research laboratories, or healthcare. The course will allow students to engage in tasks related to genetic engineering, microbiology, bioinformatics, bioprocessing, and other relevant areas of biotechnology and biological sciences.

BIOL 499: Capstone Project (3 credits)

Research will be allotted to the students on the recommendation of the Departmental Committee. It is offered in summer to the students of junior year.

BIOT 201: Introduction to Biotechnology (3 credits)

Biotechnology- definition and history; tools of biotechnology, foundations of biotechnology and interdisciplinary pursuit; branches and/or applications of biotechnology in medicine, industry, environment, agriculture (food, livestock, fisheries, algae, fungi, etc.); protection of biotechnological products; safety in biotechnology; public perception of biotechnology; biotechnology and ethics; biotechnology and the developing world.

BIOT 202 / MBIO 212: Protoplast, Cell and Tissue Culture (4 credits)

A brief history of tissue culture technique, plant growth regulators, culture media, microbial contamination and prevention procedures; plant cell, tissue and organ culture, animal cell culture, multiplication and differentiation of cells. Callus, suspension, cell line culture, isolation, culture and fusion of protoplasts and application of tissue culture techniques.

BIOT 211 / MBIO 211: Biosafety and Biosecurity (3 credits)

Prerequisite: BIOL 102 or BIOL 105 or FSc Premedical

This course provides an overview of biosecurity/biosafety, and practices, equipment, and facilities for the safe and secure handling of dangerous pathogens in a laboratory setting. Students attending this course will be able to have knowledge about the operation or maintenance of laboratories handling infectious agents.

BIOT 301 / MBIO 301: Analytical Techniques in Biology (3 credits)

Introduction, microscopy, different types of microscopes; centrifugation; types of centrifuges, principle, significance and applications; flow cytometry, spectroscopy: UV- visible, FTIR; electrophoretic separation, types of electrophoresis, SDS PAGE, DNA gel electrophoresis; western, southern and northern blotting; immunochemical techniques, ELISA; types of ELISA, principles and applications of ELISA,

chromatography: gel filtration, HPLC, FPLC, TLC, ion exchange; spectrometry: LC-MS, GC-MS, ESI-MS, MS-MS, MALDI- TOF-MS, applications; microtomy; bioanalytical techniques, types of PCR; RT- PCR, real time PCR, gradient PCR: overview of sequencing techniques.

BIOT 302: Fundamentals of Enzymology (4 credits)

Prerequisites: BIOL 313

The course covers a brief history of enzymes, the enzyme structure, an introduction to the amino acids that make up protein structure and determine function relationships, specificity of enzyme action, physical organization of enzymes (multienzyme complex), chemical and enzymatic kinetics, enzyme-substrate interaction and the roles that enzymes play as the fountain of life.

BIOT 305: Commercialization of Biotechnology Products (3 credits)

An overview of commercial products and services that Biotechnology offers. General aspects related to the quality control and criterion for industrially important bioprocesses, their management and impact on the current market. Resource planning and management of bioinoculant; antimicrobial agents; metabolites, enzymes and therapeutic proteins. Biotechnology and intellectual property rights. Industry interaction and technology transfer. Basics of effective marketing and promotion of Biotechnology products. Steps involved in commercialization of biotechnological merchandise.

BIOT 307 / MBIO 307: Molecular Immunology (3 credits)

Prerequisites: BIOL 201

Introduction to immunology; the basic processes involved in triggering the immune system and rendering it resistance or susceptibility to different infections. The study of molecular and biochemical events that influence immune responses. This course also includes innate (nonspecific) and adaptive (specific) immunity, immunoglobulins: structure and functions, antigens, antibody formation and hypersensitivity.

BIOT 309 / MBIO 309: Microbial Biochemistry (3 credits)

Prerequisites: BIOL 313

Microbial growth: mathematical expression of growth, growth curve, measurement of growth and growth yields; synchronous growth and continuous culture. Metabolic diversity among microorganisms, metabolism of carbohydrates, lipids, amino acids, purines and pyrimidines in prokaryotes. Antimicrobial agents: mode of action and resistance to antibiotics.

BIOT 313/MBIO 313: Molecular Biology (4 credits)

Prerequisite: BIOL 201

History, structure and function of DNA, DNA replication in prokaryotes and eukaryotes, structure, function and types of RNA, transcription, post transcriptional processing, translation, post translational processing in prokaryotes and eukaryotes, control of gene regulation in prokaryotes and eukaryotes, mutation and mutagens, DNA damage and repair, recombination and transposable elements.

BIOT 314: Bioenergetics and Metabolism (3 credits)

Prerequisite: BIOL 102 or BIOL 105 or FSc Prerequisite: BIOL 313

This course covers intermediate metabolism in biological systems. Pathways of breakdown and synthesis of biological molecules such as carbohydrates, lipids and nitrogenous compounds will be examined. Emphasis will be placed on the thermodynamics of the reactions and the regulatory mechanism of pathways.

BIOT 315 / MBIO 315: Genomics and Tools of Bioinformatics (4 credits)

Prerequisite: BIOT 313

Introduction to genomics, techniques involved in studying of genomes i.e. manual and automated DNA sequencing, genotyping, mapping and assembling a genome, an introduction to pharmacogenomics, personal genomics, SNP, RFPL, microsatellite DNA markers, brief history and introduction of bioinformatics, use of online data banks for nucleic acid and protein analysis, prediction of attributes of proteins and nucleic acids on the basis of tabulated data, DNA microarray, proteomics and survey of a genome of a model organism using bioinformatics tools.

BIOT 316 / MBIO 316: Fundamentals of Virology (3 credits)

Prerequisites: BIOL 201 OR BIOL 203

Origin and nature of viruses, taxonomy and classification, ultra structure of viruses, virus isolation, purification and identification, models of viral replication, viral genome analysis, chemotherapy of viral infections, virus host interaction, immunity to viral infections, important viral families of human importance; family characteristics, transmission, epidemiology and pathogenicity.

BIOT 317: Food Analysis and Sensory Evaluation (3 credits)

Prerequisites: CHEM 160 or BIOL 201

The course aims to equip students with essential skills in food analysis and sensory evaluation, preparing them for careers in food industry, quality control, research, and development. It integrates theoretical knowledge with practical experience to develop a comprehensive understanding of how to assess and ensure food quality and safety.

BIOT 407: Aquaculture Technology (4 credits)

Prerequisites: BIOL 105 OR BIOL 201 OR BIOL 222

Introduction, sources and quality of water, culture systems (open, semi-closed and closed system). Water flow and pumps, filtration and water treatment, culture methods for seaweed, molluscs, crustacean fishes and higher vertebrates, natural food, artificial feed harvesting techniques and policies on leasing.

BIOT 408 / MBIO 408: Recombinant DNA Technology (4 credits)

Prerequisites: BIOT 313

This course includes the central dogma of molecular biology and structural aspects of gene expression in prokaryotes and eukaryotes which introduces a good knowledge of promoter, CDS and terminator sequences; the laboratory bacterial strains and their genotypes, maintenance of genotypes and preservation of modified bacterial strains; isolation of DNA from different types of tissues and isolation of synthetic and natural plasmids from bacteria; different types of plasmid vectors for bacterial, animal and plant systems; restriction enzymes, their types and utilization in recombinant DNA technology, other DNA modifying enzymes and their uses with particular emphasis on gene cloning; transformation systems for bacteria, animal cells and plant tissues; screening of transgenics through various technologies, e.g., southern hybridization, PCR and RT-PCR.

BIOT 409: Food and Dairy Technology (3 credits)

Status of food industry in Pakistan & abroad; classification of foods: perishable, semi perishable and foods with longer shelf life. Processing of milk and dairy products: pasteurization, UHT for milk, cream, butter and

concentrated dried milk. Fermented dairy products: microbiology of starter cultures, cheese technology, yogurt, acidophilus milk etc. Nutritional and therapeutic benefits of fermented milk products. Food preservation by chemical & physical methods, food spoilage, food packaging, recent trends in food technology, Use of genetically modified foods, food safety; Concept of Hazard Analysis Critical Control Points (HACCP).

BIOT 411: Agriculture Biotechnology (3 credits)

Prerequisite: BIOT 201

Introduction and origin of Biotechnology; soil biotechnology; microbial interactions in agriculture, microbial control of fungal plant pathogens and importance of microorganisms for soil fertility, plant growth promoting rhizobacteria, secondary metabolites with antifungal activities, methods to produce transgenic plants with biotic and abiotic resistance, genetic manipulation of fruit ripening, engineering plant protein composition for improved nutrition, genetic manipulation of crop yield by enhancement of photosynthesis, production of high value proteins in plants, vaccines from plants and biofuels.

BIOT 412 / MBIO 412: Medical Biotechnology (3 credits)

Prerequisites: BIOT 313

Nanobiotechnology, cancer immunotherapy, gene therapy, stem cell biotechnology, knockout mice and gene inserts, siRNA, genetically engineered animals, infectious diseases, diagnostics and antibiotic resistance, biomaterials in regenerative medicine, vaccine technology, novel antimicrobial agents, their design and other future medical biotechnologies.

BIOT 413: Structural and Computational Biology (3 credits) – 46th AC

Prerequisites: BIOT 313

The course will focus on the fundamentals of nucleic acid and protein sequence analysis, introduction of analysis and modeling of complex biological systems. DNA sequencing technologies, library complexity and short read alignment (mapping), genome assembly and annotation, metagenome/microbiome data analysis, ChIP-seq analysis; DNA-protein interactions, simultaneous ChIP-seq peak discovery and motif sampling, RNA-sequence analysis: expression, isoforms, biological functions and prediction, probabilistic grammatical models of RNA structure, introduction to protein structure; structure comparison and classification, predicting protein structure, and predicting protein interactions.

BIOT 498: Internship (3 credits)

Pre-requisites: Completion of 6th semester

The internship is designed to provide students with hands-on experience in the fields of biotechnology and biological sciences. The internship aims to bridge the gap between academic knowledge and practical application, offering students an opportunity to apply theoretical concepts in real-world settings. During the internship, students will work with professionals in industries such as pharmaceuticals, agriculture, environmental sciences, research laboratories, or healthcare. The course will allow students to engage in tasks related to genetic engineering, microbiology, bioinformatics, bioprocessing, and other relevant areas of biotechnology and biological sciences.

BIOT 499: Capstone Project (3 credits)

Research will be allotted to the students on the recommendation of the Departmental Committee. Research is offered in summer to the students of junior year.

MICR 200: Microbial Taxonomy (3 credits)

This course will cover the structure of microbial communities, including the roles of microbes within an ecosystem, and the impact of environment on the community, the participation of diverse microorganisms in maintaining elemental cycles, physiological and molecular diversity within a community of microbes, methods and approaches used to study relationships and evolution (phylogeny) of microbes, particularly bacteria and archaea, and current classification of microbe groups, explore taxonomic strategies and approaches used to name microorganisms, and criteria used to define bacterial species and subspecific divisions within species, principles and methods behind studying and identifying cultured and uncultured microorganisms.

MICR 203: General Immunology (3 credits)

This course is designed for undergraduate students who are interested in developing a basic understanding of our immune system. It provides an overview of the immune system and functions of the organs involved. This course covers the basics of what goes into a functional immune system, how it has evolved to respond to our environment, and how it changes with time. Additionally, this course will give students an appreciation of how our immune system adapts to growing threats and how scientists have used this understanding to lead immunology centered disease treatments. For students who are interested in further studies, it prepares them for a more advanced molecular immunology course.

MBIO 498: Internship (3 credits)

Internships will be allotted to the students on recommendation of the Departmental Committee. Duration of internship is from 6 to 8 weeks, and this is offered in the summer to the students of junior year.

MBIO 499: Capstone Project (3 credits)

Research will be allotted to the students on the recommendation of the Departmental Committee. Research is offered in summer to the students of junior year.

*** Cross listed Courses:**

Courses with two designators (coding) are marked with a * to identify them as cross-listed courses. Students must select the correct designator for their applicable program to be counted towards the degree.

Department of
MASS
COMMUNICATION

Bachelor of Science in Mass Communication

Introduction

The Department of Mass Communication was established in 2003. It aims to equip its students with all the tools that are required to secure employment or pursue higher research degrees. This is made possible by dedicated faculty, a state-of-the-art Forman Media Center, and links with the media industry. Particular attention is paid to research/project topics offered by the department to ensure that our students learn and properly implement research methodologies suitable for their audio-video projects. Students receive individual attention from research/project supervisors at every stage. The department offers a Bachelor of Science degree and is part of the Faculty of Humanities.

Learning Objectives

- i. To provide a program that develops critical thinking among students.
- ii. To provide a program with an emphasis on ethical decision-making in a variety of media-related contexts.
- iii. To provide a program with an emphasis on communication research.
- iv. To provide a program with a focus on journalistic writing and editing.
- v. To provide students with hands-on training in media production

Program Learning Outcomes

- i. **Knowledge of communication related concepts:**
Graduates will demonstrate knowledge of communication related concepts and critically analyze media messages.
- ii. **Evaluation of legal and ethical dynamics of Mass Media:**
Graduates will be able to evaluate the legal and ethical dynamics of mass media.
- iii. **Communication Research:**
Graduates will conduct research in the field of communication to solve contemporary world problems.
- iv. **Content Production:**
Graduates will design and produce original content for different media platforms, including media messages appropriate to the audience, purpose, and context.
- v. **Market Readiness:**
Graduates will describe further opportunities in career and education related to the field of mass communication

NOTE: The Department of Mass Communication offers specializations in the following:

1. Print & Electronic Media
2. Advertising & Public Relations

However, students may opt for a simple Major in Mass Communication without specialization.

Requirements for BS Mass Communication Major

1. Major in Mass Communication WITHOUT Specialization

Core Courses	Elective Courses	Total Credits	Required MCGPA
6 (18 credits)	12 (36 credits)	54	2.0

2. Major in Mass Communication WITH Specialization

Core Courses	Specialization Courses	Elective Courses	Total Credits	Required MCGPA
6 (18 credits)	5 (15 credits)	7 (21 credits)	54	2 .0

Core Courses	Credits
MCOM 201	3
MCOM 301	3
MCOM 409	3
MCOM 414	3
MCOM 498	3
MCOM 499	3
Core Courses Total	18
Elective Courses	Total Credits
Any course offered by the department.	
WITHOUT Specialization	36
WITH Specialization	21

Print and Electronic Media	
Specialization Courses	Credits
MCOM 202	3
MCOM 302	3
MCOM 401	3
MCOM 402	3
MCOM 410	3
Specialization Courses Total	15

Advertising and Public Relations	
Specialization Courses	Credits
MCOM 200	3
MCOM 303	3
MCOM 304	3
MCOM 400	3
MCOM 415	3
Specialization Courses Total	15

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Courses recommended for Major in Mass Communication:

ECON 100, URDU 207, URDU 208, SOCL 100, CSCS 210, ENGL 207, and SOCL 355

Requirements for the Minor in Mass Communication

Core Courses	Elective Courses	Total Credits	Required MCGPA
4 (12 credits)	2 (6 credits)	18	2.0

Core Courses	Credits
MCOM 101	3
MCOM 201	3
MCOM 301	3
MCOM 310	3
Total	12
Elective Courses	Credits
Any course offered by the department.	6
Electives courses Total	6

Course Descriptions

MCOM 101: Introduction to Communication Studies (3 credits)

The course gives a brief introduction to print, electronic, and online journalism as well as advertising and public relations. It encompasses functions of news organizations, and introduces students to the basics of news writing, reporting and editing. The students are also exposed to the content of various media.

MCOM 105: Media and Information Literacy (3 credits)

This course provides a comprehensive introduction to Media and Information Literacy in the contemporary digital environment, with particular attention to the Pakistani context. It examines core concepts including information disorder and disinformation, cognitive biases and source credibility, online harms and digital safety, relevant digital laws such as the Prevention of Electronic Crimes Act (PECA), and foundational fact-checking and verification practices. Designed for undergraduate students in Mass Communication and related fields, the course emphasizes practical skill-building through case studies, digital safety exercises, and applied fact-checking assignments. It equips students to critically evaluate information and engage responsibly in digital spaces.

MCOM 110: English Writing for Journalists (3 credits)

The course deals with the use of English for specific purposes i.e., Mass Communication with a special focus on jargon used specifically in media content.

MCOM 200: Copywriting (3 Credits)

Prerequisite: MCOM 101

The course introduces copy designing and copy management skills required for the advertising and public relation industry. It also covers copywriting techniques for public relations and advertising media including print, electronic, outdoor and digital.

MCOM 201: News Reporting (3 credits)

Prerequisite: MCOM 101

The course explores the mechanics, elements, value and structure of stories for print and electronic media. It also introduces news sources for print and electronic media, as well as the qualifications and functions of a reporter. The course introduces basics of camera and microphone reporting, interpretative and investigative reporting, reporting beats and interview techniques.

MCOM 202: Sub-Editing (3 credits)

Prerequisite: MCOM 101

This course incorporates introduction, importance and process of sub-editing. It also covers functions and qualifications of a sub-editor, source of news, types and trends in headline making, techniques of news editing and caption writing for pictorial display.

MCOM 203: Media & Peace Building (3 credits)

Prerequisite: MCOM 101

The course introduces interactive communication approaches and tools of Mass Communication in conflict resolution. It also explores tools for conflict analysis, analyzes the role of theater and music in the peace building process, and further elaborates art for peace, vision and framework for social justice.

MCOM 207/ENGL 207*: Media and Literature (3 credits)

Prerequisite: WRCM 101, WRCM 102

This course centers around the role of media as a means of understanding literature; the importance of language and literature with reference to specific social and cultural issues.

MCOM 290: Communication Research I (3 credits)

Prerequisite: MCOM 201

This course provides students with a foundational understanding of the research process in media and communication. It introduces key concepts, processes, elements, measurement techniques, and research methods used in the field. Students will learn how to investigate mass media phenomena, assess media effects, and analyze the role of media and communication in society using basic research principles.

MCOM 301: Media Systems (3 credits)

Prerequisite: MCOM 201

The historical development of an institution in a society is primarily a struggle for more room to assert/express by the institution, while society plays a regulatory or punitive role through its political arm, the government. The study of media in this case means the study of the interplay of these forces. Press laws are a direct result of the performance of the media and vice versa. The media laws and media institutions form what we call the media system. It will help us discuss the rise, growth, and present status, along with the prospects for the future, in a better context. Instead of teaching linear chronologically, we will focus on a more teleological approach, looking at the interplay of power within the media and the political system (government).

MCOM 302: Opinion writing (3 credits)

Prerequisite: MCOM 201

This course covers the various aspects of newspaper content. It elaborates on the importance of editorial pages, editorial writing, and qualifications of editorial writers. It also deals with techniques of newspaper columns, feature, article, and blog writing.

MCOM 303: Principles of Public Relations (3 credits)

Prerequisite: MCOM 200

It covers various aspects of public relations in Pakistan including corporate/ organizational public relations, consumer public relations, media management, and conflict management. The students are also introduced to press release, press note, Handout, Press communiqué writing.

MCOM 304: Principles of Advertising (3 credits)

Prerequisite: MCOM 200

The course introduces scope, function, and socio-economic aspects of advertising. It also looks at advertising research, functions of advertising agencies, designing and evaluation of advertising campaigns.

MCOM 308: Online Journalism (3 credits)

Prerequisite: MCOM 201 and students majoring in Mass Communication

The field of journalism has drastically changed ever since the emergence of the Internet in the 1990s. Newsrooms of today make full use of digital tools to gather, write and present news. The ways in which news is disseminated by the media and consumed by users have also changed with the proliferation of websites, mobile apps and social networks. Web desks have become a staple in newsrooms and terminologies such as online journalism, digital journalism, multimedia journalism, computer-assisted reporting are being used extensively among journalists and academia. It has, thus, become necessary that students of mass media are equipped with the knowledge and skills of digital media.

MCOM 310: Communication Theory I (3 credits)

Prerequisite: MCOM 201

This course introduces key models of communication and core perspectives in media studies. It emphasizes understanding communication processes through various models and theoretical lenses. Students will explore the nuances and complexities of communication by engaging with theories, paradigms, and conceptual frameworks that shape the study of communication.

MCOM 350: Risk and Crisis Communication (3 credits)

Prerequisite: MCOM 303

This course focuses on two vital activities of public relations: Risk and Crisis Communication. It also provides a profound understanding of trends and developments that could have a significant impact on an organization or society.

MCOM 400: Media Management in Ad & PR (3 Credits)

Prerequisite: MCOM 304 & MCOM 303

This course covers the media management areas including managing media relations, media buying, media selling, media monitoring and audience analysis for advertising and public relations industry. This course will involve the application of strategic management principles to the development of public relations and advertising plans and programs.

MCOM 401: Radio Broadcasting: A Theoretical Introduction (3 credits)

Prerequisite: MCOM 301

The course introduces the origin and development of radio, functions of broadcasting house, distinctive features of radio news, interviewing for radio programs, duties of a radio producer, trends in FM radio in public and private sectors in Pakistan.

MCOM 402: Television Production (3 credits)

Prerequisite: MCOM 301

The course focuses on the set-up and working of television programming, duties of a TV producer, program shooting and editing. It also includes program research and script writing, and details of different phases of TV production.

MCOM 403: International Journalism (3 credits)

Prerequisite: MCOM 201

This course is an introduction to theories and research on media systems across the globe. Emphasis will be placed on the study and application of theories and research methods that have relevance for professional practice in the fields of communication and journalism. Students will learn about the structures of media systems in the U.S. and other social and political cultural contexts.

MCOM 404: Community Journalism (3 credits)

Prerequisite: MCOM 201

Community Journalism analyzes the status and issues regarding women, children, minorities and human rights in the local community with special emphasis on Pakistan. It also covers media coverage, role of NGOs and mass awareness campaigns in Pakistan, as well as the role of mass media in reporting human rights events.

MCOM 409: Communication Theory II (3 credits)

Prerequisite: MCOM 310

This course offers an advanced and comprehensive study of major theories in media and communication. It examines the dynamic relationship between theory and research and analyzes how technological advancements continue to shape, refine, and transform contemporary media and communication theories. The course further enables students to situate their own research projects within appropriate theoretical frameworks in media and communication studies.

MCOM 410: Documentary Making (3 credits)

Prerequisite: MCOM 402

The course explores the various types and modes of documentary making. It also provides a comparative analysis of national as well as international documentaries. Technical skills of script writing, cinematography and editing are also taught in this course.

MCOM 412: International Communication (3 credits)

Prerequisite: MCOM 310

This course evaluates the different approaches to international communication, with special reference to global media and its influence on different regions. It also discusses issues of free flow of information, foreign policy and trans-border data flow.

MCOM 413: Media, Society & Culture (3 credits)

Prerequisite: MCOM 409

This course aims to understand the relationship between media, society and culture. It explores the processes of cultural socialization, homogenization, acculturation, cultural diversity, pluralism and relativism.

MCOM 414: Communication Research II (3 credits)

Prerequisites: MCOM 290 and 310, only for students majoring in Mass Communication.

This course explores advanced components of media and communication research with both academic and industry relevance. Additionally, students will develop awareness of current trends in communication research, fostering deeper interest in pursuing research within the fields of mass media and communication.

MCOM 415: Communication Campaign Design (3 Credits)

Prerequisite: MCOM 303 & MCOM 304

This course will cover designing different campaigns including health communication, political communication, public awareness, capacity building and media advocacy campaigns. It will also encompass the evaluations of campaigns by international organizations like WHO, UNDP, USAID and CIDA.

MCOM 430: Media Lab (4 Credits)

Prerequisite: MCOM 304 or MCOM 402

MCOM 430 is a specialized lab course, which will cover essential media skills including layout designing for magazines & circulars as well as video editing through professional software. Students will be provided with knowledge about camera specifications, camera handling, support equipment, and techniques of videography. In addition, students will be provided with the opportunity to practice through field assignments.

MCOM X95: Themes (1-3 credits)

Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

MCOM 498: Internship (3 credits)

Only for students majoring in Mass Communication

An internship will allow students to experience the functioning of media organizations. Students will register for the internship course only after the 6th semester.

MCOM 499: Capstone Project (3 credits)

Only for students majoring in Mass Communication Prerequisite: MCOM 414 Students are required to submit one research Thesis/Project on any approved topic in the 8th semester.

Department of
MATHEMATICS

Introduction

The Department of Mathematics has established a profound reputation for providing excellent education as par international standards through earnest efforts of its faculty. This Department is an integral part of the Faculty of Computer and Mathematical Sciences. It offers a Baccalaureate degree in Mathematics that provides a broad range of introductory and advanced courses incorporating areas of pure, applied and computational Mathematics. Over the years, our highly esteemed faculty members have ensured to inculcate critical and analytical skills in our students. The department is elevating students' proficiency in the subject through cross listing of courses with departments of Physics, Statistics and Computer Sciences. Significant efforts are made for both individual and communal characteristics development of our students. We aim to nurture students in a healthy and competitive environment to enable them to have an active role in higher echelons of the mathematics community. The success and recognition of our program can be gauged through the acceptance of a large number of our students in national and international universities for advanced degrees.

Mission

The mission of the B.S Mathematics degree program is to impart computational and analytical skills to its students to cultivate an aptitude of logical and quantitative rationale towards practical and scholarly interests. The program seeks to further its mission by teaching mathematics to the students aspiring to achieve a primary degree in the subject and to the students' pursuing degrees in other subjects requiring computational and interpretive acumen for comprehensive knowledge. Our program also aims to cater to the students in pursuit of careers in academia and industry, and to enable them to employ their scientific proficiency to devise efficient solutions of problems in our community.

Learning Objectives

- i. To equip students with fundamental knowledge of mathematical principles and its development into a profound theoretical and computational framework.
- ii. To furnish students with quantitative and metaphysical skills to transmit the knowledge into reproducing, creating and assimilating the situations with an intent of significant outcome.
- iii. To reinforce the extensive capacity of mathematical education into implementation and materialization.
- iv. To integrate mathematical logic and precision into impartial conduction of affairs through contemplation of principles of ethics, integrity and accountability.
- v. To prepare students to pursue advanced studies, research and professional outlook employing mathematical qualifications and skills.

Program Learning Outcomes

- i. **Understanding:**
Students will demonstrate diverse mathematical principles and their specific employment in applied and theoretical branches of the discipline.
- ii. **Analyzing:**
Students will solve problems using analytical and computational methods.
- iii. **Evaluation:**
Students will demonstrate subject knowledge into building up a theoretical and critical premise for the solutions of problems of abstract nature.
- iv. **Application:**

Students will apply mathematical expertise to assimilate and interpret the information in other disciplines and scientific pursuits.

- v. **Human Development Element:**
Students will be able to employ logical reasoning and methodical insight into conducting their affairs in consideration with principles of ethics and integrity.
- vi. **Future:**
Students will be able to pursue advanced degrees, research and career in related domains.

Bachelor of Science in Mathematics

The 4-year degree program in Mathematics was launched in 2005. Students majoring in Mathematics take 48 credit hours in their major as well as 47 or 48 in general education and 35 or 34 respectively as free electives. Students majoring in Mathematics take courses as free electives from the mathematics department or cross-listed courses with the other departments. A degree in Mathematics develops clear logical thinking. The coursework looks at topics in Mathematics: from pure Mathematics to how Mathematics is used in the real world. Cross listing of courses with the other departments further helps students to analyze Mathematics as a real-world tool. Students obtain a secure understanding of Mathematics with a good choice of topics in pure and applied Mathematics at an advanced level.

Requirements for BS Mathematics Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
14 (42 credits)	4 (12 credits)	54	2.0

Core Courses	Credits
MATH 201/DATA 204	3
MATH 202	3
MATH 203	3
MATH 209	3
MATH 213	3
MATH 301	3
MATH 302	3
MATH 307	3
MATH 309	3
MATH 310	3
MATH 311	3
MATH 313	3
MATH 498	3
MATH 499	3
Core Courses Total	42
Elective Courses (Choose any 4 from the list)	Credits
MATH 204	3
MATH 212	3
MATH 303	3
MATH 304/STAT 313	3

MATH 312	3
MATH 314/STAT 304	3
MATH 315/STAT 311	3
MATH 316/PHYS 341	3
MATH 317	3
MATH 318	3
MATH 402	3
MATH 403/CSCS 403	3
MATH 404	3
MATH 407	3
MATH 410/PHYS 461	3
MATH 411/PHYS 462	3
Electives courses Total	12

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

See the Chair of Mathematics or the undergraduate adviser for a degree plan during the first year of study at FCC. An asterisk (*) indicates that the course is cross listed with other departments.

Requirements for the Minor in Mathematics

Core Courses	Elective Courses	Total Credits	Required Major CGPA
3 (9 credits)	3 (9 credits)	18	2.0

Core Courses	Credits
MATH 102/DATA 104 or MATH 111	3
MATH 103/DATA 203 or CSCS 202	3
MATH 201/DATA 204	3
Core Courses Total	9

Elective Courses for Non-CS students (Choose any 4 from the list)	Credits
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MATH 202	3
MATH 203	3
MATH 204	3
MATH 209	3
MATH 212	3
MATH 213	3
MATH 301	3
MATH 302	3
MATH 303	3
MATH 304/STAT 313	3
MATH 307	3
MATH 309	3
MATH 312	3
MATH 313	3
MATH 314/STAT 304	3
MATH 315/STAT 311	3
MATH 316/PHYS 341	3
MATH 317	3
MATH 318	3
MATH 402	3
MATH 403/CSCS 403	3
MATH 404	3
MATH 406	3
MATH 407	3
MATH 410/PHYS 461	3
MATH 411/PHYS 462	3
Electives courses Total	9

Elective Courses for CS students (Choose any 4 from the list)	Credits
MATH 203	3
MATH 204	3
MATH 212	3
MATH 213	3
MATH 303 or COMP 113	3
MATH 304/STAT 313	3
MATH 317	3
MATH 318	3
MATH 403/CSCS 403	3
Electives courses Total	9

Requirements for Joint minor in Mathematics and Statistics

For a Joint Minor in Mathematics and Statistics a student needs to study 6 courses; this includes 2 Statistics core courses, 2 Mathematics core courses, and 2 elective courses by choosing one elective course each from the list of elective courses listed by the departments.

Core Courses	Elective Courses	Total Credits	Required MCGPA
4 (12 credits)	2 (6 credits)	18	2.0

Core Courses		Credits
Statistics	STAT 102/MATH 105/Data 102	3
	STAT201/DATA 201	3
Mathematics	MATH 102 or MATH 111	3
	MATH 103 or CSCS 202	3
Core courses Total		12

Any 2 courses from the following elective

	Elective Courses	Credits
Choose 1 from the following		3
Statistics	STAT 208/ /DATA 208	3
	STAT 301	3
	STAT 303 /DATA 303	3
	STAT 304 /MATH314	3
	STAT 311 /MATH 315	3
	STAT 313 /MATH 304	3
	STAT 401 / MATH 408	3
Choose 1 from the following		3
Mathematics	MATH 201	3
	MATH 202	3
	MATH 203	3
	MATH 209	3
	MATH 212	3
	MATH 303 or COMP 113	3
	MATH 304 / STAT 313	3
	MATH 310 or CSCS 320	3
	MATH 312	3
	MATH 314/ STAT 304	3
	MATH 315 / STAT 311	3
	MATH 408 /STAT 401	3
Electives courses Total		3+3 = 6

Course Descriptions

MATH 100: Quantitative Skills (3 credits)

Basic Algebra and number theory, rounding, estimating, scientific notation, algebraic expressions, fractions, factoring, solving equations, two equations with two unknowns and their applications to daily life problems, quadratic equations and their applications, percentage problems (profit, loss, discount, interest, commission, taxation), ratio and proportion, work problems, distance problems (time, distance, speed), basic geometry, mean, median and mode.

MATH 101: Pre-Calculus & Trigonometry (3 credits)

Fundamentals, solution of equations and inequalities, lines, functions, linear and quadratic functions, polynomial and rational functions, operations on functions, inverse functions, synthetic division, remainder and factor theorem, partial fractions, exponential, logarithmic and trigonometric functions, trigonometric identities, solution of right and oblique triangles.

MATH 102/DATA 104 : Calculus I (3 credits)

Prerequisite: MATH 101 or A-Level Mathematics or Intermediate with Mathematics Functions, graph of functions, translation, stretching and compressing graphs, limits, continuity and differentiability, differentiation and its basic rules, indeterminate forms, L'Hopital's rule, integration and its techniques, fundamental theorem of calculus, evaluating definite integral.

MATH 103/DATA 203: Introductory Linear Algebra (3 credits)

Prerequisite: MATH 101 or A Level Mathematics or Intermediate with Mathematics Introduction to system of linear equations, matrices and matrix operations, elementary matrices, Gaussian elimination, Gauss Jordan method for solving a system of linear equation determinants and their properties, vector spaces, subspaces, linear independence, bases and dimensions.

MATH 105/STAT 102/DATA 102 *: Probability and Probability Distributions (3 credits)

Note: Please see the contents from the list of Statistics courses.

MATH 107 / STAT 101/DATA 101 *: Statistical Methods (3 credits)

Note: Please see the contents from the list of Statistics courses.

MATH 108: Quantitative Reasoning I (Basic) (3 credits)

This course develops foundational quantitative reasoning skills required for effective decision-making in academic, professional, and everyday contexts. Students will learn to interpret numerical information, apply proportional reasoning, analyze financial scenarios, construct algebraic models, understand geometric relationships, and interpret basic statistical data. The course strengthens logical thinking and builds confidence in applying quantitative methods to real-world socio-economic challenges.

MATH 109: Quantitative Reasoning I (Advanced) (3 credits)

This course develops advanced quantitative reasoning skills required for analytical decision-making in academic, professional, and real-world contexts. Students will interpret complex numerical information, apply proportional reasoning in multi-step scenarios, analyze financial models, construct algebraic and quantitative models, interpret geometric and spatial relationships, and evaluate statistical data. The course strengthens logical thinking, analytical depth, and quantitative argument evaluation in socio-economic contexts.

MATH 112: Quantitative Reasoning II (Basic) (3 credits)

Prerequisite: MATH 108

This course is a sequential course, following Quantitative Reasoning I, to fulfill the General Education requirement. The course emphasizes logical reasoning, critical thinking, mathematical modeling and mathematical statistical analysis, to develop students' analytical skills required to steer through the intricate phenomena of the modern world. The course enhances students' ability to construct and evaluate logical arguments, identify logical fallacies and think critically. It is tailored to acquaint students with quantitative concepts and techniques needed to interpret and analyze numerical data effectively.

MATH 113: Quantitative Reasoning II (Advanced) (3 credits)

Prerequisite: MATH 109

This course is a sequential course, following Quantitative Reasoning I, to fulfill the General Education requirement. The course emphasizes logical reasoning, critical thinking, mathematical modeling and mathematical statistical analysis, to develop students' analytical skills required to steer through the intricate phenomena of the modern world. The course enhances students' ability to construct and evaluate logical arguments, identify logical fallacies and think critically. It is tailored to acquaint students with quantitative concepts and techniques needed to interpret and analyze numerical data effectively.

MATH 201/DATA 204: Calculus II (3 credits)

Prerequisite: MATH 102

Applications of derivative: increasing and decreasing functions, maxima and minima of a function, concavity, inflection points of a function, Rolle's Theorem, the mean value theorem, Taylor's theorem, applications of integration: area and arc length, volume and surface of revolution, introduction to improper integrals, infinite series, power series, introduction to conic section.

MATH 202: Ordinary Differential Equations (3 credits)

Prerequisite: MATH 102

Introduction and classification of differential equations, basic concepts, formation, separable and reducible to separable equations, homogeneous and reducible to homogeneous equations, exact and inexact equations, integrating factors and linear equations, Bernoulli and Clairaut equations, second order homogeneous and non-homogeneous equations, initial and boundary value problems, homogeneous equations with constant coefficients, linear Independence and Wronskian, solving non-homogeneous equations with the methods of undetermined coefficients and variation of parameters, Cauchy-Euler equation, Laplace and inverse Laplace transformations, and their properties, to solve ordinary differential equations.

MATH 203: Vector Analysis (3 credits)

Prerequisite: MATH 102

Scalars and vectors, laws of vector algebra, scalar and vector fields, product of two vectors and their applications, scalar and vector triple products, ordinary and partial differentiation of vector fields and its use in basic differential geometry, gradient of a scalar field, divergence and curl of a vector field, ordinary integration of vector fields, line integrals, surface integrals and volume integrals, divergence, and Stokes theorem.

MATH 204: Introduction to Quantitative Modelling (3 credits)

Pre-requisites: MATH 103

This course is designed to provide a fundamental understanding of physics models, including their formulation, mathematical representation, and predictive analysis using Python. The accessibility of Python as an open-source programming platform, combined with its ability to efficiently analyze the models through robust computational libraries will enhance the learning experience in models' building and analysis. Additionally, the course aims to integrate applications of mathematical modelling from various fields, such as finance, epidemiology, engineering, and health sciences, to broaden the scope and relevance of the contents.

MATH 209: Linear Algebra (3 credits)

Prerequisites: MATH 103 and MATH 102

Review of vector spaces, subspaces and bases, row space, column space, rank, nullity, inner product spaces, orthogonal basis, Gram Schmidt process, orthogonal matrices, eigenvalues and eigenvectors, diagonalization, orthogonal diagonalization, positive definite and negative definite matrices, linear and inverse linear transformation.

MATH 210: Set Theory (3 credits)

Prerequisites: MATH 101 or A Level Mathematics or Intermediate with Mathematics Sets and basic operations on sets, relations, functions, cardinal numbers, computing cardinals, denumerable and countable sets, power of continuum, cardinal arithmetic, Cantor-Bernstein theorem, ordered sets, ordinal numbers, axioms of choice, well ordering theorem, and Zorn's lemma.

MATH 212: Elementary Number Theory (3 credits)

Prerequisite: MATH 101 or A Level Mathematics or Intermediate with Mathematics Division algorithm, divisibility properties, Euclidean algorithm, properties related to greatest common divisor and least common multiple, Diophantine linear equations, prime numbers, fundamental theorem of arithmetic, prime numbers, and divisibility, congruences, properties of congruences, Euler's Phi function, Fermat's theorem, residue classes modulo n , solution of congruences, the Chinese remainder theorem.

MATH 213: Introduction to Scientific Computing (3 credits)

Prerequisite: MATH 102 and MATH 103

This course is designed to enhance the computational and programming strengths of mathematics major students using technical computing language. However, all science majors can capitalize on this opportunity to improve their mathematical programming skills. By taking this course, students will develop skills to write moderate-sized programs for solving various mathematical problems as well as utilize built-in functions to expedite the computations. The course will consist of interactive lectures and guided workshops where students will be engaged in hands-on activities to work on building their programming command.

MATH 301: Multivariable Calculus (3 credits)

Prerequisite: MATH 201

Vectors, analytic geometry in 3-space, sphere, cylindrical surfaces, quadric surfaces, cylindrical and spherical coordinates, functions of multiple variables, limit and continuity, partial and directional derivatives, chain rule and implicit differentiation, tangent plane, normal vector and normal line, maxima and minima of function of more than one variable, Lagrange's multipliers, double integrals over rectangular and non-rectangular domains, double Integral in polar coordinates, triple integral in rectangular, cylindrical and

spherical coordinates, change of variables in multiple integrals, line integral, conservative vector field and Green's theorem.

MATH 302: Mechanics (3 credits)

Prerequisite: MATH 203

Composition and resolution of forces, friction, kinematics, and dynamics of a particle.

MATH 303: Discrete Mathematical Structures (3 credits)

Prerequisite: MATH 101 or A-Level Mathematics or Intermediate with Mathematics Propositional logic and logical connectives, logical equivalence, methods of proofs, mathematical induction, recurrence relations, relations on sets, equivalence relation and equivalence classes, partial order relations, basic counting principles, pigeonhole principle, permutation and combinations, inclusion-exclusion rule, counting techniques, introduction to graphs.

MATH 304/STAT 313*: Operations Research (3 credits)

Prerequisite: MATH 103 or STAT 102

Introduction to operations research, graphical solution, Simplex method, two phase method, M-method, sensitivity analysis, primal dual relationship, dual simplex method, transportation model, assignment models, transshipment models, network models, queuing theory.

MATH 307: Complex Analysis (3 credits)

Prerequisite: MATH 201

Algebra of complex numbers, geometric representation of complex numbers, Euler's and Demoivre's formulas, powers and roots of complex numbers, definition of a function of complex variable, limit and continuity, differentiable and analytic functions, C-R equations, entire functions and harmonic functions, exponential, trigonometric hyperbolic and logarithmic functions and inverse trigonometric and hyperbolic functions, sequence and series of complex numbers, geometric series, convergence and divergence tests of series including nth root test, ratio and comparison tests with proofs, power series and radius of convergence, complex integration, contour integrals, Green's theorem, Cauchy-Goursat theorem, deformation theorem of contours, Cauchy-Integral formulae, Taylor and Laurent's series, singularities, poles and residues, calculus of residues, Cauchy' residue theorem with applications to solve trigonometric integrals.

MATH 309: Real Analysis (3 credits)

Prerequisite: MATH 201

Sets and functions, finite, infinite, countable and uncountable sets, algebraic and order properties of set of real numbers, triangle inequality, completeness property of set of real numbers, archimedean property, limits of sequences, algebra of limits, convergent and divergent sequences, convergence of monotone sequence, Bolzano Weierstrass theorem, limits of functions, sequential criterion for limits (convergence and divergence), squeeze theorem, continuous functions, composition of continuous functions, uniform continuity, differentiable functions, and Caratheodory's theorem.

MATH 310: Numerical Analysis (3 credits)

Prerequisite: MATH 102 or MATH 103

Solution of system of linear equations, solution of nonlinear equations, error analysis, interpolation by polynomials, Lagrangian interpolation, numerical differentiation, numerical integration, computer programming will be done by using any suitable software like MATLAB, MAPLE or MATHEMATICA.

MATH 311: Topology and Metric Spaces (3 credits)

Prerequisite: MATH 210

Metric spaces, neighborhoods, open and closed sets, topological spaces, limit point, interior point, closure, dense subsets, basis and sub basis, induced topology, product and quotient topology, continuity, homeomorphism, convergence and Cauchy sequences, complete metric space, separation axioms, compactness, and connectedness.

MATH 312: Integral Equations (3 credits)

Prerequisite: MATH 202

Integral equations and their types, Abel's problem, resolvent and iterated kernels, eigenvalues and functions, Leibniz rule of differentiation under the sign of integration, conversion of multiple integral to single integral, conversion of initial and boundary value problems to Volterra and Fredholm integral equations, solution of homogeneous Fredholm integral equation of the second kind with separable kernels, Fredholm theorem, Fredholm alternative theorem and an approximate method, method of successive approximations and resolvent kernels, solution of Fredholm integral equation by successive approximation, solution of Volterra integral equation by successive approximation, Neumann series, solution of Volterra integral equation of second kind by reducing to differential equation, solution of integral equations using Hilbert Schmit method. application of Laplace transformation to solve Volterra integral equation with convolution type kernels.

MATH 313: Group Theory (3 credits)

Prerequisite: MATH 103 and MATH 210

Definition and examples of group, subgroups, cyclic groups, cosets, Lagrange's theorem, normalizer, centralizer, the center of a group, conjugacy classes, normal subgroups, quotient groups, homomorphism, isomorphism and automorphism, kernel and image of homomorphism, isomorphism theorems, permutation groups, the cycle decomposition of a permutation group, Caley's theorem, direct product of two groups.

MATH 314/STAT 304*: Distribution Theory (3 credits)

*Prerequisite: MATH 105/STAT 102**

Note: Please see the contents from the list of “Statistics” courses.

MATH 315/STAT 311*: Mathematical Statistics (3 credits)

*Prerequisite: MATH 105/STAT 102**

Note: Please see the contents from the list of “Statistics” courses

MATH 316/PHYS 341*: Methods of Mathematical Physics (3 credits)

Prerequisite: PHYS 221 or MATH 203

Note: Please see the contents from the list of “Physics” courses

MATH 317: Mathematical Foundation of Machine Learning (3 credits)

Pre-requisites: MATH 204

The course aims to introduce the mathematical foundations of machine learning algorithms and their development using tools from calculus and mathematical statistics. The course will explore the application and integration of multivariable functions, the multivariate chain rule, Taylor series approximation, gradient descent, optimization, and stochastic gradients to develop and evaluate machine learning models. Additionally, the concepts of maximum likelihood estimation and dimensionality reduction techniques for a large set of input system shall be covered.

MATH 318: Introduction to Financial Mathematics (3 credits)

Pre-requisites: MATH 102

This course provides an introduction to finance, focusing on use of mathematics to understand financial products and markets. It aims to deliver basic understanding of interest rates, financial contracts, no-arbitrage principles, and discrete-time financial models. Building on this foundation, the course introduces stochastic calculus as a tool to study continuous time financial models. Furthermore, the Black-Scholes formula and its application for options pricing shall also be discussed.

MATH 402: Ring Theory (3 credits)

Prerequisite: MATH 313

Definition and examples of rings and subrings, units, invertible element, zero-divisors, nilpotent, idempotents. Ideals, operations on ideals, prime ideals, maximal ideals, quotient ring, ring homomorphism, integral domains, the Chinese remainder theorem, divisibility in integral domains, greatest common divisor, least common multiple, Euclidean ring, principal ideal domain, unique factorization domains, finite field, polynomial ring in several variables, symmetric polynomials and its fundamental theorem.

MATH 403/CSCS 403*: Graph Theory (3 credits)

Prerequisite: MATH 303/COMP 113

Graphs and graph models, connected graphs, common classes of graphs, operations on graphs, regular graphs, degree sequence, isomorphic graphs, characterization and properties of trees, spanning trees, connectivity, Eulerian graphs, Hamiltonian graphs, planar graphs, Euler identity, Kuratowski's theorem, vertex and edge coloring, digraphs.

MATH 404: Partial Differential Equations (3 credits)

Prerequisite: MATH 202

Introduction, formation of first order partial differential equations (PDEs), Lagrange's equations, integral surfaces passing through a given curve, surfaces orthogonal to a given system of surfaces, the Cauchy's problem for quasi linear first order PDEs, classification of second order PDEs, reduction to canonical (or normal) forms, solutions of elliptic, parabolic and hyperbolic PDEs, the method of separation of variables, boundary value problems, the vibrating string, conduction of heat in solids, Laplace's Equation, introduction and properties of Laplace transform, Laplace transforms of elementary functions, inverse Laplace transform, convolution theorem, solution of PDEs by Laplace transform, Fourier integral representation, Fourier transform pair, transform of elementary functions, Fourier sine and cosine transform, solutions of PDEs by Fourier transforms.

MATH 407: Functional Analysis (3 credits)

Prerequisite: MATH 311

Normed spaces, Banach spaces, convex spaces, equivalent norms, quotient spaces, linear operators, finite dimensional normed spaces, continuous and bounded linear operators, dual spaces, inner product spaces, orthonormal sets and bases, Hilbert spaces, annihilators, projections, decomposition theorem in Hilbert spaces, linear functionals on Hilbert spaces, reflexivity of Hilbert spaces.

MATH 410/PHYS 461*: Quantum Mechanics I (3 credits)

Prerequisite: MATH 302 or PHYS 301 and PHYS 341

Note: Please see the contents from the list of “Physics” courses

MATH 411 / PHYS 462*: Quantum Mechanics II (3 Credits)

Prerequisite: MATH 410/PHYS 461: Quantum Mechanics I

Note: Please see the contents from the list of “Physics” courses

MATH 498: Internship (3 Credit Hours)

The Internship course provides students with an opportunity to apply mathematical knowledge, analytical reasoning, and computational skills in real-world institutional and industry settings. Students may work with banks, financial institutions, insurance and actuarial firms, data analytics companies, software houses, research institutes, statistical bureaus, engineering firms, NGOs working on quantitative projects, or educational institutions. The internship emphasizes professional conduct, ethical responsibility, and reflective learning while helping students explore potential career paths related to mathematics, data science, finance, research, and education.

MATH 499: Capstone Project (3 Credit Hours)

This course serves as the culminating integrative experience for BS Mathematics majors. It requires students to apply mathematical theory, analytical methods, and computational tools to real-world or theoretical problems arising in science, engineering, economics, public health, cryptography, and data-driven domains. Students will work independently under faculty supervision to develop a substantial project in either an applied and computational mathematics track or a pure and theoretical mathematics track. The course features milestone-based submissions and concludes with a formal presentation and defense of findings in a departmental capstone forum.

*** Cross listed Courses:**

Courses with two designators (coding) are marked with a * to identify them as cross-listed courses. Students must select the correct designator for their applicable program to be counted towards the degree.



Department of **PHARMACY**

Introduction

The Department of Pharmacy at Forman Christian College (A Chartered University), Lahore, established in 2015, is committed to excellence in pharmaceutical education, research and innovation. Accredited by the Pharmacy Council of Pakistan (PCP), the department offers a Doctor of Pharmacy (PharmD) program that integrates pharmaceutical sciences with clinical practice. The department further strengthened its academic portfolio with the launch of the MPhil in Pharmaceutics program in 2026.

Following the revised PharmD curriculum (2025) approved by the Pharmacy Council of Pakistan (PCP), the program equips students with knowledge, practical competencies, professional skills and ethical values. The PharmD curriculum covers the five core disciplines of Pharmaceutics, Pharmacology, Pharmaceutical Chemistry, Pharmacognosy, and Pharmacy Practice. Learning is supported by internationally qualified faculty, an outcome-based curriculum, individualized learning through small class sizes, modern teaching methods, advising and mentoring, and well-equipped laboratories.

Students gain experiential learning through clinical clerkships at Evercare Hospital, Lahore, and professional training at United Christian Hospital and Mercy Health Center. The department maintains strong collaborations with leading academic, healthcare, industrial, and regulatory organizations through Memoranda of Understanding (MoUs), including the Pakistan Council of Scientific and Industrial Research, the Punjab Agriculture, Food and Drug Authority, Chughtai Healthcare, NovaMed Pharmaceuticals, and CCL Pharmaceuticals. These partnerships provide students with opportunities for internships, industrial training, collaborative research, and professional development.

Our pharmacy graduates have established successful careers in pharmaceutical manufacturing, research and development, quality assurance and quality control, regulatory affairs, hospital and clinical pharmacy, community and retail pharmacy, academia, public health, and entrepreneurship. Many alumni have also earned competitive international scholarships for postgraduate studies and are pursuing advanced education and professional careers at leading universities and research institutions.

Committed to academic excellence, research, innovation, professional ethics, and community service, the Department of Pharmacy continues to prepare competent, compassionate, and globally competitive pharmacists who contribute to the advancement of pharmaceutical sciences and healthcare.

The Department of Pharmacy is part of the Faculty of Natural Sciences.

Vision

The vision of the Department of Pharmacy is to be the center of excellence for education and training in the life-saving profession of pharmacy, instilling FCCU core values.

Mission

The Doctor of Pharmacy Program at FCCU prepares graduates capable of providing high-quality patient-centered care in healthcare settings, striving for excellence and innovation in pharmacy practice, community wellness, product development, pharmaceutical industry, drug delivery and drug discovery by educating them in fundamental and applied concepts, knowledge, laboratory techniques and skills of pharmaceutical sciences.

Doctor of Pharmacy (PharmD) Program

Keeping in view the ever-increasing demand for professionally skilled pharmacists nationally and internationally, the Department of Pharmacy offers the Doctor of Pharmacy (PharmD), a five-year professional degree program accredited by the Pharmacy Council of Pakistan designed to prepare graduates for the evolving pharmaceutical industry, including research and development, manufacturing, quality assurance, quality control, hospital and clinical pharmacy, marketing, drug regulatory affairs, community pharmacy, and other areas of pharmaceutical care.

The Doctor of Pharmacy (PharmD) degree program at Forman Christian College (A Chartered University) follows the revised curriculum (2025) approved by the Pharmacy Council of Pakistan (PCP) and the Higher Education Commission (HEC). The curriculum is delivered over five years through a combination of theoretical instruction, laboratory training, practical learning, and clinical clerkships. Additionally, students complete General Education courses in accordance with the University's academic requirements to support their intellectual, professional, and personal development.

Program Educational Objectives

- i. Prepare trained and skilled graduates to excel in dosage form design, formulation development, manufacturing/dispensing of safe and high-quality medications for patient use, and innovations in drug delivery.
- ii. Equip graduates with the knowledge and skills to implement patient care plans, contribute to drug discovery, and develop new therapeutic agents.
- iii. Cultivate professional skills and management abilities that enable graduates to serve effectively in clinical, hospital, community, nuclear pharmacy settings, regulatory agencies, pharmaceutical marketing, and research.
- iv. Develop competencies in drug discovery, quality assurance of pharmaceutical products, and the synthesis of drug substances.
- v. Foster competencies in the isolation and characterization of phytochemicals, and the production and standardization of natural products.
- vi. Instill professional ethics, legal responsibility, and attitudes that reflect integrity, excellence, discipline, and respect in pharmacy practice.
- vii. Prepare graduates as professionals and researchers committed to lifelong learning and capable of becoming competent pharmacy practitioners.

Program Learning Outcomes

- i. **Pharmaceutics**
Graduates will demonstrate fundamental and applied knowledge about all the processes of turning pure drugs into a dosage form able to be safely and effectively used by the patients pursuing pharmaceutical formulation development and pharmaceutical technology.
- ii. **Pharmacology**
Graduates will demonstrate knowledge of basic mechanisms of drug discovery, pharmacokinetics, and pharmacodynamics of chemicals / drugs on biological systems.
- iii. **Pharmacy Practice**
Graduates will apply scientific knowledge in pharmaceutical practice to provide optimum patient care
- iv. **Pharmaceutical Chemistry**
Graduates will apply the knowledge of organic chemistry in combination with structural & chemical biology and pharmacology to design, structure, and synthesis of pharmaceuticals and biologically active molecules
- v. **Pharmacognosy**

Graduates will apply the knowledge of physical, chemical, and biological properties in discovery and development of drug substances, excipients, nutraceuticals, and cosmeceuticals of natural origin; ethno pharmacology & toxicology of crude drugs

vi. **Ethics**

Graduates will demonstrate professional ethics, attitudes, and appropriate legal conduct towards patients, colleagues and community with integrity, excellence, discipline, and respect.

vii. **Skills**

Graduates will effectively communicate by listening, speaking and writing in a manner that facilitates positive interaction with patients, health professionals and the public

viii. **Careers**

Graduates will describe future career opportunities and higher education opportunities available in the field of pharmacy.

Degree Requirements of Doctor of Pharmacy (PharmD)

- **PharmD Roadmap:** The Doctor of Pharmacy (PharmD) degree program at FCCU follows a defined roadmap as per the PharmD curriculum approved by the Pharmacy Council of Pakistan and Higher Education Commission.
- **PharmD Courses:** The PharmD roadmap includes Major Core, Allied/Interdisciplinary and General Education courses.
- **Credit Hours:** The total credit hours of the PharmD degree are 208.
- **Co-requisite Courses:** All laboratory courses as specified in the PharmD roadmap are co-requisites of their respective theory courses.
- **Pre-requisite Policy:** Students following the new PharmD roadmap shall be required to successfully complete all courses up to and including the 3rd professional year (Junior level) before enrolling in courses of the 4th and 5th professional years (Rising Seniors and Seniors).
- **Maximum Degree Completion Tenure:** As per the guidelines of the Pharmacy Council of Pakistan, the maximum duration for completion of the PharmD program shall be seven (07) years, extendable by one (01) additional year subject to approval by the relevant statutory bodies of the institution on a case-to-case basis.
- **One Year Extension in Degree Tenure:** As per the regulations of the Pharmacy Council of Pakistan, the prescribed maximum duration for completion of the PharmD program is seven years, extendable only for the exceptional cases by one additional year subject to the approval of the statutory bodies of the respective institution.
- **Independent Study Policy:** The offering of Independent Study shall not be applicable to core PharmD courses, given the structured, professional, and regulatory nature of the program.
- **Summer Semester Enrollment Policy (NS/W Cases):** As per the requirements of the Pharmacy Council of Pakistan (PCP), summer semester offerings are limited to remedial, deficiency, failure, or repetition courses only. Any student awarded a No Show (NS) or Withdraw (W) grade in any course of the PharmD program shall be considered as undertaking a fresh course attempt, and such courses shall not fall under remedial, deficiency, failure, or repetition categories. Accordingly, these courses shall not be eligible for enrollment during the summer semester.

Requirements for Doctor of Pharmacy (PharmD) Program

The Doctor of Pharmacy (PharmD) degree program at FCC follows the curriculum approved by the Pharmacy Council of Pakistan and Higher Education Commission. Courses of in this curriculum are offered each semester. Additionally, students have to take general education courses as per university policy.

Total Credits	Required MCGPA
208	2.0

Courses Semester I	Credits
UNIV 100	3
ENGL 106	3
PHRM 120	3
PHRM 121	1
PHRM 122	3
PHRM 123	1
PHRM 124	3
PHRM 125	1
PHRM 126	3
PHRM 127	1
Total Credits	22

Courses Semester II	Credits
PHRM 130	3
PHRM 131	1
PHRM 132	3
PHRM 133	1
PHRM 134	3
PHRM 135	1
PHRM 136	3
PHRM 137	1
PHRM 138	3
PHRM 139	1
ISLM 101 or CRST 105	2
Total Credits	22

Courses Semester III	Credits
PHRM 220	3
PHRM 221	1
PHRM 222	3
PHRM 223	1
PHRM 224	2
PHRM 225	1
PHRM 226	3
PHRM 227	1
PHRM 228	3
PHRM 229	1
Total Credits	19

Course Semester IV	Credits
PHRM 230	3
PHRM 231	1
PHRM 232	3
PHRM 233	1
PHRM 234	3
PHRM 235	1
PHRM 236	3
PHRM 237	1
PKST 101	2
ISLM 100A	1
URDU 101	3
Total Credits	22

Course Semester V	Credits
PHRM 320	3
PHRM 321	1
PHRM 322	3
PHRM 323	1
PHRM 324	3
PHRM 325	1
PHRM 326	2
PHRM 240	3
PKST 103	2
ISLM 100B	1
Total Credits	20

Course Semester VI	Credits
PHRM 330	3
PHRM 331	1
PHRM 332	3
PHRM 333	1
PHRM 334	3
PHRM 335	1
PHRM 336	2
CSCS 100	3
PHRM 245	3
Total Credits	20

Course Semester VII	Credits
PHRM 420	3
PHRM 421	1
PHRM 422	3

PHRM 423	1
PHRM 424	3
PHRM 425	1
PHRM 426	3
ENGL 108	3
BUSN 220	2
Total Credits	20

Course Semester VIII	Credits
PHRM 430	3
PHRM 431	1
PHRM 432	3
PHRM 433	1
PHRM 434	3
PHRM 435	1
PHRM 436	2
PHRM 437	1
PHRM 438	2
PHRM 439	2
PHRM 498	3
Total Credits	22

Course Semester IX	Credits
PHRM 441	3
PHRM 442	1
PHRM 443	3
PHRM 444	3
PHRM 445	1
PHRM 446	3
PHRM 447	1
PHRM 448	2
PHRM 449	1
UNIV 200	3
Total Credits	21

Course Semester X	Credits
PHRM 461	3
PHRM 462	1
PHRM 463	3
PHRM 464	3
PHRM 465	1
PHRM 466	3
PHRM 467	1

PHRM 468	2
PHRM 499	3
Total Credits	20

Course Descriptions

1st Semester (1st Professional PharmD)

UNIV 100 Foundation of University Education (3 credits)

This course is especially designed for freshmen to help them make the best use of their years of university study at FCCU. UNIV 100 course facilitates student integration into university life; assists them to make informed choices about their four years at FCC & beyond and equips freshmen with necessary academic skills to support their success. Each student will become more aware of his or her strengths and weaknesses in learning by working on a variety of skills. By examining their values and developing learning strategies this course will help students become successful in FCCU's liberal arts program.

ENGL 106: Functional English (3 credits)

This course is designed to develop students' functional proficiency in English through the integrated teaching of reading, listening, writing, and speaking, with systematic support for grammar, vocabulary development, and academic and real-life communication practices. The course emphasizes meaningful communication in academic, professional, and everyday contexts while fostering critical comprehension, clarity of expression, and intercultural awareness. Students will engage with authentic texts, participate in discussions and presentations, and produce a range of academic and practical written genres. The course aims to cultivate confident, ethical, and effective communicators who can navigate academic environments and real-world communication situations.

PHRM 120: Pharmaceutics-IA: Physical Pharmacy-I (3 credits)

This course introduces fundamental physicochemical principles underlying pharmaceutical sciences and drug formulation. It covers solution chemistry, solubility, dissolution, ionization, buffers, and micromeritics. Emphasis is placed on applying these principles to dosage form design and drug stability. The course also provides historical and professional context for pharmacy practice.

PHRM 121: Pharmaceutics-IA: Physical Pharmacy-I [Lab] (1 credit)

This laboratory course emphasizes experimental evaluation of key physicochemical properties relevant to pharmaceutical systems. Students engage in the preparation and analysis of solutions, assessment of solubility behavior, design and evaluation of buffer systems, and characterization of powder properties using standard pharmaceutical techniques. The course develops skills in precise measurement, instrument calibration, and quantitative data interpretation, with a focus on linking experimental observations to formulation behavior and pharmaceutical applications.

PHRM 122: Pharmaceutical Chemistry-IA: Organic Chemistry-I (3credits)

This course builds a strong foundation in organic chemistry with emphasis on concepts essential for pharmaceutical sciences. It covers bonding theories, electronic effects, stereochemistry, and conformational analysis, along with the structure, properties, and nomenclature of functional groups relevant to drug

molecules. Students explore key organic reaction mechanisms in aliphatic and aromatic systems and their applications in drug synthesis. The course integrates theoretical understanding with pharmaceutical relevance to support rational drug design and development.

PHRM 123: Pharmaceutical Chemistry-IA: Organic Chemistry-I [Lab] (1 credit)

This laboratory course develops practical skills in organic chemistry with emphasis on pharmaceutical relevance and safe laboratory practice. Students are trained in handling chemicals, identifying functional groups, synthesizing derivatives, and characterizing compounds using techniques such as melting point, boiling point, and crystallization. The course integrates classical qualitative analysis with purification methods to build competency in identifying pharmaceutically important organic substances. It also reinforces good laboratory practices, accuracy, and analytical thinking required in pharmaceutical sciences.

PHRM 124: Pharmaceutical Chemistry-IIA: Biochemistry-I (3 credits)

This course provides a comprehensive introduction to pharmaceutical biochemistry, focusing on the molecular and metabolic basis of human health and disease. It covers the structure, function, and metabolism of carbohydrates and lipids, along with bioenergetics, including oxidative phosphorylation and reactive oxygen species. The course also explores the biochemical roles of vitamins and hormones, emphasizing their physiological functions, deficiencies, and interactions with drugs. Through integration of metabolic pathways and clinical relevance, students develop an understanding of how biochemical processes influence disease states and pharmaceutical interventions.

PHRM 125: Pharmaceutical Chemistry-IIA: Biochemistry-I [Lab] (1 credit)

This laboratory course focuses on qualitative and quantitative biochemical analysis of clinically important biomolecules, with emphasis on pharmaceutical and diagnostic applications. Students perform standard tests for carbohydrates, lipids, sterols, bile pigments, and renal markers, and interpret results in the context of human physiology and disease. The course builds practical skills in clinical chemistry techniques, data interpretation, and safe laboratory practices, preparing students to connect biochemical findings with patient care and pharmaceutical decision-making.

PHRM 126: Physiology-I (3 credits)

This course provides a comprehensive introduction to human physiology, focusing on the fundamental principles governing the normal functioning of cells, tissues, and organ systems. Core topics include basic cell functions and membrane transport mechanisms, neural control mechanisms, central and autonomic nervous systems, muscle physiology, cardiovascular physiology, and gastrointestinal processes of digestion and absorption. Emphasis is placed on integrating physiological concepts to understand how different systems coordinate to maintain homeostasis.

PHRM 127: Physiology-I [Lab] (1 credit)

This laboratory course will provide hands-on training in fundamental physiological techniques and clinical procedures relevant to human body function assessment. Students will perform experiments involving measurement of vital parameters such as blood pressure, heart rate, and body mass index, as well as observing physiological processes like intestinal motility and sensory functions. The course also introduces essential clinical skills including blood sampling, injection techniques, and interpretation of electrocardiogram patterns.

Note: These courses are co-requisite with each other; PHRM 120 and PHRM 121, PHRM 122 and PHRM 123, PHRM 124 and PHRM 125, PHRM 126 and PHRM 127.

2nd Semester (1st Professional PharmD)

PHRM 130: Pharmaceutics-IB: Physical Pharmacy-II (3 credits)

This course expands on physical pharmacy concepts with focus on interfacial phenomena, rheology, disperse systems, and stability studies. It covers emulsions, suspensions, colloids, and adsorption processes in pharmaceutical systems. Students learn about drug degradation pathways and reaction kinetics. The course prepares students for formulation development and industrial applications.

PHRM 131: Pharmaceutics-IB: Physical Pharmacy-II [Lab] (1 credit)

This lab course focuses on experimental evaluation of physicochemical and kinetic principles. Students conduct studies on reaction kinetics, viscosity, surfactants, and suspension stability. Techniques such as rheogram plotting and Arrhenius analysis are introduced. Emphasis is placed on interpreting experimental data for pharmaceutical applications.

PHRM 132: Pharmaceutical Chemistry-IB: Organic Chemistry-II (3 credits)

This course provides advanced knowledge of organic chemistry with a focus on heterocyclic compounds and their significance in pharmaceutical sciences. It covers the preparation, properties, and applications of important heterocycles and fused ring systems commonly found in drug molecules. The course also explores reactive intermediates and detailed mechanisms of named organic reactions used in synthesis. Emphasis is placed on modern synthetic strategies, including green chemistry, biocatalysis, and microwave-assisted techniques, to promote efficient and sustainable development of pharmaceutical compounds.

PHRM 133: Pharmaceutical Chemistry-IB: Organic Chemistry-II [Lab] (1 credit)

This laboratory course focuses on the synthesis of pharmaceutically relevant organic compounds through classical and modern reaction pathways. Students perform a range of preparations including aromatic substitutions, condensation reactions, nucleophilic substitutions, and green synthesis approaches. Emphasis is placed on understanding reaction mechanisms, improving yield and purity, and applying safer, sustainable methodologies in organic synthesis. The course develops practical competence in multi- step synthesis, product isolation, and evaluation of drug-related compounds.

PHRM 134: Pharmaceutical Chemistry-IB: Biochemistry-II (3 credits)

This course provides an advanced understanding of pharmaceutical biochemistry with emphasis on proteins, amino acids, enzymes, and nucleic acids in relation to health and disease. It explores metabolic pathways, enzyme regulation, and molecular mechanisms underlying drug action and therapeutic interventions. The course also examines cellular signaling through secondary messengers and highlights the clinical relevance of biochemical markers used in disease diagnosis. By integrating biochemical concepts with pharmaceutical and clinical applications, the course prepares students to interpret laboratory data and apply biochemical knowledge in patient care and drug development.

PHRM 135: Pharmaceutical Chemistry-IB: Biochemistry-II [Lab] (1 credit)

This laboratory course focuses on quantitative and qualitative analysis of clinically important biomolecules with direct relevance to pharmaceutical and diagnostic practice. Students perform standard biochemical

assays for amino acids, proteins, glucose, urea, bilirubin, creatinine, calcium, and uric acid, along with urine analysis and liver and kidney function tests. Emphasis is placed on accurate measurement, standard curve construction, interpretation of clinical data, and correlation with disease states. The course develops practical competence in clinical biochemistry techniques, analytical reasoning, and professional laboratory practices.

PHRM 136: Physiology-II (3 credits)

This course provides an integrated study of advanced human physiology, focusing on the coordinated functions of major organ systems involved in homeostasis and survival. Core topics include blood physiology, cardiovascular circulation, respiratory physiology, renal physiology, endocrine and reproductive physiology, and mechanisms of temperature regulation. Emphasis is placed on understanding the interdependence of these systems in maintaining internal balance under normal and pathological conditions.

PHRM 137: Physiology-II [Lab] (1 credit)

This laboratory course will focus on experimental assessment of hematological and respiratory physiology. Students will perform essential diagnostic procedures including blood analysis, pulmonary function testing, and evaluation of hemostasis, while also gaining practical exposure to life-saving techniques such as nebulization and cardiopulmonary resuscitation. The course further emphasizes the interpretation of heart and lung sounds and highlights the role of preventive strategies in controlling vector-borne diseases.

PHRM 138: Anatomy & Histology (3 credits)

This course provides a comprehensive introduction to human anatomy and histology, focusing on the structural organization of the body at both macroscopic and microscopic levels. Core topics include the anatomy of the thorax, heart, abdominal cavity, urinary system, upper and lower limbs, nervous system, and special senses (eye and ear), along with fundamental histological techniques and their pharmacological relevance. Emphasis is placed on correlating structure with function to support understanding of normal body organization.

PHRM 139: Anatomy & Histology [Lab] (1 credit)

This lab course provides a comprehensive understanding of human gross anatomy and basic histology, focusing on the structural organization of major organ systems. Students will observe key anatomical regions including thoracic, abdominal, neurological, musculoskeletal, and reproductive systems using models and specimens. The course also introduces fundamental histological techniques and enables identification of microscopic features of tissues and organs relevant to normal physiological function.

ISLM 101: (2 credits)

This course offers students a thorough introduction to the core teachings of Islam, including its beliefs, practices, historical development, and societal impact. It is designed to equip students with a solid academic and cultural understanding of Islam, enhancing their ability to engage with Islam's diverse dimensions. By the end of the course, students will be better prepared to approach complex conversations about Islam's past, present, and future roles with empathy, respect, and well-informed perspectives.

CRST 152: Christian Ethics (2 credits)

Biblical and theological foundations of Christian ethics with special emphasis on developing the skills necessary to formulate ethical questions and find their solutions in the Bible.

Note: These courses are co-requisite with each other; PHRM 130 and PHRM 131, PHRM 132 and PHRM 133, PHRM 134 and PHRM 135, PHRM 136 and PHRM 137, PHRM 138 and PHRM 139.

3rdSemester (2nd Professional PharmD)

PHRM 220: Pharmaceutics-IIA: Drug Delivery Systems and Formulation Science-I (3 credits)

This course introduces the design and development of conventional dosage forms and drug delivery systems. It covers solid dosage forms, compounding techniques, pharmaceutical calculations, and incompatibilities. Students learn pre-formulation considerations and selection of appropriate dosage forms. The course builds foundational skills in formulation science.

PHRM 221: Pharmaceutics-IIA: Drug Delivery Systems and Formulation Science-I [Lab] (1 credit)

This laboratory course provides practical training in compounding and preparation of pharmaceutical dosage forms. Students develop skills in weighing, measuring, and preparing solid and liquid formulations. Techniques for extemporaneous compounding and handling excipients are emphasized. The course reinforces formulation principles through hands-on experience.

PHRM 222: Pharmacognosy-IA: Basic-I (3 credits)

This course explores the historical development, scope, and foundational concepts of natural medicines derived from plant, animal, mineral, and microbial sources. Key topics covered include traditional systems of medicine, preparation of crude drugs for the commercial market, detection methods for adulteration and the detailed morphological and chemical classification of secondary metabolites such as carbohydrates, lipids, Tannins and resins. Designed for early-stage pharmacy students, this foundational course establishes the core ethnopharmacological and chemical knowledge required for advanced pharmaceutical sciences.

PHRM 223: Pharmacognosy-IA: Basic-I [Lab] (1 credit)

This laboratory course provides hands-on experience in the macroscopic and microscopic evaluation of crude drugs from various natural sources. Key practical topics include the organoleptic assessment of organized and unorganized drugs, powder drug microscopy, section cutting, and the detection of physical adulterants. It equips pharmacy students with essential laboratory skills for the physical authentication and quality assessment of botanical materials.

PHRM 224: Pharmaceutics-IIIA: Basic Pharmaceutical Microbiology-I (2 credits)

This course introduces fundamental microbiology concepts relevant to pharmaceutical sciences. It covers microbial structure, classification, growth, and environmental interactions. Students explore human microbiota, infectious diseases, and pharmaceutical applications of microorganisms. The course establishes a foundation for applied microbiology and sterility concepts.

PHRM 225: Pharmaceutics-IIIA: Basic Pharmaceutical Microbiology-I [Lab] (1 credit)

This laboratory course focuses on essential microbiological techniques and aseptic practices. Students perform staining, culturing, and microbial identification methods. Experiments include microbial growth analysis and environmental monitoring. The course emphasizes laboratory safety and data interpretation in microbiology.

PHRM 226: Pharmacology and Therapeutics-IA (3 credits)

This course provides a foundational understanding of pharmacology and therapeutics, focusing on the principles governing drug action, disposition, and clinical use. Core topics include general pharmacology

(pharmacokinetics and pharmacodynamics), molecular and cellular pharmacology with receptor systems and signaling pathways, drugs acting on the autonomic nervous system, pharmacological management of gastrointestinal disorders, and an introduction to pharmaco-microbiomics. Emphasis is placed on integrating mechanistic concepts with therapeutic applications to support rational drug use.

PHRM 227: Pharmacology and Therapeutics-IA [Lab] (1 credit)

This laboratory course provides practical exposure to fundamental pharmacological principles through experimental studies on isolated tissues and animal models. Students will learn essential laboratory techniques, including preparation of physiological solutions, drug administration, and observation of drug effects on organ systems such as the gastrointestinal tract and eye. The course also emphasizes ethical handling of laboratory animals, experimental design, and interpretation of pharmacodynamic responses.

PHRM 228: Pathology (3 credits)

This course provides a comprehensive foundation in pathology, focusing on the mechanisms and structural changes underlying disease processes in the human body. Core topics include general pathology concepts such as cellular adaptations, cell injury and death, inflammation and repair, along with systemic pathology of the cardiovascular, neurological, gastrointestinal, and endocrine/metabolic systems, as well as fundamental principles of neoplasia. Emphasis is placed on understanding the pathophysiological basis of diseases and their progression at cellular and organ levels.

PHRM 229: Pathology [Lab] (1 credit)

This laboratory course focuses on the interpretation of histopathological and clinical pathology reports to understand disease mechanisms and diagnostic features. Students will analyze reports related to inflammation, neoplasia, hematological disorders, and organ-specific pathologies, with emphasis on correlating microscopic findings with clinical conditions.

Note: These courses are co-requisite with each other; PHRM 220 and PHRM 221, PHRM 222 and PHRM 223, PHRM 224 and PHRM 225, PHRM 226 and PHRM 227, PHRM 228 and PHRM 229.

4th Semester (2nd Professional PharmD)

PHRM 230: Pharmaceutics-IIB: Drug Delivery Systems and Formulation Science-II (3 credits)

This course advances knowledge of dosage forms including parenterals, semi-solids, aerosols, and novel drug delivery systems. It addresses sterile product formulation, IV admixtures, and cosmeceuticals. Students learn formulation strategies and stability considerations. The course bridges conventional and advanced drug delivery concepts.

PHRM 231: Pharmaceutics-IIB: Drug Delivery Systems and Formulation Science-II [Lab] (1 credit)

This lab course focuses on preparation and evaluation of biphasic and semi-solid dosage forms. Students formulate emulsions, suspensions, gels, and capsules. Practical skills in compounding and equipment handling are developed. Emphasis is placed on formulation consistency and quality.

PHRM 232: Pharmacognosy-IB: Basic-II (3 credits)

This course provides an in-depth exploration of the chemistry, classification, and biological significance of complex natural products derived from terrestrial and marine sources. Key topics covered include the extraction methods, structural categorization, and pharmacological applications of volatile oils, glycosides,

alkaloids, and marine bioactive agents. Designed for pharmacy students, the course bridges basic phytochemistry with advanced therapeutic applications, equipping learners with foundational knowledge essential for modern drug discovery and pharmacy practice.

PHRM 233: Pharmacognosy-IB: Basic-II [Lab] (1 credit)

This laboratory course reinforces theoretical concepts through practical identification tests, extraction protocols, and evaluation of active constituents from crude drugs, containing carbohydrates, tannins, resins, volatile oils, glycosides and alkaloids. This course builds essential laboratory competencies in the chemical profiling and quality assessment of natural products for future pharmaceutical applications.

PHRM 234: Pharmaceutics-IIIB: Pharmaceutical Microbiology-II Applied Pharmaceutical Microbiology and Immunology (3 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course covers applied microbiology, sterilization, antimicrobial agents, and immunology. It includes microbial control methods, resistance mechanisms, and viral diseases. Students also study immune responses, vaccines, and hypersensitivity reactions. The course integrates microbiology with clinical and pharmaceutical applications.

PHRM 235: Pharmaceutics-IIIB: Pharmaceutical Microbiology-II Applied Pharmaceutical Microbiology and Immunology-I [Lab] (1 credit)

This laboratory course develops skills in microbial control, sterility testing, and antibiotic sensitivity analysis. Students perform preservative efficacy testing, MIC determination, and environmental monitoring. Emphasis is placed on regulatory compliance and quality assurance. The course provides practical insights into pharmaceutical microbiology testing.

PHRM 236: Pharmacology and Therapeutics-IB (3 credits)

This course provides an advanced understanding of pharmacology and therapeutics, emphasizing individualized drug therapy and the management of major systemic disorders. Core topics include pharmacogenomics and pharmacogenetics with clinical applications, drugs acting on the cardiovascular system, drugs affecting endocrine function, and pharmacological management of respiratory disorders. Emphasis is placed on integrating genetic principles with therapeutic decision-making to optimize drug efficacy and safety.

PHRM 237: Pharmacology and Therapeutics-IB [Lab] (1 credit)

This laboratory course provides hands-on training in experimental pharmacology with emphasis on drug administration techniques, in vivo and in vitro pharmacodynamic studies, and ethical considerations in animal and human research. Students will develop practical skills in handling laboratory animals, performing dosing procedures, and evaluating drug effects on physiological systems such as cardiovascular, renal, and metabolic functions. The course also introduces international ethical guidelines governing biomedical research and promotes responsible conduct in pharmacological experimentation.

PKST 101: Pakistan Studies (2 credits)

This course examines the historical evolution, ideological foundations, constitutional development, socio-economic structure, and contemporary challenges of Pakistan. It explores key themes including the Pakistan Movement, constitutional history, governance structures, economic planning, cultural diversity, and foreign

policy orientation. The course develops students' critical understanding of Pakistan's political trajectory, regional engagements, and global positioning. It is designed as a compulsory foundational course for undergraduate students.

ISLM 100-A (Fehm-e-Qur'an I) (1 credit)

This course introduces students to the core vocabulary, common expressions, and fundamental linguistic features of Quranic Arabic. It emphasizes the practical development of skills to recognize and comprehend the language of the Holy Qur'an directly, establishing a strong foundation for further study leading to Fehm-e-Qur'an II in accordance with the HEC Model Curriculum. The course develops students' ability to recognize and comprehend the meanings of Qur'anic words and phrases directly within the text, establishing a strong linguistic foundation for further study. By taking this course, students are expected to understand approximately 30–40% of the recurring vocabulary of the Holy Qur'an, preparing them for more advanced linguistic engagement in Fehm-e-Qur'an II.

URDU 101: Communicative Urdu (3 credits)

Communication and its different means, brief introduction to Urdu language, some fundamentals of Urdu grammar, functional Urdu, creative writing and journalistic Urdu.

Note: These courses are co-requisite with each other; PHRM 230 and PHRM 231, PHRM 232 and PHRM 233, PHRM 234 and PHRM 235, PHRM 236 and PHRM 237.

5th Semester (3rd Professional PharmD)

PHRM 320: Pharmaceutical Chemistry IIIA: Pharmaceutical Analysis-I (3 credits)

This course introduces the principles and techniques of pharmaceutical analysis with emphasis on quantitative and instrumental methods used in drug evaluation. It covers titrimetric, electrochemical, and thermal analysis techniques, including their theory, operation, and applications in quality control of pharmaceutical substances. Students develop the ability to select appropriate analytical methods, interpret results, and understand sources of error. The course integrates analytical concepts with pharmaceutical relevance to support accurate drug analysis and regulatory compliance.

PHRM 321: Pharmaceutical Chemistry IIIA: Pharmaceutical Analysis-I [Lab] (1 credit)

This laboratory course provides hands-on training in pharmaceutical analysis using classical and instrumental techniques. Students perform titrimetric, electrochemical, polarimetric, polarographic, and fluorimetric methods to determine the quality, purity, and concentration of pharmaceutical substances. Emphasis is placed on method selection, accuracy, data interpretation, and quality control of APIs and excipients. The course develops practical skills required for pharmaceutical analysis, regulatory compliance, and industrial applications.

PHRM 322: Pharmacognosy-IIA: Applied (3 credits)

This course introduces students to the science of natural products through chromatographic and extraction techniques, molecular pharmacognosy, and clinical applications of herbal medicines. Core topics include methods for isolating and identifying plant constituents, molecular tools for authentication and detection, and evaluation of herbal drugs in common diseases. Designed for PharmD students, it emphasizes critical appraisal of natural medicines, preparing learners to integrate traditional knowledge with modern scientific practice.

PHRM 323: Pharmacognosy-IIA: Applied [Lab] (1 credit)

This laboratory course provides practical training in chromatographic techniques, including paper, thin layer, and column chromatography, for the separation and identification of constituents from natural products. It also introduces molecular pharmacognosy practices such as DNA extraction and barcoding for plant authentication. The course is designed to reinforce theoretical concepts through hands-on experimentation related to natural product analysis. Emphasis is placed on developing technical competence, accuracy, and interpretation of experimental results in pharmacognosy.

PHRM 324: Pharmacology and Therapeutics-IIA (3 credits)

This course provides an in-depth study of pharmacology and therapeutics, focusing on the clinical use of drugs in the management of infectious, inflammatory, and immune-related disorders. Core topics include principles of chemotherapy with antibacterial, antifungal, antiviral, antiprotozoal, and anticancer agents, anti-inflammatory and anti-rheumatic drugs, and immunopharmacology including monoclonal antibodies and immunomodulatory therapies. Emphasis is placed on rational drug selection, mechanisms of action, and the application of evidence-based treatment guidelines.

PHRM 325: Pharmacology and Therapeutics-IIA [Lab] (1 credit)

This laboratory course focuses on the experimental evaluation of analgesic, anti-inflammatory, and antioxidant drug activities using established in vivo and in vitro models. Students will perform pharmacological assays to assess drug efficacy and develop skills in data interpretation and experimental analysis. The course also integrates clinical case studies to enhance understanding of modern therapeutics, including monoclonal antibodies, immunomodulators, and rational antibiotic use in patient care.

PHRM 326: Pharmacy Practice-IA: Hospital and Community Pharmacy (2 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course provides a comprehensive overview of the essential operations and professional standards required for managing pharmacy services in both hospital and community settings within Pakistan's healthcare system. Students will examine safe medication use processes, inventory and supply chain management, regulatory compliance, and the critical role of technology in enhancing patient safety and medication management. This junior-level course emphasizes evidence-based practice and the use of clinical decision support systems to prepare students for effective participation in multidisciplinary healthcare teams

PHRM 240: Quantitative Reasoning-I (Mathematics) (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course provides a foundational understanding of mathematical concepts essential for scientific reasoning and problem-solving. Key topics include algebraic equations and progressions, trigonometric functions, analytical geometry, differential calculus, and integral calculus. The course emphasizes the application of mathematical techniques to analyze quantitative problems and develop logical reasoning skills. It is designed for undergraduate students to build a strong mathematical base for further studies in science and professional programs.

PKST 103: Ideology & Constitution of Pakistan (2 credits)

This course provides a foundational study of the ideology and constitutional development of Pakistan. It examines the historical evolution of the ideological foundations of Pakistan, the Two-Nation Theory, and the

contributions of key leaders in the Pakistan Movement. The course also analyzes the structure, evolution, and major provisions of the Constitution of Pakistan (1973), including fundamental rights, principles of policy, and federal governance. It aims to develop informed and responsible citizenship through constitutional literacy and civic awareness.

ISLM 100-B (Fehm-e-Qur'an II) (1 credit)

Prerequisite: ISLM100-A

Building upon the foundation established in Fehm-e-Qur'an I, this course advances students' proficiency in understanding the language of the Holy Qur'an through the systematic expansion of Qur'anic vocabulary and deeper study of its grammatical and linguistic structures. Emphasis is placed on strengthening students' ability to comprehend Qur'anic verses directly, enabling them to read the text with greater confidence, accuracy, and independence. Upon successful completion of the course, students are expected to understand approximately 80–85% of the recurring vocabulary of the Holy Qur'an

Note: These courses are co-requisite with each other; PHRM 320 and PHRM 321, PHRM 322 and PHRM 323, PHRM 324 and PHRM 325.

6th Semester (3rd Professional PharmD)

PHRM 330: Pharmaceutical Chemistry IIIB: Pharmaceutical Analysis-II (3 credits)

This course provides a comprehensive understanding of instrumental techniques used in pharmaceutical analysis, with emphasis on spectroscopic and chromatographic methods. It covers the principles, instrumentation, and applications of UV, IR, NMR, mass spectrometry, and atomic absorption/emission spectroscopy for drug identification and characterization. The course also introduces chromatographic techniques such as TLC, HPLC, GC, and ion exchange chromatography for separation and quantitative analysis of pharmaceutical compounds. Students develop the ability to select appropriate analytical methods and interpret data for quality control and research applications.

PHRM 331: Pharmaceutical Chemistry IIIB: Pharmaceutical Analysis-II [Lab] (1 credit)

This laboratory course focuses on instrumental techniques used in pharmaceutical analysis, with emphasis on UV–Visible spectrophotometry and chromatographic methods. Students gain hands-on experience in determining absorption spectra, λ -max, and specific absorbance, as well as performing quantitative assays of pharmaceutical substances. The course also introduces method development and validation procedures using techniques such as TLC, HPLC, and GC. Emphasis is placed on data interpretation, accuracy, and regulatory relevance in quality control of APIs and formulations.

PHRM 332: Pharmacognosy-IIB: Advanced (3 credits)

This course provides a theoretical understanding of herbal drugs used in various disease conditions, focusing on their mechanisms of action, therapeutic uses, dosage forms, and adverse effects. It covers principles of industrial pharmacognosy, including herbal product development, regulatory requirements, and formulation approaches. The course also addresses quality control, standardization, and evaluation of herbal drugs, along with key concepts of nutraceuticals and natural cosmetics. Emphasis is placed on the pharmaceutical and industrial aspects of natural products.

PHRM 333: Pharmacognosy-IIB: Advanced [Lab] (1 credit)

This practical course provides applied understanding of herbal drugs through evaluation of their physicochemical properties and quantitative determination of phytoconstituents. It focuses on standard methods used to assess the quality, purity, and consistency of crude drugs and herbal formulations. The course supports theoretical concepts of pharmacognosy by emphasizing basic quality control and standardization techniques. It prepares students to ensure safe and effective use of natural products in pharmacy practice.

PHRM 334: Pharmacology and Therapeutics-IIB (3 credits)

This course provides an advanced understanding of pharmacology and therapeutics, focusing on drugs affecting the central nervous system, principles of toxicology, and emerging technological applications in pharmacology. Core topics include pharmacological management of central nervous system disorders (such as sedatives and hypnotics, anxiolytics, antidepressants, antipsychotics, antiepileptics, antiparkinsonian drugs, opioids, anesthetics, and related agents), fundamental concepts of toxicology and its clinical management, and applications of artificial intelligence in drug discovery and personalized medicine. Emphasis is placed on integrating therapeutic principles with safety considerations and evidence-based clinical guidelines.

PHRM 335: Pharmacology and Therapeutics-IIB [Lab] (1 credit)

This laboratory course will focus on the experimental evaluation of central nervous system (CNS)-active drugs, including anesthetics, hypnotics, antidepressants, and anticonvulsants using established animal models. Students will gain hands-on experience in assessing drug effects through behavioral and physiological assays, alongside learning modern approaches such as AI- based drug-target interaction analysis and pharmacogenomics. The course also emphasizes regulatory and ethical frameworks, including OECD guidelines for toxicity testing, to ensure safe and responsible pharmacological research.

PHRM 336: Pharmacy Practice-IB: Dispensing, Social and Administrative Pharmacy (2 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course provides foundational knowledge and practical skills in the administrative and professional aspects of dispensing within the Pakistani healthcare system. Students will explore pharmaceutical calculations, aseptic preparation techniques, and the management of controlled substances, while adhering to mandatory service standards. The curriculum emphasizes patient-centered care through the development of education materials, effective counseling strategies, and the implementation of medication safety protocols like the "seven rights" of administration. This course is essential for junior-level students preparing for professional roles in diverse pharmacy settings

CSCS 100: Applications of Information and Communication Technology (2+1 credits)

This course provides a comprehensive introduction to Information and Communication Technologies (ICT) for freshman undergraduate students. The course covers fundamental ICT concepts, productivity tools, and practical applications across various domains including education, healthcare, finance, and e-commerce. Students will develop essential digital literacy skills, understand ethical considerations in technology use, and gain hands-on experience with contemporary ICT tools and platforms. The course emphasizes both theoretical understanding and practical application through a combination of lectures and laboratory sessions.

PHRM 245: Quantitative Reasoning-II (Biostatistics) (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course provides a foundational understanding of statistical methods and their applications in pharmaceutical sciences. Key topics include data organization and presentation, descriptive statistics, probability distributions, regression and correlation, hypothesis testing, analysis of variance, and use of statistical software. The course emphasizes the application of statistical tools to analyze data, interpret results, and support evidence-based decision-making. It is designed for undergraduate students to develop essential analytical skills for research and professional practice.

Note: These courses are co-requisite with each other; PHRM 330 and PHRM 331, PHRM 332 and PHRM 333, PHRM 334 and PHRM 335.

7th Semester (4th Professional PharmD)

PHRM 420: Pharmaceutics-IVA: Industrial Pharmacy-I (3 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course introduces large-scale pharmaceutical manufacturing processes and unit operations. Topics include mixing, drying, size reduction, filtration, compression, coating, and encapsulation. Students learn about industrial equipment and process optimization. The course prepares students for pharmaceutical production environments.

PHRM 421: Pharmaceutics-IVA: Industrial Pharmacy-I [Lab] (1 credit)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course provides hands-on training in pharmaceutical manufacturing processes. Students perform granulation, tablet compression, and milling operations. Evaluation of product quality and process parameters is emphasized. The course bridges theory with industrial practice.

PHRM 422: Pharmaceutics-VA: Biopharmaceutics and Pharmacokinetics-I (3 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course introduces the fundamental principles governing drug absorption, distribution, metabolism, and excretion, with emphasis on the relationship between drug formulation and bioavailability. It covers basic pharmacokinetic models, compartmental analysis, and factors influencing drug disposition in the body. Students will learn to interpret plasma concentration–time data and apply mathematical models to describe drug kinetics. The course provides a foundation for optimizing dosage regimens and understanding the impact of physiological and formulation variables on therapeutic outcomes.

PHRM 423: Pharmaceutics-VA: Biopharmaceutics and Pharmacokinetics-I [Lab] (1 credit)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This laboratory course provides practical training in the application of biopharmaceutic and pharmacokinetic principles. Students perform experiments related to drug dissolution, bioavailability assessment, and analysis of plasma concentration–time data. Emphasis is placed on calculating pharmacokinetic parameters, including rate constants, half-life, and clearance using appropriate models. The course develops skills in data interpretation, use of software tools, and application of kinetic concepts to optimize drug therapy.

PHRM 424: Pharmacy Practice-IIA: Clinical Pharmacy-I (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course introduces the fundamental principles of clinical pharmacy and the evolving role of pharmacists from dispensing-focused functions to patient-centered care within the Pakistani healthcare system. Students explore comprehensive patient assessment, medication safety through a systems approach, pharmacovigilance, and the clinical application of therapeutic drug monitoring. This course is designed for fourth-year students to establish the foundational skills necessary for evidence-based decision-making and public health promotion.

PHRM 425: Pharmacy Practice-IIA: Clinical Pharmacy-I [Lab] (1 credit)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This lab course provides hands-on experience in applying the fundamental principles of clinical pharmacy to simulated and real-world patient care scenarios. Students will practice essential skills such as collecting medication histories, interpreting laboratory data for therapy planning, performing renal dose adjustments, and analyzing medication errors and adverse drug reactions. Designed for fourth-year students, this course utilizes case studies and clinical informatics tools to bridge the gap between theoretical knowledge and professional practice in medication safety and optimization.

PHRM 426: Pharmaceutics-VI: Pharmaceutical Quality Management System (3 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course provides a comprehensive overview of quality management principles applied to pharmaceutical manufacturing and regulatory environments. It covers Good Manufacturing Practices (GMP), quality assurance (QA), quality control (QC), validation, documentation systems, and risk management. Students will learn about regulatory frameworks, auditing processes, and compliance requirements essential for ensuring product safety, efficacy, and consistency. The course prepares students to implement and manage quality systems in pharmaceutical industries and healthcare settings.

ENGL 108: Expository Writing (3 credits)

Prerequisites: ENGL 106

In the second course of the Writing and Communication Program, students will build on what they learned in the first course while learning how to research and incorporate academic sources. In the first of two major papers and presentations, students will analyze and report the history and status quo of a current controversy in Pakistan or the world. The second paper and presentation build on the first with the addition of the students' view on and possible solution for the controversy. For the second major presentation students will analyze their audience beforehand in order to craft and deliver a maximally persuasive speech in the PechaKucha style.

BUSN 220: Entrepreneurship (2 credits)

This course is designed to promote entrepreneurial spirit and outlook among students, encouraging them to think critically, identify opportunities, and transform their ideas into successful ventures. It aims at imparting them with the requisite knowledge, skills and abilities, enabling them to seize the identified opportunities for initiating ventures and successfully navigating the challenges that come with starting a business and managing it. The course covers topics relevant to entrepreneurship including setting up and initiation of business

(including requirements for registration and incorporation with regulators such as SECP and others), market research, opportunity identification, business planning, financial literacy for managing finances and securing funding, marketing and sales, team building and innovation. Overall, the course is geared towards personal growth and professional development for pursuing innovative ideas, availing opportunities and initiating start-ups.

Note: These courses are co-requisite with each other; PHRM 420 and PHRM 421, PHRM 422 and PHRM 423, PHRM 424 and PHRM 425.

8th Semester (4th Professional PharmD)

PHRM 430: Pharmaceutics-IVB: Industrial Pharmacy-II (3 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course focuses on the design, operation, and regulatory requirements of pharmaceutical manufacturing facilities. It covers facility layout, architectural considerations, and utility systems including HVAC, process piping, and environmental control for various dosage forms such as oral solids, liquids, semi-solids, and sterile products. Students will learn principles of regulatory compliance, occupational safety, and handling of hazardous materials in accordance with GMP standards. The course also introduces pressure vessels, reactors, and concepts of reliability, availability, and maintainability (RAM) in ensuring efficient and safe pharmaceutical production.

PHRM 431: Pharmaceutics-IVB: Industrial Pharmacy-II [Lab] (1 credit)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This laboratory course provides hands-on training in advanced pharmaceutical manufacturing and coating technologies. Students perform practical exercises including preparation of ointments, granules, emulsions, and lyophilized products, along with tablet coating techniques such as film, sugar, enteric, and fluid bed coating. Emphasis is placed on process optimization, equipment handling, and evaluation of product quality through dissolution testing and statistical analysis. The course also includes industrial visits to expose students to real-world pharmaceutical production environments and quality practices.

PHRM 432: Pharmaceutics-VB: Biopharmaceutics and Pharmacokinetics-II (3 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course provides an advanced understanding of pharmacokinetic principles, including linear and non-linear kinetics and their applications in clinical and research settings. It covers pharmacokinetic modeling approaches such as compartmental and non-compartmental analysis, along with concepts of half-life, volume of distribution, multiple dosing, and intravenous infusion kinetics. Students will explore pharmacokinetic variations in disease states and learn strategies for dose adjustment in renal and hepatic impairment. The course also introduces PK/PD correlations and modern approaches such as physiologically based pharmacokinetic (PBPK) modeling for rational drug therapy optimization.

PHRM 433: Pharmaceutics-VB: Biopharmaceutics and Pharmacokinetics-II [Lab] (1 credit)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This laboratory course provides practical training in pharmacokinetic parameter computation, including bioavailability (F), $AUC_{0-\infty}$, and biological half-life under zero- and first-order kinetics. Students are introduced to bioanalytical method development using HPLC for drug quantification in biological matrices. The course also covers bioequivalence study design and evaluation based on ICH and FDA guidelines. In addition, learners gain hands-on experience in computational pharmacokinetics using software such as PK-Sim, Phoenix WinNonlin, Monolix, Simcyp, and NONMEM.

PHRM 434: Pharmacy Practice-IIB: Clinical Pharmacy-II (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year

This course focuses on the advanced application of clinical pharmacy principles through the systematic development, implementation, and monitoring of patient-centered drug therapy plans. Key topics include the Pharmacist Patient Care Process (PPCP), Medication Therapy Management (MTM), antimicrobial stewardship, and the utilization of advanced digital health tools to optimize pharmacotherapy. Designed for fourth-year students, the course emphasizes independent clinical decision-making and effective communication strategies to improve medication adherence and therapeutic outcomes across diverse healthcare environments

PHRM 435: Pharmacy Practice-IIB: Clinical Pharmacy-II [Lab] (1 credit)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This laboratory course focuses on the practical application of advanced clinical pharmacy principles through the systematic development and documentation of patient-centered drug therapy plans. Students utilize standardized approaches like the Pharmacist's Workup of Drug Therapy (PWDT) and the Pharmacist Patient Care Process (PPCP) to identify drug therapy problems and formulate evidence-based interventions. The lab emphasizes skill development in patient education, adherence assessment, antimicrobial stewardship, and the use of digital health tools to prepare students for professional roles in multidisciplinary teams. This senior-level course is a co-requisite to PHRM 434 and serves to bridge the gap between theoretical knowledge and independent clinical practice.

PHRM 436: Pharmaceutical Chemistry IV: Pharmaceutical Quality Control (2 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for all the courses of 4th and 5th professional year.

This course provides an overview of quality control principles and analytical methods used in the evaluation of pharmaceutical raw materials and products. Key topics include sources of impurities, pharmacopeial standards, sampling and documentation, analytical method development and validation, and quality risk management in accordance with international guidelines. The course emphasizes the application of regulatory standards and analytical techniques to ensure product quality, safety, and compliance. It is designed to prepare students for roles in pharmaceutical quality control, assurance, and regulatory environments.

PHRM 437: Pharmaceutical Chemistry IV: Pharmaceutical Quality Control [Lab] (1 credit)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for all the courses of 4th and 5th professional year.

This course provides practical training in quality control techniques used for the evaluation of pharmaceutical raw materials and chemical substances. Key topics include calibration of laboratory equipment, pharmacopeial testing, standardization of reagents, impurity determination, analytical method development and validation, and basic quality risk management. The course emphasizes hands-on application of analytical procedures and compliance with pharmacopeial and regulatory standards to ensure product quality and safety. It is designed to prepare students for laboratory-based roles in pharmaceutical quality control and assurance.

PHRM 438: Pharmacy Practice-III: Civics & Community Engagement (2 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course explores the principles of civic responsibility and the essential role pharmacists play in advancing community welfare and social justice within the Pakistani healthcare system. Students will examine healthcare governance, pharmacist-led community engagement strategies, public health advocacy, and the ethics of digital citizenship. This fourth-year course features a unique requirement where students must complete and report on 100 hours of community service to foster professional accountability and leadership.

PHRM 439: Pharmacy Practice-IV: Pharmaceutical Marketing and Management (2 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course provides a comprehensive analysis of the marketing and management principles tailored specifically to the pharmaceutical industry within the Pakistani healthcare context. Students will examine foundational marketing theories, digital promotion strategies, pharmaceutical supply chain management, and the ethical/legal frameworks that govern the sale of prescription and over-the-counter products. This senior-level course is designed to equip students with the strategic planning, human resource management, and financial oversight skills required for leadership roles in pharmaceutical e-commerce and business development.

PHRM 498: Internship (3 credits)

Note: These courses are co-requisite with each other; PHRM 430 and PHRM 431, PHRM 432 and PHRM 433, PHRM 434 and PHRM 435, PHRM 436 and PHRM 437.

9th Semester (5th Professional PharmD)

PHRM 441: Pharmaceutics-VIIA: Pharmaceutical Technology-I (3 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course covers novel and modified drug delivery systems, including controlled, sustained, delayed, and targeted release formulations along with advanced routes such as transdermal, pulmonary, ocular, nasal, and colon-specific delivery. It also introduces polymer-based systems, nanotechnology-based carriers, and key nanocarriers such as liposomes, niosomes, dendrimers, and nanoparticles. Students gain understanding of Quality by Design (QbD) principles, risk-based formulation development, and experimental design tools. The course further highlights the role of AI/ML in drug delivery and emerging technologies like 3D printing, hot-melt extrusion, electrospinning, and computational pharmaceutics.

PHRM 442: Pharmaceutics-VIIA: Pharmaceutical Technology-I [Lab] (1 credit)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This laboratory course focuses on pre-formulation evaluation of excipients, including micromeritic properties such as bulk density, tapped density, Carr's Index, Hausner's Ratio, and particle size distribution. Students apply Quality by Design (QbD) and AI- assisted Design of Experiments (DoE) approaches, including factorial, CCD, and Box- Behnken designs for formulation optimization. The course also covers process development and validation, dissolution profiling using model-dependent and independent methods, and matrix tablet formulation for drugs with varying solubility. In addition, learners gain hands-on experience in palletization, coating, and comparative evaluation of swelling and disintegration behaviors of dosage forms.

PHRM 443: Pharmacy Practice-VA: Pharmaceutical Regulatory Sciences-I (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course provides a comprehensive introduction to the legislative processes and foundational principles of pharmaceutical regulation within the Pakistani healthcare system. Students will examine the historical evolution of drug laws, the organizational structure and functions of the Drug Regulatory Authority of Pakistan (DRAP), and the detailed provisions of the Drugs Act 1976. This senior-level core course is designed to equip students with the legal knowledge necessary to ensure professional compliance and ethical conduct in various pharmacy practice settings.

PHRM 444: Medicinal Chemistry-I (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for all the courses of 4th and 5th professional year.

This course provides a comprehensive understanding of medicinal chemistry, emphasizing drug design approaches (SBDD, LBDD, SAR/QSAR), prodrugs, and bioisosterism in modern drug discovery. It integrates chemical structure, biological activity, and therapeutic applications of major drug classes including anesthetics, CNS-active drugs, hormones, and anticancer agents. The course promotes problem-solving, innovation, and safe medication practices through clinically relevant examples. It also develops ethical responsibility, effective communication, and lifelong learning skills in alignment with the Pharmaceutical Sciences PLOs and Dee Fink's significant learning framework.

PHRM 445: Medicinal Chemistry-I [Lab] (1 credit)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for all the courses of 4th and 5th professional year.

This practical course integrates in-silico and wet lab techniques to develop hands-on skills in medicinal and pharmaceutical chemistry. Students gain experience in molecular modeling, drug-likeness prediction, and basic docking, alongside synthesis, functional group analysis, and drug assay procedures. The course bridges computational tools with experimental practices to support drug development and quality evaluation. It emphasizes problem-solving, ethical laboratory conduct, communication, and lifelong learning.

PHRM 446: Pharmacy Practice-VIA: Advanced Clinical Pharmacy-I (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course focuses on developing advanced clinical reasoning skills to optimize pharmacotherapy for complex disease states within the Pakistani healthcare context. Key topics covered every time the course is offered include the management of CNS disorders, gastrointestinal conditions, dermatological infections, and chronic metabolic diseases such as diabetes and hypertension. Designed for senior-level students, the course emphasizes the integration of pathophysiology and evidence-based practice to formulate comprehensive pharmaceutical care plans.

PHRM 447: Pharmacy Practice-VIA: Advanced Clinical Pharmacy-I [Lab] (1 credit)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This laboratory course provides practical training in advanced clinical pharmacy by applying theoretical principles to complex patient care scenarios involving CNS, gastrointestinal, and chronic metabolic disorders. Students focus on developing comprehensive pharmaceutical care plans, conducting patient counseling sessions, and performing dose adjustments to optimize therapeutic outcomes within the Pakistani healthcare context. This senior-level lab utilizes simulated and real-world case studies to bridge the gap between evidence-based guidelines and independent clinical practice.

PHRM 448: Clinical Pharmacology (2 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses.

This course provides an in-depth understanding of the principles of clinical pharmacology across special patient populations, emphasizing the safe, effective, and evidence-based use of medicines. It examines the impact of physiological changes during pregnancy, infancy, childhood, and aging on drug pharmacokinetics and pharmacodynamics, and explores the risks associated with teratogenicity and drug use during lactation. The course also addresses the influence of major co-morbid conditions, including hepatic impairment, renal dysfunction, and diabetes, on drug response and therapeutic outcomes. Through the application of clinical pharmacological principles, students will develop the knowledge and skills required to optimize individualized pharmacotherapy and promote rational drug use in diverse patient populations.

PHRM 449: Clinical Pharmacology [Lab] (1 credit)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses.

This laboratory course complements the Clinical Pharmacology theory course by providing students with Through the discussion and .case-based learning experiences involving special patient populations ,hands-on and evidence- ,pharmacodynamic ,students apply pharmacokinetic ,analysis of real-life clinical scenarios ,pediatrics ,lactation ,based pharmacotherapeutic principles to optimize medication use in pregnancy ,The course emphasizes rational drug selection .and patients with co-morbid conditions ,geriatrics and clinical ,identification of drug-related problems ,medication safety ,individualized dose calculations and ,communication ,Students will develop problem-solving .decision-making in complex patient populations and ,interpreting patient-specific clinical information ,critical thinking skills by evaluating therapeutic options recommending safe and effective pharmacotherapy in accordance with current clinical guidelines and best practices.

UNIV 200: Critical Thinking, Ethics, and Artificial Intelligence (3 credits)

This course invites students to think clearly and ethically about one of the defining technologies of our time: artificial intelligence. Through close analysis of arguments, real-world case studies, and guided discussion,

students will learn how to evaluate evidence, recognize bias, identify common logical fallacies, and construct persuasive, well-reasoned arguments of their own.

Drawing on major ethical frameworks, the course asks students to grapple with pressing questions surrounding AI – from academic integrity and authorship to privacy, fairness, and responsible decision-making. Emphasis is placed on practical application: short writing assignments and case-based discussions challenge students to apply ethical reasoning to everyday university contexts and broader public life.

By the end of the course, students will understand the tools of critical thinking and know how to use them responsibly in an AI-shaped world.

Note: These courses are co-requisite with each other; PHRM 441 and PHRM 442, PHRM 444 and PHRM 445, PHRM 446 and PHRM 447, PHRM 448 and PHRM 449.

10th Semester (5th Professional PharmD)

PHRM 461: Pharmaceutics-VIIB: Pharmaceutical Technology-II (3 credits)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This course covers pharmaceutical biotechnology with focus on development and production of biologics such as vaccines, insulin, enzymes, antibiotics, and amino acids. It introduces gene-based drug delivery systems including mRNA and siRNA therapeutics, along with their development pipelines. Students also study biosimilars and biobetters, including regulatory and commercial aspects. The course further includes monoclonal and polyclonal antibodies and the role of AI in biologics development and bioprocess optimization.

PHRM 462: Pharmaceutics-VIIB: Pharmaceutical Technology-II [Lab] (1 credit)

Pre-requisite: All courses up to and including the 3rd professional year (juniors) must be successfully completed as prerequisites for enrollment in 4th (rising seniors) and 5th professional year (senior) courses

This laboratory course provides hands-on training in advanced drug delivery systems, including formulation of transdermal patches, microemulsions, dissolving microneedles, fast dispersible tablets, and gastroretentive floating tablets. Students also prepare and characterize pharmaceutical hydrogels and evaluate key formulation performance parameters. The course further introduces molecular biology techniques such as nucleic acid extraction, PCR, gel electrophoresis, DNA quantification, and sequencing methods. In addition, learners gain understanding of antigen–antibody reactions and their applications in biotechnology-based pharmaceutical analysis.

PHRM 463: Pharmacy Practice-VB: Pharmaceutical Regulatory Sciences-II (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This course provides an advanced examination of the legal and operational frameworks governing the pharmaceutical industry in Pakistan, with a primary focus on the Drug Regulatory Authority of Pakistan (DRAP) Act 2012 and the integration of digital management systems. Students will explore global regulatory benchmarking, drug registration using Common Technical Document (CTD) requirements, Good Manufacturing Practices (GMP) compliance, and the regulation of medical devices and cosmetics. This senior-level course is essential for students preparing for leadership roles in industrial quality management, regulatory affairs, and post-market surveillance.

PHRM 464: Medicinal Chemistry-II (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for all the courses of 4th and 5th professional year.

This course provides an in-depth study of medicinal chemistry of key therapeutic classes including cardiovascular, antidiabetic, diuretic, antimicrobial, antiparasitic, antiviral, and immunomodulatory agents. It emphasizes the relationship between chemical structure, SAR, biological action, and therapeutic applications to support rational drug selection and patient safety. The course integrates pharmaceutical and clinical knowledge to address real-world healthcare challenges. It also develops problem-solving abilities, ethical practice, communication skills, and lifelong learning.

PHRM 465: Medicinal Chemistry-II [Lab] (1 credit)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for all the courses of 4th and 5th professional year.

This practical course focuses on synthetic medicinal chemistry, emphasizing the preparation, identification, and characterization of pharmaceutically important organic compounds. Students develop hands-on skills in synthesis of heterocyclic and therapeutic agents, along with qualitative identification of raw materials using standard chemical tests. The course integrates theoretical medicinal chemistry with laboratory practice to enhance problem-solving, analytical skills, and quality control awareness. It also promotes ethical laboratory conduct, effective communication, and lifelong learning in alignment with Pharmaceutical Sciences PLOs and Dee Fink's significant learning framework.

PHRM 466: Pharmacy Practice-VIB: Advanced Clinical Pharmacy-II (3 credits)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This terminal clinical course focuses on the advanced management of high-acuity disease states and emergency pharmacotherapy within the specific context of the Pakistani healthcare system. Key topics covered every time the course is offered include IV fluid therapy, oncology (breast and prostate cancer), management of infectious diseases like Malaria and Dengue, and pharmacotherapy for renal and hepatic failure. This senior-level course is designed to transition students into independent practitioners capable of optimizing therapy in emergency situations and managing complex coagulation and hematological disorders.

PHRM 467: Pharmacy Practice-VIB: Advanced Clinical Pharmacy-II [Lab] (1 credit)

Pre-requisite: All the courses till and including 3rd professional year are pre-requisites for the courses of 4th and 5th professional year.

This terminal laboratory course provides senior students with intensive practical training in the management of high-acuity disease states and emergency pharmacotherapy within the hospital and clinical environment. Students focus on developing comprehensive pharmaceutical care plans, performing complex dose adjustments for renal and hepatic failure, and conducting Therapeutic Drug Monitoring (TDM) for medications with narrow therapeutic indices. The lab serves as the final clinical bridge to independent practice, emphasizing evidence-based decision-making for oncology, regional infectious diseases, and urological disorders.

PHRM 468: Bioethics (2 credits)**PHRM 499: Capstone Project (3 credits)**

Note: These courses are co-requisite with each other; PHRM 461 and PHRM 462, PHRM 464 and PHRM 465, PHRM 466 and PHRM 467.

Department of
PHILOSOPHY

Bachelor of Studies in Philosophy

Introduction

The vision and mission of the Philosophy department at Forman Christian College (A Chartered University) is to find their fullest expression in the discipline of philosophy, one of the oldest and most enduring fields of human inquiry.

The wisdom of the sages is captured in the branches Metaphysics and Epistemology; the application of this sagacity is developed by the branches of Logic and Ethics. The wide diversity of theories and application is complemented by an in-depth study of specific thinkers from ancient medieval, modern and contemporary periods. The study of Philosophy thus is the most significant guarantor of critical and creative thinking, and an indispensable requirement of today.

The Department of Philosophy at FCC is relatively new – launched in 2010, but the rich tradition of learning and knowledge-seeking is adopted and adapted for contemporary times. It is part of the Faculty of Humanities.

Learning Objectives

- i. To provide students with methods and techniques to think critically and develop analytical skills.
- ii. To facilitate students in acquiring the knowledge and discursive thinking required to deal with basic human issues.

Program Learning Outcomes

- i. **Dialectical Thinking:**
Students will be able to demonstrate an understanding of the history of philosophical thought and assess alternatives in the solution of issues faced in human experience.
- ii. **Academic Integrity:**
Students will demonstrate and abide by research ethics and values in line with the Core Values of FCCU.
- iii. **Theory and Application:**
Students will be able to differentiate between levels of abstraction in the solving of real world problems.
- iv. **Research Methodology:**
Students will demonstrate research methodology and skills necessary for higher level philosophical enquiry.
- v. **Future Prospects:**
Students will be able to create, analyze, and distinguish valid argumentation from invalid argumentation to better understand opportunities for their professional and personal development.

Requirements for BS Philosophy Major

Core Courses	Elective Courses	Total Credits	Required MCGPA
7 (21 credits)	11 (33 credits)	54	2.0

Core Courses	Credits
PHIL 101	3
PHIL 201	3
PHIL 202	3
PHIL 221	3
PHIL 331	3
PHIL 498	3
PHIL 499	3
Core Courses Total	21
Elective Courses	Credits
Any 11 electives from the list of philosophy courses	33
Total	33

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for the Minor

Core Courses	Elective Courses	Total Credits	Required MCGPA
4 (12 credits)	2 (6 credits)	18	2.0

Core Courses	Credits
PHIL 101	3
PHIL 201	3
PHIL 202	3
PHIL 221	3
Core Courses Total	12
Elective Courses	Credits
Any 2 electives from the list of philosophy courses	6
Total	6

Course Descriptions

General Education Course

UNIV 200: Critical Thinking, Ethics, and AI (3 credits)

This course invites students to think clearly and ethically about one of the defining technologies of our time: artificial intelligence. Through close analysis of arguments, real-world case studies, and guided discussion, students will learn how to evaluate evidence, recognize bias, identify common logical fallacies, and construct persuasive, well-reasoned arguments of their own.

Drawing on major ethical frameworks, the course asks students to grapple with pressing questions surrounding AI – from academic integrity and authorship to privacy, fairness, and responsible decision-making. Emphasis is placed on practical applications: short writing assignments and case-based discussions challenge students to apply ethical reasoning to everyday university contexts and broader public life.

By the end of the course, students will understand the tools of critical thinking and know how to use them responsibly in an AI-shaped world.

Core Courses

PHIL 101: Introduction to Philosophy (3 credits)

Selection of the problems historically identified as philosophical along with the methods philosophers have used to solve these problems including justice and moral order, evaluation and justification of belief and human value and dignity, identifying the problems that have bothered critical thinkers, followed by selective philosophical solutions and their authors.

PHIL 201: Philosophy: Ancient through Medieval (3 credits)

Rise of critical thought in the pre-Socratic Greek world and its development through the issues related to deriving the morally right and individual significance by understanding the universe's structure and function,

classic Platonic and Aristotelian worldviews, evaluation of their Eastern and Western historical critics through the end of the medieval period.

PHIL 202: Philosophy: Modern to Contemporary (3 credits)

Critical thought as it develops from the 16th century CE to the present, including Eastern as well as Western thinkers, issues that captured attention because of the rise of the sciences, the development of naturalism, humanism and the challenges of 20th century social crises.

PHIL 221: Logic: How to Think Clearly (3 credits)

Examination of logic, including both Stoic contributions as well as the systematic organization of the rules of right thinking developed by Aristotle and expanded by Medieval and later thinkers, concern about the issues raised by J S Mill and others who systematized inductive logic.

PHIL 498: Internship (3 credits)

Prerequisites: (Intro to Philosophy; at least one course in Ethics or Critical Thinking)

This course provides supervised internship experience in professional settings as a core component of the philosophy major. Students engage in structured reflection on ethical, social, and conceptual issues encountered in the workplace, with emphasis on applying philosophical frameworks such as ethics, critical reasoning, and social analysis. The course includes weekly faculty-led seminars, reflective writing, and a final analytical paper connecting theory with practice. It is a mandatory experiential learning requirement aligned with HEC Undergraduate Education Policy 2023.

PHIL 499: Capstone Project (3 credits)

Prerequisites: The core courses listed in the catalog, that is, PHIL 101, PHIL 201, PHIL 202, and PHIL 221 and 18 credits of electives from the major as listed in the catalog.

Comprehensive study of contemporary issues in philosophy involving the development of a major research paper with the provision of a community internship.

Electives Courses

PHIL 231: Philosophy of Religion (3 credits)

Set of issues that has dominated modern and contemporary concerns about religious thought, problem of evil, meaningfulness of God talks, relevance of religion for moral and social justice etc.

PHIL 301/PLSC 301*: Ancient, Medieval, and Early Modern Political Philosophy (3 Credits)

A study of political thought from early Greece through the 17th Century. Analysis and evaluation of the issues and positions will be conducted using some original sources from philosophers including Plato, Aristotle, Hobbes, Machiavelli, Locke and Rousseau.

PHIL 303/PLSC 303*: Contemporary Political Philosophy (3 credits)

Debates concerning the nature of political rights and duties, and the justification of political theories such as Utilitarianism, Marxism, and Democracy. Of special interest will be the classic documents influencing contemporary discussion, such as Jean-Jacques Rousseau's *Discourse on Inequality*, J. S. Mill's *on Liberty*, Che Guevara, *Global Justice: Liberation and Socialism*, Gandhi, *All Men are Brothers*, John Rawls, *A Theory of Justice* (1971, Robert Nozick, *Anarchy, Society and Utopia* (1974).

PHIL304/HIST 301*: Philosophy of History (3 credits)

This course studies in-depth the ideas behind the study of history, the different schools of thought in history, and the underlying theories. Special attention will be given to the development of South Asian and Muslim philosophy of history.

PHIL 322: Symbolic Logic (3 credits)

Prerequisite: Phil 221 or permission from the instructor.

Logical formulation of the ideal language is the basis of modern computer language, physics, and linguistics, natural deduction using quantification, sentential calculus.

PHIL 325: Philosophy of Education (3 credits)

Prerequisite: Any Philosophy course or permission from Instructor

Key issues relating to the conveying and acquisition of knowledge, examining their resolution by those theories which have been used to direct classroom practices, evaluating these approaches in terms of contemporary methods and challenges.

PHIL 331: Theories of Ethics (3 credits)

Prerequisite: Any Philosophy course or permission from Instructor

Ethical issues of objectivity vs. subjectivity in moral judgment, relativity vs. universalizability of moral principles, the logical foundation of moral perspectives, the scope and limits of moral language etc. in light of contemporary theories designed to resolve them.

PHIL 332: Metaphysics (3 credits)

Prerequisite: Any Philosophy course or permission from Instructor

This course analyzes the nature and structure of reality with attention to issues such as physical and immaterial reality, time, the connection of mind and body, and the nature of the human person. Attention may also be given to the building of worldviews into language, the molding of perception by cultural structures, and the tools necessary to reveal how the regularities of human experience and worth are to be explained.

PHIL 341: Epistemology (3 credits)

Prerequisite: Any Philosophy course or permission from Instructor

This course provides a critical analysis of the foundations of knowledge. Classical theories will be examined but the emphasis will be on recent trends in substantiating belief and defining knowledge.

PHIL 342: Philosophy of Science (3 credits)

Prerequisite: Any Philosophy course or permission from Instructor

Rise of the scientific method from the perspective of the logic of scientific explanation, relevance of theory to experimental information, dependence of scientific explanation on a paradigm of how the world functions etc.

PHIL 401: Philosophical Investigations: The Ancient Period (3 credits); PHIL 402: Philosophical Investigations: The Ancient Period (3 credits); PHIL 403: Philosophical Investigations: The Ancient Period (3 credits)

Prerequisite: Any Philosophy course or permission from Instructor

The courses are rotated over a three-year sequence and allow the student to concentrate on careful study in the thought, context and impact of a specific ancient philosopher.

PHIL 411/CRST 452 *: Philosophical Investigations: The Medieval Period (3 credits); PHIL 412: Philosophical Investigations: The Medieval Period (3 credits); PHIL 413: Philosophical Investigations: The Medieval Period (3 credits)

Prerequisite: Any Philosophy course or permission from Instructor

The courses are rotated over a three-year sequence and allow the student to concentrate on careful study in the thought, context and impact of a specific medieval philosopher.

PHIL 421: Philosophical Investigations: The Modern and Contemporary Periods (3 credits); PHIL 422: Philosophical Investigations: The Modern and Contemporary Periods (3 credits); PHIL 423: Philosophical Investigations: The Modern and Contemporary Periods (3 credits)

Prerequisite: Any Philosophy course or permission from Instructor

The courses are rotated over a three-year sequence and allow the student to concentrate on careful study in the thought, context and impact of a specific modern or contemporary philosopher.

PHIL440/ENGL440*: Literature and Philosophy (3 credits)

Prerequisites: ENGL 201

The focus of this course is to trace the connections between two disciplines: Literature and Philosophy; students survey how writers make use of philosophical ideas, themes, vocabulary and language; study of how literary texts can focus on important philosophical issues; emphasis on both Philosophy in literature and Philosophy as Literature.

PHIL X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

*** Cross listed Courses:**

Courses with two designators (coding) are marked with a * to identify them as cross listed courses. Students must select the correct designator for their applicable program to be counted towards the degree.

Department of
PHYSICS

Bachelor of Science in Physics

Introduction

Physics has been taught at Forman Christian College (A Chartered University) since it was established in 1864. The Physics Department was instituted in 1907 by Prof. D. J. Fleming, who was its first Head. A long line of distinguished professors and prominent scientists have served at this department, including Nobel Laureate Dr. Arthur Compton, Prof. J. M. Benade and Dr. Piara Singh Gill. Dr. Compton conducted most of his research on cosmic rays while a faculty member at FCC, which led to his receiving the Nobel Prize for Physics in 1927. Prof. J. M. Benade was one of the longest serving professors in the department, eventually retiring as its Head in 1970. He was an active researcher and the Asian representative in Dr. Compton's international research team. Dr. Compton's student Dr. Piara Singh Gill was a faculty member at FCC from 1940 to 1947 and active in research. He was associated with the University of Chicago and the Georgia Institute of Technology. All these scientists and professors have made significant contributions to the field of Physics.

The Department of Physics offers a Bachelor of Science in Physics that concentrates on building its students' capacities as physicists through a variety of courses and learning activities. The department is part of the Faculty of Natural Sciences.

Vision

The department will be highly teaching and research-oriented to train our undergraduate students. It will be engaged in interdisciplinary activities of teaching and research with other departments at FCC, and other universities in Pakistan. It will have taken a lead in international research collaboration, and significant industrial partnership for mutual benefit and career opportunities for our students.

Mission

The mission of the BS Physics program is to produce competent physicists with excellent knowledge who are capable to apply practical skills in different areas of physics for employment and higher education through providing curriculum based on liberal arts education to ensure greater breadth of knowledge along with in-depth knowledge of physics offering an opportunity to learn and experience various aspects of physics under one roof.

Learning Objectives

- i. To impart fundamental knowledge and skills in physics.
- ii. To train the students to conduct and interpret scientific experiments.
- iii. To emphasize mathematical skills.
- iv. To use physics knowledge for betterment of the society.
- v. To produce competent future physicists.

Program Learning Outcomes

i. Core Knowledge

Demonstrate knowledge and skills in core physics concepts to analyze and apply a broad range of physical phenomena

ii. Analysis and Communication

Interpret, analyze and communicate the experimental data in the field of physics

iii. Proficiency in Mathematics

Demonstrate proficiency in mathematical concepts needed for a proper understanding of physics.

iv. Ethics and Values

Demonstrate knowledge of the ethical norms in the field of physics

v. Problem Solving

Apply physical and mathematical methods to a real-world problem in an application area

vi. Future

Identify opportunities for higher studies and career advancement in physics

Requirements for BS Physics Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
11 (38 credits)	16 credits	54	2.0

Core Courses	Credits
PHYS 103	4
PHYS 104	4
PHYS 221	4
PHYS 222	4
PHYS 301	3
PHYS 321	4
PHYS 334	3
PHYS 461/MATH 410	3
PHYS 462/MATH 411	3
PHYS 498	3
PHYS 499	3
Core courses Total	38
Elective Courses	Credits
Any 200 or above level courses offered by the department	3 / 4 each
Elective Courses Total	16

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Students are also recommended to take MATH 101, MATH 102, MATH 201, MATH 202 and MATH 203.

Requirements for the Minor in Physics

- A minor in Physics is open to students of all disciplines with a minimum CGPA of 2.50.

Core Courses	Elective Courses	Total Credits	Required Minor CGPA
2 (8 credits)	3(10 credits)	18	2.0

Core Courses	Credits
PHYS 103	4
PHYS 221	4
Core courses Total	8
Elective Courses	Credits
One course PHYS 102 or PHYS 104	4
At least two (3 credit hours courses) from courses of 200 or above	6
Elective Courses Total	10

Notes

The department offers these courses in accordance with available faculty and student load. Students should consult their faculty Advisor or the department Chairperson as to availability in a particular semester or academic year.

Students desirous of pursuing higher educational Physics degrees in Pakistan should ensure that their combined aggregate credits in Physics, Chemistry and Mathematics, including courses in the General Education Science and Mathematics category, should exceed 70 credits.

The courses PHYS 341 (Mathematical Methods of Physics), PHYS 461 (Quantum Mechanics I), and PHYS 462 (Quantum Mechanics II) are cross-listed with the Mathematics Department. Kindly see the Department of Mathematics pages for details.

Physics courses which appear in the Environmental Sciences section of the catalog likewise count towards the fulfillment of degree requirements in both programs.

Course Descriptions

PHYS 100: Introduction to Physics (3 credits)

Not for Students who have studied Physics at Intermediate or A-Levels or equivalent

This introductory course presents the basic principles of physics using elementary mathematics. It emphasizes conceptual understanding and practical applications that relate physics to everyday life, helping students develop scientific awareness and quantitative reasoning skills. The course covers fundamental physics concepts using algebra (calculus is helpful but not required). Concepts to be taken are scope of physics; communications; basic electricity; motion and laboratory. The laboratory component introduces students to

basic measuring instruments and simple experiments. It develops skills in accurate measurement, data recording, and interpretation of results.

PHYS 101: General Physics (3 credits)

Prerequisite: Intermediate with physics or A level Physics. This course will fulfill the requirement of the general course only. This course is not recommended for those who are planning to opt for a Bachelor of Science in Physics.

This course introduces non-science majors to the central ideas of physics through everyday phenomena and modern applications. Topics include measurement and uncertainty, motion and forces, energy and power, waves and optics, electricity and magnetism, electromagnetic waves and modern technologies, an introduction to nuclear physics and nuclear energy, and selected contemporary topics such as cosmology, spacetime, black holes, and an introductory view of relativity. The course includes laboratory experiences that reinforce core concepts through observation, measurement, and data analysis.

PHYS 102: Basic Electronics / CSCS 105 Basic Electronics * (4 credits)

Prerequisite: PHYS 100 or A-Level or Intermediate Physics

Note: The CSCS 105 course is open only to Computer Science majors.

Electricity, magnetism, DC and AC Current, modern Physics, laboratory.

PHYS 103: Mechanics (4 credits)

Prerequisite: PHYS 100 or Intermediate or A-Level Physics or equivalent

Study of physical phenomena in mathematical terms, statics and dynamics of particles and rigid bodies, oscillatory and rotary motion, gravitation and fluid mechanics, laboratory.

PHYS 104: Wave and Vibrations (4 credits)

Prerequisite: PHYS 103 or Intermediate or A-Level Physics or equivalent

Study of physical phenomena in mathematical terms, types of waves, mathematical representations, energy of waves, interference, diffraction and polarization, laboratory.

PHYS 151/ENVR 151 *: Introduction to Sources of Energy and Environment (3 credits)

Not recommended for first semester Freshmen

Conventional energy resources, fossil fuels including petroleum, natural gas, coal and tar sands, the promise and problems of nuclear energy, alternative energy sources, wind, solar, biogas, tidal etc., energy conservation, environmental pollution and its global effects.

PHYS 192/SOCL 192 *: Science and the World Around Us (3 credits)

The course is open to undergraduates in all departments.

This course bridges the divide between science and non-science students by introducing major scientific concepts using simple language and creates an understanding of important contemporary issues that simultaneously involve science, society, and politics. Main topics: the nature and history of science; great achievements of science; science and politics in the modern world; the challenge of climate change; science and war.

PHYS 221 *: Electricity and Magnetism (4 credits)

Prerequisite: PHYS 103 or PHYS 104

Electrostatics, magneto-statics, electric current, laws of magnetism, Maxwell's equations, electromagnetic energy and electromagnetic wave equations, laboratory.

PHYS 222: Modern Physics (4 credits)

Prerequisite: PHYS 221 or PHYS 103

Study of Einstein's special theory of relativity, black body radiation, the Bohr atom, elementary wave mechanics, atomic and molecular spectra, exclusion principle, periodic table, X-ray spectroscopy, introduction to lasers, laboratory.

Physics 234: Heat and Thermodynamics (3 credits)

Prerequisite: PHYS 103

Heat and Thermodynamics is one of the key undergraduate physics courses. It connects the world of everyday systems, of astronomical objects, and of chemical and biological processes with the world of molecular, atomic, and electronics systems. The course will be introduced through a unified approach to the equilibrium thermal properties of large systems based on the quantum viewpoint and statistical probability. The laws of thermodynamics and the concepts of entropy, temperature, chemical potential, free energy, and thermodynamic potential will be covered. The heat transfer, phase transition, and classical kinetic theory will be discussed.

PHYS 255/ENVR 255 *: Introduction to Meteorology (3 credits)

Prerequisite: PHYS 102 or PHYS 103 or PHYS 151

Study of the physical processes of condensation, precipitation, radiation and radiative transfer, solar radiation, atmospheric motion measuring properties of the atmosphere, ionosphere and magnetosphere, Earth's magnetic field and charge density movement in the atmosphere.

PHYS 300: Fundamental of Optics (4 credits)

Prerequisite: PHYS 104, and PHYS 221

One Dimensional Waves, Harmonic Waves, Phase and Phase Velocity, Group Velocity, laws of electromagnetic theory, Electromagnetic Waves, Poynting Vector, Rayleigh scattering, Reflection, Refraction, Fermat's Principle, Total Internal Reflection, Lenses, Aperture and field stop, Mirror, Aberration, Astigmatism, Polarizers, Dichroism, Birefringence, Application of the Fresnel equation, Liquid Crystals, Interference, Young's experiment, Michelson's Interferometer, Fabry-Perot Interferometer, Applications of Interferometry, Diffraction, Huygens-Fresnel Principle, Fraunhofer and Fresnel diffraction, The Double Slit, Grating spectroscopy, Holography, laboratory.

PHYS 301: Classical Mechanics (3 credits)

Prerequisite: PHYS 103 or MATH 302

Study of the motion of particles and system of particles, direct application of Newtonian mechanics, Lagrangian formulation, Hamiltonian formulation, motion under an inverse force field, two body problems, planetary orbital motion, Legendre transformation, canonical transformations and their properties, Poisson's brackets, theorems and invariance.

PHYS 321: Electrodynamics (4 credits)

Prerequisite: PHYS 221 and MATH 203

Emphasis on the unity of electric and magnetic phenomena, introduction of electrostatics and magneto-statics, solution of boundary-value problems, time-varying fields, gauge transformations, Maxwell's equations and wave equations, electromagnetic wave propagation in lossless, Lossy and metallic media, wave propagation through coaxial transmission lines, rectangular waveguides and radiation from oscillating dipoles, laboratory.

PHYS 331: Electronics I (4 credits)

Prerequisite: PHYS 221 or PHYS 102

Circuit analysis, Characteristics and applications of semiconductor devices and circuits, power supplies, special diodes, Bipolar junction transistors (BJTs) with biasing analysis and frequency response, BJT models, Field effect transistors (FETs), signal analysis of BJTs and FETs, Power amplifiers, Feedback concepts and types, Oscillator circuits and applications, Multivibrators, Laboratory.

PHYS 332: Electronics II (4 credits)

Prerequisite: PHYS 331

Models for active devices, single-ended and differential amplifiers, current sources and active loads, operational amplifiers, feedback, design of analogue circuits for particular functions and specifications, design of decision-making circuits, memory type circuits and digital circuits, laboratory.

Physics 334: Statistical Physics (3 credits)

Prerequisite: PHYS 234

Statistical physics is one of the key advanced undergraduate physics courses. The course will be introduced through a unified approach to the equilibrium thermal properties of large systems based on the quantum viewpoint and statistical probability. The heat transfer, phase transition, and classical kinetic theory will be discussed. Statistical physics is one of the key advanced undergraduate physics courses. It connects the world of everyday systems, of astronomical objects, and of chemical and biological processes with the world of molecular, atomic, and electronics systems. The course will be introduced through a unified approach to the equilibrium thermal properties of large systems based on the quantum viewpoint and statistical probability. The heat transfer, phase transition, and classical kinetic theory will be discussed.

PHYS 341/MATH 316 *: Methods of Mathematical Physics (3 credits)

Prerequisite: PHYS 221 or MATH 203

Vector analysis and special function curvilinear coordinates, Legendre polynomials, Bessel functions, Neumann functions, Cauchy-Riemann equations, Fourier series and Fourier transformations, Tensor analysis.

PHYS 342: Computational Physics (3 credits)

Prerequisite: PHYS 221

Introduction to scientific computing, Basic numeric tools to solve physics problems, Ordinary differential equations, Partial differential equations, Finite difference time domain method (FDTD), Monte Carlo technique.

PHYS 351 *: Environmental Physics (3 credits)

Prerequisite: PHYS 103 or PHYS 151

Introduction to environmental physics, radiation, radiation balance, heat and mass transfer, micrometeorology of crops.

PHYS 352: Plasma Physics (3 credits)

Prerequisite: PHYS 221

Plasma state; Criteria for the plasma state; Debye shielding, Gas discharge; Space physics; Controlled fusion; Modern astrophysics; MHD energy conversion; Ion/Plasma propulsion; Solid-State plasma, single particle motion in electric and magnetic fields (uniform, non- uniform, time-independent and time varying fields); Plasmas as magnetic mirror, continuity equation, Poisson's equation, Waves and wave propagations in plasmas; Phase velocity; Group Velocity; Dispersion relations.

PHYS 353: Introduction to Nonlinear Methods (3 credits)

Prerequisites: PHYS 341

Linear Differential equations; Wave amplitude dependent phase velocity and Non-Linear effect; Multiple Scale Analysis; Introduction to Chaos - One dimensional model; Discovery of the solitary waves; Introduction to Korteweg deVries equation, solitary limit; Relation between amplitude, speed and width; Sagdeev Potential; Conservation Laws; Wave equation and Dispersion Relation; Dispersive Waves, Phase velocity and group velocity; Reductive Perturbation Method; Solitary waves, Dispersion and Nonlinearity; Nonlinear Schrodinger equation; Evolution equation for envelope function.

PHYS 422: Nuclear Physics (3 credits)

Prerequisite: PHYS 222

Structure and properties of the Nucleus, Nuclear forces, nuclear phenomenology, reaction and stability of nuclear models, radiation and decay, Alpha, Beta and Gamma Decay, Nuclear Reactions and Detection of radiation.

PHYS 451 *: Sources of Energy (3 credits)

Prerequisite: PHYS 222

Study of the different alternative sources of energy including hydroelectric, wind, solar, photovoltaics, nuclear and thermonuclear (fission and fusion). New energy technologies like fuel cells and the role of hydrogen, the Physics of these sources and their environmental impact.

PHYS 460: Introduction to Medical Physics (3 Credits)

Prerequisite: PHYS 222 or PHYS 422 or Equivalent

Medical Physics is a course concerned with the application of physics in medicine, especially (but not exclusively) in radiation medicine; i.e., radiation therapy, medical imaging, and nuclear medicine. The Course aims at offering quality education in the applications of Physics and Technology in Medicine to Physicists and potential scientists of similar scientific disciplines, enabling them to participate in promoting health and research. Topics covered will include imaging metrics, ionizing radiation and radiation safety, radioactivity, radiation therapy, computed tomography, nuclear medicine, ultrasound, and magnetic resonance imaging, etc.

PHYS 461/MATH 410 *: Quantum Mechanics I (3 credits)

Prerequisite: PHYS 301 or MATH 302 and PHYS 341

Historical origination of the quantum theory, foundation of wave mechanics, Schrodinger wave equation and its solution for free particles, the hydrogen atom and the harmonic oscillator.

PHYS 462/MATH 411 *: Quantum Mechanics II (3 credits)

Prerequisite: PHYS 461

Matrix mechanics, vector spaces and linear operators, Time dependent and time independent perturbation theories, WKB approximation, Identical Particles, Scattering, Dirac equation, application of principles of quantum mechanics to solid state and nuclear systems.

PHYS 472: Lasers (3 credits)

Prerequisite: PHYS 321 or PHYS 341

Study of the concepts of laser, spontaneous and stimulated emissions, absorption, pumping process, properties of laser beams, laser resonators, matrix formulation of geometric optics, stable and unstable laser resonators, modes in a laser cavity, loop gain Q-switching, energy levels of molecules.

PHYS 481: Solid State Physics I (3 credits)

Prerequisite: PHYS 341

Study of solids, crystal structure, direct and reciprocal lattices, types of bonding, lattice vibrations, the thermal, electrical properties of solids and the effects of crystals.

PHYS 482: Solid State Physics II (3 credits)

Prerequisite: PHYS 481

Free electrons, Fermi gas, nearly free electrons, energy bands, optical transitions, superconductors and magnetic properties.

PHYS 483: Materials Science (3 credits)

Prerequisite: PHYS 221

Study of the properties of material, internal structure of materials, performance of materials during manufacture, production and processing, performance of materials during service, crystal structures, crystal geometry, solidification, crystalline imperfections, diffusion in solids, thermodynamics and phase diagrams and electrical materials.

PHYS 498: Internship (3 credits)

Students will have to work in a known industry/organization or the university/institute for six to eight weeks and will observe the timings as prescribed by the host organization. Director Internships will act as a liaison officer between the department and the industry/organization, university/institute. The student will have a supervisor from the department as well from the host organization. At the end of the completion of the training students will submit a written report to both the supervisors and will be evaluated by the departmental committee.

PHYS 499: Capstone Project (3 credits)

Each student works on an independent project under the supervision of a faculty member, with the expectations that the student will prepare a senior thesis and will present a seminar on his/her work.

PHYS X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

*** Cross listed Courses:**

Courses with two designators (coding) are marked with an (*) to identify them as cross listed courses. Students must select the correct designator for their applicable program.

Department of
**POLITICAL
SCIENCE**

Introduction

The Department of Political Science at FCCU is one of the largest departments in Social Sciences. The courses offered by the department are open to students majoring in Political Science as well as other disciplines. The department's teaching faculty had the distinction of having scholars of national and international prominence like Dr Carl W Wheelless, Prof Mary Wheelless, Dr Kitchen, Dr Ryan Brasher, Dr Anwar M Barkat, Dr Shokat Ali, Dr Hamid Kizilbash, Prof Naseem Zakariya, Dr Parveen Shaukat and Dr Shafqat Hussain Chaudhary.

The quality of instructional work in the department is indeed of a high standard. Research work is encouraged. There is a dedicated faculty available to teach various papers and supervise research. The department arranges extensive lectures, seminars, and study tours to facilitate academic excellence in students. Alumni of this department have contributed positively to national uplift and have excelled in numerous professions.

The department offers a Bachelor of Science degree in Political Science and is part of the Faculty of Social Sciences.

Bachelor of Science in Political Science

The 4-year degree in Political Science encourages students to take courses in the fields of International Relations, Politics of Pakistan, and Political Philosophy. Students are encouraged to develop a critical understanding of various phenomena of national and international politics and their abilities to critically evaluate these concepts are nurtured through active classroom discussions.

4 STREAMS in Bachelor of Science Program (core courses required for all streams bolded)

Levels	Political philosophy	International Relations	Pakistan politics	Comparative Politics
100	PLSC 101 (Introduction to Political Science)		PLSC 102 (Pakistan National Government)	
200		PLSC 203 (International Relations)	PLSC 204: Pakistan Government - Provincial/ Local	PLSC 212 Comparative Politics PLSC 201 (Government of Western Europe and United states) PLSC 202 (Gov. of Developing Countries)
300	PLSC	304 (Research Methodology)	Pre-req: STAT 100/101/103; PLSC 336 (Public Administration)	ECON 203 PLSC 340 (Politics of Afghanistan and Iran) PLSC 341 (Politics of China) PLSC 342 (Politics of India)

400	PLSC 402 (Islam and Modernity) PLSC 413 (Critical Theory and PostColonial Situation)	PLSC 400 (Current Political Problems) PLSC 401 (International Political Economy) PLSC 412 (Foreign Policy Analysis)	PLSC 400 (Current Political Problems)	PLSC 400 (Current Political Problems) PLSC 412 (Foreign Policy Analysis)
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Learning Objectives

- Demonstrate in-depth knowledge of the major concepts of politics.
- Think critically and analytically with a view to developing the habit of lifelong learning.
- Learn appropriate skills for careers in foreign and domestic service, politics, law school, teaching, research, and graduate study in Political Science.
- Function as active citizens in keeping with high ethical standards.
- Write analytically on an issue and present it effectively to an audience.

Program Learning Outcomes

- In-Depth Knowledge of Major concepts of Politics**
Demonstrate sound theoretical and practical understanding of the political concepts and epistemological debates at the forefront of the academic discipline of Comparative Politics and International Relations.
- Real World Application**
Applying the learned knowledge on contemporary issues.
- Professional Skills**
Develop competence in a range of transferable skills, including critical and analytical communication, and independent study skills.
- Written and Critical Evaluation skills**
Analyze and critically evaluate the work of other social scientists using appropriate data analysis techniques.
- Written and Critical Analytical Skills**
Apply the techniques learnt in their own research to predict and propose solutions to long-standing issues.

Requirements for BS Political Science Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
10 (30 credits)	8 (24 credits)	54	2.0

Core Courses	Credits
PLSC 101	3
PLSC 102	3

PLSC 203	3
PLSC 212	3
PLSC 301	3
PLSC 302	3
PLSC 304	3
PLSC 321	3
PLSC 498	3
PLSC 499	3
Core courses Total	30
Elective Courses	Credits
Any course offered by the department	24
Electives courses Total	24

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for the Minor in Political Science

A minor in Political Science, which is open to students from any discipline with a minimum CGPA of 2.50.

Total Credits	Required Major CGPA
18	2.0

3 courses from (100-200) level must be taken, which includes PLSC 101.

(Note: Core course required for minor in Political Science is PLSC 101). 3 courses from (300-400) level must be taken for a minor in Political Science. (Note: Any course prerequisites must be taken into consideration.)

Course Descriptions

PLSC 101: Introduction to Political Science (3 credits)

Areas covered in political science include the nature of political science, the nature and forms of the state, structure of government, political dynamics, and the development of an appropriate political science vocabulary.

PLSC 102: Pakistan Government-National (3 credits)

A history of the freedom movement and study of the main institutions of the national government and what makes the Pakistan government unique.

PLSC 201: Government of Western Europe and the United States (3 credits)

Prerequisite: PLSC 101

Parliamentary, presidential, unitary, and federal systems of major western nations.

PSC 202: Governments of Developing Countries (3 credits)

Prerequisite: PLSC 101

Unique characteristics of governments in Asia and Africa, historical development and a comparison between these nations and the rest of the world.

PLSC 203: International Relations (3 credits)

Prerequisite: PLSC 101

Theory and practice of International Relations using the distinction between realism and idealism as the basis for study, power relationships, theories of war and conflict, international morality, collective security, and terrorism.

PLSC 204: Pakistan Government-Provincial/Local (3 credits)

Government at the provincial and local level with an examination of the basic institutional arrangements of the provincial government, provincial elections, political party organization, state public policy matters and a detailed exploration of the operation of government at these levels.

PLSC 212: Theories of Comparative Politics (3 credits)

Prerequisite: PLSC 101

Political forces, institutions, and practices of state, describing, explaining, and predicting political events, importance of geographical, racial, ideological, ethnic and socioeconomic explanation of political institutions, processes and behavior, political structure, institutions, ideologies, interest groups and governmental systems, analysis of decision-making processes, political conflict and change and group interaction.

PLSC 301/ PHIL 301 *: Ancient, Medieval and Early Modern Political Theory (3 credits)

Prerequisite: PLSC 101

Political thought from early Greece through the 17th Century using original sources from philosophers including Aristotle, Plato, Hobbes, Machiavelli, Locke, Rousseau and Hegel.

PLSC 302: Modern Political Theory (3 credits)

Prerequisite: PLSC 101

Modern ideologies since the French Revolution, including liberalism, conservatism, capitalism, nationalism, fascism, and anarchism.

PLSC 303: Contemporary Political Theory (3 credits)

Prerequisite: PLSC 101 & 302

Status of rights, utilitarianism, liberalism, communitarian Marxist, libertarian and feminism using John Stuart Mill's Utilitarianism on Liberty, Essay of Bentham, Milton Friedman's Capitalism and Freedom.

PLSC 304: Research Methodology (3 credits)

Prerequisite: STAT 100 or STAT 101 or STAT 103 or ECON 203

Techniques and tools for significant research in the field of political science.

PLSC 305: Islamic Political Thought (3 credits)

Prerequisite: PLSC 101

Development of Islamic political thought from ancient times to the present, Muslim thinkers Al-Farabi, Al-Mawardi, Al-Ghazzali, Ibn Khaldun, Shah Waliullah and Allama Muhammad Iqbal.

PLSC 310/ HIST 312 *: Politics of the Middle East (3 credits)

Prerequisites: PLSC 101

Political development and advance of modernization of the area, the role of Islam, Arab- Israeli conflict, politics of Persian Gulf, politics of OPEC, political parties, military, and politics of change.

PLSC 311: Politics of Developing Areas (3 credits)

Prerequisite: PLSC 101

Examination of the development of political awareness and economic growth, assessing explanations for the failure of development of some countries and the strategies used to escape the poverty of underdevelopment. It will examine the interrelationships between 'first' and 'third' worlds, critique the major developmental theories; understand the underpinnings of development strategies, examine the complex nature of some of the major challenges facing the developing countries.

PLSC 317: Political Dynamics: Parties and Processes (3 credits)

Prerequisites: PLSC 101, PLSC 102

Two party and multi-party systems include a discussion of what parties are, history of political parties, parties and elections, parties in a federal system and parties around the world.

PLSC 321: Pakistan Foreign Policy (3 credits)

Prerequisite: PLSC 101

Status and relationships between Pakistan and the rest of the world with special emphasis upon relations with the Islamic World and the United States.

PLSC 322: International law (3 credits)

Prerequisites: PLSC 101, PLSC 203

Historical evolution of international law, coverage of classifications of states, rights and duties of jurisdiction, theories of nationalism, diplomatic relations, operation and enforcement of treaties, redress of differences by war and other methods and neutrality.

PLSC 323: International Organization (3 credits)

Prerequisites: PLSC 101, PLSC 203

Background of the United Nations Organization with an analysis of the success and failure of the League of Nations, the Security Council, the General Assembly and the organs of the United Nations and Pakistan's position on the issues.

PLSC 330: Constitutional Law in Pakistan (3 credits)

Prerequisites: PLSC 101, PLSC 102

History, formation, and implementation of each of the constitutions of Pakistan and the interpretation of test cases before the Supreme Court of Pakistan.

PLSC 331: Constitutional Law -- United States (3 credits)

Development of American federalism and national power, civil rights and civil liberties, commerce clause and nationalization of the economy, various amendments introduced in the US constitution, role of Supreme Court in American government, the controversy over the interpretation of different approaches to constitutional interpretations.

PLSC 335: Public Opinion (3 credits)

Prerequisite: PLSC 101

General nature of public opinion and its development and application to Pakistan, modern techniques of measurement.

PLSC 336: Public Administration (3 credits)

Prerequisites: PLSC 101, Junior Status

Art of administration, organizational aspects, management agencies, unity under the chief executive, departmental organization, federal-provincial and headquarters field- relationships, line functions, fiscal management, budget strategy and tactics, and government career service.

PLSC 337: Gender, War, and Security in the 21st Century (3 credits)

Prerequisites: PLSC 101

This course explores the intersections of gender, war, and security in contemporary global politics. It critically examines how gender shapes and is shaped by armed conflict, military institutions, and security policies. Through theoretical perspectives and case studies, students will analyze the roles of women and men in war, the impact of conflict on gender relations, and the contributions of feminist security studies to understanding modern security challenges.

PLSC 338: Political Marketing (3 credits)

Prerequisites: PLSC 101

This course explores the application of marketing principles to political contexts, focusing on how political actors engage citizens, build brand identity, and influence public opinion. It covers campaign strategies, media influence, voter segmentation, and digital political communication. The course features comparative case studies from Pakistan as well as a diverse range of other countries. It is suitable for students in political science, business, or liberal arts interested in campaign management, political strategy, and public communication.

PLSC 339: Gender and Politics (3 credits)

Prerequisites: PLSC 101

This course aims to introduce students to the relationship between gender and politics with conceptual and analytical tools to analyze patriarchal structures and unequal power relations that restricts human potential. It will adopt an intersectional approach toward understanding the interplay of a multiplicity of oppressions that women and men are confronted with, the course will analyze the marginalized identities stemming from social divisions based on class, caste, and religious differences.

PLSC 340: Politics of Afghanistan and Iran (3 credits)

Prerequisites: PLSC 101

Our two western neighbors are constantly in the news, particularly regarding state-to-state relations with Pakistan. But how much do we really know about the internal politics of Iran and Afghanistan? In this course, we will look at the historical development of these countries and their emergence onto the world stage in the 20th Century as non-colonized societies whose boundaries were nonetheless drawn by colonial powers. We will look at the impact of the Cold War on these countries to explain their revolutionary moments in 1978 and 1979. We will look at the impact of the Islamic revolution on Iranian political life, as well as the decades of instability in Afghanistan that followed the Soviet invasion of 1979. Lastly, we will look at the effects of 9/11 on Afghan and Iranian politics, with particular emphasis on efforts at state building in Afghanistan. The course will not simply consist of a discussion of current events. Rather each session will build on general and comparative political science theory to help illuminate events in these countries.

PLSC 341: Politics of China (3 credits)

Prerequisites: PLSC 101

From the ruins of the Cultural Revolution to Deng Xiaoping's economic reforms and market liberalization, China re-emerged as the second largest economy of the world in terms of Gross Domestic Products (GDP), and largest economy of the world in terms of Purchasing Power Parity (PPP). On its domestic front, the communist regime has performed well in curbing the growing poverty rate and lifting millions of Chinese out of extreme poverty. On the international front China emerged as one of the largest aid donor countries to the global south and has become an important actor in global politics. It is asserting its influence and trying to present herself as a model for the developing world. This course intends to explore the political history of modern China, the role of ideology particularly Marxism, Leninism, and Maoism in shaping the identity of modern China. The conception of state, the notion "one country, two systems" the political and historical background of 'Cultural Revolution', the politics of communist regime with Hong Kong, Taiwan, and Tibet, the ideas of nationalism and the issues of modern identity in China, various reforms in political, social, and economic systems, an introduction to Chinese law, and China's view on "rule of law".

PLSC 342: Politics of India (3 credits)

Prerequisites: PLSC 101

India is not only a rising global power but is also an exceptional case of a successful democracy in the Third World. In this course, we will analyze the trajectory of Indian politics since 1947. Themes for the course include electoral politics, communalism, and rising authoritarianism in India. We will also discuss resistance movements in India, including the politics of peasants, Dalits, Kashmiris, and women in the country. Finally, we will engage with theoretical work on India and see how the trajectory and practice of Indian democracy differ from hegemonic models in the West. By the end of the course, we hope that students will know some of the salient events in postcolonial India and will be familiar with contemporary debates in the country. It is expected that students will read the texts assigned for each class. We will examine the texts on Monday and Wednesday. For our Friday classes, we will discuss a contemporary issue and see how it relates to the themes discussed in the texts for the week.

PLSC 400: Current Political Problems (3 credits)

Prerequisites: PLSC 101, Junior Status

Topical issues and themes of justice, equality and liberty, women's rights, race relations, child labor, birth control and other topics as chosen by the Professor and members of the class.

PLSC 401: International Political Economy (3 credits)

Prerequisites: PLSC 101, ECON 100

Phenomena that are both political and economic in nature, substantive issues, methodological and conceptual framework: rational choice theory.

PLSC 402: Islam and Modernity (3 credits)

Prerequisite: PLSC 101

Interrelation of Islam and Modernity by deploying a multi-disciplinary perspective and addressing challenging questions about the nature of Islam's mission in the world, major accounts of modernity's genesis in the West and concomitant decline of religion, historical terrain of modern West's forays into Muslim societies in the era of imperialism and colonialism.

PLSC 405: Governance Challenges of Pakistan (3 credits)

Prerequisite: PLSC 101 and PLSC 102

This course investigates the structural, institutional, and policy-related governance issues in Pakistan. It tracks the historical evolution of governance including political institutions, political instability, corruption, and civil-military bureaucracy relations. It also covers the socio-economic disparities and their impact on governance.

PLSC 407: Poverty and Governance in South Asia (3 credits)

Prerequisite: PLSC 101

This course explores the complex relationship between poverty and governance in South Asia. It examines the role of socio-political institutions, economic policies, and governance structures in addressing or perpetuating poverty. The course also covers governance challenges such as corruption, weak institutions, policy failures, and exploring poverty alleviation strategies, social welfare programs, and international aid interventions.

PLSC 411: War Studies: Theory and Practice (3 credits)

Prerequisite: PLSC 101

This course offers a critical examination of war as a political, strategic, and socio-cultural phenomenon. It engages with classical and contemporary theories of war and the evolution of military doctrines, while analyzing key forms of conflict including insurgencies, terrorism, cyber and hybrid warfare. Ethical dimensions such as just war theory and rules of engagement are explored alongside the role of media in shaping the legitimacy and perception of warfare. Bridging scholarly frameworks with real-world case studies, the course provides a comprehensive understanding of war in theory and practice.

PLSC 412: Foreign Policy Analysis (3 credits)

Prerequisite: PLSC 203

Patterns and processes involved in the formulation of a country's foreign policy, deciphering the apparent 'black box' of state, and highlighting the actors, processes and organizations and motives that help to shape a country's foreign policy.

PLSC 413: Critical Theory and Post-Colonial Situation (3 credits)

Prerequisite: PLSC 302 or PLSC 303

Emerging trends in the contemporary theoretical field of post-colonial studies which aims to conceptualize the interrelation of culture, power and knowledge in post-colonial societies, examination of works of Aijaz Ahmed, Ashis Nandy, Partha Chatterjee, Gayatri Spivak, Homi Bhabha, Gyan Parkash, Arif Dirlik, Kwame Appiah and others to raise fundamental questions about the scope, ambitions, and epistemological transgression of post-colonial theory.

PLSC X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

PLSC 498: Internship (3 credits)

Open to Political Science majors with a grade point average of 3.00 or above

The internship course provides students with a structured professional learning experience that connects academic knowledge with practical engagement in the field and sub fields of Political Science. Through supervised placements in relevant institutions students gain hands-on experience and an opportunity to apply their theoretical knowledge acquired in the classroom to real world context

PLSC 499: Capstone Project (3 credits)

Prerequisite: PLSC 304

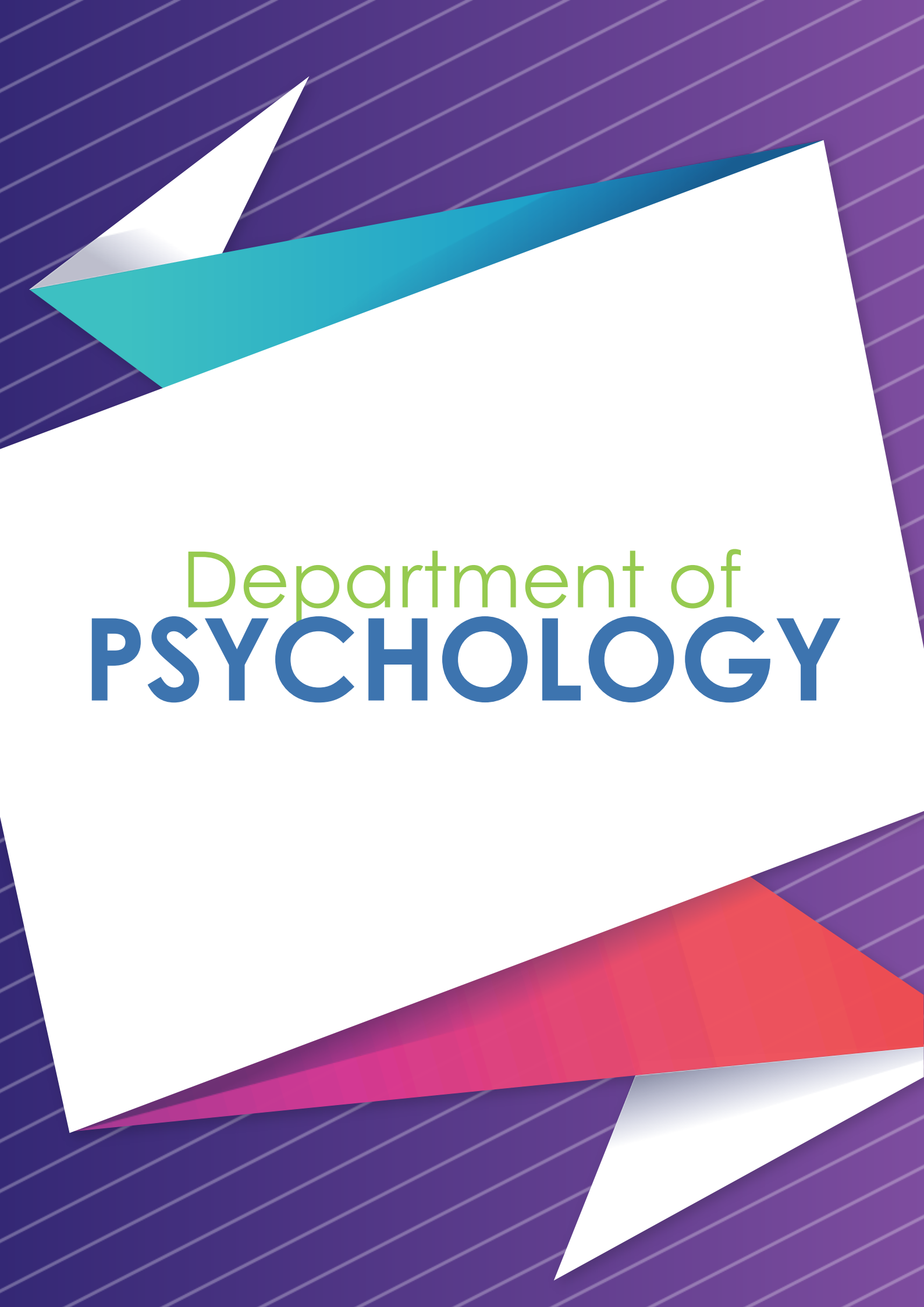
A capstone project in Political Science allows students to bring together the concepts, principles and methods that they have learned in their course of study and to apply their knowledge and acquired competencies to address the real-world political problems.

Students are required to produce an original analytical research paper of approximately 3,500–5,000 words on a significant theme within Political Science. The topic must be developed in consultation with the instructor and must address a clearly defined research problem grounded in existing scholarly debates.

Students are expected to engage with primary data (qualitative and/or quantitative) and demonstrate the ability to apply appropriate research methods to real-world political issues.

*** Cross listed Courses:**

Courses with two designators (coding) are marked with a * to identify them as cross-listed courses. Students must select the correct designator for their applicable program to be counted towards the degree



Department of **PSYCHOLOGY**

Introduction

The undergraduate program in Psychology introduces students to an understanding of the basic core of psychological knowledge, theory, methods of research and opportunities for hands-on practice of psychology. It is an interesting and challenging area of scientific enquiry with the potential to benefit both individuals and society. The Department of Psychology offers a broad range of undergraduate courses in the major areas of Psychology.

Learning Objectives

- i. Develop an understanding of major psychological theories, concepts, and best practices.
- ii. Demonstrate the ability to apply psychological concepts through critical and creative thinking in theoretical and research contexts.
- iii. Demonstrate meaningful connections between psychological principles, theories, and practices.
- iv. Develop self-awareness and social understanding by critically examining human behavior, relationships, and cultural contexts through psychological frameworks.
- v. Broaden new interests, reflect values, and inculcate ethical principles in real-world settings.
- vi. Develop self-directed learning skills and effectively choose the field of psychology to pursue higher education and career pathways.

Mission BS Psychology Program

The mission of the BS Psychology program is to equip students with the knowledge of fundamental concepts of Psychology, encourage critical thinking, and promote a sense of professional ethics to prepare them for higher education and employment.

Program Learning Outcomes

- i. **Foundational Knowledge**
Differentiate and explain major psychological theories, concepts, and evidence-based practices across diverse fields of psychology.
- ii. **Application**
Critically analyze and apply psychological constructs, principles, and theories to address individual, social, and community problems.
- iii. **Integration**
Integrate and analyze psychological constructs and theoretical frameworks to interpret human behavior and mental processes across diverse contexts.
- iv. **Human Dimension**
Demonstrate awareness of feelings, thoughts, and behaviors by interpreting them through various psychological perspectives and cultural contexts
- v. **Caring**
Exhibit ethical sensitivity, empathy, and responsibility by engaging in effective psychological practices that promote individual and community well-being.
- vi. **Learning How to Learn**
Develop and implement strategies for continuous learning, self-assessment, and professional growth in psychological research and practice.

Mission BS Applied Psychology Program

The mission of the BS Applied Psychology program is to provide students with a specialized and comprehensive education that focuses on experiential learning, critical thinking, and ethical decision-making. By equipping them with culturally sensitive knowledge and skills, the program aims to address real-world issues and promote the well-being of individuals and the community.

Program Learning Outcomes

- i. Foundational Knowledge**
Differentiate and explain key psychological principles, theories, and empirical findings across various fields and specializations of applied psychology through written and oral assessments.
- ii. Application**
Apply psychological knowledge through case-based analysis, field projects, research activities, and simulations to address culturally and contextually relevant real-world problems.
- iii. Integration**
Synthesize and evaluate psychological principles with local and global trends, linking theoretical knowledge to interdisciplinary and personal contexts.
- iv. Human Dimension**
Identify and assess emotional processes, cognitive biases, and cultural dynamics to enhance self-awareness, empathy, and effective interpersonal engagement.
- v. Caring**
Demonstrate ethical sensitivity, compassion, and inclusivity by engaging in psychological practices that foster respect and positive interpersonal relations within diverse cultural contexts.
- vi. Learning How to Learn**
Develop and implement strategies for self-directed learning, reflective practice, and continuous professional development through training and supervision in applied psychology contexts.

Learning Objectives

- i.** Identify psychological principles, key theories, and concepts in core areas of applied psychology.
- ii.** Demonstrate psychological knowledge and skills through experiential learning, critical analysis, and culturally sensitive problem-solving abilities across diverse real-world contexts.
- iii.** Integrate academic learning with lived experiences while developing meaningful connections between theory and practice.
- iv.** Enhance self-awareness and interpersonal effectiveness by applying psychological principles to promote well-being, cultural sensitivity, and constructive engagement with individuals and communities.
- v.** Cultivate a sense of responsibility, empathy, and professional interest towards individual and community well-being.
- vi.** Develop independent, responsible, and reflective learners, capable of pursuing advanced studies and professional practice.

Bachelor of Science in Psychology

Students majoring in Psychology have two options. They can study to earn a Bachelor of Science in Psychology or Bachelor of Science in Applied Psychology degree. It is recommended that students choose their courses logically and sequentially, i.e. take 300 and 400 level courses during junior and senior year.

Requirements for BS Psychology Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
12 (38 credits)	16 credits	54	2.0

Core Courses	Credits
PSYC 100	3
PSYC 150	3
PSYC 220	4
PSYC 280	3
PSYC 298	3
PSYC 305	3
PSYC 340	3
PSYC 350	3
PSYC 360	4
PSYC 415	3
PSYC 490	3
PSYC 498	3
Core courses Total	38
Elective Courses	Credits
The remaining courses can be selected from the list of courses offered by the department.	16
Electives courses Total	16

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for the Minor in Psychology

Open to students (of all disciplines) with a minimum CGPA of 2.00.

Core Courses	Elective Courses	Total Credits	Required Major CGPA
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2 (6 credits)	12 credits	18	2.0
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Core Courses	Credits
PSYC 100	3
PSYC 150	3
Core courses Total	6
Elective Courses	Credits
The remaining courses can be selected from the list of courses offered by the department.	18
Electives courses Total	18

Requirements for BS Applied Psychology Major

BS Applied Psychology offers five specializations:

- Mental Health
- Industrial and Organizational
- Leadership and Management
- Artificial Intelligence
- Forensic Psychology

Specialization Core Courses	Elective Courses	Total Credits	Required Major CGPA
14 (44 - 46 credits)	14 - 16 credits	60	2.0

Mental Health	
Specialization Courses	Credits
PSYC 100	3
PSYC 150	3
PSYC 220	4
PSYC 305	3
PSYC 340	3
PSYC 350	3
PSYC 360	4
PSYC 415	3
PSYC 430	3
PSYC 440	3
PSYC 465	4
PSYC 467	3

PSYC 490	3
PSYC 498	3
Specialization Courses Total	45

Leadership and Management	
Specialization Courses	Credits
PSYC 100	3
PSYC 140	3
PSYC 160	3
PSYC 220	4
PSYC 230	3
PSYC 245	3
PSYC 275	3
PSYC 280	3
PSYC 305	3
PSYC 350	3
PSYC 360	4
PSYC 415	3
PSYC 490	3
PSYC 498	3
Specialization Courses Total	44

Industrial and Organizational	
Specialization Courses	Credits
PSYC 100	3
PSYC 105	3
PSYC 220	4
PSYC 280	3
PSYC 290	3
PSYC 301	3
PSYC 302	3
PSYC 305	3
PSYC 350	3
PSYC 360	4
PSYC 415	3
PSYC 475	3
PSYC 490	3
PSYC 498	3
Specialization Courses Total	44

Artificial Intelligence	
Specialization Courses	Credits
PSYC 100	3
PSYC 150	3
PSYC 165	3
PSYC 220	4
PSYC 225	3
PSYC 305	3
PSYC 335	4
PSYC 350	3
PSYC 354	4
PSYC 355	3
PSYC 360	4
PSYC 415	3
PSYC 490	3
PSYC 498	3
Specialization Courses Total	46

Forensic Psychology	
Specialization Courses	Credits
PSYC 100	3
PSYC 220	4
PSYC 305	3
PSYC 340	3
PSYC 350	3
PSYC 360	4
PSYC 385	3
PSYC 390	3
PSYC 395	3
PSYC 415	3
PSYC 435	3
PSYC 465	4
PSYC 490	3
PSYC 498	3
Specialization Courses Total	45

Elective Courses	Credits
The remaining courses can be selected from the list of	14-16

courses offered by the department.	
Electives courses Total	14-16

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	60
University Electives	13-16
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Course Descriptions

PSYC 100: Introduction to Psychology (3 credits)

This course is designed to provide an overview of the historical background and evolution of Psychology as a scientific discipline. It emphasizes key areas such as research methods, biological basis of behavior, sensation and perception; learning and memory; motivation and emotion; personality, stress and illness, social behavior, psychological disorders and their treatment.

PSYC 101: Life Skills (3 credits)

Pre-requisite: None

This course applies psychological theories and principles to enhance personal and professional life skills. Through interactive discussions, self-assessment, and experiential activities, students will develop competencies such as emotional intelligence, stress management, effective communication, problem-solving, and resilience which will ensure cultivation of self-awareness, adaptability, interpersonal effectiveness and personal growth.

PSYC 105: Effective Communication, influence, and persuasion (3 credits)

Pre-requisite None

Course Type (Core for specialization in Industrial and organizational)

The course emphasizes the complexities of communication, influence, and persuasion in various human interactions both personal and professional. Concepts such as effective communication and integration of different communication styles along with the significance of empathetic active listening and adaptation of communication strategy with respect to diverse organizational frameworks will also be highlighted.

PSYC 140 Psychology of Leadership (3 credit)

Prerequisite: PSYC 100

This course serves as a foundational exploration of leadership as a psychological construct. It examines various approaches to leadership through the lens of major psychological theories, as well as the factors that influence effectiveness in leaders and the community, while highlighting the practical implications.

PSYC 150: Developmental Psychology-I (3 credits)

Prerequisite: PSYC 100

This course covers the essential concepts of human development across the lifespan. It explores major developmental theories, focusing on physical, cognitive, social, and personality aspects of development at each stage of life. Key topics include nature vs nurture; stability vs change; continuity vs discontinuity.

PSYC 160 Emotional Intelligence (3 credits)

Prerequisite: PSYC 100

This course focuses on developing the ability to recognize, understand, and manage one's own emotions, as well as those of others. Key topics include interpersonal relationships, leadership skills, and overall well-being by exploring self-awareness, self-regulation, motivation, empathy, and social skills.

PSYC 165: Artificial Intelligence in Psychology (3 credits)

Pre-requisite: PSYC 100

This course examines AI's role in psychological theories and research, including cognitive personality and developmental psychology. Further exploration of AI's application is computational social sciences for modeling mind and behavior. This course emphasizes the use and understanding of Ai tools in theory, practice and case studies.

PSYC 175/EDUC 120 *: Educational Psychology (3 credits)

This course is designed to explore the principles of Psychology as applied to the educational process. It examines the psychological foundation of teaching and learning, with a focus on the characteristics of learners, teachers and classroom environment. Key topics include characteristics of individual learners, the teacher, learning process, classroom management, and introduction to psychological measurements, and creativity in children.

PSYC-180 Psychology of Art and Aesthetics (3 credit hours)

Prerequisite: PSYC 100

This course offers an introduction to both traditional and contemporary theoretical framework concerning the perception, and the creation of art. Emphasis is placed on the psychology of creative processes, psycho-aesthetics, and the perception of art, with particular attention to the sociocultural aspects. Key topics include identifying and understanding of self-expression of art such as sculpture, painting, architecture, as well as popular arts including dance, theatre, music and film.

PSYC 220: Statistics for Psychology (4 credits)

Prerequisite: PSYC 100

This course provides exploration of the foundational statistical concepts and their practical implementation with hands-on training in the use of the latest SPSS (Statistical Package for the Social Sciences) for quantitative data analysis in experimentation and research. Emphasis is on (a) descriptive statistical techniques including frequency distributions, graphical representations, and measures of central tendency and variability, and (b) inferential statistical techniques covering key parametric and non-parametric tests.

PSYC 225 Ethics in Psychology (3 credit)

Prerequisite: PSYC 100

This course is structured to provide a comprehensive overview of the ethical principles and professional standards in Psychology. It primarily focuses on the APA Ethics Code, Pakistan Mental Health Ordinance and ethical standards set by the Pakistan Association of Clinical Psychologists in all relevant fields of psychology. It also addresses potential ethical dilemmas and contemporary issues in the field.

PSYC 230: Management and Leadership Skills (3 credits)

Pre-requisite: PSYC 140

The course bridges psychological theories with practical leadership, team management, and ethical decision-making skills, essential for psychology graduates pursuing careers in human interaction fields. This course will cover applications of psychological concepts in managing personal effectiveness, ethical dilemmas, inclusive environments, conflict resolution and fostering personal and organizational growth with inculcation of psychological concepts such as emotional intelligence, motivation and behavior and their pivotal role in navigation of organizational settings.

PSYC 240: Theories of Personality (3 credits)

Prerequisite: PSYC 100

This course focuses on understanding personality through the lens of the major theories including psychodynamic, behavioral, humanistic, trait, cognitive, biological and contemporary approaches. Key topics include biographical accounts of major theorists, theoretical analysis, current research developments and their application within the field of personality psychology.

PSYC 245: Political Psychology (3 credits)

Pre-requisite: PSYC 100

This course will explore the psychological mechanisms behind political behavior, beliefs, and attitudes, including individual and group influences on decision making, identity, ideology and group behavior. It integrates interdisciplinary theories and methods (psychology, political science, sociology, neuroscience) to examine the diverse foundations of political action and related issues of political behavior.

PSYC 275 Psychology of Peace and Conflict (3 Credits)

Prerequisite: PSYC 100 Introduction to Psychology Course

The course focuses on how people navigate conflicts and strive for peace. It will examine major sources of conflict, interpersonal disputes, and scale conflicts from the micro to the macro level and various forms of violence. The course will delve into theoretical perspectives and case studies in the light of conflict and peace. The key concepts include societal and cultural injustices and their psychological, emotional, and societal impact on individuals to promote peace and post-conflict community-building.

PSYC 280: Social Psychology (3 credits)

Prerequisite: PSYC 100

This course examines the historical development, theoretical framework, and the scope of social Psychology. Key topics include social perception, cognition and identity, interpersonal relationships, attribution processes, conformity, prosocial behaviors, group dynamics, leadership, attitudes, prejudice and aggression. It also

emphasizes critical understanding of social processes, ethical consideration and application to experimentation and real-world contexts.

PSYC 285: Spiritual Psychology (3 credits)

Pre-requisite: PSYC 100

This course explores spiritual psychology, and it distinguishes spirituality from religiosity within a modern psychological framework. It focuses on self-awareness, inner growth and the impact of spiritual insight to enhance mental and emotional wellbeing. Though these studies will examine how spiritual beliefs and practices influence psychological health, human behavior and therapeutic processes.

PSYC 290: Consumer Psychology (3 credits)

Prerequisite: PSYC 100

This course explores the psychological foundations of consumer behavior, with a focus on key concepts such as perception, cognition, learning, attitudes, cognitive dissonance, risk-taking, motivation and personality of the buyer. Emphasis on the interrelations of economics and sociocultural factors on decision-making, with particular attention to consumer psychology within the context of Pakistan is studied.

PSYC 295 Media Psychology (3 credits)

Prerequisite: PSYC 100

This course explores the fascinating connection between human behavior and its various forms of media, such as television, film, social media, and different forms of advertising. Through a blend of theory, research, and practical applications, emphasis will be on understanding how media shapes the thoughts, emotions, and behaviors of the individual and society at large.

PSYC 298: History and Systems of Psychology (3 credit)

Prerequisite: PSYC 100

This course covers historical origin of modern psychology with a focus on Greek, Muslim, European, and American contributions to philosophy and physiology. The key topics include structuralism, functionalism, associationism, psychodynamic, behaviorism, existentialism, humanistic, gestalt, cognitive, evolutionary psychology, and contemporary trends.

PSYC 300: Positive Psychology (3 credits)

Prerequisite: PSYC 100

This course explores the historical roots, empirical foundations and experiential aspects of human flourishing. Key topics include subjective well-being, the science of happiness, flow, mindfulness, spirituality, optimism, resilience, forgiveness, modesty, wisdom, courage, empathy, altruism. and practical applications of the strengths and virtues that contribute to a meaningful life.

PSYC 301: Industrial and Organizational Psychology (3 Credits)

Prerequisite: PSYC 100

This course integrates theory, research and application pertaining to Industrial and organizational psychology. Key topics include need assessment, hiring, training, evaluation of employees' motivation, performance, effectiveness and satisfaction, leadership, organizational structure and turnover.

PSYC 302: Psychology of Marketing and Selling (3 credits)

Pre-requisite PSYC 290

This course focuses on exploring the intersection of psychology, sales and marketing, emphasizing cognitive, emotional and social factors influencing decision making and consumer behavior. The course is a blend of theoretical and practical applications, aiming to discover consumer needs and developing effective strategies for marketing success.

PSYC 305: Research Methods in Psychology (3 credits)

Prerequisite: PSYC 220

This course offers a comprehensive overview of the research methodology and history of scientific inquiry pertaining to quantitative research methods. The key topics include identification of the research gap, choosing and defining the variables and the population, formulation of research questions and hypothesis, research designs, sampling strategies, data collection and analysis, ethical consideration and report writing.

PSYC 310 School Psychology (3 credit hours)

Prerequisite: PSYC 100

This course focuses on understanding the theory and practice of psychology with learners of all ages, their families and the schooling process. The key topics include psychological assessment of emotional-behavioral problems, factors contributing to student academic success, learning difficulties, intervention, prevention and health promotion services within the ecological and socio-cultural context of schools.

PSYC 320: Cross-Cultural Psychology (3 Credit hours)

Prerequisite: PSYC 100

This course delves into the diversity and the impact of culture by examining sociocultural variation in human behavior, cognition, emotions, motivation, and perception. Topics include cross-cultural knowledge applied to various settings such as health and well-being, gender, society, counseling, organization, and education. The focus is also on comparative studies between human development and cultures.

PSYC 330 Environmental Psychology (3 Credit)

Prerequisite: PSYC 100

This course offers an introduction to environmental psychology, an emerging subfield that examines the reciprocal relationships between individuals, groups and their physical environment. Key topics include the psychological impacts of climate change and ecological changes, the role of human behavior in contributing to environmental issues, natural, artificial and technological hazards and their prevention.

PSYC 335: Computational Psychology (4 Credit)

Pre-requisites: PSYC 165 and PSYC 220

This course explores the intersection of psychological theories and computational models to simulate human cognition and behavior. Major topics are algorithms, simulations, predictive models for cognitive processes, emotional responses and decision making. Applications for this course extend to artificial intelligence, human-computer interaction and biofeedback training. Key topics include algorithms, simulations, and the application of computational methods to model cognitive processes, emotional responses, and decision-making. Emphasis is placed on developing predictive models and interpreting their implications for psychological research. Students will also explore the role of computational psychology in fields such as artificial intelligence, human-computer interaction, and biofeedback training.

PSYC 340: Abnormal Psychology (3 credits)

Prerequisite: PSYC 100

This course provides an overview of the nature and conceptualization of abnormal behavior, including historical developments with particular emphasis on Pakistan. It examines the major theoretical models of psychopathology, including psychoanalytic, biological, behavioristic, humanistic, cognitive-behavioral perspectives and contemporary theories. It provides an in-depth exploration of psychological disorders and differential diagnosis as classified in the latest edition of DSM.

PSYC 350: Biopsychology (3 credits)

Prerequisite: PSYC 100

This course provides an overview of the mental processes and behavior solely through the lens of biological perspective. The key topics include subdivisions of biopsychology, research, neural communication, anatomy and physiology of the spinal cord and brain, endocrine factors, sensory processes, motivation, emotion, learning, memory, language, sleep, reproduction, and psychopathology.

PSYC 354: Cognitive Psychology (4 credits)

Prerequisite: PSYC 350

This course explores the theory, principles and empirical research that forms the foundations of cognitive psychology. Key topics include information processing, perception, attention, mental imagery, representation of knowledge, memory, reasoning, problem solving, decision making, language structures and comprehension.

PSYC 355: AI Psycholinguistics (3 credits)

Pre-requisites: PSYC 165 and PSYC 354

Intersection of psycholinguistics and AI is explored through this course looking at how models evaluate human language processing. Emphasis is on the role of Large Language Models (LLMs) and generating humans like text, applications like translation and context creation. Ethical concerns like biases in AI language models and their societal implications will also be critically examined. Through research-driven discussions, students will gain insights into the cognitive mechanisms of language and the advancements in AI that mirror these processes. Ethical considerations, including biases in AI language models and their societal implications, will also be critically examined.

PSYC 360: Psychological Testing and Measurement (4 credits)

Prerequisite: PSYC 100

This course introduces the historical foundations, fundamental concepts, and functions of psychological testing. It covers psychometric properties of psychological tests, assessment methods (scales, questionnaires, tests, batteries etc.) categories of tests such as intelligence, personality, screening, diagnostic, neuropsychological and aptitude tests. It also emphasizes administration, scoring, interpretation and report writing, in educational, occupational and clinical settings while also adhering to ethical considerations of APA.

PSYC 365: Neuropsychology (3 credits)

Pre-requisite: PSYC 350

This course provides an introduction to clinical neuropsychology, including healthy neural functioning, primarily focusing on brain damage, loss of function, clinical assessment and treatment. This course covers

functional and anatomical bases of human behavior and neuropsychological disorders, as well as the connection between assessment and brain anatomy and function. The treatment of neuropsychological dysfunction will also be covered.

PSYC 370: Geriatric Psychology (3 credits)

Pre-requisite: PSYC 150

Medical advancements, improved healthcare systems, and awareness about the significance of healthy lifestyles have led to enhanced longevity and a growing interest in factors related to aging and wellbeing. This course will introduce the students to nature and the scope of Geriatric Psychology encompassing various dimensions of adult development, especially focusing on late adulthood and accompanying challenges. The primary focus will be on healthy and successful aging.

PSYC 375: Psychology of Gender (3 credits)

Prerequisite: PSYC 150

This course focuses on the concept of gender versus sex, development of gender roles, gender stereotypes, and the role of biological, psychological, social, and cultural factors in creating and maintaining gender differences. Social roles, attitudes, and achievement in men and women are also examined across socio-psychological settings.

PSYC 380: Liberation Psychology (3 credits)

Pre-requisite: PSYC 275

Liberation psychology focuses on the marginalized population. It will cover introduction, origin, development, key concepts and principles of liberation psychology. Participatory action research, community-based interventions and clinical applications will be focused on. Working with special population, marginalized communities, issues, feminist liberation psychology and social advocacy will be a part of this course.

PSYC 385: Forensic Psychology (3 credits)

Prerequisite: PSYC 340

This course explores the intersection of psychology and the criminal justice system, focusing on the roles psychologists play in forensic settings. Topics include historical foundations, development of criminal behavior, expert testimony, criminal profiling, trial consulting, and psychological assessment of individuals with suspected deviant behavior. It also focuses on insanity and competency, media violence, sexual assault, and victimology, with emphasis on the Pakistani criminal justice system.

PSYC 390: Criminal Psychology (3 credits)

Pre-requisite PSYC 385

The course explores the role of biological, social and psychological factors contributing to criminal behavior and factors influencing criminality including personality types, mental health conditions, age, socioeconomic status and gender. Through case studies, this course will examine implications of psychological theories in various crimes, criminal profiling, victim psychology and its role in law enforcement and legal system.

PSYC 395: Correctional Psychology (3 credits)

Pre-requisite: PSYC 385

This course analyzes psychology's role within the correctional system, focusing on offender assessment, rehabilitation, and management. Special emphasis on relevant psychological theories and practices including criminal behavior evaluation, risk assessment, and incarceration impact on mental health. Topics cover forensic assessment models and tools, intervention strategies, and ethical considerations, recidivism and the effectiveness of correctional programs.

PSYC 400: Psycholinguistics (3 credits)

Pre-requisite: PSYC 354

Course Description: Psycholinguistics is an interdisciplinary field which studies the psychological and neurobiological factors that enable humans to acquire, use, comprehend, and produce language. Students will explore the cognitive processes involved in language acquisition, comprehension, and production, as well as the neural mechanisms underlying these processes.

PSYC 415: Research Methods in Psychology-II (3 credits)

Prerequisites: PSYC305

This course introduces the students to the nature, significance, and philosophical roots of qualitative research. The key topics are qualitative research designs including narrative, phenomenology, ethnography, grounded theory, and case study: data collection methods including observations, interviews, and focus groups. An orientation regarding action research, mixed methods, and software aided qualitative analysis is provided.

PSYC 425: Sports Psychology (3 credits)

Prerequisite: PSYC 100

The course covers theories and practical application of Sports Psychology, with emphasis on the core domains such as personality, the role of the nervous system in arousal, stress and anxiety, mental performance enhancements, motivation, resilience, concentration, teamwork, and conflict management. Furthermore, emphasis is placed on accepting both success and failure gracefully, motivation, focused goal setting, peak performance, and well-being.

PSYC 430: Health Psychology (3 credits)

Prerequisite: PSYC 100

This course focuses on the biopsychosocial aspects of health. The key topics include general health maintenance, recovery from disease or injury, stress and its management, immune system, personality characteristics associated with disease, healthy lifestyles, pain and its management, illness prevention, and managing chronic illness, terminal illness and death.

PSYC 435: Legal Psychology (3 credits)

Pre-requisite: PSYC 385

The course focuses on intersection between psychology and the legal system, primarily focuses on topics such as eyewitness testimony, investigative interviewing, interrogations, insanity defense, and punishment. The course inculcates psychological research into conceptualization of legal systems, integrating psychological and ethical assessment with practice and policies of legal system.

PSYC 440: Counseling Psychology (3 credits)

Prerequisite: PSYC 340

This course encompasses theories, assessment procedures, and traditional and contemporary approaches to counseling. The key topics include educational, occupational and trauma counseling, counseling for emotional and sexual problems, couples, marriage and family counseling as well as community mental health counseling in the light of ethical standards of APA.

PSYC 465: Clinical Psychology (4 credits)

Prerequisites: PSYC 340, PSYC 360

This course explores the historical development of clinical psychology, assessment methods, diagnosis and traditional and contemporary therapeutic interventions. The key topics include history taking, MSE, assessment, differential diagnosis, diagnosis, case formulation and management plans for various psychopathologies. Legal and ethical considerations in clinical practice according to APA are also covered.

PSYC 467: Tele-Mental Health (3 credit)

Pre-requisites: PSYC 440 and PSYC 465

This course provides an in-depth exploration of tele-mental health, telepsychology, ethical and legal considerations, privacy and confidentiality standards, integration of artificial intelligence in psychotherapy and its role in evolving mental health services. The effectiveness of telemental health, limitations in relation to humanitarian emergencies, along with forensic and home based telepsychiatry, AI assisted therapy, and the trajectory of digital mental health care will be discussed in detail.

PSYC 470: Community Psychology (3 credits)

Prerequisite: PSYC100

This course encompasses historical development and core concepts of Community Psychology. The key topics include dynamic relationships between individuals and their communities, community mental health and well-being, appreciation of diversity, empowerment of disadvantaged and marginalized groups, advocacy for social justice and peace, and the fostering of healthy community relationships.

PSYC 475: Entrepreneurship in Psychology (3 credits)

Pre-requisite: PSYC 301

This course will offer a flavor of various theoretical frameworks of entrepreneurship explaining nature, types, and the psychological processes involved. The course encompasses entrepreneurial skills such as assertiveness, negotiation skills, resilience, stress management, emotional intelligence, nonverbal communication, problem solving, budget development, evaluating the market and creating a niche for proposed services and their integration in practical fields through experiential learning.

PSYC 490 Capstone Project (3 credit)

Prerequisite: PSYC 415

This course involves a collaborative/group research project on diverse topics in psychology, with an emphasis on conducting research that is both ethical and evidence-based, under the guidance of Psychology faculty. It consists of the entire research process, including conducting a literature review, selecting a suitable methodology, collecting and analyzing data, and preparing a comprehensive research report.

PSYC X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

PSYC 498: Internship (3 credits)

Prerequisites: Open to psychology majors

This course is designed to act as a bridge between academic learning and professional careers. Internships offered through the Psychology Department provide meaningful training and practical work experience, creating a strong connection between theoretical knowledge and real-world application.

Note: This course consists of classroom teaching and practical experience at the internship venues. The outcome of this internship will be a portfolio showcasing the students' activities and the skills they have developed throughout the course.

PSYC 499: Senior Thesis (6 credits)

Prerequisites: PSYC 305 and PSYC 415

A student-led research project focused on a topic of the student's choice, conducted independently under the supervision of a faculty member. This course spans over two semesters. The first semester includes conceptualizing and writing of a methodically and ethically sound research proposal and getting it approved by the relevant bodies. The second semester includes data collection and analysis, finalizing and submitting the dissertation and viva-voce exam.

Department of
**RELIGIOUS
STUDIES**

Introduction

The Department of Islamic Studies was renamed in 2005 as the Department of Religious Studies to be a model of interfaith harmony and representative of the vision of the founder of Pakistan which he once stated thus, “We are starting in the days when there is no discrimination, no distinction between one community and another, no discrimination between one caste or creed and another. We are starting with this fundamental principle that we are all citizens and equal citizens of one State”.

The Department of Religious Studies contributes to fulfilling the mission, vision and values of FCC by creating an atmosphere that is conducive to character building, liberal arts education, higher-order thinking skills, and inter-faith relations. Here students have the opportunity to study in a religiously diverse environment where they work together with people of different religious faiths, ethnicities and cultures. The department adopts an interdisciplinary approach to the academic study of religion to provide students with a broader understanding of the role of religion in human life. Although most courses offered by the department are on Islam and Christianity, we do not endorse a particular creed or sectarian position.

The Department of Religious Studies offers: Bachelor of Studies in Islamic Studies, Bachelor of Studies in Christian Studies, MPhil in Biblical Studies and Certificate in Biblical Language (18 credits). The lower division courses for each concentration are designed to provide students with a fundamental understanding of their respective majors, while the upper division courses provide them with depth and breadth of knowledge. The Department of Religious Studies is part of the Faculty of Humanities.

Career Opportunities

Majoring in Religious Studies prepares students for a variety of careers such as teaching, religious leadership, counseling, motivational speaking, media, politics, research and writing.

Bachelor of Studies in Islamic Studies

Learning Objectives

- i. Teach students fundamental concepts and intellectual basis of various branches of Islamic Studies
- ii. Train students how to recite, translate, and interpret texts of the Holy Quran, Hadith and other fundamental sources of Islam.
- iii. Guide learners how to solve everyday problems in the light of Islamic teachings.
- iv. Inculcate in students the social intelligence and global mindset for interpersonal effectiveness.
- v. Develop students for effective and authentic research skills.
- vi. Edify students for personal development, moral transformation, and spiritual elevation.

Program Learning Outcomes

- i. **Knowledge & Skills**
Explain fundamental concepts and intellectual basis of various branches of Islamic Studies and understand basic Arabic Language and terminologies.
- ii. **Skills & Application**
Recite, translate, and interpret texts of the Holy Quran, Hadith and other sources of Islam
- iii. **Analysis - Critical Thinking**
Diagnose and solve everyday problems in the light of Islamic teachings, principles and values

iv. Human dimension - Problem solving

Demonstrate respect for diversity and apply Golden Rule as the fundamental principle of the actions.

v. Integration - Creativity

Describe/Explore professional and further education opportunities in the field of Islamic studies and formulate a personal plan for the inner process of spiritual development and personal growth.

Requirements for BS Islamic Studies Major

Core Courses	Elective Courses	Total Credits	Required MCGPA
10 (30 credits)	8 (24 credits)	54	2.0

Core Courses	Credits
ISLM 104	3
ISLM 201	3
ISLM 202	3
ISLM 301	3
ISLM 302	3
ISLM 304	3
ISLM 305 /SOCL 305/CRST 354	3
ISLM 401/ CRST 453	3
ISLM 498	3
ISLM 499	3
Core Courses Total	30
Elective Courses	Credits
Other Islamic studies courses	24
Elective Courses Total	24

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for the Minor in Islamic Studies

Core Courses	Elective Courses	Total Credits	Required MCGPA
3 (9 credits)	3 (9 credits)	18	2.0

Core Courses	Credits
ISLM 104	3
ISLM 202	3
ISLM 401	3
Core Courses Total	9
Elective Courses	Credits
Other Islamic studies courses	3*3
Elective Courses Total	9

Bachelor of Studies in Christian Studies

Learning Objectives

- i. To help students see the Bible's storyline as the key to understanding the Bible itself.
- ii. To teach the Bible's teachings from the Bible itself.
- iii. To build hermeneutical skills in students so that they can be confident to interpret the Bible.
- iv. To teach students how to take what they have learned in the Bible and apply it to contemporary life.
- v. To help students develop critical thinking skills so that they can understand and analyze other worldviews and explain their own.
- vi. To instill in our students a desire to learn the Bible and give them a passion in helping in their community to share that same love.
- vii. To make sure that from the start students know the opportunities and limitations inherent in graduating with a degree in Christian studies.

Program Learning Outcomes

- i. **Content of the discipline:**
Demonstrate knowledge of the central storyline of the Bible, understanding how the original authors of various biblical books presented and explained that story, and understanding as well how these original authors applied that storyline to their own historical context.
- ii. **Theory of the discipline:**
Identify what are the Bible's key teachings, displaying skill in explaining them
- iii. **Methodology and skills:**
Using proper hermeneutical practices, show biblical competence in reading and interpreting various passages of the Bible
- iv. **Higher Order Critical thinking:**
Exhibit critical thinking skills which enable the student to develop a mature Christian worldview, capable of interacting with competing worldviews
- v. **Application of knowledge to the real world:**
Demonstrate skill in applying the teachings of the Bible to the student's context and community, as well as to the greater Pakistani and global communities
- vi. **Values:**
In accordance with FCCU's core values, exhibit spiritual maturity through: Evidencing holistic growth in Christian character and spiritual formation, demonstrating their commitment to promoting reconciled relationships and restored communities, and Showing godly character and actions which reflect true Christian citizenship to the nation of Pakistan
- vii. **Future:**

Articulate how their degree can lead to opportunities for professional involvement, employment, or further education.

Requirements for BS Christian Studies Major

Core Courses	Elective Courses	Total Credits	Required MCGPA
15 33(credits)	19 credits	52	2.0

Core Courses	Credits
CRST 151	3
CRST 155	3
CRST 281	3
CRST 291	3
CRST 381	3
CRST 391	3
CRST 453/ISLM 401	3
CRST 498	3
CRST 499	3
CRST 211	1
CRST 212	1
CRST 311	1
CRST 312	1
CRST 411	1
CRST 412	1
Total	33
Elective Courses	Credits
Any CRST or ISLM course except CRST 152 (Christian Ethics)	19
Total	19

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	52
University Electives	21-24
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for Minor in Christian Studies

Minor in Christian Studies is open to students of all disciplines with a minimum of CGPA 2.00.

Core Courses	Elective Courses	Total Credits	Required MCGPA
12	6	18	2.0

Core Courses	Credits
CRST 151	3
CRST 155	3
CRST 211	1
CRST 291	3
Core Courses Total	10
Elective Courses	Credits
6 credit hours from any following course:	
CRST 281	3
CRST 381	3
CRST 391	3
CRST 451	3
2 credit hours from Applied Theology	
CRST 212	1
CRST 311	1
CRST 312	1
CRST 411	1
CRST 412	1
Elective Courses Total	8

Minor in Biblical Language (Either New Testament Greek or Biblical Hebrew)

Any regularly enrolled Forman Christian College (A Chartered University) student wishing to gain proficiency in a biblical language (either *Koine* Greek, the language of the Christian New Testament or biblical Hebrew, the language of the Christian Old Testament) may earn a minor in Biblical Language through the Department of Religious Studies. This affords opportunities for students to study the Hebrew or Greek languages without the burden of pursuing a major in Christian Studies. This might be helpful for students who plan to pursue post-graduate degrees in history, religious studies or philosophy. It would especially be helpful for students who intend after graduation to enter a graduate program in theology, perhaps in a Christian seminary. The Certificate consists of 6 courses (18 credits) in either the Hebrew or Greek language.

Learning Objectives

1. Demonstrate competence in translating Biblical texts, which enables students to interpret these texts with growing sensitivity to semantics, syntax, context, genre and manuscript evidence.
2. Demonstrate competent critical thinking skills by the application of principles of good biblical exegesis.

Requirements for Minor in Biblical Language (New Testament Greek)

Core Courses	Elective Courses	Total Credits	Required MCGPA
4 (12 credits)	6 credits	18	2.0

Core Courses	Credits
CRST 271	3
CRST 272	3
CRST 371	3
CRST 372	3
Core Courses Total	12
Elective Courses	Credits
1 Greek Reading Course (Any Greek course with the word “Reading” in its course title)	3
1 Greek Exegesis Course (Any Greek course with the word “Exegesis” in its course title).	3
Elective Courses Total	6

Requirements for Minor in Biblical Language (Biblical Hebrew)

Core Courses	Elective Courses	Total Credits	Required MCGPA
12	6	18	2.0

Core Courses	Credits
CRST 231	3
CRST 232	3
CRST 331	3
CRST 332	3
Total	12
Elective Courses	Credits
1 Hebrew Reading Course (Any Hebrew course with the word “Reading” in its course title)	3
1 Hebrew Exegesis Course (Any Hebrew course with the word “Exegesis” in its course title)	3
Total	6

Certificate in Biblical Language (18 credits)

Anyone who is not enrolled in the Baccalaureate program and wishes to gain proficiency in a Biblical Language (either *Koine* Greek, the language of the Christian New Testament or Biblical Hebrew, the language of the Christian Old Testament) may earn a Certificate in Biblical Language through the Department of Religious Studies. This affords opportunities for students and others to study the Hebrew or Greek languages without the burden of pursuing a major in Christian Studies. This might be especially helpful for post-graduate students who feel their postgraduate research would be enriched through the acquisition of Hebrew or Greek or professors of religious studies who wish to expand their knowledge of the Bible. The Certificate consists of 6 courses (18 credits) in either the Hebrew or Greek language.

Learning Objectives

1. Demonstrate competence in translating Biblical texts, which enables students to interpret these texts with growing sensitivity to semantics, syntax, context, genre and manuscript evidence.
2. Demonstrate competent critical thinking skills by the application of principles of good Biblical exegesis.

Requirements for the Biblical Language Certificate (Choose either Greek or Hebrew)

For the Greek Language

12 credits Basic Greek Language Courses: CRST 271, CRST 272, CRST 371, CRST 372

3 credits Greek Reading Courses: CRST 441, CRST 442

3 credits Greek Exegesis Courses: CRST 475, CRST 476

For the Hebrew Language

12 credits Basic Hebrew Language Courses: CRST 231, CRST 232, CRST 331, CRST 332

3 Credits Hebrew Reading Courses: CRST 421, CRST 422

3 Credits Hebrew Exegesis Course: CRST 431

Course Descriptions

ISLM 100-A (Fehm-e-Qur'an I) (1 credit)

This course introduces students to the core vocabulary, common expressions, and fundamental linguistic features of Quranic Arabic. It emphasizes the practical development of skills to recognize and comprehend the language of the Holy Qur'an directly, establishing a strong foundation for further study leading to Fehm-e-Qur'an II in accordance with the HEC Model Curriculum. The course develops students' ability to recognize and comprehend the meanings of Qur'anic words and phrases directly within the text, establishing a strong linguistic foundation for further study. By taking this course, students are expected to understand approximately 30–40% of the recurring vocabulary of the Holy Qur'an, preparing them for more advanced linguistic engagement in Fehm-e-Qur'an II.

ISLM 100-B (Fehm-e-Qur'an II) (1 credit)

Prerequisite: ISLM100-A

Building upon the foundation established in Fehm-e-Qur'an I, this course advances students' proficiency in understanding the language of the Holy Qur'an through the systematic expansion of Qur'anic vocabulary and deeper study of its grammatical and linguistic structures. Emphasis is placed on strengthening students' ability

to comprehend Qur'anic verses directly, enabling them to read the text with greater confidence, accuracy, and independence. Upon successful completion of the course, students are expected to understand approximately 80–85% of the recurring vocabulary of the Holy Qur'an.

ISLM 101: Islamic Education (2 credits)

This course offers students a thorough introduction to the core teachings of Islam, including its beliefs, practices, historical development, and societal impact. It is designed to equip students with a solid academic and cultural understanding of Islam, enhancing their ability to engage with Islam's diverse dimensions. By the end of the course, students will be better prepared to approach complex conversations about Islam's past, present, and future roles with empathy, respect, and well-informed perspectives.

ISLM 103: Islamic Ideology (3 credits)

Prerequisite: ISLM 101 or CRST 152

The course is designed to meet the requirements of those who want to enhance their fundamental understanding of Islam as a code of life. Contents include definition of ideology, functions of ideology, Islamic ideology, chief characteristics of Islamic ideology, social teachings of Islamic ideology, economic principles of Islam, political teachings of Islam, and legal thoughts of Islamic ideology.

ISLM 104: Arabic Communication Skills (3 credits)

Arabic grammar and composition, basic structure of Arabic language; everyday conversation, vocabulary enhancement, translation and composition.

ISLM 201: Tajweed-Ul-Quran (3 credits)

Prerequisite: ISLM 101

Reading and recitation of the Holy Quran, speech sounds of Arabic, Qiraat, Arabic phonology, articulation and accent, and pronunciation.

ISLM 202: Introduction to 'Ulūm al-Qur'ān (3 credits)

Prerequisite: ISLM 101 or CRST 152

Key topics include the definition of the Qur'an, the historical development of its preservation and compilation, major content types, general style, causes of revelation, selected textual readings, the inimitability (I'jāz) of the Qur'an, the essential qualifications (ādāb) of an exegete (Mufasssir), and various methodologies and types of Qur'anic interpretation.

ISLM 204: Introduction to Islamic Culture & Civilization (3 credits)

Prerequisite: ISLM 101 or CRST 152

This course examines the meaning of culture and civilization and their relationship to human beliefs, values, institutions, and everyday life. It explores the Qur'anic and Prophetic foundations of Islamic civilization and traces its development through the Rashidun, Umayyad, Abbasid, and Ottoman periods. Students are introduced to key contributions in governance, society, knowledge, arts, and trade, as well as the interaction between Islamic and Western civilizations and selected contemporary issues in the modern world.

ISLM 205: Islam and Human Rights (3 credits)

Prerequisite: ISLM 101 or CRST 152

The course investigates the relationship between Islamic theological and juridical traditions and the modern international human rights framework. Human rights have emerged as the dominant global moral language of our time. This course fills a critical gap in traditional Islamic Studies curricula by bridging classical Islamic ethics, Maqasid al-Shariah, and contemporary human rights scholarship, with direct application to issues of religious freedom, gender equity, minority rights, and social justice in Pakistan and globally.

ISLM 301: Hadith Studies (3 credits)

Prerequisite: ISLM 101 or CRST 152

Examines Hadith—the sayings, actions, and approvals of Prophet Muhammad (ﷺ)—as a source of Islamic knowledge, law, ethics, and spirituality. Covers ‘Ulūm al-Ḥadīth, authentication, classification, historical development, canonization, textual engagement with canonical collections, women’s scholarly contributions, moral reasoning, ethical citizenship, and contemporary applications in global contexts.

ISLM 302: Islamic Jurisprudence (3 credits)

Prerequisite: ISLM 101 or CRST 152

Definition of law sources of Islamic Sharia, Qur'an, Sunnah, Ijma, Qiyas, Ijtihad, types of Islamic law, Islamic injunctions and family law, Islamic law and jurisprudence with its historical development.

ISLM 303: Contemporary Muslim World (3 credits)

Prerequisite: ISLM 101 or CRST 152

Geographical, social and cultural features of the modern Muslim world, concept of Ummah, its resources, population, political and economic systems, organizations, challenges to the Muslim world and their solutions.

ISLM 304: Seerat-Un-Nabi (3 credits)

Prerequisite: ISLM 101 or CRST 152

Development of biographical studies of the Prophet Muhammad (PBUH), their influence on Islamic thought throughout Islamic history, the Prophet (PBUH) as an example to be followed, the difference between the Prophet's (PBUH) tradition (Hadith) and his biography (Seerah), the Prophet's (PBUH) life before his mission, the early period of Makkah where he faced opposition, migration to Madinah, establishment of the Islamic state in Madinah, treaties and relations with non-Muslims, Ghazwaat and conquest of Makkah and the last sermon and its impact on modern human life.

ISLM 305/ CRST 354/ SOCL 305*: Interfaith Dialogue and Relations (3 credits)

Prerequisite: ISLM 101 or CRST 152

Understanding dialogue and its process, Dialogue in everyday life, the Need for interfaith dialogues in the modern context, the Role of interfaith dialogue and relations for peace and social harmony, General and specific goals of interfaith dialogue, Challenges to interfaith relations, Effective strategies to develop interfaith relations, Historical interactions between major religious traditions, Theories and principles of effective dialogue, Social & professional significance of interfaith relations.

ISLM 306/CRST 355*: Religion and Science (3 credits)

Prerequisite: ISLM 101 or CRST 152

Relationship between modern empirical science and religion, the nature of science and the epistemologies of science and of religious belief, conflict or concord between religion and science on common domains, religion and science in search of reality.

ISLM 307: Sufism in Islam (3 credits)

Prerequisite: ISLM 101 or CRST 152

Sufism represents the spiritual and ethical dimension of Islam, emphasizing inner purification (tazkiyah), spiritual excellence (ihsan), divine love, and moral refinement. Emerging in the formative centuries of Islam, it developed as a disciplined path complementing Islamic law and theology, evolving into a rich intellectual, ethical, and institutional tradition. This course introduces Sufism as a framework for personal transformation, ethical conduct, and intercultural understanding. It explores historical evolution, doctrinal foundations, major scholars, institutional expressions, regional developments in the Indo-Pak areas, and contemporary relevance. Students critically examine Sufism's role in shaping individual character, societal harmony, and global peace.

ISLM 308: Islamic Theology (3 credits)

Prerequisite: ISLM 101 or CRST 152

This course offers a comprehensive introduction to the major schools, central doctrines, and historical development of Islamic classical theology ('Ilm al-Kalām). Students will explore the intellectual foundations of key theological concepts such as God's nature (Tawhid wa sifāt), divine justice and predestination (qadar), and the status of faith and sin. The curriculum traces the emergence of classical theological schools—including the Mu'tazila, Ash'ariyya, and Māturīdiyya—and their engagement with philosophical and interfaith challenges. By examining primary source texts from seminal Muslim theologians and their counterparts in Jewish and Christian thought, students will critically analyze theological debates and their lasting impact on Islamic intellectual history. The course culminates by relating these classical methodologies to modern discourses on faith, reason, and interreligious dialogue.

ISLM 310: Al-Da'awah wa al-Irshad (Preaching and Guidance in Islam) (3 credits)

Prerequisite: ISLM 101 or CRST 152

This course introduces students to the theory and practice of Da'awah (Islamic invitation) and Irshād (spiritual guidance) through Qur'anic teachings, Prophetic methodology, and classical Muslim scholarship. It examines ethical communication, spiritual mentorship, and reform-oriented engagement within historical and contemporary contexts. Students explore outreach in pluralistic societies, interfaith interaction, youth discourse, extremism, and digital religious communication. Through analytical study and applied learning, the course prepares students to understand religious guidance as an intellectually grounded, ethically responsible, and socially responsive activity.

ISLM 311: Classical Arabic for the Study of the Qur'an and Islamic Literature (3 credits)

Prerequisite: ISLM 101 or CRST 152

This course develops advanced skills in reading and analyzing classical Arabic texts, focusing on the Qur'an, Hadith, and early Islamic Khutbaat (sermons), proverbs, prose, and poetry. Students engage with authentic selections to strengthen grammar, vocabulary, and rhetorical understanding (balāghah). Emphasis is placed on Qur'anic style, Prophetic discourse, and foundational works (Ummahāt al-Kutub), enabling students to critically interpret classical sources and apply linguistic analysis in Islamic studies and scholarship.

ISLM 401/CRST 453*: Comparative Study of Religions (3 credits)

Prerequisites: ISLM 101 or CRST 152

Major religions of the world, cultural contexts, scriptures, fundamental beliefs, practices and sacred art are examined, various global forms of religions, South and East Asian rich traditions of Hinduism, Buddhism, Taoism and Confucianism, monotheistic religions with roots in the Middle East: Judaism, Christianity and Islam, comparative study of religions.

ISLM 402: Islam and the West (3 credits)

Prerequisite: ISLM 101 or CRST 152

Islam and the West, some general considerations; Islam and the West on peace, terrorism, democracy, and human rights; Western understanding of Islam (orientalism); Muslim perception of the Western civilization; Study and analysis of the thesis of “Clash of Civilization”

ISLM 403: Modern Islamic Thoughts (3 credits)

Prerequisite: ISLM 101 or CRST 152

Modern Islamic thoughts with the emphasis on Islamic political, economic, social and educational thoughts, political, strategic, economic and social factors underlying modern Islam, comparison of thoughts of Jamal al-din Afghani, Shah Wali Ullah, Allama Iqbal and Syed Maududi.

ISLM 406/CRST 455*: Research Methodology (3 credits)

Prerequisite: Only for Religious Studies majors

What is research? Need for research, Types and classification of research, research methodologies and frameworks, management of research, selecting research topic, research proposal development, approaches to data collection, data analysis, research in religious studies, research ethics, writing and publishing research report.

ISLM 407/CRST 454*: Teaching of Religious Studies (3 credits)

Prerequisite: ISLM 101 or CRST 152

Teaching techniques include the fundamental skills and characteristics of religious teachers, teaching strategies, classroom management, lesson planning, course designing, feedback and assessment, and materials development.

ISLM X95/CRST X95: Themes (1-3 credits) Sections:

A-E to be of 1 credit

F-J to be of 2 credits

K-Z to be of 3 credits

ISLM 498: Internship (3 credits)

Only the majoring students with CGPA 2 or above will be eligible for internship.

To graduate in Islamic Studies, students must be engaged for 135 hours over the semester (nine hours per week) internship after their fourth semester. This internship, conducted under the supervision of both a professional in the field and a departmental faculty advisor, can be at an Islamic center, madrassa, school, college, university, library, or research journal, where students work as research assistants or sub-editors, or serve the community in institutions like orphanages or NGOs that promote social harmony. Grading will be based on the successful attainment of internship objectives, assessed by a faculty member or the departmental committee.

ISLM 499*: Capstone Project (3 credits)

Prerequisite: ISLM406/CRST455

Students will undertake a supervised capstone research project on a topic approved by the department. The project is designed to enable students to synthesize and apply disciplinary knowledge, research methodologies, critical thinking, and communication skills to investigate real-world issues and propose evidence-based findings or solutions. The course includes proposal development, literature review, research implementation, report writing, and oral defense before a departmental committee.

CRST 151: Introduction to Christian Thought (3 credits)

Foundational Christian beliefs such as the nature of God, the person and work of Christ, the purpose of the church, the meaning of Christian life and growth, and the nature of God's Word as revealed in the Old and New Testaments.

CRST 152: Christian Ethics (2 credits)

Biblical and theological foundations of Christian ethics with special emphasis on developing the skills necessary to formulate ethical questions and find their solutions in the Bible.

CRST 155: Interpreting the Bible (3 credits)

Interpreting the Old and New Testaments in light of a particular passage's historical and cultural context as well as placing that passage in its context within the Bible itself. This course focuses on acquiring practical skills in identifying different types of biblical literature, skills in interpreting those types of literature and skills in employing appropriate research methods. The end goal is to enable students to grasp what impact the original writer wanted any piece of biblical writing to have on his original readers.

CRST 211: Applied Theology 1: Introduction (1 credit)

Prerequisite: CRST 152 or CRST 151

An introduction to principles governing application of biblical theology to the individual and community in a holistic approach, concerned with emotions, intellect, relationships and mental, physical and spiritual health.

CRST 212: Applied Theology 2: Prayer (1 credits)

Prerequisite: CRST 211

An introduction to principles governing application of biblical theology to prayer on an individual and corporate level.

CRST 213: Applied Theology 7: The Christian Family (1 credits)

Prerequisite: CRST 211

This course explores the biblical foundations, theological principles, and practical applications of Christian family life, guiding students to interpret key Scriptures, evaluate contemporary cultural perspectives, and cultivate spiritually mature commitments to marriage, parenting, and community within their local and global contexts.

CRST 231: Biblical Hebrew 1 (3 credits)

Introduces the student to the grammar, vocabulary and syntax of Biblical Hebrew (i.e., the original language in which the Old Testament of the Bible was written). This course seeks to help students to read, translate and interpret the ancient Hebrew Old Testament.

CRST 232: Biblical Hebrew 2 (3 credits)

Prerequisite: CRST 231

Continues the study of the grammar, vocabulary and syntax of Biblical Hebrew (i.e., the original language in which the Old Testament of the Bible was written). This course seeks to help students to read, translate and interpret the ancient Hebrew Old Testament.

CRST 252: Christian History (3 credits)

Prerequisites: CRST 151

Rise of Christianity from the period immediately following the ministry of Jesus Christ to the contemporary worldwide Christian movement, studying Christians who have made significant impact upon the Church. In addition, the course investigates different Christian movements with special emphasis on Christianity in South Asia from the missionary work of the Apostle Thomas to the present day.

CRST 271: New Testament Greek 1 (3 credits)

First half of a year's course on beginning Greek. Introduces the student to the grammar and syntax of Koine Greek (i.e., the original language in which the New Testament of the Bible was written). This course seeks to help students to read, translate and interpret the ancient Greek New Testament.

CRST 272: New Testament Greek 2 (3 credits)

Prerequisite: CRST 271

Second half of a year's course on beginning Greek. Introduces the student to the grammar and syntax of Koine Greek (i.e., the original language in which the New Testament of the Bible was written). This course seeks to help students to read, translate and interpret the ancient Greek New Testament.

CRST 281: Old Testament I- Law and Poetry (3 credits)

Prerequisite: CRST 155

An introduction to the Old Testament, examining historical, cultural, political, and religious backgrounds of the Old Testament world as well as surveying the various kinds of literature found in the Old Testament. This course will also survey the five books of the Torah (Genesis—Deuteronomy) and the five books of poetry (Job—Ecclesiastes). Special attention will be given to interpreting these books according to sound hermeneutical principles.

CRST 291: New Testament I-The Gospels (3 credits)

Prerequisite: CRST 155

A survey of the New Testament Gospels, introducing the historical, cultural, political and religious background of the New Testament in general and the four gospels in particular and examining the life, teachings and impact of Jesus Christ. Special attention will be given to interpreting these gospels according to sound hermeneutical principles.

CRST 311: Applied Theology 3: Scripture (1 credit)

Prerequisite: CRST 211

An introduction to principles governing application of biblical theology to Christian scriptures, focusing on holistic personal and corporate engagement with scripture.

CRST 312: Applied Theology 4: Servanthood (1 credit)

Prerequisite: CRST 211

An introduction to principles governing application of the biblical theology of servanthood as taught by Jesus Christ and the apostles, focusing on both personal and corporate expressions of service.

CRST 313: Applied Theology 8: The Church (1 credits)

Prerequisite: CRST 211

An introduction to Jesus' teachings about the church how Christian believers should relate to Church and be involved in their local church.

CRST 331: Biblical Hebrew 3 (3 credits)

Prerequisite: CRST 232

A deepening emphasis on syntax, translation and interpretation, continuing the study of the grammar, vocabulary and syntax of Biblical Hebrew (i.e., the original language in which the Old Testament of the Bible was written). This course seeks to help students to read, translate and interpret the ancient Hebrew Old Testament.

CRST 332: Biblical Hebrew 4 (3 credits)

Prerequisite: CRST 331

Applying principles of grammar, semantics and syntax to the translation and interpretation of Biblical Hebrew (i.e., the original language in which the Old Testament of the Bible was written). This course seeks to help students to read, translate and interpret the ancient Hebrew Old Testament.

CRST 352: Christian Theology (3 credits)

Prerequisites: CRST 151, CRST 152

Approaches to presenting a reasonable and rational basis for the Christian faith including investigation of historical evidence, evaluation of philosophical arguments, examination of biblical reliability, and explanation of key biblical teachings.

CRST 353: Jesus Christ: Life, Teachings, and Impact (3 credits)

Prerequisite: CRST 351

Life, ministry and teachings of Jesus Christ within their prophetic, cultural and historical setting as revealed in the four gospels of the Bible with a special emphasis on Bible study methods for communicating his teachings to others.

CRST 354/ISLM 305/SOCL 305 *: Interfaith Dialogue and Relations (3 credits)

Prerequisite: ISLM 101 or CRST 152

Understanding dialogue and its process, Dialogue in everyday life, the need for interfaith dialogues in the modern context, the Role of interfaith dialogue and relations for peace and social harmony, General and specific goals of interfaith dialogue, Challenges to interfaith relations, Effective strategies to develop interfaith relations, Historical interactions between major religious traditions, Theories and principles of effective dialogue, Social & professional significance of interfaith relations.

CRST 355/ISLM 306*: Religion and Science (3 credits)

Prerequisite: ISLM 101 or CRST 152

Relationship between modern empirical science and religion, the nature of science and the epistemologies of science and of religious belief, conflict or concord between religion and science on common domains, religion and science in search of reality.

CRST 371: New Testament Greek 3 (3 Credits)

Prerequisite: CRST 272

Last third of a unified program of studies for learning the ancient Greek language. Introduces the student to the grammar and syntax of Koine Greek (New Testament Greek), with a view towards the translation and exegesis of the New Testament.

CRST 372: New Testament Greek 4 (3 Credits)

Prerequisite: CRST 371

Advanced grammar and syntax to aid in the translation and interpretation of Koine Greek (i.e., the original language in which the New Testament of the Bible was written).

CRST 381: Old Testament II- History and Prophecy (3 credits)

Prerequisite: CRST 155

A survey of the Old Testament books of history (Joshua—Esther), focusing on the unified story described in those books, examined within the historical, cultural, political and religious backgrounds of the Old Testament world. This course will survey the Old Testament prophets (Isaiah—Malachi) in the context of their historical settings, with a special emphasis on messianic prophecy. Special attention will be given to interpreting these books according to sound hermeneutical principles.

CRST 391: New Testament II- The New Testament Letters and Acts (3 credits)

Prerequisite: CRST 155

A survey of the Acts of the Apostles and the New Testament letters (Romans- Revelation), examining these books within their first century context, looking at cultural, historical, political and religious issues impacting the first century church. Special attention will be given to interpreting these books according to sound hermeneutical principles.

CRST 411: Applied Theology 5: Teaching (1 credit)

Prerequisite: CRST 211

An introduction to the principles governing application of the biblical theology of teaching Christian beliefs and Scripture, focusing on transformational approaches to communicating the Christian faith.

CRST 412: Applied Theology 6: The Christian in Society (1 credit)

Prerequisite: CRST 211

An introduction to the principles governing application of the biblical theology of how a Christian must respond to living within the larger community of those who do not share the Christian faith.

CRST 413: Applied Theology 9: The Mission (1 credits)

Prerequisite: CRST 211

This 1 credit course introduces students to the biblical concept of the Mission Dei and explores how God's mission shapes the identity, purpose, and participation of believers in both local and global contexts.

CRST 421: Biblical Hebrew Reading 1: Hebrew Narrative Literature (3 credits)

Prerequisite CRST 331

Reading through Hebrew Old Testament narrative passages with a concentration on grammar and narrative flow as well as sermon and church lesson applications.

CRST 422: Biblical Hebrew Reading 2: Hebrew Narrative/Legal Literature. (3 credits)

Prerequisite: CRST 331

Reading through narrative and legal passages with a special emphasis on the Book of Deuteronomy. Students will gain skills in applying this theology to contemporary Pakistan.

CRST 425: Hebrew Reading

Prerequisite: CRST 331

Reading through Hebrew Old Testament passages with a concentration on grammar and narrative flow as well as sermon and church lesson applications.

CRST 431: Hebrew Exegesis 1(3 credits)

Prerequisite: CRST 332

An introduction to the principles and methods of exegesis in OT narrative texts. Students will be expected to prepare for class presentations and to interact critically with the materials, with the goal of applying these principles learned to church lessons and sermon preparation.

CRST 435: Hebrew Exegesis (3 credits)

Prerequisite: CRST 332

An exegetical study of a selected Old Testament book or books, using with discernment various exegetical methods, critical tools, and Bible software as part of an exegetical method. May be taken multiple times for different Old Testament books.

CRST 439 (a-e) Select English Old Testament Book Study (3 credits)

Prerequisite: CRST 155 or permission of the instructor.

Study of select Old Testament books or books in the English Bible.

CRST 441: New Testament Greek Reading 1: The Gospel of Mark (3 credits)

Prerequisite: CRST 371

Reading extensively from the Greek New Testament Gospel of Mark with grammatical analysis of the text and discussion of textual criticism issues (analysis of variant readings to determine the original wording of the text).

CRST 442: New Testament Greek Reading: Galatians (3 credits)

Prerequisite: CRST 371

Reading extensively from the Greek New Testament, Paul's letter to the Galatians with grammatical analysis of the text and discussion of textual criticism issues (analysis of variant readings to determine the original

wording of the text). Includes projects for communicating key theological points in sermons or church lessons.

CRST 445: Greek Reading: Select New Testament Book(s) (3 credits)

Prerequisite: CRST 331

Reading through Greek New Testament passages with a concentration on grammar and narrative flow as well as sermon and church lesson applications.

CRST 451: Paul's Life, Theology and Impact (3 credits)

Prerequisite: CRST 351 or CRST 391

A study of the life of Paul, including the political, cultural and religious context of Paul's first century Mediterranean world. In addition, this course will discuss Paul's theology, setting it in the context of the earliest Christian teachings, those of both Jesus Christ and his apostles. Students will survey Paul's 13 letters in light of their origin, purpose, and audience as well as focusing on two of Paul's representative letters.

CRST 452/PHIL 411*: Great Theologians (3 credits)

Prerequisite: CRST 151

The student will concentrate on careful study in the thought, context, and impact of a specific Medieval theologian, such as Augustine, Aquinas, Anselm etc.

ISLM 406/CRST 455*: Research Methodology (3 credits)

Prerequisite: Only for Religious Studies majors

What is research? Need for research, Types and classification of research, research methodologies and frameworks, management of research, selecting research topic, research proposal development, approaches to data collection, data analysis, research in religious studies, research ethics, writing and publishing research reports.

CRST 471 (a-e) Select English New Testament Book Study (3 credits) *Prerequisite: CRST 155 or permission of the instructor.*

Study of select New Testament books or books in the English Bible.

CRST 475: Greek Exegesis 1: The Sermon on the Mount.

Prerequisite: CRST 372

A verse-by-verse exposition. Emphasis includes the continued development of exegetical methodology, the investigation of interpretative options and the discovery of practical applications. Translation and a practical, exegetical sermon manuscript are required. Includes projects for communicating key theological points in sermons or church lessons.

CRST 476: Greek Exegesis: Philippians.

Prerequisite: CRST 372

A close reading of Philippians in the Greek text. As part of the in-class discussion students are required to translate, decline and conjugate. Students will produce sermons or church lessons as a part of this course.

CRST 478: New Testament Greek Exegesis (3 credits)

Prerequisite: CRST 371

An Exegetical study of a selected New Testament book or books, using with discernment various exegetical methods, critical tools, and Bible software as part of an exegetical method. May be taken multiple times for different New Testament books.

CRST 485: Themes in Christian Theology (3 credits)

Prerequisite: CRST 151

An in-depth student of a particular key theme in Christian theology, examining biblical teachings and the historical development of this theme.

CRST 498: Internship (3 credits)

Prerequisite: 2.00 GPA & 70 completed credits.

A supervised internship in a Christian NGO, ministry, or church setting. The internship must span a minimum of six weeks, five days per week, with students spending 4–5 hours per day at the organization. Students are jointly supervised by an on-site supervisor and a faculty supervisor. Students are responsible for ensuring that the Internship Evaluation Form is completed by their on-site supervisor.

CRST 499: Capstone Project (3 credits)

Prerequisite: Minimum GPA of 2.0. At least 70 completed credit hours, including at least 9 upper division CRST or ISLM credits.

A culminating assessment of the Christian Studies major. Students complete ten comprehensive exams, taken without prior study, to evaluate cumulative knowledge in biblical studies, theology, Christian history, and applied ministry, not counted in the student's final grade for the course. In addition, students write a senior research paper of no more than 10,000 words demonstrating advanced skills in analysis, interpretation, and theological reflection, supervised by a Christian Studies faculty member.



School of **MANAGEMENT**

Introduction

School of Management at Forman Christian College (A Chartered University) was established in 2005 and has already gained a reputation for the quality and diversity of its programs. SOM's mission is to provide students with modern holistic business education with a strong experiential component blended in with the latest learning technologies. The programs have been designed by a team that has rich experience of developing and leading high-quality management programs in the country at undergraduate, graduate, and at Executive levels. The Department of Business is part of the School of Management.

Bachelor of Business Administration

The program has been designed to ensure that students understand best practice in business and function effectively in the practical world. The course contents intend to provide the right balance between academic and real-world applications. The 4-year degree has a strong emphasis on developing skill and confidence and currently offers specializations in

- Accounting and Finance
- Operations Management
- Marketing & Sales Management
- Human Resource Management
- Business Analytics

Business students are expected to maintain a minimum CGPA of 2.0 during the program. However, business students are expected to maintain a higher CGPA in their major. Students take core (required) courses throughout the four years of the program and begin to take courses in their areas of specialization from the third year. Students take 75 credit hours in their business major (72 in taught courses and 3 in the mandatory Internship). The choice of specialization is made in the second year.

School of Management Vision

To be recognized as a leading business school in Pakistan, integrating business education with liberal arts.

School of Management Mission

To provide quality business education integrated with liberal arts to develop ethical, professional, and socially responsible leaders who contribute to society through critical thinking, holistic skills, and impactful service.

Program Objectives

- Describe key concepts and theory of best practice in management, finance, and marketing.
- Demonstrate technical rigor for analysis and decision making in business.
- Apply leadership skills for effective future business management.
- Analyze the demands of the local environment and apply business skills and tools to the real world.
- Employ interpersonal and communications skills required by effective managers.
- Consistently use values and ethics in a business environment.
- Describe entrepreneurial career opportunities that serve the needs of Pakistan and the international community.

Total Degree Requirements

Requirement Category	Total Credits
General Education	45-48
Major Requirements	75
University Electives	1-4
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

For Business Students UNIV 200 “Critical Thinking, Ethics, and Artificial Intelligence” Removed from the General Education requirements and the same course included in the roadmap as a core business course BUSN 371 “Critical Thinking, Ethics, and Artificial Intelligence in Business.”

Requirements for Business courses Program

Core Courses	Specialization Courses	Total Credits	Required MCGPA
19 (57 credits)	6 (18 credits)	75	2.0

Note: All students must ensure they only register for courses corresponding to their academic year, i.e. 100 level for Freshmen, 200 level for Sophomore, 300 level for Junior and 400 level for Senior year students.

Core Courses	Credits
BUSN 101	3
BUSN 121	3
BUSN 160	3
BUSN 170	3
BUSN 201	3
BUSN 206	3
BUSN 225	3
BUSN 250	3
BUSN 260	3
BUSN 280	3
BUSN 321	3
BUSN 330	3
BUSN 370	3
BUSN 371	3
BUSN 460	3
BUSN 491	3
BUSN 497	3
BUSN 498	3
BUSN 499	3
Core courses Total	57

Specializations

The Business department offers 5 specializations whose requirements are listed below.

Human Resource Management	
Specialization Courses	Credits
BUSN 350	3
BUSN 353	3
BUSN 354	3
BUSN 450	3
BUSN 451	3
BUSN 453	3
Specialization Courses Total	18

Business Analytics	
Specialization Courses	Credits
BUSN 341	3
BUSN 344	3
BUSN 347	3
BUSN 441	3
BUSN 444	3
BUSN 447	3
Specialization Courses Total	18

Accounting & Finance	
Specialization Courses	Credits
BUSN 301	3
BUSN 306	3
BUSN 322	3
BUSN 402	3
BUSN 405	3
BUSN 410	3
Specialization Courses Total	18

Marketing and Sales	
Specialization Courses	Credits
BUSN 382	3
BUSN 383	3
BUSN 385	3
BUSN 480	3
BUSN 484	3
BUSN 485	3
Specialization Courses Total	18

Operations Management	
Specialization Courses	Credits
BUSN 361	3
BUSN 364	3
BUSN 368	3
BUSN 461	3
BUSN 464	3
BUSN 469	3
Specialization Courses Total	18

Course Descriptions

General Education Course

BUSN 220: Entrepreneurship (2 credits)

This course is designed to promote entrepreneurial spirit and outlook among students, encouraging them to think critically, identify opportunities, and transform their ideas into successful ventures. It aims at imparting them with the requisite knowledge, skills and abilities, enabling them to seize the identified opportunities for initiating ventures and successfully navigating the challenges that come with starting a business and managing it. The course covers topics relevant to entrepreneurship including setting up and initiation of business (including requirements for registration and incorporation with regulators such as SECP and others), market research, opportunity identification, business planning, financial literacy for managing finances and securing funding, marketing and sales, team building and innovation. Overall, the course is geared towards personal growth and professional development for pursuing innovative ideas, availing opportunities and initiating start-ups.

Core Courses

BUSN 101: Principles of Accounting (3 credits)

This course introduces students to the fundamental principles, concepts, and practices of financial accounting. It covers the accounting cycle, recording of business transactions, preparation and analysis of financial statements, adjusting entries, internal controls, and ethical considerations in accounting. Students develop the ability to interpret financial information and understand the role of accounting in business decision-making.

BUSN 121: Microeconomics (3 credits)

This course introduces students to key microeconomic principles and their application to business and policy decisions. It covers scarcity, opportunity cost, demand and supply, consumer behavior, elasticity, production and cost analysis, market structures, pricing decisions, labor markets, and market failures. Students learn to use microeconomic tools to analyze individual, firm-level, and market-level decisions in local and global contexts.

BUSN 160: Management Communications (3 credits)

This course develops students' professional written and oral communication skills for business, academic, and career-related contexts. It covers business message planning, email and memo writing, negative and persuasive

messages, resume development, interviews, elevator pitches, presentation design, formal dressing, non-verbal communication, and ethical communication. Students learn to communicate clearly, professionally, and confidently across diverse formal situations.

BUSN 170: Principles of Management (3 credits)

This course introduces students to the fundamental principles and functions of management, including planning, organizing, leading, controlling, decision-making, organizational design, and employee motivation. It connects key management theories with practical business situations while addressing contemporary issues such as organizational culture, diversity, ethics, social responsibility, and environmental change. Students learn how managers apply management concepts to lead organizations effectively in dynamic business environments.

BUSN 201: Intermediate Accounting 1 (3 credits)

Prerequisite: BUSN 101

This course builds upon the foundational concepts introduced in Principles of Accounting by developing students' understanding of intermediate accounting theories, standards, and financial reporting practices. It covers preparation and analysis of financial statements, accounting for assets and liabilities, revenue recognition, corporate reporting, and the application of accounting standards within business environments. Students learn to apply accounting principles to more complex financial transactions, reporting requirements, and professional accounting practices.

BUSN 206: Management Accounting and Control (3 credits)

Prerequisite: BUSN 101

Topics include cost behavior, budgeting, variance analysis, performance measurement, responsibility accounting, and strategic cost management. Students learn how accounting information supports operational efficiency and managerial decision-making in business organizations. This course introduces students to the use of accounting information for managerial planning and decision.

BUSN 225: Economic Applications for Business (3 credits)

Prerequisite: BUSN 121

This course introduces students to key macroeconomic concepts and their relevance to business decision-making. It covers GDP, inflation, employment, interest rates, savings, aggregate demand and supply, business cycles, economic growth, and fiscal and monetary policy. Students learn to apply the economic way of thinking to understand macroeconomic conditions and their impact on businesses, markets, and society.

BUSN 250: Organizational Behavior (3 credits)

Prerequisite: BUSN 170

Impact of individuals, groups and structures on behavior within a formal organizational context and applying knowledge to improve the effectiveness of the organization. Concepts include leadership skills, team structures, managing interpersonal relationships and conflicts.

BUSN 260: Operations & Project Management (3 credits)

For Junior year students only Prerequisite: BUSN 170

Operations and Project Management is a discipline and profession that studies and practices the process of planning, designing, and operating production systems and subsystems, in both manufacturing and services,

to achieve the goals of the organization. To successfully deal with today's complex businesses, the operations management professionals and the project management professionals need an understanding of conventional organizational functions and the association of these functions to relevant stakeholders. This course covers key concepts including process design, capacity planning, forecasting, inventory management, quality management, supply chain integration, project management, productivity improvement, and material requirement planning. This course targets a systematic way of controlling and managing the inputs and outputs within the various business functions of an integrated supply chain, shaping the whole practice being socially, environmentally, and economically sustainable. Students will gain a thorough perspective of problems involved in valuable planning and execution of the supply chain, resulting in enhanced competitiveness for organizations.

BUSN 280: Marketing & Selling Skills (3 credits)

Prerequisite: BUSN 170

This course develops an understanding of key marketing concepts. It starts by introducing the marketing process. Subsequently, its different parts are individually discussed. Finally, all aspects of the marketing process are integrated. With short cases and relevant examples, this course provides a unique opportunity to equip students with the necessary background to undertake advanced courses in marketing and strategy.

BUSN 321: Financial Management (3 credits)

For Junior year students only Prerequisite: BUSN 201

This course introduces students to the fundamental concepts and techniques of financial management used in business decision-making. It covers financial statement analysis, time value of money, risk and return, capital budgeting, working capital management, financing decisions, and the role of financial markets and institutions. Students learn how financial managers evaluate investment opportunities, manage organizational resources, and make strategic financial decisions to maximize firm value.

BUSN 330: Entrepreneurship and Innovation (3 credits)

Prerequisite: BUSN 170

This course builds upon BUSN 220 by focusing on the development, validation, feasibility assessment, and presentation of entrepreneurial ventures. While BUSN 220 introduces students to the foundations of entrepreneurship, this course emphasizes advanced application through opportunity refinement, innovation, business model development, market research, strategic analysis, financial feasibility, pre-feasibility studies, and venture pitching. Students will work progressively on a business idea throughout the semester and will develop it through structured assignments, including innovation analysis, entrepreneur interviews, market research, business problem analysis, community engagement, feasibility work, and a final business plan presentation.

BUSN 370: Management Information Systems (3 credits)

For Junior year students only Prerequisite: BUSN 170

This course introduces students to the strategic role of information systems in organizations. It covers how information systems support business strategy, competitive advantage, decision-making, knowledge management, and organizational performance. The course emphasizes the managerial use of technology to align business needs with IT solutions and to understand the relationship between information systems, organizational structures, and business processes.

BUSN 371: Critical Thinking, Ethics, and Artificial Intelligence in Business (3 credits)

For Junior year students only Prerequisite: BUSN 170

This course provides students with a rigorous understanding of the role of critical thinking, ethical reasoning, and Artificial Intelligence (AI) in contemporary business environments. The course introduces students to key concepts including logical reasoning, decision-making, ethical frameworks, responsible AI, data ethics, bias in algorithms, digital transformation, and the societal and managerial implications of AI-driven business practices. Through case studies, discussions, and applied business scenarios, students will develop analytical and ethical decision-making skills necessary for responsible leadership in the age of artificial intelligence. The course is particularly suitable for business students seeking to understand how emerging technologies are reshaping organizations, markets, and professional responsibilities.

BUSN 460: Business Law (3 credits)

For Senior year Business students only

This course introduces students to the Pakistani legal system and its relevance to business. It covers general principles of law, dispute resolution, law and ethics, contracts, sales, torts, company law, consumer law, and business law in the global context. Students learn how legal frameworks affect business decisions, responsibilities, and organizational practices.

BUSN 491: Corporate Governance and Business Ethics (3 credits)

This course is designed to provide students with a comprehensive understanding of corporate governance and business ethics principles, including the roles and responsibilities of various stakeholders, the evolution of governance codes globally, and the application of governance in specific contexts such as corporate institutions and emerging markets. Through case studies and ethical analysis, students will develop critical thinking and ethical decision-making skills essential for responsible leadership in business.

BUSN 497: Business Strategy (3 credits)

For Senior year Business students only

This course develops students' understanding of how firms achieve competitive advantage and long-term success in dynamic business environments. It covers market analysis, strategic positioning, resource allocation, strategy formulation, implementation, and evaluation. Students learn to analyze business situations and apply strategic thinking to support effective organizational decision-making.

BUSN 498: Internship (3 credits)

All business students must undertake the internship in the summer after the Junior year.

It is a supervised internship course that provides students with an opportunity to apply business knowledge and skills in a professional workplace environment. The internship enables students to gain practical experience in organizational settings, develop professional competencies, and critically reflect on how academic learning in the business discipline informs real-world business practices. The course includes faculty supervision, workplace engagement, reflective learning, and academic assessment through activity logs, written reports, and a final viva or presentation.

BUSN 499: Capstone Project (3 credits)

For Senior year Business students only

Business Capstone Project is an integrative course that allows students to apply the knowledge, skills, and competencies developed throughout the BBA program. The course emphasizes applied learning, analytical

thinking, interdisciplinary integration, professional judgment, ethical awareness, and independent academic work. Students are expected to demonstrate the ability to connect business theory with practical and academic business problems.

BUSN X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

Specializations

The Business department offers 5 specializations whose requirements are listed below.

1. Specialization in Operations Management:

Program Objectives

- i. **Foundation:** Gaining foundational knowledge in business management to understand key concepts and best practices.
- ii. **Application:** Applying business management knowledge to address local and global business needs.
- iii. **Integration:** Utilizing decision-making tools and techniques to apply knowledge in emerging business environments, integrating various business disciplines.
- iv. **Human Dimension:** Developing interpersonal, communication and leadership skills that enhance effective business management.
- v. **Caring:** Encouraging ethical decision-making and fostering a caring attitude toward society.
- vi. **Learning to Learn:** Developing self-directed learners for dynamic business environments.

Program Learning Outcomes

- i. **Foundational Knowledge**
Demonstrate a thorough understanding of key concepts and best practices in business Management.
- ii. **Application**
Apply business management knowledge and analytical skills to solve real-world business challenges in local and global contexts.
- iii. **Integration**
Utilize decision-making tools and interdisciplinary approaches to analyze and adapt to emerging business environments.
- iv. **Human Dimension**
Exhibit effective interpersonal, communication, and leadership skills necessary for managing teams and organizations successfully.
- v. **Caring**
Foster ethical decision-making in business and a commitment to positive societal impact.
- vi. **Learning to Learn**
Develop self-directed learning skills, embracing continuous personal and professional growth in a dynamic business landscape.

BUSN 361: Operational Excellence (3 credits)

Prerequisite: BUSN 360

Operational Excellence is a course designed to develop students' understanding of continuous improvement, efficiency enhancement, and sustainable business performance. The course focuses on the principles, tools, and techniques used to optimize organizational processes, improve productivity, reduce waste, and enhance quality and customer satisfaction. Students will explore concepts such as Lean Management, Six Sigma, Total Quality Management, process improvement, performance measurement, and operational strategy. Emphasis is placed on problem-solving, innovation, teamwork, and data-driven decision-making to achieve competitive advantage. Through case studies and practical applications, students will gain the skills required to create efficient, agile, lean and high-performing organizations.

BUSN 364: Production, Planning & Control (3 credits)

Must be taken at the same time as BUSN 360 for students specializing in Operations Management.

Production Planning is a course that focuses on the effective planning, scheduling, coordination, and control of production activities within manufacturing and service organizations. The course provides students with knowledge of forecasting, capacity planning, inventory management, material requirements planning, Distribution requirement planning, aggregate planning, and production scheduling techniques to ensure efficient utilization of resources. Students will learn how to balance demand and supply, minimize production costs, and timely delivery. Emphasis is placed on decision-making, process optimization, and integration of modern technologies in production systems. Practical applications and skill development exercises help students to develop analytical and managerial skills for efficient production planning and control.

BUSN 368: Productivity Management (3 credits)

Prerequisite: BUSN 360

Productivity Management is a course that examines the principles and techniques used to improve organizational efficiency, effectiveness, and performance. This course focuses on measuring productivity, optimizing resource utilization, reducing waste, and enhancing quality through continuous improvement practices. Students will explore productivity tools, performance measurement systems, process improvement strategies, and workforce management techniques. Emphasis is placed on innovation, operational efficiency, and sustainable growth. This course aims to develop the analytical and managerial skills needed to enhance productivity and organizational competitiveness through experiential learning.

BUSN 461: New Product Development (3 credits)

Prerequisite: BUSN 361

New Product Development is a course that focuses on the systematic process of idea conceiving, creating, designing, and launching innovative products to meet customer needs and achieve competitive advantage. This course covers key concepts like idea generation, market research, product design, feasibility analysis, prototyping, testing, commercialization, and product life cycle management. Students will explore creativity techniques, innovation strategies, cross-functional teamwork, and project management practices used in successful product development. Emphasis is placed on customer value, sustainability, quality, and market responsiveness.

BUSN 464: Total Quality Management (3 credits)

Prerequisite: BUSN 361

Total Quality Management is a course that focuses on the principles, philosophies, and practices used to achieve continuous improvement in organizational performance and customer satisfaction. TQM course

emphasizes quality planning, quality control, process improvement, teamwork, leadership, and customer-focused management strategies. Students will explore key concepts such as Quality Assurance, Six Sigma, Lean Management, benchmarking, statistical quality control, and ISO standards. Emphasis is placed on developing a culture of continuous improvement and operational excellence across all organizational functions through innovative kaizen circles, Lean Six Sigma and Lean Production. Applied industry-led projects and research articles will enhance the knowledge of students of TQM.

BUSN 469: Project Management Processes (3 credits)

Prerequisite: BUSN 361

Project Management Processes is a course that provides students with a comprehensive understanding of the systematic processes involved in planning, executing, monitoring, controlling and closing projects successfully. The course focuses on key project management areas including scope, time, cost, quality, risk, scope creep, communication, procurement, human resource management and stakeholder management. Students will learn how to apply project management tools, techniques like Gantt Chart, CPM, PERT, and methodologies to achieve project objectives within defined constraints. Project Activity Crashing is one of the powerful techniques to set the project within timelines with increased budget cost. Emphasis is placed on leadership, teamwork, decision-making, and problem-solving in dynamic project environments. Through practical exercises, case studies, and project simulations, students will develop the project management skills required to manage projects effectively and efficiently. Learning of Project Management softwares likes MS Project or JIRA is the added feature.

2. Specialization in Marketing and Sales:

Program Objectives

- i. **Foundation:** Gaining foundational knowledge in business management to understand key concepts and best practices.
- ii. **Application:** Applying business management knowledge to address local and global business needs.
- iii. **Integration:** Utilizing decision-making tools and techniques to apply knowledge in emerging business environments, integrating various business disciplines.
- iv. **Human Dimension:** Developing interpersonal, communication and leadership skills that enhance effective business management.
- v. **Caring:** Encouraging ethical decision-making and fostering a caring attitude toward society.
- vi. **Learning to Learn:** Developing self-directed learners for dynamic business environments.

Program Learning Outcomes

- i. **Foundational Knowledge**
Demonstrate a thorough understanding of key concepts and best practices in business Management.
- ii. **Application**
Apply business management knowledge and analytical skills to solve real-world business challenges in local and global contexts.
- iii. **Integration**
Utilize decision-making tools and interdisciplinary approaches to analyze and adapt to emerging business environments.
- iv. **Human Dimension**

Exhibit effective interpersonal, communication, and leadership skills necessary for managing teams and organizations successfully.

v. Caring

Foster ethical decision-making in business and a commitment to positive societal impact.

vi. Learning to Learn

Develop self-directed learning skills, embracing continuous personal and professional growth in a dynamic business landscape.

BUSN 382: Marketing Communications (3 credits)

Prerequisite: BUSN 280

The course entails an in-depth study of the key concepts of integrated marketing communications with a focus on modern communication tools and practices. Students will be able to comprehend how marketing communications are used to support strategic business goals in various organizations, particularly in the local/Pakistani context.

BUSN 383: Sales Management (3 credits)

Prerequisite: BUSN 280

The course equips students with the knowledge and skills necessary to manage a sales force effectively within a competitive business environment. Students will learn how to develop, implement, and evaluate strategic and tactical sales initiatives to drive performance. A key focus is placed on the integration of sales with marketing functions, customer relationship management (CRM), and the use of technology to enhance efficiency. Students will also explore leadership and motivational strategies, sales force design, forecasting, and budgeting techniques that contribute to the creation of high-performing teams.

BUSN 385: Consumer Behavior (3 credits)

Prerequisite: BUSN 280

The course offers an in-depth examination of consumer behavior, exploring the psychological, social, and cultural factors that influence consumer decisions. Students will understand how motivations, attitudes, perceptions, personality, and social influences shape consumer behavior. By the end of this course, students will be equipped with the analytical skills to interpret and predict consumer actions, essential for customer-driven marketing strategies.

BUSN 480: Marketing Research (3 credits)

Prerequisite: BUSN 383

The course introduces the principles, methods, and applications of marketing research to support business decision-making. Students gain skills to design, conduct, and interpret research for real-world marketing problems, covering the entire process from problem definition to data analysis and reporting. Both qualitative and quantitative techniques, including surveys, focus groups, experiments, and analytics, are emphasized through practical learning.

BUSN 484: Brand Management (3 credits)

Prerequisite: BUSN 383, BUSN 385

This intensive course equips marketing professionals with strategic tools to build, manage, and scale brands using a customer-based cognitive perspective. By moving through four core modules, defining, building, tracking, and advancing- participants master fundamental concepts such as brand identity, positioning, and

architecture. The curriculum bridges the gap between theory and practice by integrating digital storytelling, ethical management, and data-driven decision-making. By blending industry expert insights with personal branding sessions, the program prepares the next generation of leaders to steward brands that resonate deeply and maintain long-term equity in competitive global markets.

BUSN 485: E-Business (3 credits)

Prerequisite: BUSN 370

In today's fast-paced digital landscape, businesses must constantly evolve to remain competitive. This course aims to provide a comprehensive understanding of the relationship between e-business, e-commerce, and digital transformation. Students will explore strategic tools for online marketing, customer relationship management, and supply chain integration, learning to develop innovative digital business models and strategies.

3. Specialization in Accounting & Finance:

Program Objectives

- i. **Foundation:** Gaining foundational knowledge in business management to understand key concepts and best practices.
- ii. **Application:** Applying business management knowledge to address local and global business needs.
- iii. **Integration:** Utilizing decision-making tools and techniques to apply knowledge in emerging business environments, integrating various business disciplines.
- iv. **Human Dimension:** Developing interpersonal, communication and leadership skills that enhance effective business management.
- v. **Caring:** Encouraging ethical decision-making and fostering a caring attitude toward society.
- vi. **Learning to Learn:** Developing self-directed learners for dynamic business environments.

Program Learning Outcomes

- i. **Foundational Knowledge**
Demonstrate a thorough understanding of key concepts and best practices in business Management.
- ii. **Application**
Apply business management knowledge and analytical skills to solve real-world business challenges in local and global contexts.
- iii. **Integration**
Utilize decision-making tools and interdisciplinary approaches to analyze and adapt to emerging business environments.
- iv. **Human Dimension**
Exhibit effective interpersonal, communication, and leadership skills necessary for managing teams and organizations successfully.
- v. **Caring**
Foster ethical decision-making in business and a commitment to positive societal impact.
- vi. **Learning to Learn**
Develop self-directed learning skills, embracing continuous personal and professional growth in a dynamic business landscape.

BUSN 301: Financial Reporting (3 credits)

Prerequisite: BUSN 201

This course focuses on advanced financial reporting concepts and the preparation of financial information for various stakeholders. It covers financial statement analysis, reporting standards, stakeholder equity, leases, earnings per share, financial instruments, and disclosure requirements. Students develop the ability to interpret and evaluate financial reports within professional and regulatory contexts.

BUSN 306: Islamic Finance (3 credits)

This course explores the foundations, instruments, and innovations of Islamic finance, covering Shariah principles, financial mathematics, and contract structures like Mudarabah, Musharakah, Murabaha, and Sukuk. It emphasizes risk-sharing, ethical finance, and Fintech integration through case studies and project-based learning, preparing students for real-world Islamic financial product development.

BUSN 322: Corporate Finance (3 credits)

Prerequisite: BUSN 321

This course develops students' understanding of corporate financial decision-making and value maximization. Topics include capital budgeting, capital structure, dividend policy, risk and return analysis, valuation techniques, working capital management, and financing decisions. Students learn how organizations make strategic financial decisions in domestic and international financial environments.

BUSN 402: Auditing and Taxation (3 credits)

Prerequisite: BUSN 301

This course provides a comprehensive understanding of auditing and taxation principles as they apply in the business environment. Students will explore statutory and internal auditing processes, assurance engagements, and ethical considerations, alongside a detailed study of income tax laws applicable to individuals and businesses in Pakistan. The course aims to develop students' ability to apply audit techniques, assess internal controls, interpret tax laws, and prepare tax returns. With a strong emphasis on analytical thinking, regulatory compliance, and ethical judgment, students will gain the essential skills needed to operate effectively in accounting and assurance services.

BUSN 405: Investment and Portfolio Analytics (3 credits)

Prerequisite: BUSN 321

Investment Portfolio Analytics introduces students to the principles and practices of investment decision-making with a strong emphasis on analytical tools and data-driven strategies. The course explores the fundamentals of financial markets, security analysis, portfolio construction, and performance evaluation. Students will learn how to assess risk and return, apply modern portfolio theory, and use quantitative methods and financial technologies to analyze and optimize investment portfolios. Through practical case studies and the use of analytical software, students will develop hands-on skills to make informed investment decisions in dynamic financial environments.

BUSN 410: Financial Technology (3 credits)

Prerequisite: BUSN 301

This course introduces students to the rapidly evolving world of Financial Technology (Fintech), exploring how innovation is transforming traditional financial services. Students will examine core Fintech areas including digital payments, blockchain, crowdfunding, robo-advisory, InsurTech, and RegTech. Through lectures, case studies, and a final project, students will develop interdisciplinary skills combining finance,

technology, and business strategy. By the end of the course, students will be prepared to engage critically with emerging Fintech trends and their impact on global finance.

4. Specialization in Human Resource Management:

Program Objectives

- i. **Foundation:** Gaining foundational knowledge in business management to understand key concepts and best practices.
- ii. **Application:** Applying business management knowledge to address local and global business needs.
- iii. **Integration:** Utilizing decision-making tools and techniques to apply knowledge in emerging business environments, integrating various business disciplines.
- iv. **Human Dimension:** Developing interpersonal, communication and leadership skills that enhance effective business management.
- v. **Caring:** Encouraging ethical decision-making and fostering a caring attitude toward society.
- vi. **Learning to Learn:** Developing self-directed learners for dynamic business environments.

Program Learning Outcomes

- i. **Foundational Knowledge**
Demonstrate a thorough understanding of key concepts and best practices in business Management.
- ii. **Application**
Apply business management knowledge and analytical skills to solve real-world business challenges in local and global contexts.
- iii. **Integration**
Utilize decision-making tools and interdisciplinary approaches to analyze and adapt to emerging business environments.
- iv. **Human Dimension**
Exhibit effective interpersonal, communication, and leadership skills necessary for managing teams and organizations successfully.
- v. **Caring**
Foster ethical decision-making in business and a commitment to positive societal impact.
- vi. **Learning to Learn**
- vii. Develop self-directed learning skills, embracing continuous personal and professional growth in a dynamic business landscape.

BUSN 350: Human Resource Management (3 credits)

Prerequisite: BUSN 170

This course introduces students to the fundamental concepts and practices of Human Resource Management. It examines how organizations manage people to achieve individual, team, and organizational goals. Key areas include human resource planning, recruitment and selection, training and development, performance management, compensation, employee relations, and workplace ethics. Through theory, practical examples, and case discussions, students will understand the strategic role of HRM in improving organizational effectiveness.

BUSN 353: Human Resource Practice (3 credits)

Prerequisite: BUSN 350

This course exposes students to the practical application of Human Resource Management in different organizational settings in Pakistan and abroad. It focuses on how HR systems, policies, and practices are designed and implemented in real workplace situations. Through extensive use of case studies, students will analyze HR challenges related to recruitment, training, performance management, compensation, employee relations, and organizational change. The course helps students connect HR theory with practice and develop problem-solving skills for managing people effectively.

BUSN 354: Performance Management (3 credits)

Prerequisite: BUSN 350

This course introduces students to the principles and practices of managing employee performance in organizations. It covers performance appraisal systems, annual performance reviews, goal setting, feedback, performance standards, and the connection between performance and rewards. Students will also examine how clear roles, responsibilities, and work expectations help organizations improve employee performance in a fair and systematic manner.

BUSN 450: Labor Law (3 credits)

Prerequisite: BUSN 350

This course introduces students to important areas of labor law, with reference to Pakistan. It covers basic legal concepts related to employment, contracts, workplace rules, wages, benefits, incentive schemes, pension and provident fund regulations, and minimum wage laws. The course helps students understand how labor laws shape employment relationships and workplace practices.

BUSN 451: Strategic HRM (3 credits)

Prerequisite: BUSN 350

This course focuses on the strategic role of human resource management in achieving organizational goals. It examines how HR policies and practices can be aligned with business strategy, and how people can be developed and managed as a source of long-term value. Topics include human capital, workforce planning, talent management, employee development, and the role of HR in organizational performance.

BUSN 453: Change Management (3 credits)

Prerequisite: BUSN 350

This course introduces students to the process of managing change in organizations. It examines why organizations need to change, how change can be planned and implemented, and why employees often resist change. The course also looks at the role of leadership, communication, motivation, teamwork, and organizational culture in helping change efforts succeed.

5. Specialization in Operations Business Data Analytics

BUSN 341: Statistics for Business Data Analytics (3 credits)

Prerequisite: MATH 100 or another MATH or STAT course

This course provides a foundational understanding of statistical methods and their application in business data analysis. Students will learn to collect, analyze, and interpret data to inform strategic business decisions. Topics include descriptive statistics, probability theory, hypothesis testing, and regression analysis, with an emphasis on using statistical software for practical application.

BUSN 344: Programming Language for Business Data Analytics (3 credits)

Prerequisite: CSCS 100

This course introduces students to the fundamentals of a modern programming language (e.g., Python) and its application in business data analysis. Students will learn to write code for data manipulation, cleaning, exploratory analysis, and visualization. The course emphasizes practical, hands-on application to prepare students for a variety of data-driven roles.

BUSN 347: Database Systems and Cloud Computing (3 credits)

Prerequisite: CSCS 100

This course equips students with the essential knowledge of modern data management. Students will learn about relational and non-relational database systems, data modeling, and the use of SQL to query and manipulate data. The course also introduces cloud computing concepts and services, focusing on how these platforms are used to store, process, and manage large-scale business data.

BUSN 441 Business Modelling and Intelligence (3 credits)

Prerequisite: BUSN 341

This course combines the strategic art of business modeling with technical practice of business intelligence. Students will learn to use data to build quantitative models for forecasting, what-if analysis, and optimization. The course then focuses on translating these models and their outcomes into compelling data visualizations and interactive dashboards using leading BI tools, enabling more effective and data-driven decision-making.

BUSN 444: Machine Learning for Business

Prerequisite: BUSN 341 and BUSN 344

This course introduces the concepts and applications of machine learning, focusing on its use as a strategic tool for business decision-making. Students will learn the fundamentals of supervised and unsupervised learning algorithms and their practical application to business problems such as customer churn prediction, fraud detection, and demand forecasting. The course will emphasize hands-on projects using a programming language (e.g., Python) and its machine learning libraries.

BUSN 447 AI: Fundamentals and Business Applications

Prerequisite: BUSN 370

This course provides a comprehensive overview of Artificial Intelligence (AI), focusing on its core concepts, technologies, and strategic applications in business. Students will explore areas such as natural language processing (NLP), generative AI, and large language models (LLMs). The course will emphasize the ethical considerations, implementation strategies, and future implications of AI for organizations, preparing students to become AI-literate business leaders.

Department of
SOCIOLOGY

Bachelor of Science in Sociology

Introduction

Sociology is the systematic study of human society. It looks at social behavior, culture, social institutions, grouping within a society, and the relationships between different groups in society. Sociology helps us to identify social issues within our society and gives us conceptual tools to understand those issues so that we can have a society that is better for all of its members. Students who study Sociology will learn to look at their own society and other societies in new ways, to question assumptions, to understand the society from the research of others, to record observations from a neutral position and to analyze them objectively, and to interpret their findings, all within an ethical framework. Sociology helps orient students in terms of their future life, so that they learn skills helpful in many areas of business, government, and nongovernmental organizations, as well as their personal lives. The Department of Sociology's goal is to have students understand themselves and society at a family, group, local, regional, national, and global level.

The Sociology department is part of the Faculty of Social Sciences.

Degrees Offered

The Department of Sociology at Forman Christian College (A Chartered University) offers, a Bachelor of Science in Sociology degree. The emphasis is on the development of sociological skills, observing, analyzing and interpreting the events of human behavior of individuals and groups at the family, group, societal, national and global levels. In addition, the Department of Sociology also offers three minors: a Minor in Sociology, a Minor in Criminology and a minor in Demography and Population Studies.

Mission of BS Sociology Program

The Mission of the Department of Sociology, BS Sociology degree program is to prepare students in various professional occupations in Sociology and/ or the pursuit of advanced degrees in Sociology by educating them in the fundamental concepts, theories, research, service, and ethics of Sociology.

Program Objectives BS Sociology

- i. To teach students concepts and theories in sociology, anthropology, gender studies, development, and cultural studies
- ii. To develop students' understanding of research methods, analysis techniques and communication skills used in Sociology and other Social Sciences
- iii. To teach students about the importance of values such as community service, integrity, respect and dignity of humans, and discipline in addition to professional ethics in sociology and other social sciences
- iv. To facilitate students in developing and recognizing various professional skills and opportunities in Sociology and other social sciences.

Program Learning Outcomes

- i. Students will demonstrate understanding of core sociological and/or anthropological concepts and theories by critically conceptualizing, theorizing, and objectively evaluating social patterns and problems
- ii. Apply sociological knowledge and thinking to real-world situations.

- iii. Students will demonstrate understanding of how to formulate a research project, and to identify social problems and issues, using sociological theories
- iv. Students will plan and execute a quantitative and/or qualitative research methods project in sociology, including all elements of a social research to draw evidence-based results, and present theoretically informed recommendations to address current social problems.
- v. Students will demonstrate effective communication to convey sociological research including literature, data, and results to a broader audience
- vi. Students will demonstrate, practice and display general, professional, and research ethics and values
- vii. Students will describe and list various educational and professional opportunities available to sociologists and social scientists, and be able to identify a career path for themselves

Requirements for BS Sociology Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
9 (27 credits)	27 credits	54	2.0

At least 8 Sociology courses must be at the 300/400 level.

Core Courses	Credits
SOCL 100	3
SOCL 201/ CRIM 201	3
SOCL 202	3
SOCL 301	3
SOCL 325	3
SOCL 350	3
SOCL 494	3
SOCL 498	3
SOCL 499	3
Core courses Total	27
Elective Courses	Credits
Any course offered by the department	27
Electives courses Total	27

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Important Note:

Students majoring in Sociology can opt for Criminology and/or Demography and Population Studies as a minor.

Requirements for the Minor in Sociology

Core Courses	Elective Courses	Total Credits	Required Major CGPA
2 (6 credits)	4 (12 credits)	18	2.0

At least 2 courses must be at the 300/400 level.

Core Courses	Credits
SOCL 100	3
SOCL 301	3
Core courses Total	6
Elective Courses	Credits
Any course offered by the department	12
Electives courses Total	12

Requirements for the Minor in Criminology

Core Courses	Elective Courses	Total Credits	Required Major CGPA
2 (6 credits)	4 (12 credits)	18	2.0

At least 2 courses must be at the 300/400 level.

Core Courses	Credits
CRIM 110	3
CRIM 220 / SOCL 220	3
Core courses Total	6
Elective Courses	Credits
Any course offered by the department	12
Electives courses Total	12

Requirements for the Minor in Demography and Population Studies

Core Courses	Elective Courses	Total Credits	Required Major CGPA
2 (6 credits)	4 (12 credits)	18	2.0

At least 2 courses must be at the 300/400 level

Core Courses	Credits
SOCL 110	3
STAT 103	3
Core courses Total	6
Elective Courses	Credits
ECON 210	3
GEOG 233	3
GEOG 323	3
SOCL 335	3
GEOG 424	3
SOCL 435	3
Electives courses Total	12

Course Descriptions

SOCL 100: Introduction to Sociology (3 credits)

Presents fundamental concepts of sociology. Helps students to observe and understand the actions, beliefs, and interactions of people in their own and other societies and to think critically about themselves in relation to social structures in their own and other societies at the individual, group and societal levels. Analyzes current social issues in Pakistan and other countries in terms of sociological concepts. Assignments focus on analyzing and interpreting social issues in societies around the world in sociological terms with practical assignments that reinforce classroom learning.

SOCL 101: Introduction to Social Work (3 credits)

Introduces the values, ethics, history and methods of professional social work practice with particular emphasis on the profession in Pakistan. The course helps students understand the impact of social and economic problems on individuals, families, and communities. The course introduces basic social work principles and techniques to help people help themselves and improve the quality of their lives when experiencing problems due to societal and economic factors such as poverty, homelessness, social discrimination, substance abuse, or involving family issues like parent-child conflict, marital conflict, or caring for aged relatives.

SOCL 103: Introduction to Social Policy (3 credits)

Social policy is the response of government and society to challenges rooted in culture, economics, work, and health. The objective of social policy is to promote holistic wellbeing across society, especially in consideration of marginalized groups such as the impoverished, elderly, women, children, refugees and racial and religious minorities. This course will aim to identify the social factors that shape the process through which social policies are made.

Theories of social policy and evidence of policy implementation which promote wellbeing will be discussed, along with their limitations and strengths. The difficulties in simulating social policy across countries due to differences in social and environmental backgrounds and the importance of developing region-specific policies will also be deliberated. Some of the specific topics that will be covered during the course include ethics and risk; employment and poverty reduction; access to education and healthcare; housing and equality; and globalization and welfare.

SOCL 104: Civics and Community Engagement (2 credits) General Education course

This course is designed to provide students with fundamental knowledge about civics, citizenship, and community engagement. Students will learn about the essentials of civil society, government, civic responsibilities, inclusivity, and effective ways to participate in shaping the society which will help them apply theoretical knowledge to real-world situations to make a positive impact on their communities.

SOCL 110: Introduction to Demography (3 credits)

Demography is the study of human populations and the processes through which populations change. The course will familiarize students with conceptual frameworks and important issues within the discipline of demography and population studies. The relationship between social context and individuals' demographic patterns will be established through existing population based research. The course will also explore the key concepts and measures used to study population dynamics. The course will also identify and apply major theoretical perspectives in demography and assess the conceptual differences among them.

SOCL 115: Anthrozoology (3 credits)

This course aims to provide students with fundamental knowledge about sociological theoretical frameworks which are used to examine how law is shaped by society and how laws affect society and human rights. Students will also learn about how the law influences and is influenced by social change and social realities-including gender, class, ethnicity, politics, and the economy. Specific issues such as social control and the enforcement of law, Pakistani laws, globalization of law and human rights in context of international law will also be explored.

SOCL 118: Sociology of Law and Human Rights (3 credits)

This course aims to provide students with fundamental knowledge about sociological theoretical frameworks which are used to examine how law is shaped by society and how laws affect society and human rights. Students will also learn about how the law influences and is influenced by social change and social realities-including gender, class, ethnicity, politics, and the economy. Specific issues such as social control and the enforcement of law, Pakistani laws, globalization of law and human rights in context of international law will also be explored.

SOCL 150: Sociology of Globalization (3 credits)

Presents theories of globalization and the history of globalization, various ways in which societies and cultures have been transformed by this phenomenon, analysis of globalization's impact on Pakistani society.

SOCL 170: Environmental Sociology (3 credits)

There exists a dialectic relationship between society and our natural environment. The course will be an attempt to introduce the field of environmental sociology enabling students to see the environment in the light

of sociological perspectives. The central aim of this course is to illuminate the students about the relationship that they have with their environment, highlighting how ecological issues are social problems.

SOCL 180 Education, Equality and Educational Policy (3 Credits)

The study of the sociology of education is a service sociology, providing practical knowledge for advancing solutions for social inequality and educational policy. This course will study sociological theories which explain the role of education in society and the causes of differential achievement. One of the aims will be to challenge social assumptions of school success and failure, in recognition of socio-demographic differences. The relationship between educational institutes and wider social structures will be explored, such as the family, classroom, teachers, culture, and the economy. The educational system and educational policies of Pakistan will also be covered.

SOCL 192/PHYS 192 *: Science & Society (3 credits)

Presents theories of globalization and the history of globalization, various ways in which societies and cultures have been transformed by this phenomenon, analysis of globalization's impact on Pakistani society.

SOCL 201/ CRIM 201 *: Sociological Research (3 credits)

Prerequisites: SOCL 100 or another introductory course in Social Sciences.

This is the basic research course in Sociology. Students learn the comparative advantages and limitations of different research orientations, the ethics of research, and strategies and techniques including experiments, field observations, interviewing, unobtrusive research and surveys. There will also be a brief introduction to qualitative research. In addition to lectures and discussions, students will learn basic research methods and techniques by doing a social research project of their own. This will help to develop skills in observation, interviewing, hypothesis building, theory building, questionnaire construction, some basic statistical tests, computer data manipulation, data interpretation, and research report writing.

SOCL 202: Qualitative Methodology (3 credits)

Prerequisites: SOCL 100

The course will present various qualitative methods of social research. It will use readings that describe methods, writings (articles and chapters from books) based on qualitative research. Students will carry out exercises that practice these methods such as Participant Observation, open-ended interviewing; photography, and possibly a focus (focal) group situation.

SOCL 203: Social Statistics (3 credits)

Prerequisites: SOCL 100

This course will provide an introduction of statistical concepts and methods to students majoring in the social sciences, enabling them to investigate social issues and discover empirical evidence through statistics. The course provides students with expertise and tools to conduct research quantitatively, from the stage of planning of data analysis to data presentation. Different statistical tools and techniques will be covered including parametric and non-parametric tests. Classes will mainly be held in computer labs in order to do practical exercises using the statistical software SPSS. Students will become familiar with generating descriptive statistics, tests of association and significance, and regression tests.

SOCL 220/CRIM 220 *: Introduction to Criminology (3 credits)

Introduces the Sociological basis to understand deviancy and criminal behavior, causes and consequences of crime, responses to crime and historical transition of ideas about crime. It also presents tools for the scientific investigation of criminal behavior.

SOCL 223: Social and Cultural Anthropology (3 credits)

Studies cultures and social groups throughout the world with an emphasis on looking at a culture from the perspective of someone in that culture. It introduces tools for more effective intercultural communications and a mirror in which to see our own cultural group more clearly. The course covers the basics of cultural concepts and ethnographic description.

SOCL 270: Sociology of Inequality (3 credits)

Prerequisite: SOCL 100

Investigates the concept of social inequality, concentrating on class, gender, religion and ethnicity as relations of domination along with a structural analysis of these social relationships, their links with each other, and their effects on societies and individuals.

SOCL 290: Political Sociology (3 credits)

Prerequisite: SOCL 100

This course explores the relation between politics and society to understand how they shape and are in turn shaped by each other. It is based on the assumption that the political field does not exist independently of social institutions. The course is organized in five thematic areas. First, we will study the modern state, paying particular attention to the process of its formation. Then we consider various theories of power. The third theme is the relation of identity and politics which we will investigate by looking at nationalism, gender and ethnicity as well as religion. We then shift our focus to social movements and their relevance for politics. The last part of the course is on the global dimensions of politics. We will cover a wide range of theories and empirical cases to make sense of these social dimensions of politics.

SOCL 291: Economic Sociology (3 credits)

Prerequisite: SOCL 100

Explores the sociology of market and economic activity and the ways that economic activities are modified or impeded by social relations and social institutions. Explores the influence of social institutions and processes on economies and economic decisions.

SOCL 301: Theoretical Perspectives in Sociology (3 credits)

Prerequisite: SOCL 100

Examines the structure and scope of sociological theorizing, how to “use information and develop reason in order to achieve lucid summations of what is going on in the world and of what may be happening within ourselves” (Wright Mills 1959:5); it presents the theory, method and object of investigation of some masters of Sociological thought. It also explains Sociological theory as the basis for Sociological research.

SOCL 302: Contemporary and Postcolonial Social Theory (3 credits)

Prerequisite: SOCL 301

This course provides advanced study of contemporary social theory alongside selected post-colonial and decolonial perspectives. Core themes include modernity and social transformation, race and capitalism, gender and representation, coloniality of power, and the global formation of sociological knowledge. The course

situates classical and contemporary arguments within broader historical trajectories and examines how key sociological categories have been reformulated across different intellectual contexts. Through sustained engagement with primary texts, students develop greater precision in theoretical analysis and a deeper understanding of the evolving contours of sociological thought.

SOCL 305/ISLM 305/CRST 354 *: Religion and Inequality (3 credits)

This is a seminar course which focuses on dialogue about religion with a focus on learning what people believe and what people of different religious traditions have in common to work toward a peaceful society. It also speaks to inequalities between religious communities and seeks ways to provide religious freedom and civil rights to all religious communities.

SOCL319: South Asian Languages and Culture (3 credits)

South Asia is a region characteristic of language diversity. This course seeks to understand the relationship between language and cultures of South Asia. Major intellectual works and currents of classical times which created traditions of literature, philosophy, and religion that continue to inspire modern South Asia will be explored. The course will also critically analyze issues related to language identity and the state role in language policies.

SOCL 325: Sociology of Gender (3 credits)

Prerequisite: SOCL 100

Analyzes cultural values, social institutions and theories in the construction of gender. The course analyzes gender inequality in contemporary societies and explores the social experience of gender across different cultures and societies. It discusses legal statutes and the social placement of gender as an outcome of socioeconomic and cultural environment.

SOCL 335: Demography and Public Health (3 credits)

Demography and public health have been closely related for over three centuries, as the health and healthcare needs of a population cannot be measured or met without knowledge of its size and characteristics. Demography is concerned with this and with understanding population dynamics - how populations change in response to the interplay between fertility, mortality, and migration. This understanding is a prerequisite for making forecasts about future population size and structure which should underpin healthcare planning. The purpose of this course will be to extend the students' knowledge of how demographic factors impact basic health issues, of social and economic changes affecting and resulting from health events, and the utility of demographic methodological applications for understanding and managing health-related events and outcomes. The effects of demographic events such as aging, childbearing, changing family structures, racial/ethnic differentials, migration, urbanization, poverty, and mortality etc. on public health issues such as health expenditures, health outcomes, prevention and reduction of diseases and disorders, and improving health care will be explored. The course through the use of rigorous research readings will also attempt to introduce students to the role of applied demography in informing and supporting public health issues.

SOCL 350: Sociology of Development (3 credits).

Prerequisite: SOCL 100

Looks in-depth at development, including theories of development, the impact of development assistance programs, and insights from a wide variety of development models and experiences.

SOCL 355: Sociology of Media (3 credits)

Prerequisite: SOCL 100

Investigates the social institution of the media and its impact on society, the analysis of the social structure of media organizations, major theories of media effects and their application to Pakistani and other societies. It examines the representation of different social groups by media and teaches methods appropriate for media research.

SOCL 363 Linguistic Anthropology (3 credits) *Prerequisites: SOCL 100 and SOCL 223.*

Linguistic Anthropologists see 'language' as a cultural resource and 'speaking' as a cultural practice. Language is, therefore, both a mode of thinking and a means of practice. According to Alessandro Duranti, language is active. It is a form of action that presupposes and at the same time brings about ways of being in the world, including what is not said.

This course looks at some common ways that language is socially situated and culturally mediated. We will begin with a review of some tools used by linguists, create a baseline of what is meant by 'cultural mediation', and then explore a selection of writing and case studies produced by Linguistic Anthropologists. Our focus is speech communities distinguished by focal activity, as in the case of Sign Language and Hip Hop, and by social practices linked to race, social class, and gender.

SOCL 370: Visual Anthropology: South Asia (3 Credits)

Prerequisites: SOCL 100 and SOCL 223

The well-known anthropologist Claude Levi-Strauss, who in 1935-39 took more than 3,000 photographs during his teaching years at Sao Paulo University (Brazil), is credited with marginalizing photography in anthropology. As David Sapir explains, he devalued visual images for their imaginary dimension and their subjectivity (Sapir 1994, p. 875). Levi-Strauss privileged writing, which is no less an image construed as thought in scripted form, over other visual representations to support his theory of structural relationship. The argument is complex, but our class will consider one particular issue for still photography.

Is there a different process of reading/comprehending visuals than there is in reading/comprehending written text? Viewed in another way: Does photography entail different interpretive skills than written materials? In exploring this prompt, consider how "Culture" mediates visualization.

SOCL 390: Cities and Urban Lives (3 credits)

Prerequisite: SOCL 100

This course approaches cities and urban lives from three different perspectives. First, it looks at the city as an infrastructure, both material and non-material, that shapes social organization and behavior. Classic Sociological questions of the relation of the city to modernity and space and social control will be addressed here. Secondly, the course traces ongoing urban transformations with a special focus on cities of the Global South. Issues taken up here include migration, marginalization, and globalization.

The final section of the course focuses on public practices in the city by looking at the overlapping realms of religion, market, culture, and community life. An understanding of these three distinct approaches will enable students to understand the city from a range of viewpoints. In focusing on a diverse range of urban forms, trends, and practices, this course also goes beyond the limitations imposed by focusing on cities only through the lens of development and progress.

SOCL 410: Sociology of Art & Culture (3 credits)

Prerequisites: SOCL 100

Investigates cultural issues sociologically using a range of theoretical approaches to the Sociology of culture, exploration of Sociological viewpoints on the nature of artistic creation and other forms of cultural activity. It analyzes what the terms 'high culture' and 'popular culture' may mean, and the stakes that are involved in their use in different social contexts.

SOCL 423: Ethnography of Pakistan (3 credits)

Prerequisites: SOCL 100, SOCL 223

Ethnography of Pakistan deals with the many cultures and societies of Pakistan. It is a reading intensive course, with readings selected to cover different geographic regions of Pakistan, different topics, and methodologies. All have been written after Partition (1947) but only a few of the writings concern that event. Indeed, Partition may not mean the same thing to everyone within the nation-state. A few readings touch on current issues of political unrest, but considered together, they provide a rich and diverse account of life in Pakistan.

SOCL 425: Sociology of Work (3 credits)

Prerequisites: SOCL 100

Explores how social groups exercise control over the work environment and make sense of their work experience; relates work to central issues of personal identity and social standing, distribution of social power and organization of work. It examines concepts of efficiency, performance, productivity and quality of work from a Sociological lens.

SOCL 430: Sociology of Consumption (3 credits)

Prerequisites: SOCL 100

Examines consumption and consumer behavior, trends in consumption by emphasizing sociocultural aspects of consumption, goods, meaning of signs, political economy of consumption, welfare and well-being and social stratification of consumption. It considers development of consumer behavior by focusing on production, marketing, distribution, sale and appropriation of goods and products having various social identities.

SOCL 435: The Sociology of Aging and Policy Choices (3 credits)

Examines biological, psychological and sociological theories and the changes that occur with aging in Pakistan and other cultures and discusses culture's influence on aging policies and practices in Pakistan and elsewhere; it presents an introduction to social policy generally and then works with students to analyze the situation of older people in Pakistan and Pakistan's policies that relate to aging and/or affect older people. Students gain the ability to identify possible solutions at an individual and policy level and experience field research with older persons.

SOCL 445: Sociology of South Asia (3 credits)

Prerequisites: SOCL 100

An exploration of the cultures and societies in South Asia is essential to develop a broad understanding of the social problems in Pakistan. This required a worldview that focuses on the commonalities in cultural heritage, colonial legacy, interaction with global and transnational forces and their influence on the social, political, social and economic structures and processes in contemporary South Asian societies. The worldview must also appreciate and look to learn from differences, with the overall aim of proposing culture- specific solutions

to region-specific problems. The aim of this course is to facilitate students to understand their society from the south Asian perspectives. In other words, students will learn to make sense of the local social trends and social problems from a comparative regional perspective.

SOCL 450: Health and Health Systems (3 credits)

Prerequisites: SOCL 100

The course will help students to: (i) understand health, illness, biomedicine and care management, and (ii) assess health systems in developed and developing countries in a comparative manner. The aim is to deconstruct medical knowledge as we know it and elaborate on how socio-economic factors such as gender, race and income impact health and inequalities in access to health. Main topics that will be covered include the sociological approaches to health and illness, social factors and illness, the meaning and experience of illness and health care systems and the connection between social structures and health outcomes.

SOCL 455: Sociology of Religion (3 credits)

Prerequisites: SOCL 100

Explores the development of religion in human history; presents the functions of religions for the individual, the religious group and for the society as a whole; organizational structures of religious groups; role of religion in social cohesion and social conflict; fundamentalism in all religions; the social organization of rituals and religious practices and the phenomenology of religious experiences.

SOCL 465: Sociology of Sport and Leisure (3 credits)

Prerequisites: SOCL 100

Presents cultural and economic relations in sport and leisure including outdoor recreation, spectator sport, informal play, tourism and other entertainment activities from contemporary and historical perspectives with emphasis on the dynamics of power and identity.

SOCL 494: Senior Sociology Seminar on Writing a Research Proposal (3 credits)

Prerequisites: SOCL 100, SOCL 201, SOCL 301

Students review critical elements of Sociological theory and select an independent research topic along with a theoretical framework for exploring their issue or question. They conduct a literature review, and then develop a plan for a proposed research project that includes its methodology, plan for analyzing the data, ethical issues to consider, permissions as needed, and a budget. Work will be done under the supervision of a faculty member(s) in Sociology in a seminar setting with other students developing their own research projects.

SOCL X95: Themes (Special Seminar in Sociology) (1-3 credits)

Prerequisites: SOCL 100, SOCL 201, SOCL 301

Provides an opportunity to explore some current issues of Sociology in some depth and integrates learning from other courses in relation to the special topic.

Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

SOCL 498: Internship (3 credits)

Prerequisites: Sociology majors and minors who have taken SOCL 100,

An internship in an organization which utilizes the skills learned in the study of Sociology. Students must develop learning goals for their internship and write a report describing their experience and what they have learned in sociological terms.

SOCL 499: Capstone Project (3 credits)

Prerequisites: SOCL 201,301

The senior capstone seminar provides sociology majors an opportunity to integrate and apply sociological theories, concepts, and analytical skills developed throughout the program. The course focuses on the sociological analysis of contemporary social issues using scholarly literature, policy documents, and secondary data. Students will complete an applied sociological project that presents evidence-based analysis and recommendations

CRIM 110: Criminal Law & Justice (3 credits)

This is a survey course designed to introduce the students to the criminal law and the justice system of Pakistan. Beginning with the incidence and nature of crime in Pakistan and the theories regarding the causes of crime, throughout the duration of this semester the three institutions that constitute the criminal justice system namely police, courts and corrections, shall be examined. The course shall include a description of the historical foundations, functions, constitution, powers (especially issues regarding discretion), and procedures and processes followed within each institution. The course will also explore the foundations, implementation and problems (substantive, procedural or administrative), related to criminal law and the criminal procedure code. Issues like discretion, constitutional issues etc. that arise when these institutes come in contact with other state/non state actors or the society at large in the routine execution of their functions and duties shall also be discussed. Upon successful completion of this course, it would be expected that the students will be familiar with the rudiments of Pakistan's criminal law and criminal justice system.

CRIM 118: Private Security (3 credits)

The course will introduce students to the private security industry in Pakistan and other countries. A detailed overview of principal duties and responsibilities necessary to work effectively in the private security organizations will be given. The course will also address issues such as policies of private security organizations, legal and ethical issues involved, and standards for security evaluation.

CRIM 201: Criminological Research (3 credits)

The objective of this course is to familiarize the students with basic research design, research methodology and data analysis in the field of criminal justice. The content of this course has been organized to render a sequential description of the various processes involved in social science research. Beginning with the formulation of research questions and their relationship to criminological theory, over the course of the semester the students will develop a working knowledge of the various research methodologies used in criminal justice research and their respective strengths and weaknesses, along with the various ethical concerns annexed to them. The final portion of this course has been designed to equip students with basic level data analysis skills by using SPSS (Statistical Package for the Social Sciences).

CRIM 220/SOCL 220 *: Criminological Theory (3 credits)

The course will introduce students to influential theories in the field of criminology. In addition to advanced study of criminology theories, various theories of crime causation such as physiological, psychological,

economic, and social perspectives will be critically appraised. Tracing the historical roots of the theories of criminology that emanated from different fields in social sciences, the course will also explore integrative theories and theoretical models that emerged as Criminology developed into an applied, multidisciplinary and problem- focused discipline. Other theories related to control and prevention of crime will also be discussed. Critical and postcolonial criminology perspectives in criminology will also be discussed. Students taking this course will be able to understand criminology theory couched within the historical development of the discipline and will be able to critically analyze major criminological theoretical perspectives.

CRIM 225: Juvenile Delinquency (3 credits)

The course shall provide the students with conceptual understanding of Juvenile Delinquency, and its causes. The course will provide theoretical insights in comprehending the phenomenon of juvenile delinquency. We will examine how society defines juvenile delinquency, theories that identify factors that push or pull minors towards delinquency, and how societies formally and informally deal with this moral and administrative paradox (including efforts aimed at prevention and rehabilitation). The course will specifically focus on prevention and control over Juvenile Delinquency through the Juvenile Justice System.

CRIM 230: Mass Media and Crime (3 credits)

The course will critically explore the emergent and ever evolving relationship between crime and mass media. Students will understand the theoretical foundations of this relationship and review research that identifies local and cross-cultural trends. The course will traverse through theoretical perspectives that explain how crime and the criminal justice system are represented in the media and the forces that influence their representation. The course will also explore how the representation of crime and the criminal justice system influence public opinion and the social construction of crime, criminality and criminal justice, and in turn criminal justice and administrative systems. The role of the media in constructing a global understanding of crime, criminality and punishment will also be critically analyzed.

Specific focus will be made on the countries in the Global South and especially Pakistan.

CRIM 310: Penology and Prison Administration (3 credits)

This course has been developed to enable the student to understand and critically evaluate the Pakistani penal system after developing an insight into the science of penology and the comparative penal systems. The course will introduce students to the various theories and perspectives that explain punishment and its role in societies. The main focus of the course will be on the prisons, both as a social institution and as a society of captives. Students will also be introduced to models of prison management, administration and reform, and will analyze case studies from different countries that highlight specific issues. The course will survey the legal and institutional framework of prisons in Pakistan and introduce students to the scant scholarly literature that analyzes the present conditions of, and issues related to prisons in Pakistan.

CRIM 360: Organized Crime and Transnational Criminal Networks (3 credits)

This course explores the triad of criminal organizations, transnational crime and global terrorism. We will examine and make theoretical sense of transnational crime in its various forms and organized criminal organization and explore its symbiotic relationship with terrorism. The social, political, and economic impact of organized crime, especially in weak democracies, will also be analyzed. We will specifically focus on the organized crime prevalent in Pakistan and the Southeast Asia region to understand how transnational criminal organizations strengthen terrorist organizations by sharing their cross jurisdictional smuggling routes and money laundering networks. The course will also look at local and international law enforcement agencies countering the threat of transnational criminal organizations.

CRIM 460: Comparative Criminal Justice Systems (3 credits)

In this course the students will develop a theory based comparative understanding of criminal justice organizations and processes. The main purpose of this course is to apply this knowledge in the context of Pakistan to better understand the Pakistani criminal justice system. The comparative approach is used to help students develop a) a theoretical foundation for understanding how the criminal justice systems operate and b) a knowledge base that would allow them to critically analyze and compare the different criminal justice systems.

CRIM 463: Radicalization and De-radicalization (3 credits)

The course will introduce students to the topic of radicalization and its importance in contemporary social science research. The students will analyze exploratory research on various radical groups throughout the world and the theories of radicalization generated from such research.

*** Cross listed Courses:**

Courses with two designators (coding) are marked with a * to identify them as cross-listed courses. Students must select the correct designator for their applicable program.



Department of **STATISTICS**

Bachelor of Science in Statistics

Introduction

Since its inception in 1974 the Department of Statistics has been providing quality statistics education, and now with a renewed vision to provide the best statistics and Data Science education with high quality statistical services across all disciplines, and ensuring international recognition; It aims to devise and develop statistical, computational and analytical techniques, and to disseminate statistical knowledge through teaching, advising, and outreach programs, by making our students responsible citizens and responsible users of statistics in order to serve the needs of the broader community. Our programs not only offer mathematical foundation of Statistics and Data Analytics but also thorough application of data analysis tools to real-life problems, this prepares students to take part in data revolution.

At BS level the Department of Statistics offers various programs:

1. BS Statistics
2. BS Data Analytics
3. Minor in Statistics
4. Minor in Data Analytics
5. Joint Minor in Mathematics and Statistics

Program Objectives:

- i. To teach students fundamental concepts of practical and applied statistics to improve statistical literacy.
- ii. To prepare students who can reflect critically on real-world problems to find solutions through interdisciplinary approaches by using statistical methods and Software.
- iii. To provide a market-oriented program that focuses on experiential learning by providing them with opportunities to gain research experience and to interact with industry professionals that prepare them to become ethical evidence-based decision makers.
- iv. To prepare our students to grow professionally and pursue higher education in statistics nationally and internationally.

Program Learning Outcome:

- i. **Disseminate knowledge in Statistics:**
Students will be able to classify data and identify appropriate statistical methods to summarize the facts.
- ii. **Critical thinkers and informed learners:**
Students will be able to Demonstrate understanding fundamental statistical theories and apply these concepts to draw valid conclusion of data
- iii. **Data Analytics:**
Students will be able to Analyze and interpret real world data problems in multidisciplinary fields by effectively using statistical software
- iv. **Evidence based decision makers:**
Students will be able to Formulate a problem and communicate a solution clearly and professionally in written and oral form.
- v. **Career and further opportunities:**

Describe future career and education opportunities in the field of statistics.

Requirements for BS Statistics Major

Core Courses	Elective Courses	Total Credits	Required Major CGPA
11 courses (33 credits)	7 courses (21 credits)	54	2.0

Core Courses	Credits
STAT 101/ MATH 107/ DATA 101	3
STAT 102 / MATH 105/ DATA 102	3
STAT 201/DATA 201	3
STAT 301	3
STAT 302	3
STAT 303/DATA 303	3
STAT 304/ MATH 314	3
STAT 305	3
STAT 403	3
STAT 498	3
STAT 499	3
Total	33
Elective Courses (Any 7)	Credits
STAT 103	3
STAT 205	3
STAT 206/DATA 206	3
STAT 207/DATA 207	3
STAT 208/DATA 208	3
STAT212/ENVR 212/BIOL212/MBIO 212	3
STAT 310/DATA 310	3
STAT 311/ MATH 315	3
STAT 313/ MATH 304	3
STAT 314	3
STAT 315/DATA 315	3
STAT 317/DATA 317	3
STAT 400	3
STAT 405/DATA 405	3
STAT 406/DATA 406	3
STAT 407	3
STAT 408	3
STAT 414/DATA 414	3
Total	21 credits

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

A student can elect both for the Minor in Data Analytics and Minor in Statistics.

Requirements for the Minor in Statistics

Courses	Total Credits	Required MCGPA
6 courses	18 credits	2.0

Courses (Choose any 6 from the list)	Credits (18)
STAT 101/ MATH 107/DATA 101	3
STAT 102/MATH 105/DATA 102	3
STAT 103	3
STAT 201/DATA 201	3
STAT 205	3
STAT 208/ /DATA 208	3
STAT 301	3
STAT 302	3
STAT 303/DATA 303	3
STAT 304/ MATH 314	3
STAT 305	3
STAT 310/ DATA 310	3
STAT 311/MATH 315	3
STAT 314	3
STAT 315/DATA 315	3
STAT 400	3
STAT 403	3
STAT 405 /DATA 405	3
STAT 406/ STAT 406	3
STAT 407	3

Bachelor of Science in Data Analytics

Program Objectives:

The objectives of the Data Analytics Program are as follows:

- i. Provide Students with a strong understanding of statistical analysis and computational tools used in data analytics.
- ii. Equip students with the ability to manage and wrangle large datasets effectively.
- iii. Develop students' expertise in programming languages such as R, Python, SQL and others commonly used in the field.
- iv. Teach students how to visualize complex data and present findings clearly to both technical and non-technical stakeholders.
- v. Promote a strong foundation in statistics, mathematics, machine learning, predictive analytics, and artificial intelligence.
- vi. Prepare students to become entrepreneurs and equipping the for internship and job placements in the growing field of data analytics.

Program Learning Outcomes

After completing Major in Data Analytics, students should be able to:

- i. Demonstrate foundational knowledge in data analytics, statistics and mathematics.
- ii. Apply programming and analytical methods to solve real world data problems.
- iii. Integrate interdisciplinary perspectives in analyzing and interpreting data.
- iv. Communicate analytical findings effectively to technical and non-technical audiences.
- v. Evaluate ethical, social and cultural implications of data driven decisions.
- vi. Demonstrate independent learning and adaptability to new tools and technologies.

Requirements for BS Data Analytics Major

Core Courses	Elective Courses	Total Credits	Required MCGPA
13 (39 credits)	5 (15 credits)	54	2.0

Core Courses	Credits
DATA101/ STAT 101/ MATH 107	3
DATA102 /STAT 102/ MATH 105	3
DATA 104/ MATH 102	3
DATA 201/ STAT 201	3
DATA 202	3
DATA 206/STAT 206	3
DATA 207/STAT 207	3
DATA 300	3
DATA 301	3
DATA 317/STAT 317	3
DATA 414/STAT 414	3
DATA 498	3
DATA 499	3
Core courses Total	39
Elective Courses (Choose any 5 from the list)	Credits

DATA 103/COMP 102	4
DATA 203/MATH 103	3
DATA 204/MATH 201	3
DATA 205	3
DATA 208/STAT 208	3
DATA 209	3
DATA 302	3
DATA 303/ STAT 303	3
DATA 304	3
DATA 305	3
DATA 310/ STAT 310	3
DATA 315/ STAT 315	3
DATA 401	3
DATA 402	3
DATA 403	3
DATA 404	3
DATA 405/STAT 405	3
DATA 406/STAT 406	3
DATA 407	3
Electives courses Total	15

Total Degree Requirements

Requirement Category	Total Credits
General Education	48-51
Major Requirements	54
University Electives	19-22
Upper Level (300-400 Level) courses	12 courses
Required Total & Major CGPA	2.0 CGPA
Total Credits for Graduation	124

Requirements for the Minor in Data Analytics

A minor in Data Analytics is open to students of all disciplines.

Core Courses	Elective Courses	Total Credits	Required MCGPA
4 (12 credits)	2 (6 credits)	18	2.0

Core Courses	Credits
DATA 206/STAT 206	3
DATA 207/STAT 207	3
DATA 317/STAT 317	3
DATA 414/ STAT 414	3

Core courses Total	12
Elective Courses (Choose any 2 from the list)	Credits
DATA 101/ STAT 101/ MATH 107	3
DATA 102/ STAT 102/MATH 105	3
DATA 103/COMP 102	4
DATA 201	3
DATA 202	3
DATA 205	3
DATA 208/ /STAT 208	3
DATA 209	3
DATA 300	3
DATA 301	3
DATA 302	3
DATA 303	3
DATA 304	3
DATA 305	3
DATA 310/ STAT 310	3
DATA 315/ STAT 315	3
DATA 401	3
DATA 402	3
DATA 403	3
DATA 404	3
DATA 405/ STAT 405	3
DATA 406/ STAT 406	3
DATA 407	3
Electives courses Total	6

Requirements for Joint minor in Mathematics and Statistics

For a Joint Minor in Mathematics and Statistics a student needs to study 6 courses; this includes 2 Statistics core courses, 2 Mathematics core courses, and 2 elective courses by choosing one elective course each from the list of elective courses listed by the departments.

Core Courses	Elective Courses	Total Credits	Required MCGPA
4 (12 credits)	2 (6 credits)	18	2.0

Core Courses		Credits
Statistics	STAT 102/MATH 105/Data 102	3
	STAT201/DATA 201	3
Mathematics	MATH 102 or MATH 111	3
	MATH 103 or CSCS 202	3
Core courses Total		12

Any 2 courses from the following elective

	Elective Courses	Credits
Choose 1 from the following		3
Statistics	STAT 208/ /DATA 208	3
	STAT 301	3
	STAT 303 /DATA 303	3
	STAT 304 /MATH314	3
	STAT 311 /MATH 315	3
	STAT 313 /MATH 304	3
	STAT 401 / MATH 408	3
Choose 1 from the following		3
Mathematics	MATH 201	3
	MATH 202	3
	MATH 203	3
	MATH 209	3
	MATH 212	3
	MATH 303 or COMP 113	3
	MATH 304 / STAT 313	3
	MATH 310 or CSCS 320	3
	MATH 312	3
	MATH 314/ STAT 304	3
	MATH 315 / STAT 311	3
	MATH 408 /STAT 401	3
Electives courses Total		3+3 = 6

Course Descriptions

STAT 100: Statistics for Everyone (3 credits)

Basic definitions, types of variables, scales of measurement, sample and population, collection and presentation of data, measures of central tendency, measures of dispersion for ungrouped data, index numbers, correlation and scatter plot.

STAT 101/MATH 107/ DATA 101 *: Statistical Methods (3 credits)

Nature and scope of Statistics, scales of measurements, measure of central tendency and dispersion for grouped data, moments, skewness and kurtosis, fundamental rules of counting, basic probability, moments in probability context.

STAT 102/MATH 105/ DATA 102 *: Probability and Probability Distributions (3 credits)

Basic set theory, different approaches and laws of probability, conditional probability and independence, Bayes' rule, random variables, mathematical expectations, mean & variance of random variables, joint (bivariate) probability distributions and properties, applications of some well-known discrete and continuous distributions.

STAT 103: Quantitative Methods in Social Sciences (3 credits)

Organizing and summarizing data using IBM-SPSS, correlation and regression analysis, reliability testing of items, application of parametric tests in single and several populations such as t, and ANOVA, chi square test of association, introduction to non-parametric tests and their application.

STAT 201/ DATA 201: Statistical Inference (3 credits)

Prerequisite: STAT 101 or STAT102

Introduction to basic terminologies of statistical inference, population and sample; Introduction to sampling distributions for single mean, difference between two means, proportion and difference between two proportions, properties of sampling distributions; point and interval estimation for means and proportions; testing of hypotheses about means, and proportions.

STAT 206/ DATA 206: Fundamentals of Data Analytics (3 credits)

This course covers the following topics that includes Introduction to programming with R, Operators, random number generators, importing and exporting data files from various database software, creating objects, matrices, lists, data frames, function in R, Loops, apply family of functions, base Graphs in R, working with strings, parallel computing with R, using the R commander Graphical User Interface, creating dashboards with R Shiny and also developing web apps with R, creating and maintaining R Packages.

STAT 207/ DATA 207: Business Analytics (3 credits)

This course includes performing exploratory data analysis and studying trends between variables by scatter plots, correlation and by fitting simple linear trends with their application on statistics software. Microsoft Power Business Intelligence services (Power BI) will be taught for data warehousing, data preparation, data visualization and creating dashboards. Performing reliability analysis of system life, application of different forecasting and smoothing methods in time related data. Further topics include work measurements: time studies and work sampling, introduction to statistical process control: concept and terms of quality, rolled throughput yield, process capability analysis, process sigma level and basic Define, Measure, Analyze, Improve and Control (DMAIC), control charts.

STAT 208/ DATA 208: Survey Methodologies (3 credits)

Population and sample, concepts of target population, survey estimation and inference, sample size determination, sampling, sampling and non-sampling errors, sample design and survey types, survey interviewing, cognitive process in answering survey questionnaire, construction of survey instruments, benchmarking questionnaires, choosing appropriate research designs, non-response, non-response errors, survey weights, analysis of data from complex surveys, accessing, preparing and working with secondary data from social surveys, comparative and longitudinal surveys, pilot survey, preliminary and technical reports.

STAT 212/BIOL 212/ENVR 212/ MBIO 212: Research Methods and Statistics (3 Credits)

This course is open to students majoring in Statistics, in addition to students from the Faculty of Natural Sciences, subject to the prerequisite requirements.

Research and its types, overview of research study designs, planning and implementation, ethics in research, selection of research topics, hypothesis formation, effective literature review and citation using software tools, Descriptive and Inferential Statistical methods, data collection and analysis tools, hypothesis testing, ANOVA, design of experiments and its significance.

Training on software (e.g. SPSS, Mendeley, Qualtrics).

STAT 301: Sampling Techniques (3 credits)

Prerequisite: STAT 201

Techniques of statistical sampling in finite populations with applications in analysis of sample survey data, simple random sampling, systematic sampling, stratified sampling, and cluster sampling, ratio and regression methods of estimation in simple random sampling, implementation of designs for the analysis of survey data, with application of techniques using R programming language and other software packages.

STAT 302: Experimental Design (3 credits)

Prerequisite: STAT 201

Principles of design of experiments. Model and analysis of completely randomized design randomized complete block design, Latin square design, relative efficiency(s), Graeco Latin and cross-over design, underlying assumptions of ANOVA in fixed effect models, multiple comparison tests, parameter estimations, estimation of missing observations.

STAT 303/ Data 303: Regression Modeling for Discovery (3 credits)

Prerequisite: STAT 201/ DATA 201

This course offers an introduction into linear regression analysis and emphasizes data analysis by using statistical software, such as R and SPSS, and results interpretation. Course topics include fundamental context for measuring relationships between variables, by using modeling approach, this includes introduction to simple and multiple linear regression; parameter estimation and inference for linear regression model; model diagnostic; prediction; strategies for building regression models; linear regression model with categorical predictors, logistic regression models and regression for count data.

STAT 304/MATH 314 *: Distribution Theory (3 credits)

Prerequisite: STAT 102/MATH 105

Expectations of functions of random variables, probability function, probability generating function and moment generating function. Theory and application of important discrete and continuous distributions and their properties.

STAT 305: Statistical Quality Control (3 credits)

Prerequisite: STAT 101

Control charts for attributes and variables. Acceptance sampling plan, quality improvement procedures, Taguchi method of online or offline approach to quality improvement; signal- noise ratios using orthogonal arrays.

STAT 310/ DATA 310: Time Series Analysis (3 credits)

Prerequisite: STAT 206

Types of time series data, trends, seasonal and cyclical analysis of data, irregular series, short term forecasting, ARMA and ARIMA models, diagnostic checking, forecasts, Box- Jenkins approach, spectral analysis.

STAT 311 /MATH 315 *: Mathematical Statistics (3 credits)

Prerequisite: STAT 102/MATH 105

Transformations and distributions of functions of random variables, linear transformations, transformations of univariate and bivariate random variables, polar transformations, derivation of chi-square, t, F distributions and their properties, non-central chi, t, F distributions and their properties, order statistics; distributions of r th order statistics, its mean and variance, distributions of minimum and maximum order statistics, joint distribution of order statistics, sampling distribution of range, mid-range and median.

STAT 313 /MATH 304 *: Operations Research (3 credits)

Prerequisite: STAT 102/MATH 103

Introduction to operations research, graphical solution, simplex method, two phase method, M-method, sensitivity analysis, primal dual relationship, dual simplex method, transportation model, assignment models, transshipment models, network models, queuing theory.

STAT 314: Reliability Analysis (3 credits)

Prerequisite: STAT 102/MATH 105

Review of probability functions, basic reliability definitions, failure time distribution, exponential time-to-failure models, hazard rates, life testing, and reliability estimation of parameters. System reliability.

STAT 315/ DATA 315: Statistical Packages and Data Analysis (3 credits)

Prerequisite: STAT 201

Introduction to data analysis using software packages, applications of parametric and non- parametric tests, model fitting, probability distribution fitting, basic multivariate analysis of survey data.

STAT 317/ DATA 317: Data Visualization with R (3 credits)

Prerequisite: STAT 206

This course covers the fundamental concept of data visualization with R: data presentation and introduction to ggplot2 for data visualization. Graphical methods to present univariate and bivariate data, visual elements like charts, graphs, and interactive maps for categorical and quantitative variables. This includes visualizations for statistical models like heat maps for relationship, linear regression, logistic regression, survival plots; Information Visualization Types: multi-dimensional, grouping, mapping, and time dependent graphs. Further topics include Design Principal of data Visualization: customizing graphs by attractive color schemes, annotations, interactive graphs and best practices in visualization.

STAT 400: Acceptance Sampling (3 credits)

Prerequisite: STAT 305

Introduction to acceptance sampling plans, classification of sampling plans, probability and operating characteristic curves, probability functions, single, double, multiple sampling and sequential sampling by attribute.

STAT 403: Point Estimation (3 credits)

Prerequisite: STAT 304

Properties of estimators: unbiasedness, consistency, sufficiency, efficiency, completeness. Methods of estimation: moments, ML, LS, minimum chi-squares, Bayes method of estimation.

STAT 405/ DATA 405: Predictive Analytics (3 credits)

Prerequisite: STAT 206

Introduction to predictive analytics, key concepts, techniques, and methods. This course will cover methods and tools for data pre-processing for forecasting tasks, techniques for selecting well-suited models for analysis. The major topics include predictive modeling process, statistical concepts for predictive modeling, regression, time series and machine learning predictive models. Predictive analytics can be applied in industries ranging from financial services to healthcare.

STAT 406/ DATA 406: Multivariate Data Analysis (3 credits)

Prerequisite: STAT 201

Multivariate data, review of multiple regression analysis, PC analysis and factor analysis, canonical correlation, hotelling T procedures, MANOVA, discriminant analysis.

STAT 407: Estimation and Hypothesis Testing (3 credits)

Prerequisite: STAT 304

Interval estimation, Neyman-Pearson Lemma, power functions, uniformly most powerful test. Deriving tests of hypotheses for parameters in normal, exponential, gamma and uniform distributions.

STAT 408: Biostatistics (3 credits)

Prerequisite: STAT 102

Introduction, probability distributions of biological variables, probit and logit transformations, ANOVA in biostatistics, developing G test, R x C test of independence.

STAT 414/ DATA 414: Statistical Machine Learning (3 credits)

Prerequisite: STAT 206

This course includes application knowledge of linear, polynomial regression, logistic regression and linear discriminant analysis. A model validation technique, cross validation and resampling method bootstrap, model selection and regularization methods (Ridge and Lasso). This course also covers nonlinear models, splines and generalized additive models, tree-based methods, random forests and boosting, support vector machines, principal components and clustering methods.

STAT X95: Themes (1-3 credits) Sections:

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

STAT 498: Internship (3 credits)

The internship is offered to students in their junior or senior year.

This course provides students with supervised practical training in a professional organization, industry, or research centers / institutions. The internship bridges academic knowledge and real-world application by allowing students to apply statistical techniques in workplace settings. Students will complete six to eight weeks of structured professional engagement under joint supervision of a faculty member and an industry supervisor. Evaluation is based on professional performance, written report, and final presentation.

STAT 499: Capstone Project (3 credits)

The Capstone Project is offered to students in their senior year.

The Capstone Project is independent supervised research or applied statistics project designed to integrate knowledge and skills acquired throughout the degree program. Students will identify a research problem or applied data challenge, conduct a literature review, design methodology, analyze data using appropriate statistical techniques, and present findings in the form of a written report and oral defense. Emphasis is placed on critical thinking, ethical data use, reproducibility, and professional reporting.

DATA 103 / COMP 102 Fundamental of Programming (4 credits)

Prerequisite: None

Basic skills of problem solving and programming, problem analysis, algorithm design, program development and testing, structured design techniques, object-oriented thought process and basic tools.

DATA 104 / MATH 102 Calculus I (3 credits)

Prerequisite: MATH 101 or A-Level Mathematics or Intermediate with Mathematics Functions, graph of functions, translation, stretching and compressing graphs, limits, continuity and differentiability, differentiation and its basic rules, indeterminate forms, L'Hopital's rule, integration and its techniques, fundamental theorem of calculus, evaluating definite integral.

DATA 202 Python for Data Analytics (3 credits)

This course introduces students to the fundamentals of Python programming for data analytics. Students will learn core programming concepts, data wrangling, and visualization techniques. The course emphasizes the use of Python libraries (NumPy, pandas, matplotlib, seaborn) for real-world statistical analysis, exploratory data analysis (EDA), and reporting. Through labs and projects, students will complete the full data analysis pipeline — from raw data import to insight generation and visualization.

DATA 203 / MATH 103 Introductory Linear Algebra (3 credits)

Prerequisite: MATH 101 or A Level Mathematics or Intermediate with Mathematics

Introduction to system of linear equations, matrices and matrix operations, elementary matrices, Gaussian elimination, Gauss Jordan method for solving a system of linear equation determinants and their properties, vector spaces, subspaces, linear independence, bases and dimensions.

DATA 204 / MATH 201 Calculus II (3 credits)

Prerequisite: MATH 102 / Data 104

Applications of derivative: increasing and decreasing functions, maxima and minima of a function, concavity, inflection points of a function, Rolle's Theorem, the mean value theorem, Taylor's theorem, applications of integration: area and arc length, volume and surface of revolution, introduction to improper integrals, infinite series, power series, introduction to conic section.

DATA 205 Data ethics, privacy and society (3 credits)

This course introduces students to the ethical, legal, and societal dimensions of data collection, analysis, and governance in the digital age. It explores foundational ethical theories, privacy principles, data protection laws, and regulatory frameworks governing statistical practice, artificial intelligence, and data-driven decision-making. Students develop an understanding of how legal compliance, ethical reasoning, and social responsibility shape modern data ecosystems. The course emphasizes real-world case studies, legal precedents, regulatory compliance, and applied statistical ethics to prepare students for responsible professional practice in data science, statistics, law, and public policy.

DATA 209 Data Journalism (3 credits)

Prerequisite: MCOM 101 and DATA 101 / STAT101

This course introduces students to real-world examples of data-driven journalism across a variety of subject areas. It focuses on understanding different types of data, methods of data collection, data cleaning and preparation, and loading datasets from diverse sources. Students will learn to analyze numerical information through visualization techniques and transform findings into clear, informative, and engaging journalistic stories for print and digital platforms.

DATA 300 Statistical Methods for Data Mining (3 credits)

Prerequisite: DATA-206 /STAT-206

This course introduces the fundamentals of Data Mining, a field of computational statistics that converts data into knowledge by identifying patterns and relationships in large datasets. Students will gain hands-on experience using different software, including R, a widely used open-source software, to apply data mining techniques for real-world problem-solving.

DATA 301 Database Systems and SQL (3 credits)

Prerequisite: DATA-206 /STAT-206

The course Database Systems and SQL provide students with a comprehensive understanding of how data is organized, stored, and managed in modern information systems. It introduces fundamental concepts of database design, including relational models, normalization, and entity-relationship diagrams, while emphasizing practical skills in SQL for creating, querying, and manipulating data.

DATA 302 Bayesian Statistics for Data Analytics (3 credits)

Prerequisite: DATA-102/STAT-102

This course provides an introduction to Bayesian thinking and its applications in data analytics. It begins with fundamental probability concepts and distributions, then advances to Bayesian inference, prior and posterior analysis, and decision-making under uncertainty. In the later part of the course, students learn to apply Bayesian analysis using R, Python, and JASP to solve real-world data problems.

DATA 304 Data Analytics in Governance and Policy (3 credits)

Prerequisite: STAT 206 and CPPG 111

This course integrates data analytics with public policy analysis to equip students with the skills needed to address real-world governance challenges using quantitative evidence. It introduces foundational concepts in public policy, including the policy cycle, identification of policy problems, and the economic and social determinants of public issues. Students learn to apply cost-benefit analysis, cost-effectiveness analysis, and contingent valuation methods to evaluate policy alternatives. The course further develops competencies in data cleaning, visualization, regression, classification, machine learning, and causal inference methods for policy evaluation. Ethical, legal, and governance implications of data-driven decision systems—including algorithmic bias, transparency, accountability, and privacy regulations. Applications draw on policy practices and evaluation frameworks used by institutions. The course concludes by training students to translate analytical findings into actionable policy recommendations and professional policy briefs.

DATA 305 Spatial Analytics (3 credits)

Prerequisite: DATA-206 and GEOG-201

This course introduces concepts and methods of spatial data analysis using R. Topics include spatial computing, handling spatial data in R (structures, import, export), spatial relationships and data modelling, analysis of point patterns, geostatistics, and areal data and geo-visualization. Students will also explore trends and innovations in spatial applications across environmental, health, urban, and business domains. Emphasis is placed on using R packages to manage, analyze, and visualize spatial data. Emphasis is placed on analytical thinking, practical skills, and effective communication of spatial insights.

DATA 401 Big Data Analytics (3 credits)

Prerequisite: DATA-301

This course introduces students to the concepts, tools, and techniques of big data analytics. It focuses on the storage, management, processing, and analysis of large-scale data using distributed systems such as Hadoop and Spark. Students will apply machine learning and graph analytics methods to real-world big data problems and develop an end-to-end analytics solution through a project.

DATA 402 Statistics for Deep Learning (3 credits)

Prerequisite: DATA 202

A Statistics for Deep Learning course provides the essential probabilistic and statistical foundation required to understand, design, and train modern neural networks. It bridges the gap between theoretical statistics and practical deep learning application, covering how models handle uncertainty, learn from data distributions, and optimize performance.

DATA 403 Health Data Analytics (3 credits)

Prerequisite: DATA-206 /STAT-206

This course introduces students to analytical methods used to transform healthcare and public-health data into actionable insights. It integrates statistical reasoning, computational tools, and domain knowledge to support evidence-based decision-making in health systems. Students will learn how to acquire, clean, analyze, model, and visualize health data using modern analytics software. Emphasis is placed on ethical data handling, data quality assessment, exploratory and predictive analytics, and effective communication of findings to both technical and non-technical audiences.

DATA 404 Sports Analytics (3 credits)

Prerequisite: DATA-206 /STAT-206

This course introduces the principles and methods of sports analytics with an emphasis on statistical analysis and data-driven decision making. It focuses on the collection, management, visualization, and analysis of sports data using the R programming language. Through real-world sports applications, students learn to evaluate player and team performance, apply statistical models, and communicate analytical findings effectively.

DATA 407 Entrepreneurship and Data Analytics (3 credits)

Prerequisite: DATA-207

This course integrates entrepreneurial principles with data analytics to equip students to make data-driven business decisions. Students will develop skills to identify opportunities, validate business ideas, analyze markets, and communicate insights using analytical tools. Emphasis is placed on creativity, problem-solving, and ethical use of data in entrepreneurial contexts.

DATA X- 95 Themes

DATA 498 Internship (3 credits)

Prerequisite: Internship to Senior students (7 semester) with CGPA 2.50 or above

This course provides students with supervised practical training in a professional organization, industry, research center, or public sector institution. The internship bridges academic knowledge and real-world application by allowing students to apply data analytical techniques in workplace settings. Students will complete six to eight weeks of structured professional engagement under joint supervision of a faculty member and an industry supervisor. Evaluation is based on professional performance, written report, and final presentation.

DATA 499 Capstone Project (3 credits)

Prerequisite: Internship to Senior students (8 semester) with CGPA 2.50 or above

The Capstone Project is an independent supervised research / applied analytics project designed to integrate knowledge and skills acquired throughout the degree program. Students will identify a research problem and applied data challenge, conduct literature review, design methodology, analyze data using appropriate data analytical techniques, and present findings in the form of a written report and oral defense. Emphasis is placed on critical thinking, ethical data use, reproducibility, and professional reporting.

Cross listed Courses:

Courses with two designators (coding) are marked with a (*) to identify them as cross-listed courses. Students must select the correct designator for their applicable program to be counted towards the degree.



Department of **URDU**

Bachelor of Studies in Urdu

Introduction

The Department of Urdu is one of the oldest at Forman Christian College (A Chartered University) and has had renowned faculty such as Maulana Farzand Ali, Dr Agha Suhail and Prof Iqbal Ahmed Khan. Both teachers and students take an active interest in the fields of research, creative writing and literary criticism. The department has also produced some fine writers. The Department of Urdu offers a Bachelor of Studies degree and is part of the Faculty of Humanities.

Learning Objectives:

1. To create informed learners in the field of Urdu literature.
2. To encourage students to cherish the rich cultural heritage of the Urdu language in Pakistan and the subcontinent.
3. To enable students to critically analyze the available Urdu literature.
4. To enable students to create and critique different artifacts in the Urdu literature.
5. To expand the area of research and innovation in the field of Urdu literature.
6. To make students fluent and proficient in Urdu language.

Program Learning Outcomes

1. Understanding of Urdu language and literature

Students will demonstrate an understanding of Urdu literature from Pakistan and the subcontinent

2. Civilized Citizen

Students' learning of Urdu literature will help them become better and informed citizens

3. Knowledge of Urdu language and literature

Students will use their knowledge of Urdu literature to develop their aesthetic and literary abilities

4. Research and critical thinking

Students will exhibit their knowledge of research practices, values, and ethics

5. Fluency and proficiency in Urdu Language

Students will exhibit fluency and proficiency in Urdu language that they can use effectively in their academic work and everyday matters

Requirements for BS Urdu Major

Core Courses	Elective Courses	Total Credits	Required MCGPA
10 (30 Credits)	8 (24 credits)	54	2.0

Core Courses	Credits
URDU 103	3
URDU 104	3
URDU 201	3
URDU 208	3
URDU 308	3
URDU 309	3

URDU 404	3
URDU 405	3
URDU 498	3
URDU 499	3
Core courses Total	30
Elective Courses	Credits
Students may choose remaining 8 courses according to their personal interest and goals	3*8
Elective Courses Total	24

Requirements for the Minor in Urdu

Courses	Credits	Required MCGPA
Students can choose any 6 courses according to their interest and goals	3*6	
Courses Total	18	2.0

Course Descriptions

URDU 101: Communicative Urdu (3 credits)

Communication and its different means, brief introduction to Urdu language, some fundamentals of Urdu grammar, functional Urdu, creative writing and journalistic Urdu.

URDU 102: A Foundational Study of Iqbal (3 credits)

This course aims to develop a basic understanding of Allama Iqbal's poetic and philosophical contributions to Urdu literature. It enables students to interpret Iqbal's Urdu poetry, particularly in the light of *Bang.e.Dra* (بانگ درا), through the different Poetic concepts of Iqbal. Students will learn to connect Iqbal's ideas in poetic and intellectual perspective.

URDU 103: A Selection of Urdu Verse (3 credits)

Ghazal (Ghalib, Mir and Iqbal), Nazam (Nazeer Akbar Abadi, Akbar Allah Abadi, Majeed Amjad and Syed Zamir Jafri).

URDU 104: A Selection of Urdu Prose (3 credits)

Letters (Ghalib), essays (Sir Syed Ahmad Khan, Wazir Agha and Mushtaq Ahmad Yousfi), short story (Premchand), character sketch (M Abdul Haque), extract of travelog (Begum Akhtar Riaz-ud-Din).

URDU 105: An Introduction to Selected Genres (3 credits)

Poetic and prose genres of Urdu literature: ghazal, nazam, rubai, qata, haiku, dastan, novel, drama and character sketch.

URDU 106: A Study of Manto (3 credits)

This course describes life sketch of Manto, analytical study of selected Urdu short stories as well as character sketches of Manto. The course has been designed to give a brief overview of different phases of Manto's art and trends developed by that time.

URDU 201: A Brief History of Urdu Language (3 credits)

This course introduces students to the linguistic development, and socio-cultural background of the Urdu language. It traces Urdu's emergence from early Indo-Aryan roots to its modern form, exploring regional influences, major literary centers, and the role of Urdu in shaping South Asian identity. Students will engage with key texts, movements, and personalities that contributed to Urdu's growth as a language of expression, culture, and communication.

URDU 202: Classical Urdu Poetry (3 credits)

Introduction to classicism, study of classical ghazals (Mir Taki Mir, Khawaja Mir Dard, Haider Ali Atish, Momin, Asadullah Khan Ghalib), masnawi (Mir Hassan) and marsya (Mir Anees).

URDU 203: Introduction to Satire & Humor in Urdu Literature (3 credits)

Difference between satire and humor, a brief history and importance of satire and humor, prose (Patras Bukhari, Ibn-e-Insha, Mushtaq Ahmad Yousfi, Col Muhammad Khan), poetry (Akbar Illahabadi, Syed Muhammad Jaffri, Syed Zamir Jafri, Anwar Masood).

URDU 204: Urdu Grammar and Literary Terms (3 credits)

Ilm-ul-Bayan, Ilm-ul-Badih, Adabi Istalahat.

URDU 205: Pakistani Poetry (3 credits)

Pakistani poetry (Munir Niazi, Shahzad Ahmad and Ahmad Faraz), nazam (Munir Niazi, Anwar Masood, Parveen Shakir and Amjad Islam Amjad).

URDU 206: Pakistani Prose (3 credits)

Pakistani fiction and prose: novel by Abdullah Hussain and short stories by Ahmad Nadeem Qasmi, Mumtaz Mufti and Bano Qudsia.

URDU 207: Literary Journalism (3 credits)

Difference between journalistic and literary use of language, evolution of literary journalism in Urdu, leading literary journals (Tehzeeb-ul-Ikhlaq, Awadh Puch, Saqi, Adbi Dunya, Naqoosh and Fanoon).

URDU 208: Script Writing in Urdu (3 credits)

Documentary writing: program scripts, journalistic scripts, business scripts, drama and film scripts.

URDU 209: Urdu and Media (3 credits)

This course explores the mutual relationship between Urdu language, literature, and media. Students will study how Urdu has served as a tool of communication, creativity, and cultural identity across various media platforms including newspapers, radio, television, film, and digital media. It also examines how literary themes, styles, and expressions of Urdu have been adapted in journalistic writing, drama, advertising, and online discourse.

URDU 210: Urdu Translation (3 credits)

This course introduces students to the principles and practice of translation with a focus on Urdu. Students will explore fundamental concepts of translation, key challenges in translating between Urdu and other languages (primarily English), and strategies to overcome them. Through practical exercises, students will translate texts of literary writings from English to Urdu and Urdu to English.

URDU 212: Applied Study of Urdu Drama (3 credits)

This course describes students in the world of Urdu drama, exploring its definition, characteristics, and various types. Students will learn the art and craft of drama, including character development, plot structure, and dialogue writing. The course emphasizes close reading, critical analysis, and appreciation of performance elements, preparing students for further study in Urdu literature and media.

URDU 301: Modern Urdu Poetry (3 credits)

Modernity and Modernism, ghazal (Hasrat Mohani, Faiz Ahmad Faiz, Nasir Kazmi), nazam (Majeed Amjad, Faiz Ahmad Faiz), lyrics (Hafeez Jalandhry), an analysis of Urdu poetry in the 20th century in a nutshell.

URDU 304: Biographical Literature in Urdu (3 credits)

Evolution of biographical literature in Urdu, biography of Khawaja Altaf Hussain Haali.

URDU 305: Autobiographical Literature in Urdu (3 credits)

Evolution of autobiographical literature in Urdu, autobiographers: Abdul Majeed Salik, Rashid Ahmed Siddiqui, Ihsan Danish and Qudrat Ullah Shahab.

URDU 306: Travelogs in Urdu (3 credits)

Evolution of travelogs in Urdu, selected extracts from Mahmood Nizaami, Begum Akhtar Riazud-Din, Ibn-e-Insha and Mustansar Hussain Tarrar.

URDU 309: A Study of Classical Urdu Ghazal (3 credits)

This course explores the Classical Urdu ghazal as a cultural, social and intellectual artifact within the literary tradition. It explores how themes of love, mysticism, identity, ethics and social values were shaped by Urdu Ghazal. Students will analyze selective ghazals through cultural, comparative and critical lenses.

URDU 401: Study of Iqbal (3 credits)

Life sketch of Iqbal, Iqbal as a poet and selected Urdu ghazals and nazams.

URDU 402: A Study of Urdu Drama (3 credits)

Art and evolution of Urdu drama, selected extracts from Anarkali, Mirza Ghalib Bandar Road Per and Man Chalay Ka Soda.

URDU 403: Modern Literary Movements in Urdu (3 credits)

Literary movements, modernity and modernism, important movements of the 20th century, romanticism, progressive movement, symbolism, modernism.

URDU 404: Criticism (3 credits)

Basic principles and definition of criticism, oriental criticism, western criticism, practical criticism.

URDU 405: Principles of Literary Research (3 credits)

Importance of literary research, evolution of Urdu research up till Aab-e-Hayat by M. Hussain Azad, principles and resources of research, terminology and preparation of research paper.

URDU 406: A Study of Ghalib (3 credits)

This course offers an in-depth study of Mirza Asadullah Khan Ghalib (1797–1869), one of the greatest poets of Urdu and Persian literature. It focuses on his poetic philosophy, linguistic mastery, philosophical themes, and the socio-cultural environment of 19th-century Delhi. Students will explore Ghalib's ghazals, letters, and prose writings, examining his contribution to Urdu poetics, aesthetics, and modern sensibility. The course combines literary criticism, historical background and interpretative reading to help students appreciate the complexity and universality of Ghalib's thought

URDU X95: Themes (1-3 credits)

A-E of 1 credit

F-J of 2 credits

K-Z of 3 credits

URDU 498: Internship (3 credit)

Prerequisite Urdu 208

The students would be required to take up an internship with any local media house for script writing and news editing. It would be a standard internship that will typically be in the summer and after the 4th semester. It will be at least 9 weeks in duration.

URDU 499: Capstone Project (3 credits)

A research paper of 50-100 pages on any topic regarding Urdu language and literature.

A research paper of 50-100 pages on any topic regarding Urdu language and literature.